

Holonomic Rigidity for Entire Hahn Functions

A sharp order-unit criterion for dilations,
with uniform exclusion bounds,
and the exact boundary of nonlinear differential rigidity

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Abstract

Fix a nonzero set-sized ordered abelian group Γ and the complex Hahn field $K_\Gamma = \mathbb{C}((t^\Gamma))$. *Entire* means strong Hahn summability of the evaluation family at every point of this one field. We prove that every strongly entire D-finite power series is a polynomial, in one or in finitely many variables, and that a fixed differential operator already excludes every nonterminating formal solution on an explicit exterior valuation region, with a common degree bound.

For dilations there is an exact alternative. If $q \in K_\Gamma^\times$ has infinite multiplicative order, a nonpolynomial strongly entire solution of a nonzero linear q -difference equation with polynomial coefficients exists if and only if $v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ . A nonzero valuation is not enough: its integer multiples must be cofinal among all scales. The proof covers units whose residues are roots of unity, and coefficients and parameters with arbitrary well-ordered Hahn supports. A partial theta series proves sharpness, and its exact strong evaluation domain is computed for an arbitrary Hahn parameter. Quantitatively, the recurrence supplies an exclusion bound that is attained on its own boundary, and a coarsened exclusion bound valid for every noncofinal dilation. A mixed differential–dilation extension is proved for nonzero noncofinal dilation valuations.

The common mechanism is an elementary finite-step escape chain: bounded valuation costs in a coefficient recurrence force a subsequence of evaluated leading exponents that never increases. This contradicts strong summability without replacing an arbitrary-rank valuation by a real-valued norm. In an explicit surreal workspace with countable cofinality and no order unit, the series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is strongly entire yet satisfies no nonzero linear differential equation and no nonzero linear dilation equation for any nontorsion parameter of that workspace.

Over $k((t^\Gamma))$, k any field of characteristic zero, a third source shows that strongly entire solutions of nonzero first-order algebraic differential equations are polynomials, and two further sources that, when Γ has no order unit, every nonpolynomial strongly entire series is differentially transcendental, jointly with its dilates by parameters whose quotients are not roots of unity. A sixth source shows that the hypothesis is necessary, with a Chebyshev-descended theta series of minimal differential order three, and a seventh that no such series has an equation of order at most two, for any value group. Γ is *not* assumed divisible; divisibility enters only in the three threshold constructions of Section 1.4, each of which states it.

Status. An AI-assisted research report; priority and independent refereeing are not certified. Lean covers support and escape lemmas, exponential and partial-theta domains, polynomial-orbit valuations, constant-point avoidance, inward stability and torsion covariance. The full classifications and the nonlinear, coefficient-field, theta-descent and order-three-threshold theorem packages remain pending. The variable derivative is not the Berarducci–Mantova derivation on surreal coefficients.

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1 The question and the principal results

1.1 Why the question is scale sensitive

A familiar way to specify a special function is to give a finite linear differential equation. Another is to give a finite relation between its values at $z, qz, q^2z, \dots$. Over the complex numbers both descriptions produce nonpolynomial entire functions. Over a surreal Hahn workspace, “entire” asks for evaluation not only at ordinary complex constants but at every infinite and infinitesimal scale available in that workspace. In a Hahn field an infinite expression is legal only when its supports satisfy an order condition and a finiteness condition, and an equation can force either condition to fail at a sufficiently large argument. The question of this report is what a finite linear equation can accommodate under that stronger global requirement. Sections 14–17 ask the same of a finite *nonlinear* algebraic differential equation, and answer it in order one; Sections 20–22 answer it in every order when the value group has no order unit; and Sections 23–25 show that with an order unit the answer changes: an explicit nonpolynomial strongly entire series satisfies an algebraic differential equation of order three. Sections 26–27 show that three is the least such order: for every value group, no nonpolynomial strongly entire series satisfies an equation of order at most two.

The answer depends decisively on which operator is used, and for dilation equations it depends further on whether the valuation of the particular parameter reaches every scale of the value group.

1.2 Conventions

We work with a fixed ordered abelian group Γ and

$$K_\Gamma = \mathbb{C}((t^\Gamma)).$$

An element of K_Γ is a formal sum $a = \sum_\gamma a_\gamma t^\gamma$ with $a_\gamma \in \mathbb{C}$ whose support $\text{supp}(a) = \{\gamma : a_\gamma \neq 0\}$ is a well-ordered set. For $a \neq 0$ put

$$v(a) = \min \text{supp}(a), \quad \text{lc}(a) = a_{v(a)} \in \mathbb{C}^\times, \quad \text{res}(a) = \text{lc}(a) \text{ when } v(a) = 0,$$

and put $v(0) = +\infty$ only where convenient; this symbol is never subtracted. Products and sums are the usual Hahn operations, with

$$v(ab) = v(a) + v(b), \quad v(a + b) \geq \min\{v(a), v(b)\}, \tag{1.1}$$

with equality in the second law when the two valuations differ. The coefficient field $\mathbb{C}$ is trivially valued, so every nonzero ordinary integer and every factorial has valuation zero. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $|\lambda|$ means $\max(\lambda, -\lambda)$ in the ordered group.

Convention 1.1 (Standing hypothesis on the value group). Γ is a nonzero set-sized ordered abelian group. Divisibility is *not* part of this standing hypothesis. Every statement below that needs divisibility says so in its own statement; those statements are collected in Section 1.4.

Definition 1.2 (Strong summability, evaluation domain, entireness). A set-indexed family $(b_i)_{i \in I}$ in K_Γ is *strongly Hahn summable* if $\bigcup_i \text{supp}(b_i)$ is well ordered and each $\gamma \in \Gamma$ lies in the supports of only finitely many members. Its sum is then defined coefficientwise, by a finite sum in $\mathbb{C}$ at each exponent. For $f = \sum_{n \geq 0} a_n z^n \in K_\Gamma[[z]]$ let $\text{Dom}_{\text{str}}(f)$ be the set of $x \in K_\Gamma$ for which $(a_n x^n)_{n \geq 0}$ is strongly summable, and let

$$\mathcal{E}_\Gamma = \{f : \text{Dom}_{\text{str}}(f) = K_\Gamma\}.$$

In d variables the family to be tested is $(a_\alpha x^\alpha)_{\alpha \in \mathbb{N}^d}$.

The word *entire* always means Definition 1.2 relative to one fixed workspace. It does not mean that a complex coefficient function is entire, it does not mean convergence in the fine topology inherited from the full surreal class, and it is never a claim about all of $\mathbf{No}[i]$. The definition also concerns the family *before* cancellation between different indices: one cannot first cancel a nonsummable family and then declare it summable, and convergence of a numerical coefficient series does not license infinitely many contributions at one Hahn exponent.

Definition 1.3 (Order unit). A positive $\mu \in \Gamma$ is an *order unit* if for every $\gamma \in \Gamma$ there is $N \in \mathbb{N}$ with $|\gamma| \leq N\mu$; equivalently, if the positive integer multiples of μ are cofinal in Γ . Zero is not an order unit.

Definition 1.4 (The two operator classes). Let D_z be the formal derivative in z , so $D_z(a) = 0$ for every $a \in K_\Gamma$, and for $q \in K_\Gamma^\times$ let $\sigma_q f(z) = f(qz)$. A series $f \in K_\Gamma[[z]]$ is *D-finite over $K_\Gamma(z)$* if

$$\sum_{r=0}^R p_r(z) D_z^r f(z) = 0, \quad p_r \in K_\Gamma[z], \quad (p_0, \dots, p_R) \neq 0, \quad (1.2)$$

and *linearly q -difference finite* if

$$\sum_{\ell=0}^r p_\ell(z) f(q^\ell z) = 0, \quad p_\ell \in K_\Gamma[z], \quad (p_0, \dots, p_r) \neq 0. \quad (1.3)$$

These are formal identities in $K_\Gamma[[z]]$, not assertions about limits of partial sums. Rational coefficients give the same notions after denominators are cleared.

We say “linearly q -difference finite” rather than silently identifying several meanings of “ q -holonomic”. In the sequence literature q is often an indeterminate with coefficients in $\mathbb{Q}(q)$; here q is a *fixed* Hahn element, possibly with infinitely many terms. The differential terminology is the usual characteristic-zero one [1]. *Nontorsion* and “of infinite multiplicative order” are used interchangeably.

1.3 The principal results

Theorem A (Differential rigidity, with a uniform exterior obstruction). *Let Γ be as in Convention 1.1 and let $0 \neq L = \sum_{r,k} c_{r,k} z^k D_z^r$ be a nonzero linear differential operator with coefficients in $K_\Gamma[z]$. There are $B_L \in \Gamma$ with $B_L \geq 0$ and $N_L \in \mathbb{N}$, both determined by the finitely many coefficients of L , such that every formal solution of $Lf = 0$ that is strongly evaluable at even one $x \neq 0$ with $v(x) \leq -B_L$ has degree less than N_L . Consequently*

$$\{f \in \mathcal{E}_\Gamma : f \text{ is } D\text{-finite over } K_\Gamma(z)\} = K_\Gamma[z].$$

Theorem B (Sharp dilation criterion). *Let Γ be as in Convention 1.1 and let $q \in K_\Gamma^\times$ have infinite multiplicative order. The following are equivalent.*

- (i) *There is a nonpolynomial $f \in \mathcal{E}_\Gamma$ that is linearly q -difference finite over $K_\Gamma(z)$.*
- (ii) *$v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ .*

When (ii) fails, every fixed equation (1.3) has a uniform exterior obstruction and a uniform polynomial degree bound, exactly as in Theorem A. When (ii) holds, an explicit partial theta series with parameter q or q^{-1} is a nonpolynomial entire solution.

Theorem C (Several-variable rigidity). *Let Γ be as in Convention 1.1 and let $F \in K_\Gamma[[z_1, \dots, z_d]]$ be strongly Hahn summable at every point of K_Γ^d . If the mixed partial derivatives of F span a finite-dimensional vector space over $K_\Gamma(z_1, \dots, z_d)$, then F is a polynomial.*

The fourth principal theorem comes from the third source manuscript and is proved in Sections 14–15. It allows an arbitrary coefficient field of characteristic zero; see Convention 14.1, which also explains why that generality is not transferred to Theorems A–C.

Theorem D (First-order nonlinear rigidity). *Let Γ be as in Convention 1.1, let k be any field of characteristic zero, and let $\mathbb{K} = k((t^\Gamma))$, with strong evaluation and entireness defined as in Definition 1.2 with k in place of $\mathbb{C}$. For every nonzero $P \in \mathbb{K}[z, Y_0, Y_1]$, every strongly entire formal solution $f \in \mathbb{K}[[z]]$ of*

$$P(z, f, D_z f) = 0$$

is a polynomial. More strongly, if a nonpolynomial formal solution exists, there is $\delta_0 \in \Gamma_{>0}$, depending on the equation and on finitely many coefficients of that solution, such that $t^{-\delta} \notin \text{Dom}_{\text{str}}(f)$ for every $\delta \geq \delta_0$. Consequently, for every nonpolynomial strongly entire f , the series f and $D_z f$ are algebraically independent over $\mathbb{K}(z)$.

Theorem D has no separant, irreducibility or solved-form hypothesis on P . Its exclusion bound depends on the solution, and Theorem 15.10 shows that this cannot be avoided, in contrast with the operator-only bound of Theorem A. By itself it does not say that a nonpolynomial entire function is differentially transcendental in all orders: independence of f and $D_z f$ does not exclude a relation involving $D_z^2 f$. Theorem E below says this when Γ has no order unit, and Theorem H shows that the hypothesis cannot be dropped: with an order unit a nonpolynomial strongly entire series can satisfy a relation of order three. Theorem I below extends Theorem D to order two for every Γ , so that three is the least order; Question 19.1, once re-scoped to order exactly two, is now re-scoped to the classification of the series of minimal order three.

The last three principal theorems come from sources 11 and 12 and are proved in Sections 20–22, over the field $\mathbb{K} = k((t^\Gamma))$ of Convention 14.1 for an arbitrary field k of characteristic zero; $\mathcal{E}_\Gamma(k)$ is its set of strongly entire series, $\text{Frac } \mathcal{E}_\Gamma(k)$ its fraction field inside $\mathbb{K}((z))$, and $\mathcal{J}_{\lambda,j} f = (D_z^j f)(\lambda z)$ (Convention 20.1).

Theorem E (All-order rigidity without an order unit). *Let Γ be as in Convention 1.1, and suppose that Γ has no order unit.*

- (a) (Sources 11 and 12.) *Every nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ is differentially transcendental over $\mathbb{K}(z)$: the series $f, D_z f, D_z^2 f, \dots$ are algebraically independent over $\mathbb{K}(z)$. Equivalently, every strongly entire solution of a nonzero algebraic differential equation $P(z, f, D_z f, \dots, D_z^r f) = 0$ over $\mathbb{K}(z)$, of any finite order and with any residue corner, is a polynomial.*
- (b) (Source 12.) *Let $\Lambda \subseteq \mathbb{K}^\times$ be such that λ/λ' is not a root of unity for distinct $\lambda, \lambda' \in \Lambda$. For every nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ the family $\{\mathcal{J}_{\lambda,j} f : \lambda \in \Lambda, j \geq 0\}$ is algebraically independent over $\mathbb{K}(z)$. In particular, for every $q \in \mathbb{K}^\times$ of infinite multiplicative order, including $v(q) = 0$, the series $(D_z^j f)(q^i z)$, $i \in \mathbb{Z}, j \geq 0$, are jointly algebraically independent over $\mathbb{K}(z)$.*
- (c) (Source 11.) *The elements of $\text{Frac } \mathcal{E}_\Gamma(k)$ that are differentially algebraic over $\mathbb{K}(z)$ are exactly the elements of $\mathbb{K}(z)$.*

For the coefficient-value rank $\text{cvr}(f)$, the dimension of the rational span of the values of the nonzero Taylor coefficients (20.4), there is a test for a single series that needs neither entireness nor a hypothesis on Γ .

Theorem F (Coefficient-value-rank criterion; source 12). *Let Γ be as in Convention 1.1 and $f \in \mathbb{K}[[z]]$. If $\text{cvr}(f)$ is infinite, the family of Theorem E (b) is algebraically independent over $\mathbb{K}(z)$ for every Λ as there.*

Theorem G (Order-unit dichotomy for mixed jets; source 12). *For Γ as in Convention 1.1, the following are equivalent.*

- (i) Γ has an order unit.
- (ii) Some nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ has all its coefficients in a subfield $L \supseteq k$ of $\mathbb{K}$ with $\text{trdeg}_k L < \infty$.
- (iii) For some nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ and some $q \in \mathbb{K}^\times$ of infinite order, the family $\{(D_z^j f)(q^i z) : i \in \mathbb{Z}, j \geq 0\}$ is algebraically dependent over $\mathbb{K}(z)$.
- (iv) (Added by the merge of source 13.) Some nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ is differentially algebraic over $\mathbb{K}(z)$.

If μ is an order unit, the partial theta series Θ_p of (8.1) with $p = t^\mu$ is a witness for (ii) and (iii): its coefficients lie in $k(p)$, and $\Theta_p(z) = 1 + z\Theta_p(pz)$. The series $\mathcal{T}_p$ of Theorem H is a witness for (ii)–(iv), and for (iii) with every q of infinite order.

As proved by source 12, Theorem G concerned the *mixed* conclusion: its witness Θ_p satisfies a dilation equation, and conditions (i)–(iii) do not say that an order unit produces a nonpolynomial strongly entire *differentially* algebraic series. Clause (iv) says this; it is added by the merge from Theorem H below, and its proof is in Section 25.1. Theorem E answers Question 19.1, as it then stood, in every order for groups without an order unit, including the equations with vanishing residue corner, and its part (b) answers the same case of Question 19.4. Both sources prove part (a) by one argument, printed once: a finite relation, linearized along its solution, puts all Taylor coefficients into one finitely generated field (Theorem 21.2); such a field has a value group of finite rational rank, which lies in a proper convex subgroup when there is no order unit, while a nonpolynomial entire series needs coefficient values at cofinally many scales. Part (b) adds dilations through the classical Skolem–Mahler–Lech theorem. Source 11 also proves a coarsening descent for arbitrary coefficient fields (Theorem 21.11), corollaries for rational systems and for Painlevé I in $\text{Frac } \mathcal{E}_\Gamma(k)$, a joint-independence criterion, and, over $\mathbb{C}((t^{\Gamma_\infty}))$, a continuum-sized family of strongly entire series with jointly independent jets, so that the fraction field has differential transcendence degree $2^{\aleph_0}$ (Theorem 22.6). The series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ satisfies no algebraic differential equation of any order (Theorem 22.1).

The eighth principal theorem comes from source 13 and is proved in Sections 23–25, over $\mathbb{K} = k((t^\Gamma))$ for an arbitrary field k of characteristic zero (Convention 23.1). For $\mu > 0$ in Γ put $p = t^\mu$, let C_n be the monic Chebyshev polynomials, $C_0 = 2$, $C_1 = z$, $C_{n+1} = zC_n - C_{n-1}$, so that $C_n(u + u^{-1}) = u^n + u^{-n}$, and put

$$\mathcal{T}_p(z) = 1 + \sum_{n \geq 1} p^{n^2} C_n(z). \quad (1.4)$$

Theorem H (Exact boundary of differential algebraicity; source 13). *For Γ as in Convention 1.1 and every field k of characteristic zero, the following are equivalent.*

- (i) Γ has an order unit.
- (ii) Some nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ is differentially algebraic over $\mathbb{K}(z)$.

- (iii) *Some nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ satisfies a nonzero algebraic differential equation of order three over $\mathbb{K}(z)$, while $f, D_z f, D_z^2 f$ are algebraically independent over $\mathbb{K}(z)$ and $D_z^3 f$ has degree two over $\mathbb{K}(z)(f, D_z f, D_z^2 f)$.*

If μ is an order unit, $\mathcal{T}_p$ is a witness for (iii); it is not D -finite over $\mathbb{K}(z)$. For every $\mu > 0$ the strong domain of $\mathcal{T}_p$ is $\{0\} \cup \{x : v(x) \geq -N\mu \text{ for some } N \in \mathbb{N}\}$, so $\mathcal{T}_p \in \mathcal{E}_\Gamma(k)$ exactly when μ is an order unit.

Theorem H answers Question 19.1 as it stood after the merge of sources 11 and 12: with an order unit a nonpolynomial entire series can satisfy an algebraic differential equation of order at least two, a differentially algebraic element of $\text{Frac } \mathcal{E}_\Gamma(k)$ need not be rational, and the absence of an order unit is necessary for differential algebraic rigidity. With Theorem D, the least order of such an equation over all examples was then two or three, and the question was re-scoped to order two; Theorem I below (source 14) shows that it is three (Section 19). The equation of $\mathcal{T}_p$ is explicit (Theorem 24.1); its residue corner polynomial vanishes identically while its full corner polynomial is a nonzero constant (Proposition 24.2). The direction (ii) $\Rightarrow$ (i) is Theorem E (a), which source 13 proves again; so Theorem E (a) and (c) hold exactly when Γ has no order unit (Corollary 25.2). The series $\mathcal{T}_p$ is a Chebyshev descent of the bilateral theta series $\sum_{n \in \mathbb{Z}} p^{n^2} u^n$ along $z = u + u^{-1}$; the classical triple product gives all its zeros, and in the surreal specialization $p = \omega^{-1}$ none of them is an omnific integer (Corollary 25.3).

The ninth principal theorem comes from source 14 and is proved in Sections 26–27, again over $\mathbb{K} = k((t^\Gamma))$ for an arbitrary field k of characteristic zero (Convention 26.1).

Theorem I (Second-order rigidity and the order-three threshold; source 14). *Let Γ be as in Convention 1.1 and k any field of characteristic zero.*

- (a) *Every $f \in \mathcal{E}_\Gamma(k)$ that satisfies a nonzero algebraic differential equation $P(z, f, D_z f, D_z^2 f) = 0$ with $P \in \mathbb{K}(z)[Y_0, Y_1, Y_2]$ is a polynomial. Equivalently, for every nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ the series $f, D_z f$ and $D_z^2 f$ are algebraically independent over $\mathbb{K}(z)$.*
- (b) *If Γ has an order unit, the least order of a nonzero algebraic differential equation over $\mathbb{K}(z)$ satisfied by a nonpolynomial element of $\mathcal{E}_\Gamma(k)$ is exactly three. If Γ has no order unit, no nonpolynomial element of $\mathcal{E}_\Gamma(k)$ satisfies one of any order.*

Part (a) needs no divisibility, rank, algebraic closedness or order-unit hypothesis, and no condition on the residue corner. Its polynomiality conclusion contains those of Theorem D and of Theorem 16.1 in orders at most two, but not their explicit exclusion bounds, and it holds in particular for equations with $\chi_P = 0$, such as (16.1). The new ingredient is the lower bound three; the witness of order three and the no-order-unit direction in part (b) are Theorems H and E (a). The proof keeps the whole initial polynomial at the infinitely many scales where it is not a monomial, reduces the equation there to one fixed homogeneous polynomial over k , and shows with logarithmic Euler derivatives that both endpoints of every such initial polynomial lie in a finite set, while a nonpolynomial entire series has such scales with arbitrarily large endpoints (Section 26.3). At order three an extra integer, the gap next to an endpoint, survives (Theorem 27.2); for the equation of $\mathcal{T}_p$ it is always one, the initial polynomials of every nonpolynomial entire solution are eventually $bX^m(1 + aX)$, and the coefficient values of such a solution are eventually strictly convex (Theorems 27.4 and 27.5). Consequences include rational two-state systems and quotients with a polynomial denominator (Corollaries 26.10 and 26.11).

For the linear theorems, the hypotheses separate two obstructions. Differential equations have bounded valuation costs after passing to their coefficient recurrence. Dilation equations can have

costs increasing linearly with the index, but only along the single scale $v(q)$; a scale that is not an order unit cannot reach all of Γ , even if its multiples are unbounded within their own Archimedean component. Strong entireness detects that failure.

Beyond the first three, the report proves: a mixed differential–dilation rigidity theorem at every nonzero noncofinal dilation scale (Theorem 10.2); a recurrence exclusion bound that is *attained*, boundary included, by an explicit formal exponential (Corollary 3.5 and Proposition 5.3); a coarsened exclusion bound valid at every nonzero noncofinal dilation valuation (Theorem 7.4); and the exact strong evaluation domain of the partial theta series for an arbitrary Hahn parameter, including arbitrary tails (Theorem 8.1).

1.4 Where divisibility of Γ is used

The core arguments — the escape mechanism, the orbit valuation lemmas, the two conversions, the theta domain theorem, and all of Theorems A–C — use only Convention 1.1. Divisibility of Γ enters in exactly three threshold constructions, with the hypothesis stated where it is used.

- (1) The refined recurrence threshold B_{rec} of (3.4), whose defining maximum divides a valuation difference by the jump length j . Corollary 3.5 states it under divisibility; Corollary 3.4 is the divisibility-free predecessor, with a coarser threshold and the same qualitative conclusion.
- (2) The refined coarsened threshold $\overline{B}_{\text{rec}}$ of (7.4), for the same reason, in the quotient group. Theorem 7.4 states the divisibility-free form and the divisible refinement separately.
- (3) The worked sparse threshold in Example 13.1, an instance of (1).

The torsion eigenspace equivalence in Proposition 9.1 needs no divisibility: inward stability of strong evaluation replaces the root-extraction argument of the source manuscripts.

The nonlinear part, Sections 14–17, uses no divisibility of Γ anywhere: the degree cutoff (15.11) divides ordinary integers in $\mathbb{Q}$, and its exclusion bounds are chosen from finitely many strict inequalities with positive integer multipliers.

The coefficient-field part, Sections 20–22, uses no divisibility either. It works in $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$ only to measure rational rank, and the principal convex bound of Lemma 20.7 uses integer multiples only.

The theta-descent part uses no divisibility. The order-three-threshold part, Sections 26–27, states its theorems for Γ as in Convention 1.1; to find scales at which the initial polynomial is not a monomial it divides in the divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, after extending the field by Lemma 26.2, and transfers the polynomiality and coefficient conclusions back to Γ , in which the hull is cofinal (Lemmas 26.2 and 26.4).

No theorem in this report states a conclusion under Convention 1.1 while resting on a proof step that divides by an integer in Γ ; the divisions of the order-three-threshold part take place in $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$. The division-free predecessors in (1) and (2) are weaker in the constant only, not in the conclusion.

1.5 Position relative to the collection and the literature

Provenance. This report is built from seven manuscripts, numbered as they arrived in the collection. Sources 08 and 09 were written independently on 22 September 2026 and both prove the differential rigidity theorem and the order-unit dilation criterion (Theorems A and B); source 08 is the base, because it assumes only $\Gamma \neq 0$ while source 09 assumed divisibility, and every proof taken from 09 was re-read for a hidden division (Section 1.4). Together they supply Sections 1–13.

Source 10, delivered later the same day, supplies the nonlinear part, Sections 14–17 and Theorem D. The repository snapshot consulted by sources 08 and 09 is

a3124af79f66b8b9c196d76b4cbc5ac3938907c4,

and the one consulted by source 10 is

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0.

At the second pin this report consisted of sources 08 and 09 alone and its text was identical to the text into which source 10 was merged, so source 10’s references to “Question 15.1” and “Theorems A–C” of this report were accurate there; that question is now Question 19.1, re-scoped in Section 19. The choices the merge made (renamed symbols, results printed once, the placement of the nonlinear part after the linear theory) are recorded in Section 14.1.

Sources 11 and 12, delivered together in a later batch and written independently, are titled *Coarsening Descent and Nonlinear Differential Rigidity for Entire Hahn–Surcomplex Functions* and *Coefficient-Field Rigidity at Unbounded Surreal Scales*. Both answer the no-order-unit case of Question 19.1 by the same argument, and together they form the coefficient-field part, Sections 20–22, with Theorems E–G. Source 12 is the base of that part: it adds dilations, answers the no-order-unit case of Question 19.4, and gives the order-unit dichotomy. Source 11 adds coarsening descent, rationality in the fraction field, the rational-system and Painlevé corollaries, the joint-independence criterion and the continuum family. Both consulted the snapshot

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at which this report was exactly Sections 1–19 and the conclusion, so their references to Questions 19.1 and 19.4 and to the README were accurate. The coefficient-field part is placed after Section 19 so that the numbering of Sections 1–19, of Questions 19.1–19.6 and of Theorems A–D, which the shipped audit files cite, is unchanged; the new questions are Questions 19.7–19.9. The merge’s choices for this part are recorded in Section 20.1.

Source 13, *Theta Descent and the Exact Boundary of Differential Rigidity*, delivered in a later batch, answers the order-unit case of Question 19.1 by an explicit series of minimal differential order three, and so shows that the order unit is the exact boundary. It forms the theta-descent part, Sections 23–25, with Theorem H, and adds clause (iv) to Theorem G. It consulted the snapshot

4cdeaec43ff1692ed2ad9f5bdf761a41872975a5,

at which this report was exactly the five-source text, so its references to Question 19.1 and Theorem E were accurate. The new part is placed after Section 22, so that the numbering of Sections 1–22, of Questions 19.1–19.9 and of Theorems A–G is unchanged; its one new question is Question 19.10. The merge’s choices for this part, including two steps it writes out, are recorded in Section 23.1.

Source 14, *The Order-Three Threshold for Entire Hahn–Surcomplex Functions*, delivered in a later batch, answers the order-two question that source 13 left open: no nonpolynomial strongly entire series satisfies an algebraic differential equation of order at most two, for every value group. It forms the order-three-threshold part, Sections 26–27, with Theorem I. It consulted the snapshot

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at which this report’s text was still the five-source text and source 13 was known to it only as a manuscript; its description of this report was accurate there. The new part is placed after Section 25, so that the numbering of Sections 1–25, of Questions 19.1–19.10 and of Theorems A–H is unchanged; the conclusion moves to the end. Its new questions are Questions 19.11–19.18. The merge’s choices for this part are recorded in Section 26.1.

The companion entire-function report. The collection already contains a merged report on cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank [12]. It fixes the same class $\mathcal{E}_\Gamma$ over the same kind of workspace, classifies the ring structure of $\mathcal{E}_\Gamma$, and computes the exact strong evaluation domain after an enlargement of the value group. None of its ring-theoretic conclusions is used to prove anything here: the present proofs start from Hahn support arithmetic. Two points of contact matter.

First, we do not reprove the order-unit coarsening machinery. That report’s Lemma 9.1, labelled `ent:lem:coarsening`, states that if $h > 0$ is an order unit of Γ there is an order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ whose kernel is the convex subgroup $H = \{\gamma : n|\gamma| < h \text{ for all } n \geq 1\}$, so that $|a|_\pi = \exp(-\pi(v(a)))$ is a nontrivial real-valued non-Archimedean absolute value inducing the intrinsic valuation topology. That is exactly the sense in which sources 08 and 09 of this report use the phrase “coarsening”, and we cite it rather than repeat it; see Remark 2.6.

Second, and this is easy to get wrong: *the two order-unit dichotomies are not the same theorem* — the companion report’s order unit decides Hermite interpolation and the Bézout property of the ring $\mathcal{E}_\Gamma$ (its Theorem 9.3), whereas the order unit here decides which annihilating operators a nonpolynomial member of $\mathcal{E}_\Gamma$ can satisfy. Neither dichotomy implies the other, and the coincidence of the invariant is not a coincidence of statement. In particular the existence of nonpolynomial entire functions at countable cofinality, and the fact that $\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}\omega^j$ has countable cofinality and no order unit, are background from that report [12], not targets of this one.

The collection’s differential-equations report uses a derivation on surreal *scalars* [13]; here D_z kills every scalar and differentiates only the formal variable, and no theorem about that scalar derivation is restated as ordinary D-finite rigidity. The rank-one analytic report [14] already contains the quadratic-exponent example $\sum t^{n^2} z^n$; our partial theta witness is explicitly classical and is not counted as a new function.

External literature. Stanley’s D-finite/P-recursive correspondence is the standard algebraic starting point [1]. We make the coefficient conversion explicit because the valuations of the resulting recurrence, rather than ordinary coefficient growth, drive the proof. Ramis treats growth of ordinary complex entire solutions of linear q -difference equations and also studies simultaneous differential and difference constraints [2]; the arbitrary-rank strong summability condition here is a different global condition, and none of its order-unit conclusions is inferred from an ordinary complex growth estimate. Garoufalidis proves eventual quadratic quasi-polynomial degree behaviour for q -holonomic sequences in a rational-function and Puiseux setting [3]; that is an important antecedent to the idea that dilation equations constrain coefficient scales, we do not claim to originate it, and we do not strengthen its quadratic quasi-polynomial conclusion. Di Vizio’s ultrametric Maillet–Malgrange theorem treats Gevrey behaviour of formal solutions of nonlinear ultrametric q -difference equations, including the valuation-zero case with Diophantine conditions [4]; we make no nonlinear generalization of it. The partial theta series belongs to the classical partial-theta literature [5], and Higman’s finite-word theorem is a standard external ingredient in the positive-support monoid lemma [6].

Terminology needs care in two further directions. Hahn meromorphic functions in the analytic setting of Müller and Strohmaier involve a convergence framework different from the formal strong evaluation used here [7]. The Berarducci–Mantova derivation differentiates surreal coefficients themselves and is not D_z [8]. Work on operation-closed surreal subfields gives context for set-sized workspaces [9], and the Conway normal-form conventions used in Section 2.4 follow current surreal field work [10]; closure under a scalar derivation is assumed in no theorem below.

The inspections of sources 08 and 09 were targeted: repository documentation, the report catalogue, the entire-function report, and the primary sources named above. Neither was a line-by-line audit of every repository manuscript or an exhaustive bibliography search, and a repository content search for `holonomic` returning no matches is not proof of absence. Source 10’s repository and literature comparison, equally targeted, is recorded in Section 17.4, that of sources 11 and 12 in Section 22.6, that of source 13 in Section 25.5, and that of source 14 in Section 27.5. Section 19 and Sections 22.7, 25.6 and 27.6 record exactly what is and is not claimed.

2 Hahn support and what evaluation means

2.1 Support calculus

Lemma 2.1 (Hahn–Neumann support calculus and the positive monoid). *Let A, B be well-ordered subsets of an ordered abelian group.*

- (1) *$A + B$ is well ordered, and each γ has only finitely many representations $\gamma = a + b$ with $(a, b) \in A \times B$.*
- (2) *If S is a well-ordered subset of $\Gamma_{>0}$, the additive monoid*

$$M(S) = \{s_1 + \cdots + s_k : k \geq 0, s_i \in S\}$$

is well ordered, and each of its elements has only finitely many representations as a finite word in letters from S .

- (3) *Multiplication by a fixed Hahn element preserves strong summability, and multiplication by a nonzero one preserves and reflects it.*

Proof. An infinite sequence in a well-ordered set has an infinite nondecreasing subsequence. Given a strictly decreasing sequence $a_n + b_n$, pass first to a subsequence with a_n nondecreasing and then to one with b_n nondecreasing, a contradiction. If infinitely many distinct pairs sum to a fixed γ , the distinct first coordinates contain an increasing sequence, forcing a decreasing sequence of second coordinates. This is part (1).

For part (2), order finite words over S by subsequence embedding with coordinatewise increase. Higman’s theorem says that words over a well-quasi-ordered alphabet are well quasi ordered [6]. An embedding cannot decrease the sum, and increases it strictly unless the two words are identical, since inserted letters are positive and a strict coordinate increase is positive. A descending sequence of sums therefore gives a bad sequence of words, and so do infinitely many distinct words with one fixed sum. Both are impossible.

For part (3), let A be the union of the supports of a strongly summable family and B the support of the multiplier. The new supports lie in $A + B$. Part (1) leaves finitely many admissible pairs at a specified exponent, and for each first coordinate only finitely many family members contribute. For a nonzero multiplier apply the same to its inverse. $\square$

In part (2) words carry their order; a fixed finite multiset has only finitely many permutations, so the commutative monoid version follows. The argument does not assume that S is bounded below by an order unit, and remains valid at arbitrary valuation rank.

Lemma 2.2 (The nonincreasing leading-exponent obstruction). *A family of nonzero Hahn elements possessing an infinite subsequence whose leading exponents are nonincreasing is not strongly summable.*

Proof. If that sequence of leading exponents takes infinitely many distinct values it has an infinite strictly decreasing subsequence, so the union of the supports is not well ordered. Otherwise it is eventually constant, and one exponent lies in the supports of infinitely many members, so local finiteness fails. $\square$

The second alternative is essential and is what makes the exclusion boundary below a *closed* region. Merely excluding a descending support does not establish strong summability: the family $(1 \cdot z^n)_{n \geq 0}$ at $z = 1$ has support union $\{0\}$, yet every summand contributes at exponent zero.

Lemma 2.3 (Two sufficient certificates). *Let $(b_n)_{n \geq 0}$ be a family in K_Γ .*

- (a) *If $v(b_n)$ tends cofinally to $+\infty$, meaning that for each $\beta \in \Gamma$ all but finitely many nonzero b_n satisfy $v(b_n) > \beta$, then (b_n) is strongly summable.*
- (b) *If $\text{supp}(b_n) \subseteq h_n + M(S)$ for a well-ordered $S \subseteq \Gamma_{>0}$, where the set of values of h_n is well ordered and each value is taken by only finitely many indices, then (b_n) is strongly summable.*

Moreover, grouping a strongly summable family into finite blocks preserves strong summability and its sum.

Proof. For (a) fix β . Only finitely many supports meet $(-\infty, \beta]$, and a finite union of well-ordered subsets of a linear order is well ordered. A descending sequence in the full union would lie below its first member, hence inside such a finite union. The same finiteness bounds the multiplicity of an exponent. For (b) use parts (1) and (2) of Lemma 2.1: a fixed target exponent admits finitely many pairs $(h, m) \in \{h_n\} \times M(S)$ with $h + m$ equal to it, and finitely many indices carry each such h ; actual supports can only be smaller than the indicated sets. For the last assertion, each block support is contained in the union of the old supports and the blocks contributing at any exponent form a finite set, while coefficientwise associativity is finite. $\square$

Criterion (a) and criterion (b) are genuinely different tools and both are used below. Criterion (a) is the shorter route whenever the term valuations really do escape every bound. Criterion (b) is what survives at the boundary, where the term valuations increase strictly but remain trapped below a fixed element of Γ ; Theorem 8.1 is exactly such a case, and no leading-valuation argument alone can decide it.

Lemma 2.4 (Inward stability of strong evaluation). *Let $f = \sum_n a_n z^n \in K_\Gamma[[z]]$, and let $x, y \in K_\Gamma^\times$ with $v(y) \geq v(x)$. If $x \in \text{Dom}_{\text{str}}(f)$, then $y \in \text{Dom}_{\text{str}}(f)$. Thus membership of a nonzero argument in $\text{Dom}_{\text{str}}(f)$ depends only on its valuation, and entireness can be tested on monomial arguments.*

Proof. Put $b_n = a_n x^n$ and $u = y/x$, so (b_n) is strongly summable and $v(u) = \eta \geq 0$. Write $u = c t^\eta (1 + w)$, where $c \in \mathbb{C}^\times$ and w is zero or has positive valuation. Let S be $\text{supp}(w)$, with η adjoined when $\eta > 0$, and put $M = M(S)$. This is a well-ordered positive alphabet, so Lemma 2.1 makes M well ordered. Every finite power u^n has support in M .

Let $A = \bigcup_n \text{supp}(b_n)$. The supports of $a_n y^n = b_n u^n$ lie in $A + M$, a well-ordered set. For a fixed exponent γ there are only finitely many pairs $(\alpha, \mu) \in A \times M$ with $\alpha + \mu = \gamma$. For each such α , strong summability of (b_n) leaves only finitely many indices n . Hence γ occurs in only finitely many supports of $b_n u^n$, proving strong summability. Equal valuations allow the argument in both directions; compare an arbitrary nonzero y with $t^{v(y)}$ to obtain the monomial test. Evaluation at zero is always finite. $\square$

The statement fixes the full coefficient series a_n . It does not say that their leading valuations alone determine the domain: higher coefficient tails can still matter at a boundary scale.

2.2 Positive valuation is not cofinal decay

For $\beta > 0$ write

$$\Delta_\beta = \{\gamma \in \Gamma : |\gamma| \leq N\beta \text{ for some } N \in \mathbb{N}\}, \quad (2.1)$$

the smallest convex subgroup containing β . It is all of Γ exactly when β is an order unit. If it is proper, there is $\gamma > 0$ outside it, and then

$$\gamma > N\beta \quad \text{for every ordinary } N \in \mathbb{N}, \quad (2.2)$$

since a positive element bounded by some $N\beta$ would lie in Δ_β .

Remark 2.5 (Sufficiency is not necessity). Let $0 < \epsilon < H$ in Γ with $n\epsilon < H$ for every ordinary integer n ; for instance $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ ordered by the highest nonzero coordinate. The family $(t^{n\epsilon})_{n \geq 0}$ is strongly summable, since its support is an increasing sequence and each exponent occurs once, and its sum is the Hahn element $(1 - t^\epsilon)^{-1}$. Yet its valuations do not tend cofinally to $+\infty$. This is the most elementary warning against identifying positive valuation with topological nilpotence, and it is why criterion (b) of Lemma 2.3 is needed at all. No appeal to convergence of partial sums at the H scale is involved.

Remark 2.6 (The real-valued coarsening is imported, not reproved). When Γ does have an order unit h , the companion report [12, Lemma 9.1, `ent:lem:coarsening`] supplies an order-preserving $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ and kernel the convex subgroup $H = \{\gamma : n|\gamma| < h \text{ for all } n \geq 1\}$, hence a nontrivial real-valued non-Archimedean absolute value $|a|_\pi = \exp(-\pi(v(a)))$ inducing the intrinsic topology. We cite that lemma and do not reprove it. No proof in this report uses it: every argument below is carried out in Γ itself, which is what makes the arguments valid when no order unit exists. The coarsening is mentioned twice, as an optional rank-one picture of the order-unit case (Remark 8.3) and for the warning of Section 1.5 that the interpolation/Bézout dichotomy of that report and the annihilating-operator dichotomy of this one share an invariant but are different theorems.

2.3 The entire class and its closure properties

Polynomials are entire, because all the evaluation families are finite. An ordinary complex entire Taylor series need not be entire in the sense of Definition 1.2: at a monomial argument $x = t^\gamma$ the family

$$\left(\frac{x^n}{n!}\right)_{n \geq 0} \quad (2.3)$$

is strongly summable exactly when $\gamma > 0$, since $\gamma = 0$ puts infinitely many contributions at exponent zero and $\gamma < 0$ gives a descending sequence. Proposition 5.3 computes the exact domain of the corresponding family with an arbitrary Hahn argument.

Proposition 2.7 (Closure). $\mathcal{E}_\Gamma$ is a K_Γ -algebra, $\sigma_q(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$ for every $q \in K_\Gamma^\times$, and $D_z(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$.

Proof. For addition, the evaluation supports lie in the union of two well-ordered sets with finite exponent multiplicities. For multiplication, the doubly indexed product family is strongly summable by Lemma 2.1 (1) and (3), and grouping by ordinary total degree preserves strong summability by the last clause of Lemma 2.3. Dilation is evaluation at qx . For $x \neq 0$ the derivative family is

$$(n+1)a_{n+1}x^n = x^{-1} \cdot (n+1)a_{n+1}x^{n+1}.$$

Discarding one term and multiplying each term by a nonzero ordinary integer enlarges neither supports nor multiplicities, and multiplication by x^{-1} preserves summability by Lemma 2.1 (3). At $x = 0$ only the constant term contributes. Iterate. $\square$

Proposition 2.8 (Faithfulness of evaluation). *If $f \in \mathcal{E}_\Gamma$ vanishes at infinitely many distinct ordinary complex arguments, then $f = 0$ as a formal series. Consequently a linear equation with rational-function coefficients that holds pointwise wherever its coefficients are defined is also a formal equation in the sense of Definition 1.4.*

Proof. Strong evaluation at 1 says that the coefficient family (a_n) has a well-ordered union of supports and is locally finite at every exponent. Hence for each γ

$$P_\gamma(T) = \sum_n (a_n)_\gamma T^n \in \mathbb{C}[T]$$

is a polynomial, and its value at an ordinary complex c is the coefficient of t^γ in $f(c)$. If f vanishes at infinitely many such c , every P_γ has infinitely many roots and is zero, so every a_n is zero. For the last assertion, clear denominators; the left-hand side is in $\mathcal{E}_\Gamma$ by Proposition 2.7, and nonzero polynomial denominators exclude at most finitely many ordinary complex arguments. $\square$

The nonexistence proofs may therefore be run on formal coefficients without weakening their reading as functional obstructions, and no interchange of infinite limits is needed anywhere.

The theorems below use only the valuation laws (1.1), the existence of every monomial t^γ , triviality of the valuation on $\mathbb{C}^\times$, and the support facts of this section. They do not require algebraic closedness, spherical completeness, Weierstrass preparation, or an embedding of Γ into $\mathbb{R}$.

2.4 Embedding the workspace into the surcomplex numbers

If Γ is an additive ordered subgroup of $\mathbf{No}$, the Conway normal-form map

$$\sum_{\gamma \in S} c_\gamma t^\gamma \mapsto \sum_{\gamma \in S} c_\gamma \omega^{-\gamma} \tag{2.4}$$

identifies the real-coefficient Hahn field with a subfield of $\mathbf{No}$ and its complexification with a subfield of $\mathbf{No}[i]$; a well-ordered S becomes a reverse-well-ordered set of Conway exponents. This viewpoint and its set-sized support discipline are standard [9, 10]. Here t^γ is a formal monomial and $\omega^{-\gamma}$ a Conway monomial; the notation does not silently identify every formal power with a choice of surreal exponentiation.

The results may therefore be read twice. Abstractly they concern Hahn fields and need no surreal construction. Concretely they restrict the exact linear descriptions available for surcomplex-valued power series on specified surreal scales. Formula (2.4) is the only bridge used, and no global surcomplex exponential is invoked.

3 The finite-step escape mechanism

The following isolates the mechanism used throughout. It needs no formula for the recurrence coefficients, no asymptotic slope for the valuations, and no assumption that cancellations are rare.

Lemma 3.1 (Escape chain). *Suppose a sequence (a_n) in K_Γ satisfies, for every $n \geq N$,*

$$c_0(n)a_n + \sum_{j=1}^s c_j(n)a_{n+j} = 0, \quad c_0(n) \neq 0, \quad (3.1)$$

and suppose $a_{n_0} \neq 0$ for some $n_0 \geq N$. Then $s \geq 1$ and there is an infinite sequence $n_0 < n_1 < n_2 < \dots$ with $j_r := n_{r+1} - n_r \in \{1, \dots, s\}$, with $a_{n_r} \neq 0$ and $c_{j_r}(n_r) \neq 0$ for every r , and with

$$v(a_{n_{r+1}}) - v(a_{n_r}) \leq v(c_0(n_r)) - v(c_{j_r}(n_r)). \quad (3.2)$$

Proof. Let $n \geq N$ with $a_n \neq 0$. The leftmost term of (3.1) is nonzero, so $s \geq 1$ and at least one later term is nonzero. If every nonzero later term had valuation strictly larger than $v(c_0(n)a_n)$, the leading monomial of the leftmost term could not cancel, contradicting (1.1). So some $j \in \{1, \dots, s\}$ has $c_j(n)a_{n+j} \neq 0$ and

$$v(c_j(n)a_{n+j}) \leq v(c_0(n)a_n),$$

which is (3.2). Taking the least eligible j makes the recursion definite. The successor coefficient is nonzero and its index is again at least N , so the construction never stops. $\square$

Remark 3.2 (The direction of the inequality matters). Solving the recurrence for its *last* coefficient typically yields a *lower* bound on a valuation, which obstructs nothing. The comparison above is made from the first nonzero coefficient forward and produces a later coefficient whose valuation is controlled from above. Cancellation among the other terms can raise the valuation of their sum, but cannot make every term on the right have valuation strictly larger than the single nonzero term on the left. Zeros and arbitrarily large cancellations elsewhere therefore do not invalidate the chain; the chain records only that finite inequality, repeatedly.

Theorem 3.3 (Evaluation exclusion criterion). *In the situation of Lemma 3.1, let $x \in K_\Gamma^\times$ and suppose that for every $n \geq N$ and every $j \in \{1, \dots, s\}$ with $c_j(n) \neq 0$,*

$$v(c_0(n)) - v(c_j(n)) + jv(x) \leq 0. \quad (3.3)$$

If $a_{n_0} \neq 0$ for some $n_0 \geq N$, then $(a_n x^n)_{n \geq 0}$ is not strongly summable. Equivalently: if $(a_n x^n)_{n \geq 0}$ is strongly summable, then $a_n = 0$ for every $n \geq N$.

Proof. Build the chain of Lemma 3.1 from n_0 and add $n_{r+1}v(x)$ to each side of (3.2), using $n_{r+1} = n_r + j_r$:

$$v(a_{n_{r+1}} x^{n_{r+1}}) \leq v(a_{n_r} x^{n_r}) + v(c_0(n_r)) - v(c_{j_r}(n_r)) + j_r v(x) \leq v(a_{n_r} x^{n_r}).$$

The evaluated leading exponents along the chain are therefore nonincreasing, and Lemma 2.2 applies. $\square$

Note that (3.3) is a weak inequality. This is the difference between an open and a closed excluded region, and it is available only because Lemma 2.2 also rules out a constant leading exponent. Section 5.2 exhibits an equation whose excluded region is exactly the closed one, and Section 8.5 exhibits a coarsened comparison where the boundary genuinely must be left in.

Corollary 3.4 (Uniform bounded cost; no divisibility). *Suppose in (3.1) there is a single $B \in \Gamma$ with $v(c_0(n)) - v(c_j(n)) \leq B$ for all $n \geq N$ and all $j \geq 1$ with $c_j(n) \neq 0$. Put $B^+ = \max(B, 0) \in \Gamma$. Then for every $x \neq 0$ with $v(x) \leq -B^+$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. Such x exist: $x = t^{-B^+}$, and $x = t^{-\delta}$ for every $\delta \geq B^+$.*

Proof. Since $B^+ \geq 0$ we have $v(x) \leq 0$, so $jv(x) \leq v(x) \leq -B^+ \leq -B$ for $j \geq 1$. Hence $v(c_0(n)) - v(c_j(n)) + jv(x) \leq B - B = 0$, and Theorem 3.3 applies. Every ordered abelian group admits a maximum of finitely many elements and the monomial t^{-B^+} exists, so no cofinal cyclic subgroup is needed. $\square$

The absence of a cofinality requirement here is exactly why the mechanism proves rigidity in groups of very large rank. It is also why the refined bound below, which divides by j , is a refinement of the constant and never of the conclusion.

Corollary 3.5 (Refined periodic threshold; requires Γ divisible). *Assume in addition that Γ is divisible. Assume each slot of (3.1) is either identically zero for $n \geq N$ or nonzero for all $n \geq N$ with periodic valuation there of common period $m \geq 1$; enlarge N past any initial nonperiodic range first. Let $\alpha_{j,r}$ be the valuation of slot j on the residue class $r \bmod m$, the identically zero slots being omitted. Put*

$$B_{\text{rec}} = \max_{\substack{0 \leq r < m \\ 1 \leq j \leq s, c_j \neq 0}} \frac{\alpha_{0,r} - \alpha_{j,r}}{j} \in \Gamma. \quad (3.4)$$

If there is no active forward slot $j \geq 1$, set $B_{\text{rec}} = 0$; the recurrence then forces $a_n = 0$ for all $n \geq N$ directly. Then for every $x \neq 0$ with $v(x) \leq -B_{\text{rec}}$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. In particular every solution with $a_n \neq 0$ for some $n \geq N$ — so in particular every nonpolynomial solution — satisfies

$$\text{Dom}_{\text{str}}(f) \subseteq \{0\} \cup \{x \in K_{\Gamma}^{\times} : v(x) > -B_{\text{rec}}\}, \quad f = \sum_n a_n z^n. \quad (3.5)$$

Proof. The case with no forward slot was handled in the statement. Otherwise divisibility makes each quotient $(\alpha_{0,r} - \alpha_{j,r})/j$ an element of Γ , so B_{rec} is defined. If $v(x) \leq -B_{\text{rec}}$ then $jv(x) \leq -j(\alpha_{0,r} - \alpha_{j,r})/j = -(\alpha_{0,r} - \alpha_{j,r})$, because multiplying a weak inequality by the positive integer j preserves it. Hence (3.3) holds and Theorem 3.3 applies. $\square$

The inequality in (3.5) is strict on the right: at $v(x) = -B_{\text{rec}}$ the chain may keep a constant leading exponent rather than descend, which is precisely why both halves of Definition 1.2 are needed. The bound is an *exclusion* bound. It does not assert that every point on the other side lies in the domain.

4 Valuations on integer and geometric orbits

4.1 Ordinary integers cannot create unbounded valuation loss

Lemma 4.1 (Integer evaluation). *Let $0 \neq P(X) = \sum_k p_k X^k \in K_{\Gamma}[X]$ and put $\beta(P) = \min_{p_k \neq 0} v(p_k)$. For all but finitely many nonnegative ordinary integers n ,*

$$P(n) \neq 0 \quad \text{and} \quad v(P(n)) = \beta(P). \quad (4.1)$$

Proof. The polynomial $t^{-\beta(P)}P$ has coefficients of nonnegative valuation and at least one of valuation zero, so its coefficientwise reduction

$$\overline{P}(X) = \sum_{v(p_k)=\beta(P)} \text{lc}(p_k)X^k \in \mathbb{C}[X]$$

is nonzero and has finitely many roots. The ordinary integers have distinct images in $\mathbb{C}$. At every integer n with $\overline{P}(n) \neq 0$, the Hahn coefficient of $t^{-\beta(P)}P(n)$ at exponent zero is nonzero, and no smaller exponent occurs. $\square$

Residue characteristic zero does real work here. The proof does not apply to p -adic integer inputs, where positive integers can have arbitrarily large valuation. Conversely no Diophantine approximation over $\mathbb{C}$ is relevant to v : every nonzero complex number, however small in ordinary modulus, has valuation zero.

4.2 A geometric orbit of nonzero valuation

Lemma 4.2 (Eventually affine valuation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(X) = \sum_k p_k X^k \in K_\Gamma[X]$. There are an index k_P with $p_{k_P} \neq 0$, the value $\beta_P = v(p_{k_P})$, and $N_P \in \mathbb{N}$, such that*

$$v(P(q^n)) = \beta_P + k_P n \lambda \quad (n \geq N_P). \quad (4.2)$$

In particular $P(q^n)$ is eventually nonzero.

Proof. The nonzero terms have valuations $v(p_k) + kn\lambda$. For two distinct indices the difference $v(p_k) - v(p_l) + (k - l)n\lambda$ is strictly monotone in n , so its sign is eventually constant and it vanishes for at most one integer, whether or not its values are cofinal in any convex subgroup. Finitely many pairs give one common threshold beyond which every pairwise comparison is strict and fixed, hence a unique minimizing index k_P . Its term cannot cancel against terms of strictly larger valuation. $\square$

Remark 4.3 (A rank-one shortcut that fails). The minimizing index need not be the smallest when $v(q) > 0$. In $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, take $q = t^\epsilon$ and $P(X) = X + t^{-H}$: the constant term has smaller valuation than q^n for every n , however large. Conversely with $P(X) = 1 + t^{-H}X$ the nonconstant term is the minimum for every n . The same phenomenon in the other sign convention: in $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, $q = t^{e_0}$ and $P(X) = t^{e_1} + X$ give $v(P(q^n)) = ne_0$ for every $n \geq 0$, so the nonzero constant coefficient never becomes the least-valued term. The proof above uses the eventual ordering of finitely many affine functions, not the Archimedean assertion that $n\epsilon$ eventually exceeds H , and not the rank-one slogan that q^n eventually beats every coefficient ratio.

4.3 Valuation-zero dilations, including residues that are roots of unity

Distinct powers of a valuation-zero q all have the same valuation, so the previous lemma says nothing. Polynomial evaluation nevertheless has only a finite periodic valuation pattern.

Theorem 4.4 (Unit-orbit valuations). *Let $q \in K_\Gamma^\times$ have $v(q) = 0$ and infinite multiplicative order, and let $0 \neq P \in K_\Gamma[X]$. Then $v(P(q^n))$ is eventually finite and periodic. More precisely, with $\zeta = \text{res}(q) \in \mathbb{C}^\times$:*

- (a) if ζ is not a root of unity, $v(P(q^n))$ is eventually constant, equal to the minimum valuation of the coefficients of P ;
- (b) if ζ has exact order h , then $v(P(q^n))$ is eventually constant on each residue class modulo h , so its eventual period divides h .

Proof. For (a) write $P(X) = \sum_k p_k X^k$ and $\beta = \min_k v(p_k)$. The coefficient of $P(q^n)$ at exponent β is $\bar{P}(\zeta^n)$ with $\bar{P}(X) = \sum_{v(p_k)=\beta} \text{lc}(p_k) X^k$, a nonzero complex polynomial. The points ζ^n are distinct, so this vanishes only finitely often, and no exponent below β occurs.

For (b) write

$$q = \zeta(1 + u), \quad \eta = v(u) > 0, \quad d = \text{lc}(u).$$

Here $u = q/\zeta - 1$ has positive valuation because q/ζ has valuation zero and residue 1, and $u \neq 0$ because otherwise $q = \zeta$ would have finite order. For every positive ordinary integer n the finite binomial expansion gives

$$y_n := (1 + u)^n - 1, \quad v(y_n) = \eta, \quad \text{lc}(y_n) = nd, \quad (4.3)$$

the leading coefficient being nonzero in characteristic zero. There is no convergence assertion: for each fixed n only finitely many powers occur.

Fix $r \in \{0, \dots, h-1\}$. For $n \equiv r \pmod{h}$ we have $\zeta^n = \zeta^r$, hence $q^n = \zeta^r(1 + y_n)$. Expand the nonzero polynomial

$$P(\zeta^r(1 + Y)) = \sum_{k=0}^{\deg P} b_{r,k} Y^k \quad (4.4)$$

and set

$$\beta_r = \min_{b_{r,k} \neq 0} (v(b_{r,k}) + k\eta), \quad J_r = \{k : v(b_{r,k}) + k\eta = \beta_r\}. \quad (4.5)$$

Then $P(q^n) = \sum_k b_{r,k} y_n^k$, and its coefficient at exponent β_r is the value at n of

$$R_r(T) = \sum_{k \in J_r} \text{lc}(b_{r,k}) d^k T^k \in \mathbb{C}[T]. \quad (4.6)$$

This polynomial is nonzero: its terms have distinct degrees and all the indicated coefficients are nonzero. Outside its finitely many integer roots the coefficient at β_r survives, so $v(P(q^n)) = \beta_r$ eventually on that class. Finitely many classes give one finite exceptional range. $\square$

Remark 4.5 (A second normalization of the same proof). Sources 08 and 09 both prove Theorem 4.4(b), by two slightly different normalizations, and we record the other one because it is occasionally more convenient. Instead of factoring out the complex root of unity, put $q^h = 1 + u$ directly, with $\eta = v(u) > 0$ and $u \neq 0$ for the same reason. Write $n = r + hk$ and expand the polynomial $Q_r(Y) = P(q^r Y)$ as $Q_r(1 + V) = \sum_k b_{r,k} V^k$. The finite binomial expansion gives $V_k = (1 + u)^k - 1$ with $v(V_k) = \eta$ and $\text{lc}(V_k) = k \text{lc}(u)$, and $P(q^n) = Q_r(1 + V_k)$. The minimum construction (4.5) and the corresponding nonzero residue polynomial (4.6), now in the variable k rather than n , finish the argument. Neither version sums a binomial series to a noninteger exponent; every expansion is finite, and arbitrary supports in q and in the coefficients of P enter only through the leading valuation and leading coefficient of finitely many Hahn elements.

Example 4.6 (A genuine two-phase pattern). Let $q = -(1 + t^\eta)$ with $\eta > 0$. Then q is nontorsion, $\text{res}(q) = -1$ has order two, and for $P(X) = X - 1$ and $n \geq 1$

$$v(P(q^n)) = v(q^n - 1) = \begin{cases} 0, & n \text{ odd}, \\ \eta, & n \text{ even}. \end{cases}$$

For odd n the residue is -2 ; for even positive n the leading term is nt^η . The valuation sequence is periodic, not the dilation itself, and a claim of eventual *constancy* without splitting residue classes would be false. The escape argument uses the bounded valuation costs; it does not replace q by its residue.

Example 4.7 (Finite exceptional indices really occur). For $q = 1 + t^\eta$ and $P(X) = X - 1 - 7t^\eta$, the leading coefficient of $P(q^n)$ at exponent η is $n - 7$, so $v(P(q^n)) = \eta$ for positive $n \neq 7$ while $v(P(q^7)) = 2\eta$. With $P(X) = X - 1 - 3t^\eta$ the exceptional index is $n = 3$, where $P(q^3) = 3t^{2\eta} + t^{3\eta}$. The exceptions are harmless for the escape argument, since it needs only an eventual range, but they cannot be ignored in a statement asserted for all indices. This is why Lemma 4.1 and Theorem 4.4 both allow finitely many exceptions even when $\text{res}(q) = 1$.

For an ordinary complex q with $|q|_\mathbb{C} \neq 1$, the complex modulus does not alter these conclusions: as an element of the Hahn coefficient field, q still has valuation zero. The ordinary modulus and the scale valuation answer different questions.

5 Differential rigidity

5.1 From an operator to a forward recurrence

A formal series is D-finite exactly when its derivatives span a finite-dimensional space over $K_\Gamma(z)$. The D-finite/P-recursive equivalence is classical [1, Theorem 1.5]; the following records the exact finite conversion, because the valuations of the resulting recurrence, not ordinary coefficient growth, are what the proof consumes.

Lemma 5.1 (Differential coefficient recurrence). *Write*

$$L = \sum_{r=0}^R \sum_{k=0}^m c_{r,k} z^k D_z^r, \quad c_{r,k} \in K_\Gamma, \quad (5.1)$$

with at least one $c_{r,k} \neq 0$, and let $f = \sum_{n \geq 0} a_n z^n$. For $n \geq m$ the coefficient of z^n in Lf is

$$\sum_{s=-m}^R Q_s(n) a_{n+s}, \quad Q_s(X) = \sum_{r-k=s} c_{r,k} (X+s)_\Gamma, \quad (5.2)$$

where $(Y)_\Gamma = Y(Y-1) \cdots (Y-r+1)$ and $(Y)_0 = 1$. At least one Q_s is nonzero. Writing s_0 for the least active shift and $u = n + s_0$, every formal solution of $Lf = 0$ satisfies, for all sufficiently large u ,

$$\sum_{j=0}^s P_j(u) a_{u+j} = 0, \quad P_j(X) = Q_{s_0+j}(X - s_0), \quad P_0 \neq 0, \quad (5.3)$$

with $s = R - s_0$. If $s = 0$, every formal solution is a polynomial.

Proof. The operator $z^k D_z^r$ carries the term of index $n+r-k$ to degree n and multiplies it by $(n+s)_r$ with $s = r - k$, which gives (5.2). For a fixed shift s the distinct derivative orders r contribute polynomials of distinct degrees in X , so a nonempty group containing a nonzero coefficient cannot cancel identically and some Q_s is nonzero. Reindexing gives (5.3). If $s = 0$ the recurrence reads $P_0(u)a_u = 0$ with $P_0 \neq 0$, so by Lemma 4.1 all but finitely many a_u vanish. $\square$

Proposition 5.2 (Uniform obstruction for P-recursive coefficients). *Suppose (a_n) satisfies (5.3) for all $n \geq N_1$, with $P_j \in K_\Gamma[X]$ and $P_0 \neq 0$. Let $\beta_j = \beta(P_j)$ for the nonzero P_j and put*

$$B = \max(\{0\} \cup \{\beta_0 - \beta_j : j \geq 1, P_j \neq 0\}) \geq 0.$$

There is $N \geq N_1$, beyond the finitely many exceptional integers of Lemma 4.1 for the finitely many nonzero P_j , such that for every $x \neq 0$ with $v(x) \leq -B$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. If Γ is divisible one may replace B by the smaller threshold

$$B_{\text{rec}} = \max\left(\left\{\frac{\beta_0 - \beta_j}{j} : j \geq 1, P_j \neq 0\right\}\right), \quad (5.4)$$

which is (3.4) with period $m = 1$. If the displayed set is empty, take $B_{\text{rec}} = 0$; then every solution terminates beyond N without an evaluation hypothesis.

Proof. Discard the identically zero P_j . By Lemma 4.1 there is N beyond which $v(P_j(n)) = \beta_j$ for all the finitely many nonzero P_j simultaneously. Apply Corollary 3.4 with this B , or Corollary 3.5 with $m = 1$ when Γ is divisible. If no $j \geq 1$ remains, the recurrence forces the same vanishing directly. $\square$

These are certified exterior obstructions, not a computation of the exact evaluation domain of each solution. Individual solutions can have stricter restrictions or can terminate much earlier.

Proof of Theorem A. Lemma 5.1 converts L into (5.3). Take $B_L = B$ from Proposition 5.2 and take N_L to be an positive integer beyond the recurrence's starting index and beyond every nonnegative integer root of the finitely many residue polynomials $\overline{P_j}$. If a formal solution is strongly evaluable at one $x \neq 0$ with $v(x) \leq -B_L$, then $a_n = 0$ for all $n \geq N_L$, so its degree is less than N_L . An entire solution is in particular strongly evaluable at the available monomial t^{-B_L} , hence is a polynomial. Conversely a polynomial is entire and is annihilated by a sufficiently high derivative, which gives the displayed equality. $\square$

The closed region $v(x) \leq -B_L$, rather than an open one, comes from the weak inequality in Theorem 3.3, hence from Lemma 2.2. A polynomial solution of degree at least N_L would in any case contradict the recurrence at its last nonzero coefficient, since $P_0(n)$ no longer vanishes there.

Theorem A is stronger than merely excluding a globally extended exponential. The operator coefficients may themselves be arbitrary Hahn series carrying infinite surreal scales. There are nevertheless only finitely many of them, so their leading valuations give a finite obstruction that one additional exterior scale defeats.

5.2 An exact boundary, not only a nonexistence result

Proposition 5.3 (Formal exponential domain). *For $\eta \in \Gamma$ let*

$$E_\eta(z) = \sum_{n \geq 0} \frac{t^{n\eta}}{n!} z^n.$$

Then $D_z E_\eta = t^\eta E_\eta$ and

$$\text{Dom}_{\text{str}}(E_\eta) = \{0\} \cup \{x \in K_\Gamma^\times : v(x) > -\eta\}. \quad (5.5)$$

Proof. The differential identity is a formal coefficient comparison. Put $y = t^\eta x$ for $x \neq 0$, so the n th evaluation term is $y^n/n!$.

If $v(y) > 0$, write $y = ct^{v(y)}(1+u)$ with $c \in \mathbb{C}^\times$ and u zero or of positive valuation, and put $S = \text{supp}(u) \subseteq \Gamma_{>0}$. Since $(1+u)^n$ is a finite integer power, $\text{supp}(y^n) \subseteq h_n + M(S)$ with $h_n = nv(y)$, and (h_n) is strictly increasing, so its values are well ordered and each is taken once. Lemma 2.3 (b) makes the family strongly summable; the factor $1/n!$ is a nonzero complex scalar and changes no support.

If $v(y) < 0$ the leading exponents $nv(y)$ strictly decrease. If $v(y) = 0$, then $y^n/n!$ has the nonzero coefficient $\text{lc}(y)^n/n!$ at exponent zero for every n , so local finiteness fails. These cases exhaust $x \neq 0$, and $v(y) > 0$ is $v(x) > -\eta$. $\square$

The recurrence of E_η is $(n+1)a_{n+1} - t^\eta a_n = 0$, with one active shift $j = 1$, so $\alpha_0 = \eta$, $\alpha_1 = 0$, and (3.4) gives $B_{\text{rec}} = \eta$ with no division by an integer at all. Comparing with (5.5): the excluded region $v(x) \leq -\eta$ of Corollary 3.5 is here *exactly* the complement of the domain, boundary included. The general exclusion bound can therefore be attained, and the closed form of the boundary is not an artefact of a lossy estimate. This example is the reason for keeping Lemma 2.2 rather than the strictly descending version alone.

There is no conflict with any scalar exponential on a surreal field: E_η is an ordinary power series in z , not a total exponential evaluated by an independent construction. Taking $\eta = 0$ recovers the elementary observation (2.3).

5.3 Algebraic and system consequences

Corollary 5.4 (Algebraic entire series). *An element of $\mathcal{E}_\Gamma$ algebraic over $K_\Gamma(z)$ belongs to $K_\Gamma[z]$.*

Proof. Let $F = K_\Gamma(z, f)$ be the finite extension generated by the algebraic series. In characteristic zero its minimal polynomial is separable, and differentiating that polynomial expresses $D_z f$ inside F , because the derivative with respect to the algebraic variable is nonzero at f and hence invertible in F . So D_z preserves F , all derivatives of f lie in the finite-dimensional $K_\Gamma(z)$ -space F , and f is D-finite. Apply Theorem A. $\square$

This is not offered as a new algebraic rigidity theorem; it records that the linear result subsumes the algebraic case without any factorization theorem for $\mathcal{E}_\Gamma$.

Corollary 5.5 (Rational differential systems). *If a finite vector Y of strongly entire formal series satisfies $D_z Y = A(z)Y$ with $A \in M_r(K_\Gamma(z))$, every component of Y is a polynomial. The same holds for an inhomogeneous system with a rational forcing vector, provided all its solution components are strongly entire.*

Proof. The $K_\Gamma(z)$ -span of the components is stable under D_z , so the successive derivatives of each component span a finite-dimensional space. For $Y' = AY + b$ append the constant component 1 and use $\begin{pmatrix} A & b \\ 0 & 0 \end{pmatrix}$. $\square$

Proposition 5.6 (Rational inhomogeneous equations). *The polynomiality conclusions of this report remain valid when the right-hand side of a linear equation is a rational function of z , provided the corresponding homogeneous hypothesis on L or on q holds.*

Proof. Clear denominators in the coefficients and in the right-hand side. The result has a polynomial right-hand side, so the coefficient recurrence is homogeneous for all sufficiently large indices, and the escape statements were formulated for eventual recurrences. The left-hand operator is still nonzero, having been multiplied by a nonzero polynomial. $\square$

This applies, for instance, to the inhomogeneous first-order theta identity (8.2) when the parameter valuation is noncofinal: it is not necessary to homogenize it in order to see the obstruction.

Example 5.7 (A sparse recurrence does not escape the obstruction). The formal Airy-type equation $f'' - zf = 0$ has the recurrence

$$a_{n+3} = \frac{a_n}{(n+3)(n+2)} \quad (n \geq 0), \quad a_2 = 0.$$

With $a_0 = 1$ and $a_1 = 0$ there are infinitely many nonzero coefficients, at indices $3n$, all of valuation zero. At $x = t^{-\delta}$ their leading exponents are $-3n\delta$, strictly decreasing for every $\delta > 0$. Gaps between nonzero coefficients do not help: the escape chain follows only the surviving congruence class.

Remark 5.8 (What transcendence has actually been proved). Every nonpolynomial member of $\mathcal{E}_\Gamma$ is transcendental over $K_\Gamma(z)$ and satisfies no nonzero *linear* differential equation over $K_\Gamma(z)$, and no finite family $f, D_z f, \dots, D_z^R f$ is linearly dependent over $K_\Gamma(z)$. This does not imply differential transcendence in the nonlinear sense. A relation

$$P(z, f, D_z f, \dots, D_z^R f) = 0$$

involving products of f and its derivatives lies outside the present proofs unless it can separately be reduced to a finite linear system. The distinction is not cosmetic: the escape mechanism exploits a finite *linear* cancellation in which one nonzero coefficient forces a later nonzero coefficient, whereas nonlinear coefficient equations involve convolutions of many earlier coefficients and supply no such forward step without further work.

That further work is the nonlinear part of this report (Sections 14–17, from source 10). It does not repair the escape chain; it replaces the recurrence by a finite initial polynomial at one large scale. Every relation of order $R \leq 1$ is thereby excluded (Theorem D), and so is every relation of higher order whose residue corner polynomial is nonzero (Theorem 16.1). Relations of order at least two with a vanishing residue corner remain outside the proofs of Sections 14–17. When Γ has no order unit they are excluded in every order by a third, different argument, which bounds the field generated by the Taylor coefficients (Theorem E, sources 11 and 12). With an order unit they are not always excluded: the series $\mathcal{T}_p$ of Theorem H (source 13) is a nonpolynomial strongly entire solution of a relation of order three with vanishing residue corner. Relations of order two are excluded for every Γ by a fourth argument, through initial polynomials at infinitely many scales (Theorem I, source 14), so three is the least order.

6 Generic lines and several-variable rigidity

A single differential equation in several variables is not enough to force polynomiality: it could ignore most of the variables. The correct hypothesis is finite-dimensionality of the span of *all* mixed partial derivatives, which is what Theorem C assumes.

Lemma 6.1 (Avoiding countably many polynomial conditions). *For any countable family of nonzero polynomials in $K_\Gamma[z_1, \dots, z_d]$ there is $c \in \mathbb{C}^d$ at which none of them vanishes.*

Proof. For one polynomial H , take the least Hahn exponent among its finitely many nonzero coefficients; the coefficient at that exponent is a nonzero $\overline{H} \in \mathbb{C}[z_1, \dots, z_d]$, and $\overline{H}(c) \neq 0$ implies $H(c) \neq 0$. The coefficients of the countably many polynomials $\overline{H}$ form a countable subset of $\mathbb{C}$, and the field they generate over $\mathbb{Q}$ is countable. Choose $c_1, \dots, c_d$ algebraically independent over that field, successively outside the algebraic closure of a countable field, which remains countable while $\mathbb{C}$ is uncountable. $\square$

The same countable-field device is used in the companion report held in `docs/surcomplex/entire-functions-at-arbitrary-rank/` (`ent:mv:thm:exceptional`) to show that the set of polynomial directions of a nonpolynomial series is not all of projective space.

Proof of Theorem C. Suppose F is not a polynomial and write it by total degree,

$$F(z) = \sum_{n \geq 0} F_n(z), \quad F_n \text{ homogeneous of degree } n, \quad (6.1)$$

so infinitely many F_n are nonzero. Let $g_1, \dots, g_\sigma$ be a basis of the finite-dimensional $K_\Gamma(z_1, \dots, z_d)$ -span of all mixed derivatives of F , taken to be mixed derivatives themselves with $g_1 = F$. The span is stable under differentiation, so there are rational matrices A_i with

$$\partial_i g = A_i(z)g \quad (1 \leq i \leq d). \quad (6.2)$$

These matrices have finitely many nonzero denominator polynomials; choose one nonzero homogeneous component of each. Apply Lemma 6.1 to every nonzero F_n , to those finitely many chosen components, and to the coordinate polynomials z_i , obtaining $c \in \mathbb{C}^d$ with $F_n(c) \neq 0$ whenever $F_n \neq 0$ and with no denominator becoming the zero polynomial under $z = wc$. Then

$$h(w) = F(wc) = \sum_{n \geq 0} F_n(c)w^n$$

is nonpolynomial.

For each $x \in K_\Gamma$ the family $(a_\alpha c^\alpha x^{|\alpha|})_{\alpha \in \mathbb{N}^d}$ is strongly summable, by entireness of F at $(c_1 x, \dots, c_d x)$. Group it into the finite blocks of fixed total degree; by the last clause of Lemma 2.3, $\sum_n F_n(c)x^n$ is strongly summable, so $h \in \mathcal{E}_\Gamma$.

The formal chain rule, applied after clearing denominators in (6.2), gives

$$\frac{d}{dw} g(wc) = \left(\sum_{i=1}^d c_i A_i(wc) \right) g(wc). \quad (6.3)$$

Every denominator here is a nonzero element of $K_\Gamma[w]$. It may vanish at $w = 0$, which is harmless: read the identity in the Laurent series field $K_\Gamma((w))$, with matrices in $M_\sigma(K_\Gamma(w))$. The finite $K_\Gamma(w)$ -span of the components of $g(wc)$ is preserved by d/dw , so its first component h is D-finite over $K_\Gamma(w)$, and Theorem A forces h to be a polynomial. This contradicts its construction. $\square$

Corollary 6.2 (Multivariate algebraic rigidity). *If a strongly entire $F \in K_\Gamma[[z_1, \dots, z_d]]$ is algebraic over $K_\Gamma(z_1, \dots, z_d)$, it is a polynomial.*

Proof. Each partial derivation extends to the finite separable extension generated by F , by differentiation of its minimal polynomial, so all mixed partial derivatives lie in one finite-dimensional space. Apply Theorem C. $\square$

The generic line is an existence device, not a promised effective choice of ordinary complex constants. The proof does not say that one prescribed line detects every nonpolynomial multivariate series; the line is chosen only after the countably many nonzero homogeneous parts and the finitely many rational connection matrices are known.

7 Dilation equations and the necessity of a cofinal scale

7.1 Exact coefficient conversion

Lemma 7.1 (Dilation coefficient recurrence). *Let $q \in K_\Gamma^\times$ have infinite multiplicative order and let $f = \sum a_n z^n$ satisfy (1.3). Write $p_\ell(z) = \sum_k p_{\ell,k} z^k$ and $Q_k(X) = \sum_\ell p_{\ell,k} X^\ell$, let d and e be the largest and smallest k with $Q_k \neq 0$, and put $s = d - e$. Then for every $n \geq 0$*

$$\sum_{h=0}^s R_h(q^n) a_{n+h} = 0, \quad R_h(X) = Q_{d-h}(q^h X), \quad (7.1)$$

with $R_0 = Q_d \neq 0$ and $R_s = Q_e(q^s \cdot) \neq 0$. Each nonzero R_h satisfies $R_h(q^n) \neq 0$ for all but finitely many n . If $s = 0$, every formal solution is a polynomial.

Proof. The coefficient of z^{n+d} in $\sum_\ell p_\ell(z) f(q^\ell z)$ is $\sum_{k=e}^d Q_k(q^{n+d-k}) a_{n+d-k}$; substituting $h = d - k$ and using $Q_{d-h}(q^{n+h}) = Q_{d-h}(q^h \cdot q^n)$ gives (7.1). Since q has infinite order the points q^n are distinct, and a nonzero univariate polynomial vanishes at only finitely many of them; this already gives eventual nonvanishing, independently of the finer valuation results. If $s = 0$ the relation reads $R_0(q^n) a_n = 0$ for every n , so all but finitely many a_n vanish. $\square$

The powers q^h in (7.1) matter: dropping them changes the recurrence and can change its valuation constants. Equivalently, taking $m = \max_\ell \deg p_\ell$ and reading off the coefficient of z^{n+m} gives $\sum_j P_j(q^n) a_{n+j} = 0$ with $P_j(X) = \sum_\ell p_{\ell, m-j} q^{\ell j} X^\ell$ and $P_0 = Q_m \neq 0$; both conversions are checked on finite exact examples by the programs of Section 18. If $m = 0$ the equation already reads $P_0(q^n) a_n = 0$ and all solutions are polynomials with no entireness assumption at all, so we may assume $s \geq 1$ below.

7.2 The valuation-zero case

Corollary 7.2 (Every unit dilation is globally rigid). *Let $q \in K_\Gamma^\times$ have $v(q) = 0$ and infinite multiplicative order. Every linearly q -difference finite member of $\mathcal{E}_\Gamma$ is a polynomial, and each fixed equation (1.3) has a uniform exterior obstruction and a uniform degree bound. If Γ is in addition divisible, every nonpolynomial formal solution has its domain bounded as in (3.5) by the constant (3.4) formed from the eventual residue-class valuations of (7.1).*

Proof. Apply Theorem 4.4 simultaneously to the finitely many nonzero R_h of Lemma 7.1. Their valuations take only finitely many values after a common starting index, with common period one or the exact order h_0 of $\text{res}(q)$. All differences $v(R_0(q^n)) - v(R_h(q^n))$ are therefore bounded above by a single element of Γ , and Corollary 3.4 gives the exterior obstruction and, via the residue polynomials, the degree bound. When Γ is divisible, Corollary 3.5 applies with $m = 1$ if the residue has infinite order, and $m = h_0$ otherwise. In both cases take the starting index beyond the exceptional range. $\square$

This includes parameters arbitrarily close to a root of unity without being one. The finite exceptional cancellations of Theorem 4.4 can move the starting index but do not remove the bound.

7.3 Nonzero but noncofinal valuation

Proposition 7.3 (Noncofinal dilation scales force polynomiality). *Let $q \in K_\Gamma^\times$ with $\beta = v(q) \neq 0$ and suppose $|\beta|$ is not an order unit of Γ . Every linearly q -difference finite member of $\mathcal{E}_\Gamma$ is a polynomial, with a uniform exterior obstruction and a uniform degree bound for each fixed equation. Divisibility of Γ is not used.*

Proof. By Lemma 4.2 applied to the finitely many nonzero R_h of (7.1), there is a common index beyond which

$$v(R_h(q^n)) = \alpha_h + k_h n \beta, \quad k_h \in \mathbb{N}.$$

Put $V = |\beta| > 0$, choose $B \in \Gamma$ with $B \geq 0$ and $B \geq |\alpha_0 - \alpha_h|$ for all relevant h , and put $M = \max_h |k_0 - k_h| \in \mathbb{N}$. Then for $h \geq 1$

$$v(R_0(q^n)) - v(R_h(q^n)) = \alpha_0 - \alpha_h + (k_0 - k_h)n\beta \leq B + MnV. \quad (7.2)$$

Because $|\beta|$ is not an order unit, (2.2) supplies $\gamma > 0$ with $\gamma > NV$ for every $N \in \mathbb{N}$; in particular $MnV < \gamma$ for all n , trivially so if $M = 0$. Hence the right side of (7.2) is bounded above by the single element $B + \gamma$. Corollary 3.4 with this bound excludes every $x \neq 0$ with $v(x) \leq -(B + \gamma)$, and the explicit witness $x = t^{-(B+\gamma)}$ is available. Since an entire solution is evaluable there, it satisfies $a_n = 0$ beyond the common starting index, hence is a polynomial of degree less than that index. Nothing in this argument divides an element of Γ by an integer. $\square$

The proof covers two different higher-rank situations. The group may have no order unit at all, so every nonzero dilation valuation is noncofinal; alternatively the group may have an order unit while the chosen q lives on a smaller scale. Global q -difference finiteness depends on the valuation of that particular dilation, not only on the group. The argument also requires no asymptotic formula for $v(a_n)$ and no eventually quadratic coefficient valuation: a single infinite escape chain suffices, which is why cancellations and sparse zero coefficients do not force a separate Skolem–Mahler–Lech analysis.

7.4 An intrinsic coarsened exclusion bound

The previous proof produces more than one bad point. Let $\beta = v(q) \neq 0$, let $\Delta = \Delta_{|\beta|}$ be the convex subgroup (2.1), and assume $\Delta \neq \Gamma$. Let $\pi : \Gamma \rightarrow \Gamma/\Delta$ be the ordered quotient and put $w = \pi \circ v$. With the eventual constants α_h of the previous proof, put

$$\overline{B}_{\text{rec}}^\flat = \max(\{0\} \cup \{\pi(\alpha_0) - \pi(\alpha_h) : 1 \leq h \leq s, R_h \neq 0\}), \quad (7.3)$$

and, when Γ is divisible so that Γ/Δ is a divisible ordered group,

$$\overline{B}_{\text{rec}} = \max_{1 \leq h \leq s, R_h \neq 0} \frac{\pi(\alpha_0) - \pi(\alpha_h)}{h}. \quad (7.4)$$

Theorem 7.4 (Coarsened exclusion). *Let $f = \sum a_n z^n$ be a nonpolynomial formal solution of (1.3) with $v(q) \neq 0$ and $\Delta = \Delta_{|v(q)|}$ proper.*

- (a) *Under Convention 1.1 alone, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}^p$ lies outside $\text{Dom}_{\text{str}}(f)$.*
- (b) *If Γ is in addition divisible, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}$ lies outside $\text{Dom}_{\text{str}}(f)$.*

Proof. Coarsening by Δ kills $n\beta$, since $n\beta \in \Delta$ for every n , so the projected valuations of the recurrence coefficients are the constants $\pi(\alpha_h)$. In case (b), $-w(x) > \overline{B}_{\text{rec}}$ gives $hw(x) < -(\pi(\alpha_0) - \pi(\alpha_h))$ for every relevant $h \geq 1$. In case (a) we have $w(x) < -\overline{B}_{\text{rec}}^p \leq 0$, hence $hw(x) \leq w(x) < -\overline{B}_{\text{rec}}^p \leq -(\pi(\alpha_0) - \pi(\alpha_h))$. Either way the projected leading valuation along an escape chain strictly decreases at every step. A strict decrease in the ordered quotient is a strict decrease in Γ , so the chain produces an infinite strictly descending sequence in the union of the supports. $\square$

The boundary is deliberately *not* excluded here. Equal leading values in the quotient can correspond to a strictly increasing sequence in Γ , and hence to a strongly summable family; Section 8.5 exhibits exactly that. Unlike Corollary 3.5, a coarsened comparison alone cannot decide local finiteness at an exponent of Γ .

8 Partial theta series: sharp witness and exact domain

8.1 The classical identities

For $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$ set

$$\Theta_p(z) = \sum_{n \geq 0} p^{\binom{n}{2}} z^n = \sum_{n \geq 0} p^{n(n-1)/2} z^n. \quad (8.1)$$

This is a standard partial theta series up to a harmless sign convention [5]. None of its coefficients vanishes, so it is not a polynomial. The coefficient identity $p^{n(n-1)/2} = p^{n-1} p^{(n-1)(n-2)/2}$ for $n \geq 1$ gives

$$\Theta_p(z) = 1 + z\Theta_p(pz), \quad (8.2)$$

and applying $\sigma_p - 1$ to $(1 - z\sigma_p)\Theta_p = 1$, with $\sigma_p z = pz\sigma_p$, yields the homogeneous equation

$$\Theta_p(z) - (1 + z)\Theta_p(pz) + pz\Theta_p(p^2z) = 0, \quad (8.3)$$

which is also obtained by substituting pz for z in (8.2) and subtracting the two inhomogeneous identities. The existence of these identities is not the new part. What matters is precisely where the series is strongly evaluable, for an arbitrary-rank value group and an arbitrary Hahn parameter.

8.2 Exact domain with arbitrary coefficient tails

Theorem 8.1 (Exact partial theta domain). *Let Γ be as in Convention 1.1 and let $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$, and let Δ_β be as in (2.1). Then*

$$\begin{aligned} \text{Dom}_{\text{str}}(\Theta_p) &= \{0\} \cup \{x \in K_\Gamma^\times : v(x) \geq -N\beta \text{ for some } N \in \mathbb{N}\} \\ &= \{0\} \cup \{x \in K_\Gamma^\times : (v \bmod \Delta_\beta)(x) \geq 0\}. \end{aligned} \quad (8.4)$$

In particular $\Theta_p \in \mathcal{E}_\Gamma$ if and only if β is an order unit of Γ . No monomial hypothesis on p or on x is needed, and Γ need not be divisible.

Proof. Evaluation at 0 is finite. For $x \neq 0$ put $\delta = v(x)$ and write

$$p = ct^\beta(1+u), \quad x = dt^\delta(1+w), \quad c, d \in \mathbb{C}^\times,$$

where each of u, w is zero or has strictly positive valuation. Let $S = \text{supp}(u) \cup \text{supp}(w) \subseteq \Gamma_{>0}$ and $M = M(S)$. The n th evaluation term is

$$p^{(n)}x^n = c^{(n)}d^n t^{h_n} (1+u)^{(n)}(1+w)^n, \quad h_n = \binom{n}{2}\beta + n\delta,$$

and the last two factors are finite integer powers, so

$$\text{supp}(p^{(n)}x^n) \subseteq h_n + M, \quad v(p^{(n)}x^n) = h_n. \quad (8.5)$$

Suppose first that $\delta \geq -N\beta$ for some $N \in \mathbb{N}$. Then

$$h_{n+1} - h_n = n\beta + \delta \geq (n - N)\beta > 0 \quad (n > N),$$

so (h_n) is eventually strictly increasing. Its set of values is therefore well ordered, including the finite initial segment, and each value is taken by finitely many indices. Lemma 2.3 (b) applied to (8.5) proves strong summability.

If no such N exists, then $\delta < -N\beta$ for every N , so $h_{n+1} - h_n = n\beta + \delta < 0$ for every n . By (8.5) the leading exponents are exactly h_n , and they form a strictly decreasing sequence, so strong summability fails by Lemma 2.2.

Finally, $\delta \geq -N\beta$ for some N says exactly that the image of δ in Γ/Δ_β is nonnegative: a nonnegative coset either is positive, whence $\delta > 0$, or is zero, whence $\delta \in \Delta_\beta$ and $\delta \geq -N\beta$ for some N ; a negative coset lies below every element of Δ_β . The last assertion is the case $\Delta_\beta = \Gamma$. $\square$

This proof controls the whole support, not only the leading valuation. Leading valuation formulas by themselves do not in general determine a Hahn evaluation domain, because hidden tails can matter; here the common positive monoid M is the extra certificate that makes the formula exact. Note also that the entire calculation is carried out in Γ : nothing is divided by an integer, and no real-valued surrogate for v is introduced.

Remark 8.2 (A second, shorter proof of the order-unit half). When $\mu = \beta$ is an order unit, entireness also follows from the cofinal-valuation criterion Lemma 2.3 (a) alone, and the routes of sources 08 and 09 are both kept here because they are genuinely different. For $x \neq 0$ with $\alpha = v(x)$, the n th evaluated term has valuation

$$\frac{n(n-1)}{2}\mu + n\alpha. \quad (8.6)$$

Given a target $\beta' \in \Gamma$, choose ordinary nonnegative integers M', B' with $\alpha \geq -M'\mu$ and $\beta' \leq B'\mu$, which is possible exactly because μ is an order unit. Then (8.6) is at least $(\frac{n(n-1)}{2} - M'n)\mu$, which exceeds $B'\mu$ for all sufficiently large ordinary n . So the term valuations tend cofinally to $+\infty$ and Lemma 2.3 (a) applies. Only leading term valuations are used, so unrestricted higher Hahn tails in p do not invalidate the conclusion. This shorter route is available only in the order-unit case: at the boundary $v(x) = -N\beta$ with β noncofinal, the term valuations increase strictly but stay trapped below a fixed element of Γ , so criterion (a) says nothing and the monoid certificate is essential.

Remark 8.3 (The order-unit case has an optional rank-one picture). When β is an order unit, the real-valued coarsening $\pi : \Gamma \rightarrow \mathbb{R}$ of [12, Lemma 9.1], normalized by $\pi(\beta) = 1$, gives the same field K_Γ a rank-one absolute value. For $x \neq 0$, its theta terms have coarsened valuations $\binom{n}{2} + n\pi(v(x))$, tending to $+\infty$. This is an optional normed-field picture of the order-unit case, not a map that replaces the Hahn exponents by their images in $\mathbb{R}$.

Such a coefficientwise replacement need not define a Hahn series. In $\mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating all $n\epsilon$, the coarsening normalized by H kills ϵ . The valid series $\sum_{n \geq 0} t^{n\epsilon}$ would then place infinitely many coefficients at exponent zero. Coarsening a valuation retains the field and changes the value map; it is not an embedding into a complex-coefficient Hahn field with smaller exponent group. No proof here requires this optional coarsening.

8.3 Completion of the sharp criterion

Proof of Theorem B. Assume (i) and suppose (ii) fails. If $v(q) = 0$, Corollary 7.2 makes every entire solution a polynomial. If $v(q) \neq 0$ and $|v(q)|$ is not an order unit, Proposition 7.3 does the same. Either way (i) fails, which proves (i) $\Rightarrow$ (ii) and supplies the stated uniform obstruction and degree bound.

Assume (ii). If $v(q) > 0$ is an order unit take $p = q$: by Theorem 8.1, $\Theta_q \in \mathcal{E}_\Gamma \setminus K_\Gamma[z]$, and (8.3) is a nonzero linear q -difference equation with polynomial coefficients. If $v(q) < 0$ and $-v(q)$ is an order unit take $p = q^{-1}$, so $v(p) = -v(q) > 0$ is an order unit and $\Theta_{q^{-1}} \in \mathcal{E}_\Gamma$ by the same theorem. Substituting q^2z for z in (8.3), with $p = q^{-1}$ so that $pz \mapsto qz$ and $p^2z \mapsto z$, gives

$$\Theta_{q^{-1}}(q^2z) - (1 + q^2z)\Theta_{q^{-1}}(qz) + qz\Theta_{q^{-1}}(z) = 0, \quad (8.7)$$

which uses only nonnegative powers of σ_q and has polynomial coefficients that are not all zero. Both signs of $v(q)$ are covered. $\square$

The partial theta function is used as an established type of construction, not advertised as a newly discovered special function. Its role is precise: it realizes the positive side of a necessary and sufficient criterion, including nonmonomial q and arbitrary valuation rank.

8.4 Two scales in the same rank-two field

Example 8.4 (One field, opposite answers). Let $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, ordered by the highest nonzero coordinate. Then e_1 is an order unit and e_0 is not.

For $q = t^{e_0}$, no nonpolynomial strongly entire series satisfies a nonzero linear q -difference equation. The series Θ_q itself is not entire: by Theorem 8.1 its exact domain is the valuation ring obtained by forgetting the e_0 coordinate, which contains every x with $v(x) = -Ne_0$ and excludes $x = t^{-e_1}$.

For $q = t^{e_1}$, Θ_q is strongly entire, though the value group still has rank two. The two conclusions occur in the *same* Hahn field: the distinction is the dilation scale, not the rank of the field.

Under $e_0 = 1$, $e_1 = \omega$ and $t^\gamma = \omega^{-\gamma}$, the first dilation is multiplication by ω^{-1} and the second by $\omega^{-\omega}$. Quadratic multiples of the first valuation never reach the second scale; quadratic multiples of the second dominate every scale in this workspace. A fuller table for the same group, including unit dilations and the torsion exception, appears in Section 11.3.

8.5 The coarse boundary really can be included

For the theta series the forward recurrence is $a_{n+1} = p^n a_n$, read off from (8.2), whose inhomogeneity affects only the index $n = 0$ and is removed by Proposition 5.6. Its two slots have valuations $n\beta$ and 0, both of which project to 0 modulo Δ_β , so the projected gap is zero and $\overline{B}_{\text{rec}} = \overline{B}_{\text{rec}}^\flat = 0$. Theorem 7.4 therefore excludes the points of negative coarsened valuation, while Theorem 8.1 includes *all* points of zero coarsened valuation. The strict inequality in the coarsened exclusion theorem cannot be replaced by a weak one in general. This is the exact mirror image of Section 5.2, where the uncoarsened bound was attained with its boundary excluded from the domain. The two boundaries behave differently because one comparison is made in Γ and the other in a quotient that has forgotten the scale doing the work.

9 Torsion parameters and why they are excluded

If $q^h = 1$ then $\sigma_q^h - 1$ annihilates *every* formal power series, so the unrestricted definition of linear q -difference finiteness is vacuous for such a parameter. This is not a technical inconvenience removable by a better proof; it is a genuine counterexample to dropping the nontorsion hypothesis in Theorem B.

The difficulty persists even after excluding operators with zero action. For $h > 1$ and

$$f(z) = z^r g(z^h),$$

one has $f(qz) = q^r f(z)$, and the operator $\sigma_q - q^r$ is not zero on all formal series when $h > 1$. A nonpolynomial entire g produces a nonpolynomial entire f , since evaluation of f at x is evaluation of g at x^h followed by multiplication by x^r . Finite-order dilations therefore have a real periodic-support exception and are not a limiting case of Theorem B. In particular “the residue of q is a root of unity” and “ q is a root of unity” are fundamentally different hypotheses: Example 4.6 is covered by the theorem and this section is not.

There is nevertheless a meaningful eigenspace statement.

Proposition 9.1 (Torsion covariance). *Let $q \in \mathbb{C}^\times$ have exact finite order m and let $0 \leq r < m$.*

- (a) *A formal series satisfies $f(qz) = q^r f(z)$ if and only if $f(z) = z^r G(z^m)$ for a unique $G \in K_\Gamma[[z]]$.*
- (b) *If $G \in \mathcal{E}_\Gamma$ then $f = z^r G(z^m) \in \mathcal{E}_\Gamma$.*
- (c) *The converse of (b) also holds, so $f \in \mathcal{E}_\Gamma$ if and only if $G \in \mathcal{E}_\Gamma$, without divisibility of Γ .*

Moreover, every root of unity of K_Γ lies in $\mathbb{C}$, so the hypothesis $q \in \mathbb{C}^\times$ loses no torsion element of the Hahn field. All parts use only Convention 1.1.

Proof. For (a), coefficient comparison gives $(q^n - q^r)a_n = 0$, so only degrees congruent to r modulo m occur, and G is read off uniquely.

For (b), evaluation of $z^r G(z^m)$ at x is x^r times the family for G at x^m , reindexed; multiplication by the fixed element x^r preserves strong summability by Lemma 2.1 (3), and at $x = 0$ the evaluation is finite.

For (c), suppose $f \in \mathcal{E}_\Gamma$ and let $y \in K_\Gamma^\times$. Set $\delta = v(y)$, $\gamma = \min(\delta, 0)$, and $x = t^\gamma$. Since $m \geq 1$, we have $v(x^m) = m\gamma \leq \delta$. Strong evaluation of f at x , restricted to the degrees $r + mn$ and multiplied by x^{-r} , gives strong evaluation of G at x^m . Lemma 2.4 then gives strong evaluation at y . At $y = 0$ evaluation is finite. No root of y is extracted.

Finally, let $\zeta \in K_\Gamma$ be a root of unity. Then $v(\zeta) = 0$; dividing by $\text{res}(\zeta)$ gives $1 + u$ with u zero or of positive valuation, and $(1 + u)^{m'} = 1$ for some $m' \geq 1$. If $u \neq 0$ then $(1 + u)^{m'} - 1$ has valuation $v(u)$ and leading coefficient $m' \text{lc}(u) \neq 0$ in characteristic zero, a contradiction. So $u = 0$ and $\zeta = \text{res}(\zeta) \in \mathbb{C}$. $\square$

The converse uses only that the values $m\gamma$ are coinital in Γ , as the explicit choice $\gamma = \min(\delta, 0)$ shows. Actual m th roots need not exist in the original Hahn field; this stronger algebraic requirement is unnecessary for the entireness equivalence.

10 An extension to mixed differential–dilation equations

For a nonzero noncofinal dilation valuation, derivatives do not repair the missing-scale problem. The extension is useful when a symbolic system allows both differentiation and rescaling of the argument.

Lemma 10.1 (Polynomial dependence on the index as well as the dilation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(U, X) \in K_\Gamma[U, X]$. There are $\beta \in \Gamma$, $k \in \mathbb{N}$ and $N \in \mathbb{N}$ with*

$$v(P(n, q^n)) = \beta + kn\lambda \quad (n \geq N).$$

Proof. Write $P(U, X) = \sum_k C_k(U)X^k$. By Lemma 4.1 each nonzero $C_k(n)$ has the fixed valuation $\beta_k = \beta(C_k)$ for all sufficiently large n . The finitely many summands then have valuations $\beta_k + kn\lambda$, and the pairwise-order argument of Lemma 4.2 produces a unique eventual minimizing index, whose term cannot cancel. $\square$

Theorem 10.2 (Mixed rigidity at a noncofinal nonzero scale). *Let Γ be as in Convention 1.1, let $q \in K_\Gamma^\times$ with $v(q) \neq 0$, and suppose $|v(q)|$ is not an order unit of Γ . If a strongly entire f satisfies a nonzero equation*

$$\sum_{r,\ell} p_{r,\ell}(z) D_z^r \sigma_q^\ell f(z) = 0, \quad p_{r,\ell} \in K_\Gamma[z], \quad (10.1)$$

where the sum is finite and the coefficients are not all zero, then f is a polynomial. Each fixed equation has a uniform exterior obstruction and a uniform degree bound.

Proof. Expand $p_{r,\ell}(z) = \sum_k c_{r,\ell,k} z^k$. With $s = r - k$, the term of coefficient index $n + s$ contributes to degree n the quantity $c_{r,\ell,k}(n + s) \underline{r} q^{\ell(n+s)} a_{n+s}$, so the coefficient recurrence has shift polynomials

$$P_s(U, X) = \sum_{r-k=s} \sum_{\ell} c_{r,\ell,k} q^{\ell s} (U + s) \underline{r} X^\ell, \quad (10.2)$$

evaluated at $(U, X) = (n, q^n)$. For fixed s and ℓ , distinct r give distinct degrees in U , so a nonempty group of nonzero coefficients cannot cancel identically and some P_s is nonzero. Reindex from the

smallest active shift; replacing U by $U - s_0$ and X by $q^{-s_0}X$ preserves polynomiality and nonzeroness, so the leftmost coefficient is $A_0(n) = P'_0(n, q^n)$ for a nonzero bivariate P'_0 , and Lemma 10.1 makes it eventually nonzero.

Lemma 10.1 gives eventual affine valuations for all the nonzero coefficients, so their differences obey a bound $B + Mn|v(q)|$ exactly as in (7.2). Since $|v(q)|$ is not an order unit, (2.2) bounds this by one element of Γ , and Corollary 3.4 finishes the proof. $\square$

The hypothesis $v(q) \neq 0$ is essential to *this* proof. For $v(q) = 0$ the separate pure differential and pure dilation conclusions have been proved above, but they are not a proof of the mixed conclusion. When Γ has no order unit, the mixed conclusion at unit valuation, for nonlinear relations as well, is Theorem E (b) of source 12, proved by a different method; with an order unit no mixed unit-valuation classification is asserted in this report; see Question 19.4.

11 Surreal workspaces that separate the theories

The results so far are stated in set-sized Hahn fields, which keeps the foundations transparent and allows a literal surreal reading without quantifying over a proper class as though it were a set.

11.1 The workspace without an order unit

Take

$$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}e_j, \quad 0 < e_0 \ll e_1 \ll e_2 \ll \cdots, \quad (11.1)$$

ordered so that the highest index with a nonzero coordinate determines the sign, equivalently $e_{j+1} > Ne_j > 0$ for all $j \geq 0$ and $N \geq 1$. Every element has only finitely many nonzero coordinates. One may identify e_j with the surreal ω^j .

The group has no order unit: if $\gamma > 0$ has largest active index j , then $e_{j+1} > n\gamma$ for every ordinary n . The sequence (e_n) is nevertheless cofinal in the whole group, so this is a countable cofinality that cannot be generated by repeatedly adding one positive element. This group and this distinction appear in the companion entire-function report [12] and are background here, not a new observation.

Corollary 11.1 (Universal nontorsion dilation rigidity). *Suppose Γ has no order unit. Then every nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfies, simultaneously for all nontorsion $q \in K_\Gamma^\times$ and all $r \geq 0$,*

$$f(z), f(qz), \dots, f(q^r z) \quad \text{are linearly independent over } K_\Gamma(z).$$

Such fields can contain nonpolynomial strongly entire series, as (11.1) and Theorem 11.2 show.

Proof. A linear dependence over $K_\Gamma(z)$ becomes, after clearing denominators, a nonzero equation (1.3). Since Γ has no order unit, no $|v(q)|$ is an order unit, so Theorem B forces f to be a polynomial. $\square$

Theorem 11.2 (An explicit universally nonholonomic entire series). *In K_{Γ_∞} the series*

$$F(z) = \sum_{n \geq 0} t^{e_n} z^n = \sum_{n \geq 0} \omega^{-\omega^n} z^n \quad (11.2)$$

is strongly entire and nonpolynomial. It satisfies no nonzero linear differential equation over $K_{\Gamma_\infty}(z)$ and, for every nontorsion $q \in K_{\Gamma_\infty}^\times$, no nonzero polynomial-coefficient linear q -difference equation. For every q of nonzero valuation it satisfies no nonzero mixed equation (10.1) either.

Proof. Fix $x \neq 0$ and write $x = ct^\delta(1+w)$ with $c \in \mathbb{C}^\times$ and w zero or of positive valuation. Choose M with δ involving only the coordinates $e_0, \dots, e_M$. The n th evaluation term has leading exponent $h_n = e_n + n\delta$ and support contained in $h_n + M(\text{supp } w)$. For $n > M$ the highest coordinate of h_n is e_n with coefficient 1, so $h_n > 0$, and the highest coordinate of

$$h_{n+1} - h_n = e_{n+1} - e_n + \delta$$

is e_{n+1} with coefficient 1, so (h_n) is eventually strictly increasing. Lemma 2.3 (b) proves strong summability; at $x = 0$ only the constant term remains. All coefficients are nonzero, so F is nonpolynomial.

Alternatively, for any $\beta \in \Gamma_\infty$ choose j larger than every active coordinate index of δ and of β ; for $n > j$ the largest active coordinate of $h_n - \beta$ is n with coefficient 1, so $h_n > \beta$, and Lemma 2.3 (a) applies for arbitrary w . Cofinal leading valuations control the whole support, since every exponent of a Hahn term is at least its leading exponent.

The differential conclusion is Theorem A. Since Γ_∞ has no order unit, Theorem B — equivalently Corollary 11.1 — excludes every nontorsion dilation equation, and Theorem 10.2 excludes the mixed equations at every q with $v(q) \neq 0$. Rational coefficient equations reduce to the polynomial-coefficient case by clearing denominators. $\square$

The universal quantifier over q matters. The parameter may be chosen after F is fixed and may itself have infinite support; each individual $v(q)$ still has finitely many coordinates in Γ_∞ , hence is noncofinal. So the theorem excludes a linear relation for *each* such parameter, not merely for an advance list of simple dilations. The series has a simple explicit coefficient rule, and this is a statement about finite linear annihilating equations, not a claim that the rule is uncomputable or lacks a finite description.

11.2 Why the same formula is not entire on all surcomplex numbers

Let $e_* = \omega^\omega$, which dominates every integer multiple of every positive element of Γ_∞ , and enlarge the group to $\Gamma_\infty + \mathbb{Q}e_*$. At the surcomplex point $x = \omega^{e_*} = t^{-e_*}$ the evaluated leading exponents are $e_n - ne_*$, whose successive differences $e_{n+1} - e_n - e_*$ are negative. The defining family is therefore not strongly summable, by Lemma 2.2.

This explicit obstruction prevents the fixed-workspace theorem from being misread as an assertion of a nonpolynomial power series entire on all of $\mathbf{No}[i]$. The example illustrates two independent restrictions: a workspace can be large enough that no single dilation scale is cofinal, yet small enough to admit a countable sequence of coefficient scales escaping all its arguments, and enlarging it further can destroy the very entireness on which the theorem operates. The companion report studies this kind of boundary in general [12]; the calculation here is included only to state the domain of the present example unambiguously, and no new universal scalar-extension theorem is claimed.

11.3 Two classification tables

Let $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, so that H is an order unit and ϵ is not.

Dilation q	Its valuation	Nonpolynomial entire solution of a nonzero linear q -difference equation?
t^H	H	Yes: Θ_{t^H} , by (8.3).
t^{-H}	$-H$	Yes: Θ_{t^H} , by (8.7).
t^ϵ	ϵ	No: positive, but noncofinal.
$1 + t^H$	0	No: an infinite-order unit.
2	0	No: complex size does not change the Hahn valuation.
-1	0	Outside the theorem: a torsion parameter, with periodic-support exceptions.

Workspace group	Parameter	Solution exists?	Reason
$\mathbb{Q}$	$v(q) \neq 0$	Yes	Every positive nonzero scale is an order unit.
$\mathbb{Q}$	$v(q) = 0$, nontorsion	No	Unit-orbit valuations are eventually periodic.
$\mathbb{Q}e_0 \oplus \mathbb{Q}e_1$, $e_1 \gg e_0$	$v(q) = e_0$	No	A lower scale is not cofinal.
the same group	$v(q) = e_1$	Yes	The top scale is an order unit.
$\bigoplus_{n \geq 0} \mathbb{Q}e_n$	any nontorsion q	No	No scale is an order unit, although F is entire.

In every row, nonpolynomial strongly entire D-finite series are impossible. Nonpolynomial strongly entire differentially algebraic series exist exactly in the rows whose group has an order unit, that is, in all rows except the last row of the second table (Theorem H), and their least differential order is then three (Theorem I); this does not depend on the parameter. Torsion parameters are deliberately absent from the second table: their annihilating operator may be an identity on every series. Finite valuation rank alone is therefore not the criterion, and the distinction persists for nonmonomial dilations, because the theorem uses the valuation rather than a requirement that q be a monomial.

Corollary 11.3 (Detecting order units by functional equations). *For a nonzero set-sized ordered abelian group Γ the following are equivalent:*

- (i) Γ has an order unit;
- (ii) some nontorsion $q \in K_\Gamma^\times$ admits a nonpolynomial strongly entire solution of a nonzero polynomial-coefficient linear q -difference equation.

Proof. Theorem B gives (ii) $\Rightarrow$ (i). For the converse take $q = t^\mu$ for an order unit μ ; then $v(q) = \mu$ is an order unit and q is nontorsion. $\square$

This recovers a property of the ordered group from a global functional equation existence question. Theorem H recovers the same property from algebraic differential equations, with no dilation. The workspace (11.1) shows why the existence of nonpolynomial entire functions is by itself strictly weaker than the existence of an order unit.

11.4 Several meanings of transcendence, kept apart

For the explicit F the established conclusions are: transcendence over $K_{\Gamma_\infty}(z)$ by Corollary 5.4; linear differential independence of every finite family of derivatives; and linear independence along every single nontorsion geometric dilation orbit. These are substantial restrictions, and they do *not* imply that no polynomial relation $P(z, F, D_z F, \dots, D_z^R F) = 0$ exists. Nonlinear differential equations lie outside these proofs, and a relation involving $F(q_1 z)$ and $F(q_2 z)$ for multiplicatively independent parameters is not covered merely by applying the single- q theorem twice.

The nonlinear part adds one more conclusion by a separate proof: F and $D_z F$ are algebraically independent over $K_{\Gamma_\infty}(z)$ (Corollary 17.2), so every nonlinear relation with $R \leq 1$ is excluded. The methods of that part exclude relations with $R \geq 2$ only when their residue corner polynomial is nonzero, and give no statement about several dilations. Because Γ_∞ has no order unit, the coefficient-field part removes both limits for this F : Theorem 22.1 (source 12) makes all $(D_z^j F)(\lambda z)$, for dilations λ with pairwise quotients that are not roots of unity, algebraically independent over $K_{\Gamma_\infty}(z)$. In particular F satisfies no algebraic differential equation of any order, and no nonzero relation, linear or not, links $F(q_1 z)$ and $F(q_2 z)$ for multiplicatively independent q_1, q_2 (Corollary 22.8, a merge observation).

12 Coefficient fields, degenerate value groups, characteristic

The one-variable proofs work over any characteristic-zero coefficient field with trivial valuation: the root-of-unity argument uses only finitely many ordinary polynomial roots and needs no algebraic closedness, and the integer-evaluation lemma needs only residue characteristic zero. All theorems also hold over $\mathbb{R}((t^\Gamma))$, the generic-line proof of Theorem C using uncountability of $\mathbb{R}$ in place of uncountability of $\mathbb{C}$. The results therefore concern surreal as well as surcomplex workspaces. More generally the core versions work over $k((t^\Gamma))$ for any characteristic-zero field k , and, as Section 1.4 records, without divisibility of Γ .

No statement here is a theorem for arbitrary valued fields of positive residue characteristic. In that setting Lemma 4.1 fails and the valuations of near-torsion geometric powers can behave differently, so any comparison with the ultrametric literature must keep the coefficient field and the residue characteristic visible. Example 17.3, from source 10, shows that characteristic zero is needed even for the single linear equation $D_z f = 0$: over $\mathbb{F}_p((t^\mathbb{Q}))$ a nonpolynomial strongly entire series has zero derivative.

The nonlinear part (Convention 14.1) is proved by its source over an arbitrary field of characteristic zero. That is a statement about Sections 14–17 only; the generality of the linear theorems is governed by the first paragraph of this section and its own argument. The same holds for the coefficient-field part (Convention 20.1), whose sources also work over an arbitrary field of characteristic zero.

We assumed $\Gamma \neq 0$ in order to speak of positive exterior scales. When $\Gamma = 0$ the field is $\mathbb{C}$ with the trivial valuation, strong evaluation at $x = 1$ already permits only finitely many nonzero coefficients because every nonzero term has support $\{0\}$, and entire series are polynomials for that immediate support reason.

13 Quantitative certificates and symbolic representation

13.1 What one can extract from an annihilating equation

The classification has a direct representational consequence. A system whose nonpolynomial special functions are represented only by finite linear differential equations cannot cover the nonpolynomial strong entire functions of a Hahn workspace. Allowing linear dilation equations enlarges the class only when a cofinal dilation scale is available. In the no-order-unit example, an exact coefficient rule such as (11.2) lies outside both pure equation formats.

This does not rule out exact symbolic computation with such functions. It specifies what a faithful representation must keep separate: the coefficient-generation rule, the exponent group, the

strong evaluation domain, and the type of finite equation being asserted. A field-independent label such as “entire theta function” discards relevant information; for Θ_p the proof certificate is not merely $v(p) > 0$, it includes that $v(p)$ is an order unit of the selected group.

For a differential operator the reduction yields a concrete certificate. Compute the finite shift polynomials Q_s of (5.2) and shift the least active index to zero. For each resulting P_j record its least coefficient valuation β_j and the corresponding nonzero complex residue polynomial $\overline{P}_j$. Record an integer N beyond their nonnegative integer roots and beyond the recurrence’s starting index. Record the threshold B of Proposition 5.2, or (5.4) when Γ is divisible. The escape statements then certify that every formal solution strongly evaluable at some $x \neq 0$ with $v(x) \leq -B$ has degree less than N .

For a valuation-zero dilation the same plan uses (7.1): if $\text{res}(q)$ has infinite order use the minimum coefficient valuation, and if it has order h use (4.4)–(4.6) on each residue class. For a nonzero-valuation dilation, finitely many affine comparisons select the eventual winner of each $R_h(q^n)$; project their constants modulo $\Delta_{|v(q)|}$ to obtain the coarsened bound (7.3), or (7.4) when Γ is divisible. When the parameter scale is noncofinal one can instead display $\gamma \gg |v(q)|$ and the explicit bad argument $t^{-(B+\gamma)}$ of Proposition 7.3.

These are mathematical certificate schemas relative to exact coefficient data and order comparisons. They are *not* a terminating algorithm for arbitrary encoded surreals. Deciding whether an arbitrary complex residue is a root of unity, comparing arbitrary surreal values, extracting arbitrary supports, or deciding whether an abstract scale is cofinal can require information not effectively available in a chosen representation, and none of this has been made computable by the proofs.

13.2 A valuation recurrence with zero coefficients allowed

Example 13.1 (Jump lengths, not just jumps). Suppose a sequence that is not eventually zero satisfies

$$t^A a_n + (n-2)t^B a_{n+1} + t^C a_{n+3} = 0 \quad (n \geq 0),$$

with $A, B, C \in \Gamma$. The middle coefficient vanishes at $n = 2$, a finite exceptional index; beyond it the three valuations are A, B, C . The escape chain may move forward by one or by three, and no assumption that all a_n are nonzero is needed. Corollary 3.4 excludes the region $v(x) \leq -\max\{A - B, A - C, 0\}$ under Convention 1.1 alone. If Γ is divisible, Corollary 3.5 gives the sharper excluded region

$$v(x) \leq -\max\left\{A - B, \frac{A - C}{3}\right\}.$$

The maximum must be divided by the actual jump length: replacing every jump by one need not preserve the intended threshold, especially when the valuation gaps have different signs. This is a worked instance of item (1) of Section 1.4.

13.3 Current Lean coverage and remaining dependency order

The ledger `docs/FORMALIZATION.md` records the formalized escape mechanism in `Surreal/HahnSeries/EscapeChain.lean`: the nonincreasing-order obstruction, escape step and infinite chain, evaluation exclusion, and the uniform bounded-cost corollary. It also covers the univariate strong-domain definitions and the formal exponential identity with its exact domain and attained exclusion boundary. Part (2) of Lemma 2.1 is mapped separately to `Surreal/HahnSeries/NeumannWords.lean`.

`Surreal/HahnSeries/PartialThetaDomain.lean` covers both sufficient criteria of Lemma 2.3, the scale-subgroup facts, the exact partial-theta domain and the theta identities. It also proves (ii) $\Rightarrow$ (i) of Theorem B and (i) $\Rightarrow$ (ii) of Corollary 11.3; the converse implications remain pending.

`Surreal/HahnSeries/UnitOrbitPolynomials.lean` covers integer valuation stabilization, eventually affine geometric-orbit valuations, the unit-orbit theorem and the bivariate index–dilation lemma. `Surreal/Algebra/GenericConstantPoint.lean` proves Lemma 6.1, including its least-exponent reduction, for uncountable constants. `Surreal/HahnSeries/TorsionCovariance.lean` covers inward stability, all of Proposition 9.1, and the preservation half of Lemma 2.1 (3). These results require no divisibility of the value group.

The remaining dependencies include the refined periodic threshold, part (1) and the reflection half of part (3) of Lemma 2.1, finite-block grouping in Lemma 2.3, Proposition 2.7, and the several-variable strong-domain definition. The operator-to-recurrence conversions and full differential and dilation classifications remain pending. For the several-variable classification, constant-point avoidance is proved; generic-line restriction, restriction of a rational differential system, and the resulting D-finiteness argument are pending.

The nonlinear rigidity theorems in Sections 14–17, the coefficient-field theorem package in Sections 20–22, the theta-descent part in Sections 23–25, and the order-three-threshold part in Sections 26–27 remain unformalized; the shared support and inward-stability prerequisites above do not prove them. For the latter source 12 suggests the order: valuation-rank inequality, proper-convex-subgroup obstruction, formal coefficient recurrence, all-order differential corollary; the mixed statements would also need a formal theorem implying the zero-set lemma for nondegenerate exponential polynomials. For the order-three-threshold part source 14 suggests two nearly independent layers: finite algebra (dehomogenization, the logarithmic Euler identities, constants of rational differentiation in characteristic zero and the two endpoint evaluations, which give the finite endpoint lemma without surreal infrastructure), then the Hahn layer (the all-scale criterion, transport through the divisible hull, finite active-index sets and the estimate (26.4)). The build and axiom audit cover only imported declarations recorded in the ledger. The Python checks are finite checks of formulas and examples; they do not establish any remaining infinite statement.

14 Nonlinear equations: finite initial polynomials

This section and the next three contain the material of source 10. They answer the first-order case of Question 19.1, which sources 08 and 09 had left open, together with a nondegenerate higher-order class and positive-weight Euler equations in several variables. The method is not the escape chain of Section 3. The warning in Remark 5.8 stands: a nonlinear equation gives coefficient convolutions, not a bounded-step linear recurrence. Instead, a whole scaled series is reduced to a finite polynomial at one large scale, and an exact leading-term computation gives the contradiction.

14.1 Coefficient field, conventions and the merge’s choices

Convention 14.1 (Coefficient field of the nonlinear part). In Sections 14–17, k is an arbitrary field of characteristic zero with the trivial valuation, Γ is as in Convention 1.1, and

$$\mathbb{K} = k((t^\Gamma)).$$

The valuation v , leading coefficient lc , residue res , strong summability, $\text{Dom}_{\text{str}}(f)$ and entireness are defined exactly as in Section 1.2 and Definition 1.2, with k in place of $\mathbb{C}$, except that, following

source 10, $\text{res}(a)$ is defined for every a of nonnegative valuation as its coefficient at exponent zero, so that $\text{res}(a) = 0$ when $v(a) > 0$. Write $\mathcal{E}_\Gamma(k)$ for the strongly entire series in $\mathbb{K}[[z]]$. In several variables, strong entireness of $f = \sum_\alpha a_\alpha x^\alpha$ means strong summability of the *joint* family $(a_\alpha b^\alpha)_{\alpha \in \mathbb{N}^m}$ at every $b \in \mathbb{K}^m$. For $k = \mathbb{C}$ one has $\mathbb{K} = K_\Gamma$ and $\mathcal{E}_\Gamma(k) = \mathcal{E}_\Gamma$. The derivative D_z kills $\mathbb{K}$, and $f' = D_z f$; no scalar surreal derivation, analytic continuation or fine-topology limit is involved.

Why the coefficient field differs. Source 10 proves every statement of the nonlinear part over an arbitrary characteristic-zero field k , and it uses neither algebraic nor real closedness. Its hypothesis on the coefficient field is therefore weaker than the hypothesis $k = \mathbb{C}$ under which Theorems A–C and the rest of Sections 1–13 are stated. This generality is recorded for the nonlinear part only. It is *not* transferred to the linear theorems: they remain stated over $\mathbb{C}$, and their possible extension to other coefficient fields is the separate remark of Section 12, resting on its own argument. In the other direction, the nonlinear part borrows from Section 2 only facts about well-ordered subsets of Γ and finite coefficient sums (Lemmas 2.1 and 2.3), whose proofs do not use any property of $\mathbb{C}$; source 10 proves the facts it needs over k itself, except the positive-support lemma used in Theorem 15.10, which it imports as standard. The group Γ is not assumed divisible, and no order unit, no countable cofinal subset and no real-valued norm is used anywhere in the nonlinear part.

Renamed symbols. Several symbols of source 10 would collide with symbols of Sections 1–13; they are renamed here, in the new material only.

Source 10	Here	Reason
$D = \max_\nu d_\nu, W$	d_P, w_P	D_z is the derivative.
H_P (residue corner polynomial)	χ_P	H is an element of Γ in Sections 2–11.
B_P (degree cutoff, in $\mathbb{Q}$)	κ_P	B, B_L, B_{rec} are elements of Γ .
B (set of right sides)	$\mathcal{R}$	The same.
r, r_0 (scale, “radius”)	δ, δ_0	r indexes derivative orders in (1.2); $t^{-\delta}$ is the notation of Corollary 3.4.
h_r, p_r (Gauss value, initial polynomial)	$\gamma_\delta, \phi_\delta$	h is a torsion order, p_r an operator coefficient.
M (an integer bound)	n_*	$M(S)$ is a support monoid.
E_q (Euler operator)	ϑ_q	E_η is the formal exponential of Proposition 5.3.
Theorems A, B, C	Theorems D, 16.5, 16.1	Theorems A–C of this report are different theorems.

The shipped audit files `10-nonlinear-rigidity-PROOF_AUDIT.md` and `10-nonlinear-rigidity-SOURCES_AND_SCOPE.md` keep the source’s notation and numbering.

Words with two meanings. *Escape* keeps the meaning of Section 3 (a chain through a linear recurrence); source 10’s lemma “escape of active degree” is a different statement and is called Lemma 14.5 here. An *exclusion bound* is, as in Section 3, an element $\delta_0 \in \Gamma$ such that evaluation fails at the arguments $t^{-\delta}$ with $\delta \geq \delta_0$; the source’s word “radius” is not used. A *corner* is always the finite set of terms of Section 14.4, never a full differential Newton polygon. The *residue polynomials* P and R_r of Section 4 are unrelated to the residue corner polynomial χ_P .

Results printed once. Source 10 reproves several facts that Sections 2 and 11 already contain. They are printed once and cited: the support calculus inside Lemma 14.2 (Lemma 2.1 (1),(3) and

Proposition 2.7); its “cofinal escape of coefficient minima” lemma, which is Lemma 2.3 (a); the positive-support geometric-series lemma, which source 10 imports and Lemma 2.1 (2) proves; the surreal embedding, which is Section 2.4 with the same monomial convention $t^\gamma \mapsto \omega^{-\gamma}$; strong entireness of the series (11.2), which is Theorem 11.2; and the remark on $\Gamma = 0$, which is in Section 12. Everything else in source 10 is printed in Sections 14–17, with its proofs.

14.2 The coefficient-family algebra

For a finite tuple $X = (X_1, \dots, X_m)$ of indeterminates put

$$\mathcal{S}_m = \left\{ F = \sum_{\alpha} b_{\alpha} X^{\alpha} \in \mathbb{K}[[X]] : (b_{\alpha})_{\alpha} \text{ is strongly summable} \right\},$$

and let $\mathcal{S}_m^+$ be the subset whose coefficients all have nonnegative valuation. The condition concerns the family of coefficients, not convergence in the X -adic topology.

Lemma 14.2 (Hahn-family algebra and its residue). *$\mathcal{S}_m$ is a $\mathbb{K}$ -subalgebra of $\mathbb{K}[[X]]$, stable under the formal partial derivatives and under the Euler operators $\sum_{\nu} q_{\nu} X_{\nu} \partial_{\nu}$ with $q_{\nu} \in k$. Coefficientwise residue is a ring homomorphism*

$$\rho : \mathcal{S}_m^+ \longrightarrow k[X], \quad \rho\left(\sum_{\alpha} b_{\alpha} X^{\alpha}\right) = \sum_{\alpha} \text{res}(b_{\alpha}) X^{\alpha},$$

whose values are polynomials, not arbitrary formal power series; ρ commutes with these differential operators.

Proof. Closure under sums, scalar multiples and products is the support calculation of Lemma 2.1 (1) and (3), as in the proof of Proposition 2.7. For strongly summable (b_{α}) and (c_{ϵ}) a fixed exponent has finitely many representations as a sum of two support exponents, each carried by finitely many indices, so the double family $(b_{\alpha} c_{\epsilon})$ is strongly summable, and regrouping by $\alpha + \epsilon$ gives the Cauchy product: regrouping can remove supported exponents by cancellation, but cannot introduce new ones or new contributors at a fixed exponent. A scalar is a one-member Hahn family, so the same argument gives scalar multiplication; a finite union handles addition. Source 10 proves this over k by the same two-coordinate sequence argument; like the Hahn-field construction itself, that argument uses ordinary set-theoretic choice. A formal derivative or an Euler operator with coefficients in k reindexes a subfamily and multiplies its members by elements of the trivially valued field k ; it creates no new support exponent. For $F \in \mathcal{S}_m^+$ the exponent zero lies in only finitely many coefficient supports, so $\rho(F)$ is a polynomial. Each X -coefficient of a product is a finite sum of products of coefficients, the residue is additive and multiplicative on elements of nonnegative valuation, and the local finiteness just proved justifies every interchange; so ρ is a ring homomorphism. Differentiation commutes with residue coefficientwise. $\square$

For $0 \neq F = \sum_{\alpha} b_{\alpha} X^{\alpha} \in \mathcal{S}_m$ put

$$\gamma(F) = \min_{b_{\alpha} \neq 0} v(b_{\alpha}), \quad \text{in}(F) = \rho(t^{-\gamma(F)} F).$$

The minimum exists because it is the least element of the nonempty well-ordered union of supports. Then $t^{-\gamma(F)} F \in \mathcal{S}_m^+$ and the *initial polynomial* $\text{in}(F)$ is nonzero.

Corollary 14.3 (Gauss multiplicativity). *For nonzero $F, G \in \mathcal{S}_m$,*

$$\gamma(FG) = \gamma(F) + \gamma(G), \quad \text{in}(FG) = \text{in}(F) \text{in}(G).$$

Proof. Normalize both factors and apply Lemma 14.2. The product of the two nonzero residues is nonzero because $k[X]$ is an integral domain, so some coefficient of $t^{-\gamma(F)-\gamma(G)}FG$ has valuation exactly zero, and none has negative valuation. This proves the asserted minimum. Taking the residue of that normalization then gives the initial-polynomial identity. $\square$

14.3 Scaled initial polynomials

Let $f = \sum_{n \geq 0} a_n z^n \in \mathbb{K}[[z]]$ and $\delta \in \Gamma$ with $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$. Then

$$F_\delta(X) = f(t^{-\delta}X) = \sum_{n \geq 0} a_n t^{-n\delta} X^n \in \mathcal{S}_1.$$

For $f \neq 0$ put

$$\gamma_\delta = \min_{a_n \neq 0} (v(a_n) - n\delta), \quad g_\delta = t^{-\gamma_\delta} F_\delta, \quad \phi_\delta = \rho(g_\delta), \quad N_\delta = \deg \phi_\delta. \quad (14.1)$$

The *active* indices are the n attaining γ_δ ; there are finitely many, N_δ is the largest, and $A_\delta = \text{lc}(\phi_\delta) \in k^\times$. More explicitly,

$$\phi_\delta(X) = \sum_{v(a_n) - n\delta = \gamma_\delta} \text{lc}(a_n) X^n.$$

Here only nonzero a_n are included. Each active index contributes at the same Hahn exponent γ_δ in the scaled coefficient family, so local finiteness makes this sum finite. Distinct indices remain distinct powers of X ; their nonzero leading coefficients cannot cancel as a polynomial.

The element γ_δ is a *coefficient Gauss value*. It need not equal $v(f(t^{-\delta}))$: $\phi_\delta(1)$ can vanish although $\phi_\delta \neq 0$. The proofs below use identities in $\mathcal{S}_1$ and in $k[X]$ only, never an identification of these two quantities, and never replace the initial polynomial by its value at 1. For example, with $\eta > 0$ the finite family $F(X) = 1 - X + t^\eta X^2$ has $\gamma(F) = 0$ and $\text{in}(F) = 1 - X$, but $F(1) = t^\eta$ has valuation $\eta > 0$.

Lemma 14.4 (Exact derivative scaling). *For every $j \geq 0$,*

$$(D_z^j f)(t^{-\delta}X) = t^{\gamma_\delta + j\delta} g_\delta^{(j)}(X), \quad \rho(g_\delta^{(j)}) = \phi_\delta^{(j)}. \quad (14.2)$$

If $N_\delta \geq j$, then $\phi_\delta^{(j)}$ has degree $N_\delta - j$ and leading coefficient $A_\delta(N_\delta)_j$.

Proof. The scaling identity is a formal coefficient computation: $D_X^j F_\delta(X) = t^{-j\delta} (D_z^j f)(t^{-\delta}X)$. The residue identity is Lemma 14.2. The falling factorial is a nonzero element of k for $N_\delta \geq j$ because k has characteristic zero. $\square$

Lemma 14.5 (Large active degrees). *Let f be nonpolynomial, let $n_* \in \mathbb{N}$, and choose $m > n_*$ with $a_m \neq 0$. There is $\delta_1 \in \Gamma_{>0}$ such that, whenever $\delta \geq \delta_1$ and $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$, every active index exceeds n_* . Moreover $\gamma_\delta \leq v(a_m) - m\delta$.*

Proof. Choose $\delta_1 > 0$ with

$$(m - n)\delta_1 > v(a_m) - v(a_n) \quad (0 \leq n \leq n_*, a_n \neq 0). \quad (14.3)$$

There are finitely many inequalities, and they need no division in Γ : take δ_1 positive and larger than the absolute values of all right-hand sides, since each multiplier $m - n$ is a positive integer. Such an

element exists: add any positive element of the nonzero ordered group to the maximum of zero and those finitely many absolute values. This also covers an empty list of low-index coefficients. The inequalities persist for $\delta \geq \delta_1$ and say that each such low-index term has scaled valuation $v(a_n) - n\delta$ strictly larger than that of the m th term. The last assertion is the definition of γ_δ . $\square$

This is not a limit $N_\delta \rightarrow \infty$ along a countable cofinal sequence, which need not exist; it concerns a final segment of $\Gamma_{>0}$ and only those scales at which strong evaluation is available. It is also unrelated to the escape chain of Lemma 3.1: no recurrence is used.

14.4 The finite corner of a differential polynomial

Let P be a nonzero differential polynomial of order at most s , written in combined-monomial form

$$P(z, Y_0, \dots, Y_s) = \sum_{\nu \in J} c_\nu z^{k_\nu} \prod_{j=0}^s Y_j^{m_{\nu j}}, \quad (14.4)$$

with J finite, every $c_\nu \in \mathbb{K}^\times$, and identical monomials already combined. Write $P[f] = P(z, f, D_z f, \dots, D_z^s f)$. Define the integers

$$d_\nu = \sum_{j=0}^s m_{\nu j}, \quad w_\nu = k_\nu - \sum_{j=0}^s j m_{\nu j}, \quad d_P = \max_\nu d_\nu, \quad w_P = \max_{d_\nu = d_P} w_\nu, \quad (14.5)$$

and

$$\mathcal{C}_P = \{\nu : d_\nu = d_P, w_\nu = w_P\}, \quad \beta_P = \min_{\nu \in \mathcal{C}_P} v(c_\nu), \quad \bar{c}_\nu = \text{res}(t^{-\beta_P} c_\nu) \in k \quad (\nu \in \mathcal{C}_P).$$

Thus (d_P, w_P) is chosen lexicographically: first the largest total differential degree, then the largest shift w . This is a finite corner selection, not the construction of a differential Newton polygon. The *residue corner polynomial* and the *full corner polynomial* are

$$\chi_P(T) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu \prod_{j=0}^s (T)_j^{m_{\nu j}} \in k[T], \quad I_P(T) = \sum_{\nu \in \mathcal{C}_P} c_\nu \prod_{j=0}^s (T)_j^{m_{\nu j}} \in \mathbb{K}[T]. \quad (14.6)$$

In general χ_P can vanish although some $\bar{c}_\nu \neq 0$, because different products of falling factorials can be linearly dependent. If $\chi_P \neq 0$, then $I_P \neq 0$, and

$$I_P(n) = 0 \implies \chi_P(n) = 0 \quad (n \in \mathbb{N}), \quad (14.7)$$

because $t^{-\beta_P} I_P$ has coefficients of nonnegative valuation and residue commutes with substitution of an ordinary integer.

Lemma 14.6 (The exact corner identity). *Let $\phi \in k[X]$ have degree $N \geq s$ and leading coefficient $A \neq 0$, and put*

$$\Omega_\phi(X) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{k_\nu} \prod_{j=0}^s (\phi^{(j)}(X))^{m_{\nu j}}.$$

Then $\deg \Omega_\phi \leq d_P N + w_P$ and

$$[X^{d_P N + w_P}] \Omega_\phi = A^{d_P} \chi_P(N). \quad (14.8)$$

In particular $\chi_P(N) \neq 0$ implies $\Omega_\phi \neq 0$.

Proof. Each corner summand has degree at most $k_\nu + \sum_j m_{\nu j}(N - j) = d_P N + w_P$. This common degree is a nonnegative integer because $N \geq s \geq j$ and $k_\nu, m_{\nu j} \geq 0$, even if $w_P < 0$. The coefficient of the summand in that degree is $\bar{c}_\nu A^{d_P} \prod_j (N)_j^{m_{\nu j}}$. Summing gives the formula. $\square$

Only finitely many nonnegative integers are roots of a nonzero χ_P , even when k is not algebraically closed. The active degree is an ordinary integer, not an element of Γ , and no root extraction or division in Γ occurs in this step.

15 First-order algebraic differential equations

15.1 A finite certificate excluding all large scales

The following theorem carries the proof of Theorem D and is also a practical audit object: its scale inequalities use only finitely many coefficients of the solution. It applies to equations of any order.

Theorem 15.1 (Nonlinear finite exclusion certificate). *Let P be as in (14.4) with $d_P \geq 1$ and $\chi_P \neq 0$, and let $f = \sum_n a_n z^n$ be a nonpolynomial formal solution of $P[f] = 0$. Choose an integer $n_* \geq s$ at least as large as every nonnegative integer root of χ_P , and choose $m > n_*$ with $a_m \neq 0$ and*

$$(d_P - d_\nu)m + w_P - w_\nu > 0 \quad (d_\nu < d_P). \quad (15.1)$$

Every $\delta \in \Gamma_{>0}$ satisfying the finitely many strict inequalities

$$(m - n)\delta > v(a_m) - v(a_n) \quad (0 \leq n \leq n_*, a_n \neq 0), \quad (15.2)$$

$$(w_P - w_\nu)\delta > \beta_P - v(c_\nu) \quad (d_\nu = d_P, w_\nu < w_P), \quad (15.3)$$

$$((d_P - d_\nu)m + w_P - w_\nu)\delta > \beta_P - v(c_\nu) + (d_P - d_\nu)v(a_m) \quad (d_\nu < d_P) \quad (15.4)$$

has $t^{-\delta} \notin \text{Dom}_{\text{str}}(f)$. Such m and δ exist, and the inequalities persist when δ increases. Consequently $\text{Dom}_{\text{str}}(f)$ omits $t^{-\delta}$ for every δ in a final segment of $\Gamma_{>0}$, and every strongly entire solution of $P[f] = 0$ is a polynomial.

Proof. An integer n_* exists because $\chi_P \neq 0$. For $d_\nu < d_P$ the coefficient $d_P - d_\nu$ of m in (15.1) is a positive integer, so all sufficiently large m satisfy it, and a nonpolynomial f has nonzero coefficients of arbitrarily large index. Every multiplier on the left of (15.2)–(15.4) is then a positive integer, so any δ that is positive and larger than the absolute values of all right-hand sides satisfies them, as does every larger δ .

Suppose $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$ for such a δ , and form $\gamma = \gamma_\delta$, $g = g_\delta$, $\phi = \phi_\delta$, $N = N_\delta$ as in (14.1). Each inequality in (15.2) puts that low-index term strictly above the m th term in scaled valuation. Thus no index at most n_* is active, by the comparison in the proof of Lemma 14.5, and

$$N > n_*, \quad \gamma \leq v(a_m) - m\delta. \quad (15.5)$$

All derivatives of g lie in $\mathcal{S}_1^+$. Substituting $z = t^{-\delta}X$, a ring homomorphism $\mathbb{K}[[z]] \rightarrow \mathbb{K}[[X]]$, Lemma 14.4 turns the ν th summand of $P[f]$ into

$$c_\nu t^{d_\nu \gamma - w_\nu \delta} X^{k_\nu} \prod_j (g^{(j)})^{m_{\nu j}}.$$

Multiply the whole identity by $t^{-\lambda}$ with $\lambda = \beta_P + d_P\gamma - w_P\delta$. The scalar in front of the ν th product then has valuation

$$v(c_\nu) - \beta_P + (d_\nu - d_P)\gamma + (w_P - w_\nu)\delta. \quad (15.6)$$

For $\nu \in \mathcal{C}_P$ this is $v(c_\nu) - \beta_P \geq 0$. For $d_\nu = d_P$ and $w_\nu < w_P$ it is positive by (15.3). For $d_\nu < d_P$, multiplying the bound on γ in (15.5) by the negative integer $d_\nu - d_P$ reverses it and gives the lower bound

$$v(c_\nu) - \beta_P - (d_P - d_\nu)v(a_m) + ((d_P - d_\nu)m + w_P - w_\nu)\delta > 0$$

by (15.4). Every polynomial product of the $g^{(j)}$ has coefficients of nonnegative valuation. Multiplication by a positive-valuation scalar makes its residue zero, including when the product itself is zero. Thus the normalized identity lies in $\mathcal{S}_1^+$, and only corner summands can survive the residue map. Applying ρ to the formal identity gives

$$0 = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{k_\nu} \prod_j (\phi^{(j)})^{m_{\nu j}} = \Omega_\phi(X). \quad (15.7)$$

But $N > n_* \geq s$ and $\chi_P(N) \neq 0$, so by Lemma 14.6 the coefficient of $X^{d_P N + w_P}$ in Ω_ϕ is $\text{lc}(\phi)^{d_P} \chi_P(N) \neq 0$, a contradiction. The last assertion follows because a strongly entire series is evaluable at every monomial argument. $\square$

An explicit, division-free exclusion bound. Let $\mathcal{R}$ be the finite set of right-hand sides of (15.2)–(15.4); an empty list contributes nothing. For any $\epsilon \in \Gamma_{>0}$,

$$\delta_0 = \epsilon + \max(\{0\} \cup \{|b| : b \in \mathcal{R}\}) \quad (15.8)$$

satisfies all of them, since their multipliers are positive integers. This is an explicit certificate; it is not claimed to be optimal. The proof uses the full supports to form the initial polynomial but only finitely many leading valuations to choose δ_0 . Leading valuations alone do *not* in general decide strong evaluation at a boundary argument, and the certificate has no converse.

For $k = \mathbb{C}$, Lemma 2.4 (inward stability) turns the monomial conclusion into the exterior form used in Theorem A: a nonpolynomial formal solution is then not strongly evaluable at any $x \neq 0$ with $v(x) \leq -\delta_0$. This combination of the two statements is made by the merge; source 10 states only the monomial form.

Corollary 15.2 (Cofinal scales suffice). *If $\chi_P \neq 0$, a formal solution of $P[f] = 0$ strongly evaluable at $t^{-\delta}$ for a cofinal set of $\delta \in \Gamma_{>0}$ is a polynomial.*

Proof. If $d_P = 0$, then P is a nonzero polynomial in z alone and has no formal solution. Otherwise a cofinal subset meets the final segment excluded by Theorem 15.1 for a nonpolynomial solution. $\square$

Dependency audit. The only infinite-support step is Lemma 14.2, with the existence and finiteness of a leading coefficient layer; the rest is finite polynomial algebra. The proof makes none of the following replacements: a higher-rank group by a real-valued norm; strong summation by convergence of finite subsums; a nonsummable family by a cancelled expression; the residue polynomial by its value at 1; a nonlinear coefficient recurrence by a linear one. The three critical checkpoints are the minimum in (14.1), the order reversal when (15.5) is multiplied by $d_\nu - d_P < 0$, and the nonvanishing of $\chi_P(N)$.

15.2 Why the corner never degenerates in order one; proof of Theorem D

Write a first-order equation in combined-monomial form as

$$P(z, Y, Y') = \sum_{k,i,j} c_{kij} z^k Y^i (Y')^j. \quad (15.9)$$

For a term, $d = i + j$ and $w = k - j$. At the corner, fixing j forces $i = d_P - j$ and $k = w_P + j$, so there is at most one corner term for each j , and (14.6) becomes

$$\chi_P(T) = \sum_{\substack{i+j=d_P \\ k-j=w_P}} \bar{c}_{kij} T^j, \quad I_P(T) = \sum_{\substack{i+j=d_P \\ k-j=w_P}} c_{kij} T^j. \quad (15.10)$$

At least one $\bar{c}_{kij}$ is nonzero by the choice of β_P , and its power T^j cannot cancel against another summand. Hence $\chi_P \neq 0$.

Proof of Theorem D. If $d_P = 0$, then P is a nonzero polynomial in z alone and has no formal solutions. Otherwise the preceding observation and Theorem 15.1 give polynomiality and the exclusion statement, with δ_0 as in (15.8). A nonzero algebraic relation between f and $D_z f$ over $\mathbb{K}(z)$ becomes a nonzero polynomial of the form (15.9) after denominators are cleared. Multiplication by a common nonzero denominator preserves both the nonzero relation and its vanishing in $\mathbb{K}((z))$, even if that denominator vanishes at $z = 0$. Thus the polynomiality conclusion contradicts nonpolynomiality of f , proving the independence statement. $\square$

The uniqueness of a corner term for fixed j is a structural fact of order one. It is exactly what fails when several derivative orders can contribute the same falling-factorial polynomial (Section 16.2).

15.3 Finite candidate degrees and the entire-solution locus

Once polynomiality is known, ordinary polynomial degree gives a sharper finite candidate set that uses the *full* coefficients. Assume $d_P \geq 1$ and define the nonnegative rational number

$$\kappa_P = \max\left(\{0\} \cup \left\{ \frac{w_\nu - w_P}{d_P - d_\nu} : d_\nu < d_P \right\}\right) \in \mathbb{Q}. \quad (15.11)$$

This divides ordinary integers in $\mathbb{Q}$, not elements of Γ .

Theorem 15.3 (Finite candidate degrees). *For the first-order polynomial (15.9) with $d_P \geq 1$, every nonconstant strongly entire solution has degree in the finite set*

$$\mathcal{D}_P = \{n \in \mathbb{Z}_{>0} : n \leq \kappa_P\} \cup \{n \in \mathbb{Z}_{>0} : I_P(n) = 0\}. \quad (15.12)$$

The constant solutions are exactly the $c \in \mathbb{K}$ with $P(z, c, 0) = 0$ as a polynomial in z . The set (15.12) is necessary; it does not promise that each candidate occurs.

Proof. Polynomiality is Theorem D. Let $f = az^n + \dots$ with $n \geq 1$ and $a \neq 0$. A term of (15.9) has degree $k + in + j(n-1) = dn + w$ and leading coefficient $c_{kij} a^d n^j$. If $n > \kappa_P$, every term with $d < d_P$ has degree below $d_P n + w_P$, and so do the terms with $d = d_P$, $w < w_P$. So the coefficient of $z^{d_P n + w_P}$ in $P[f]$ is $a^{d_P} I_P(n)$, which must vanish. I_P is nonzero by (15.10), so its set of ordinary integer roots is finite. The assertion on constants is immediate by substitution. $\square$

Remark 15.4 (What “finite” does and does not compute). The set of ordinary integer roots of I_P is well defined and finite for arbitrary Hahn coefficients, but its existence is not a uniform algorithm for arbitrary coefficient names: extracting a leading exponent, deciding coefficient equality, or testing an arbitrary complex coefficient for exact integrality can need information that no numerical approximation supplies. For rational coefficients, or other exact representations with the needed operations, (15.12) is an actual finite algebraic search. The shipped program uses only exact rational data.

Replacing $I_P(n) = 0$ by $\chi_P(n) = 0$ gives a coarser necessary set, by (14.7). Keeping I_P can be genuinely sharper.

Example 15.5 (A resonance visible only in the first layer). Let $c = t^\eta$ with $\eta > 0$, $N \in \mathbb{Z}_{>0}$, and $P = zY' - (N + c)Y$. Then $\chi_P(T) = T - N$ but $I_P(T) = T - N - c$ has no ordinary nonnegative integer root. The only formal solution in $\mathbb{K}[[z]]$ is zero, since coefficient comparison gives $(n - N - c)a_n = 0$ for every n . A residue resonance alone is not a polynomial solution.

Corollary 15.6 (An algebraic entire-solution locus). *For a first-order equation P with $d_P \geq 1$ put $n_P = \max(\{0\} \cup \mathcal{D}_P)$. There is an explicitly defined affine algebraic set $\mathcal{V}_P \subseteq \mathbb{A}_{\mathbb{K}}^{n_P+1}$ whose $\mathbb{K}$ -points are in bijection with all strongly entire formal solutions of $P[f] = 0$. The same equations parametrize all strongly entire solutions over every Hahn-field extension $\mathbb{K} \hookrightarrow \mathbb{L} = k'((t^{\Gamma}))$ induced by an embedding of coefficient fields $k \hookrightarrow k'$ and an embedding of ordered groups $\Gamma \hookrightarrow \Gamma'$.*

Proof. With indeterminates $u_0, \dots, u_{n_P}$ put $F_u(z) = \sum_{n \leq n_P} u_n z^n$ and expand $P(z, F_u, F'_u) = \sum_{\ell} Q_{\ell}(u) z^{\ell}$ with $Q_{\ell} \in \mathbb{K}[u_0, \dots, u_{n_P}]$. Let $\mathcal{V}_P$ be the common zero locus of these finitely many Q_{ℓ} . Its points are exactly the polynomial solutions of degree at most n_P , each of which is strongly entire, and Theorem 15.3 shows there are no others. Under the stated extension the monomial degrees and κ_P are unchanged, and for an ordinary integer n the element $I_P(n)$ vanishes in $\mathbb{L}$ if and only if it vanishes in $\mathbb{K}$. Theorem D applies over $\mathbb{L}$, whose coefficient field again has characteristic zero and whose group is again nonzero, so the same degree bound and equations remain exhaustive there. $\square$

This is a finite-dimensional algebraic parametrization, not a claim that the number of solutions is finite, and not a moduli statement over coefficient rings with nilpotents. The extension class is exactly the one stated.

15.4 Worked first-order equations

All identities use the formal derivative D_z . For $k = \mathbb{C}$ they are statements about surcomplex Hahn workspaces; for $k = \mathbb{R}$ the real versions follow with the non-real constant roots omitted. Consider

$$D_z f = f^2 + p(z), \quad p \in \mathbb{K}[z]. \quad (15.13)$$

The corner is the term $-Y^2$, so $d_P = 2$, $w_P = 0$ and $I_P = -1$. If p has degree $e \geq 0$, then $\kappa_P = \max(0, e/2)$, so a nonconstant entire solution has degree at most $\lfloor e/2 \rfloor$. More precisely, a solution of degree $n \geq 1$ requires $\deg p = 2n$ and the leading coefficient of p to be minus the square of that of f , since the top term of f^2 cannot be cancelled by $D_z f$. These conditions are necessary only; the remaining coefficient equations must still hold.

Proposition 15.7 (Two exact Riccati classifications). *Over $\mathbb{K} = \mathbb{C}((t^{\Gamma}))$, the strongly entire solutions of $D_z f = f^2 + 1$ are exactly $f = i$ and $f = -i$, and the only strongly entire solution of $D_z f = f^2 + 1 - z^2$ is $f = z$.*

Proof. In the first equation the degree bound excludes nonconstant solutions, and a constant must satisfy $c^2 + 1 = 0$; in the field $\mathbb{K}$, which contains i , the roots are $\pm i$. In the second the degree bound is one and constants are impossible. For $f = az + b$ coefficient comparison gives $a^2 = 1$, $2ab = 0$ and $a = b^2 + 1$, so $b = 0$ and $a = 1$, and $f = z$ is verified directly. $\square$

The first equation is also `diff:cor:riccati` of the collection's differential-equations report [13], with the same two solutions, but there $\partial u = 1 + u^2$ is solved in $\mathbf{No}[i]$ for a derivation on surreal *scalars*. The two statements concern different derivations and different objects, and neither implies the other.

Example 15.8 (The degree cutoff is attained). For every integer $m \geq 1$ the equation $D_z f = f^2 + mz^{m-1} - z^{2m}$ has the strongly entire solution $f = z^m$. Here $I_P = -1$ and $\kappa_P = m$ exactly, so the low-degree part of (15.12) cannot in general be replaced by $n < \kappa_P$.

Example 15.9 (Arbitrarily large resonant degrees). For $N \geq 1$ the equation

$$zf D_z f - Nf^2 = 0 \quad (15.14)$$

has $d_P = 2$, $w_P = 0$ and $I_P(T) = T - N$. Since $\mathbb{K}[[z]]$ is an integral domain, a nonzero solution satisfies $zD_z f = Nf$, and coefficient comparison gives exactly $f = cz^N$ with $c \in \mathbb{K}$ (including $c = 0$). The exceptional integer root is an actual degree, and no degree bound can depend only on the positions of the monomials of P : the coefficient N matters.

15.5 The exclusion bound cannot be uniform

Theorem A has an exclusion bound depending only on the operator. For nonlinear equations this is false, already for one fixed equation of first order.

Theorem 15.10 (A fixed Riccati equation defeats every uniform bound). *For $\lambda \in \Gamma$ let $c = t^\lambda$ and*

$$f_\lambda(z) = \frac{c}{1 - cz} = \sum_{n \geq 0} c^{n+1} z^n \in \mathbb{K}[[z]].$$

Then $D_z f_\lambda = f_\lambda^2$, f_λ is nonpolynomial, and

$$\text{Dom}_{\text{str}}(f_\lambda) = \{0\} \cup \{x \in \mathbb{K}^\times : v(x) + \lambda > 0\}. \quad (15.15)$$

Consequently, for every $\delta \in \Gamma$ some nonpolynomial solution of the fixed equation $D_z f = f^2$ is strongly evaluable at $t^{-\delta}$, and no exclusion bound depending on the equation alone excludes all its nonpolynomial solutions.

Proof. The derivative identity is a coefficient comparison. For $x \neq 0$ put $u = cx$, so the n th evaluation term is cu^n . If $v(u) > 0$, write $u = bt^{v(u)}(1 + w)$ with $b \in k^\times$ and w zero or of positive valuation; then $\text{supp}(u^n) \subseteq nv(u) + M(\text{supp } w)$, the leading exponents $nv(u)$ increase strictly, and Lemma 2.3 (b), resting on the positive-support monoid of Lemma 2.1 (2), gives strong summability; multiplication by c preserves it. (Source 10 cites the Hahn–Neumann positive-support lemma here instead of proving it; no Archimedean convergence of $nv(u)$ is used.) If $v(u) = 0$, every cu^n has the nonzero coefficient $\text{lc}(u)^n$ at exponent λ , so local finiteness fails. If $v(u) < 0$, the leading exponents $\lambda + nv(u)$ strictly decrease. At $x = 0$ only one member is nonzero. This proves (15.15). Given δ , choose $\lambda = \delta + \epsilon$ with any $\epsilon \in \Gamma_{>0}$, available because $\Gamma \neq 0$; no order unit is required. Then $v(t^{-\delta}) + \lambda = \lambda - \delta = \epsilon > 0$. $\square$

There is no conflict with Theorem 15.1, whose finite data, and hence whose bound, may depend on the solution; for f_λ the exact boundary moves with λ . At the boundary $v(x) = -\lambda$ geometric-series algebra cannot legalize infinitely many contributions at one exponent. Nor is there a conflict with Theorem A: each f_λ also satisfies the linear equation $(1 - t^\lambda z)D_z f - t^\lambda f = 0$, whose operator depends on λ , exactly as the operator $D_z - t^\eta$ of the formal exponential of Proposition 5.3 depends on η . (This comparison is the merge's; it is a direct computation.)

16 Higher order and positive-weight Euler equations

16.1 The nondegenerate higher-order class

Theorem 16.1 (Higher-order corner criterion and its boundary). *Let P be a nonzero differential polynomial (14.4) of any finite order with $d_P \geq 1$ and $\chi_P \neq 0$. Then every strongly entire formal solution of $P[f] = 0$ is a polynomial, and every nonpolynomial formal solution fails strong evaluation at $t^{-\delta}$ for all δ in a final segment of $\Gamma_{>0}$, with the finite certificate of Theorem 15.1. The condition $\chi_P \neq 0$ holds automatically in order one, but not in order two: a fixed second-order equation can have polynomial solutions of every degree (Proposition 16.4).*

Proof. The first two assertions are Theorem 15.1; the third is Section 15.2 and Proposition 16.4. $\square$

Proposition 16.2 (Higher-order degree candidates). *Suppose $d_P \geq 1$ and $\chi_P \neq 0$ for an equation of order at most s . Every nonzero strongly entire solution has degree in the finite set*

$$\{0, \dots, s-1\} \cup \{n \in \mathbb{N} : n \geq s, n \leq \kappa_P\} \cup \{n \in \mathbb{N} : n \geq s, I_P(n) = 0\},$$

with κ_P as in (15.11). When $s = 0$ the first set is empty. Constant solutions can always be checked by substitution.

Proof. Polynomiality is Theorem 15.1. For degree $n \geq s$ every derivative leading-term formula of Lemma 14.6 applies, and if also $n > \kappa_P$ the coefficient of $z^{d_P n + w_P}$ in $P[f]$ is $\text{lc}(f)^{d_P} I_P(n)$, which must vanish; $I_P \neq 0$ because $\chi_P \neq 0$. Degrees below s , where derivative leading terms can vanish, are listed separately. $\square$

Proposition 16.3 (The first two Painlevé polynomial equations). *In characteristic zero, the equation $D_z^2 f = 6f^2 + z$ has no strongly entire formal solution, and the equation $D_z^2 f = 2f^3 + zf + \alpha$ with $\alpha \in \mathbb{K}$ has a strongly entire solution exactly when $\alpha = 0$, the only one then being $f = 0$.*

Proof. The first equation has $d_P = 2$, $w_P = 0$, $\chi_P = -6$; the second has $d_P = 3$, $w_P = 0$, $\chi_P = -2$. Theorem 15.1 makes every entire solution a polynomial. In the first equation a polynomial of degree $n \geq 1$ gives f^2 of degree $2n$, larger than $\deg D_z^2 f$ (when nonzero) and than $\deg z = 1$, so no cancellation is possible, and a constant cannot cancel z . In the second, for $n \geq 1$ one has $3n > n + 1$ and $3n > \deg D_z^2 f$, so $2f^3$ cannot cancel; a constant c must satisfy $2c^3 + zc + \alpha = 0$ identically, which forces $c = 0$ and $\alpha = 0$, and substitution proves sufficiency. $\square$

The names only identify the displayed equations; no classification of their classical meromorphic solutions is imported. The examples illustrate a class beyond first order: a unique term of greatest total differential degree has a nonzero corner whenever its falling-factorial product is nonzero. In particular an equation $D_z^s f = Q(z, f)$ with Q of degree at least two in f has a nonzero corner and hence only polynomial strongly entire solutions; the coefficient of the top power of f may be a nonconstant polynomial in z , whose highest z -term then selects the corner.

16.2 A vanishing corner

Consider

$$z^2 f D_z^2 f + z f D_z f - z^2 (D_z f)^2 = 0. \quad (16.1)$$

All three terms have $d = 2$ and $w = 0$, and the residue corner polynomial is

$$T(T - 1) + T - T^2 = 0, \quad (16.2)$$

an identity in $k[T]$, not an exceptional integer resonance.

Proposition 16.4 (Complete formal classification of the example). *The formal solutions of (16.1) in $\mathbb{K}[[z]]$ are exactly 0 and cz^n with $c \in \mathbb{K}^\times$ and $n \in \mathbb{N}$. All are strongly entire, and their degrees are unbounded for this single equation.*

Proof. With $\vartheta = zD_z$, equation (16.1) reads $f \vartheta^2 f - (\vartheta f)^2 = 0$. For $f \neq 0$ work in the Laurent series field $\mathbb{K}((z))$ and divide by f^2 : $\vartheta(\vartheta f/f) = 0$. The constants of ϑ in $\mathbb{K}((z))$ are exactly $\mathbb{K}$, since $\vartheta \sum b_n z^n = \sum n b_n z^n$ and characteristic zero kills no nonzero integer. Hence $\vartheta f/f = c_0 \in \mathbb{K}$; if the first nonzero term of f has exponent n , comparing lowest terms gives $c_0 = n$. Then $\vartheta f = n f$ kills every coefficient except the n th. Substitution verifies the monomials and zero. $\square$

The example has two messages. A universal higher-order degree bound cannot be obtained by declaring χ_P nonzero: it may vanish identically, and a fixed equation can then have unbounded degrees. And the example is *not* a nonpolynomial entire solution, so it does not refute polynomial rigidity in higher order; it locates a failure of this proof and of a stronger uniform conclusion.

The corner is also not an invariant of the solution set. Multiplying an equation by another differential polynomial, or passing to a differential consequence, can change the corner data. For a concrete example put

$$Q = z^2 Y_0 Y_2 + z Y_0 Y_1 - z^2 Y_1^2, \quad P = Y_1, \quad \tilde{P} = Y_1(1 + zQ).$$

For every formal f , the factor $1 + zQ[f]$ has constant coefficient one and is a unit of $\mathbb{K}[[z]]$. Hence $P[f] = 0$ and $\tilde{P}[f] = 0$ have exactly the same solutions, the constants. But $\chi_P(T) = T$, whereas the degree-three corner of $\tilde{P}$ gives $\chi_{\tilde{P}}(T) = T(T(T - 1) + T - T^2) = 0$. The criterion $\chi_P \neq 0$ is a sufficient condition on the displayed combined polynomial: when $\chi_P = 0$ another equation satisfied by the same function may have a nonzero corner, or a logarithmic-derivative reduction as in Proposition 16.4 may be available. A zero corner cannot be repaired by retaining only monomial degrees, since the residue coefficients and their exact cancellation are essential. The methods of this section do not decide whether a higher-order equation with vanishing residue corner has nonpolynomial strongly entire solutions. When Γ has no order unit, Theorem E (sources 11 and 12) decides it negatively in every order; Example 21.6 computes the nonzero linearization multiplier of (16.1) along its solutions. With an order unit the answer is different: the cleared third-order equation of source 13 has residue corner $\chi_P = 0$ and the nonpolynomial strongly entire solution $\mathcal{T}_p$ (Theorem H); its full corner polynomial I_P is a nonzero constant of positive valuation, so the hypothesis $\chi_P \neq 0$ of Theorems 15.1 and 16.1 cannot be weakened to $I_P \neq 0$ (Proposition 24.2 and Remark 24.3). In order two the vanishing-corner case is decided negatively for every Γ by source 14 (Theorem I (a)): its endpoint lemma uses the whole initial polynomial at infinitely many scales, and for (16.1) it permits no nonmonomial initial polynomial at all (Example 26.7).

16.3 Positive-weight Euler equations in several variables

Fix positive integers $q = (q_1, \dots, q_m)$, write $\text{wt}(\alpha) = q \cdot \alpha$ for $\alpha \in \mathbb{N}^m$, and let

$$\vartheta_q = \sum_{\nu=1}^m q_\nu x_\nu \frac{\partial}{\partial x_\nu}, \quad \vartheta_q x^\alpha = \text{wt}(\alpha) x^\alpha.$$

For $N \geq 0$, a multi-index of weight at most N satisfies $0 \leq \alpha_\nu \leq \lfloor N/q_\nu \rfloor$ in each coordinate. Thus at most $\prod_\nu (1 + \lfloor N/q_\nu \rfloor)$ multi-indices have that bound. This finiteness, with the eigenvalue identity, is the essential feature of positive weights. Write a nonzero equation in combined-monomial form

$$P(x, Y, Z) = \sum_{\nu \in J} c_\nu x^{\alpha_\nu} Y^{i_\nu} Z^{j_\nu}, \quad (16.3)$$

to be read as $P(x, f, \vartheta_q f) = 0$, and set $d_\nu = i_\nu + j_\nu$, $w_\nu = \text{wt}(\alpha_\nu)$, with d_P , w_P , $\mathcal{C}_P$, β_P and $\bar{c}_\nu$ as in Section 14.4. The corner polynomials now keep the spatial variables:

$$\chi(T, X) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{\alpha_\nu} T^{j_\nu}, \quad I(T, X) = \sum_{\nu \in \mathcal{C}_P} c_\nu X^{\alpha_\nu} T^{j_\nu}. \quad (16.4)$$

Both are nonzero: for fixed (α_ν, j_ν) the condition $d_\nu = d_P$ forces $i_\nu = d_P - j_\nu$, so no two combined corner terms give the same monomial in (X, T) . Only finitely many integers n make $\chi(n, X)$ the zero polynomial, since they must all be roots of one nonzero X -coefficient of χ , and likewise for I .

Theorem 16.5 (Positive-weight Euler rigidity). *Let $f \in \mathbb{K}[[x_1, \dots, x_m]]$ satisfy $P(x, f, \vartheta_q f) = 0$ for a nonzero $P \in \mathbb{K}[x_1, \dots, x_m, Y, Z]$. If f is nonpolynomial, it fails joint strong evaluation at $(t^{-q_1\delta}, \dots, t^{-q_m\delta})$ for every sufficiently large $\delta \in \Gamma_{>0}$. Consequently, if f is strongly entire, or merely jointly strongly evaluable at these points for a cofinal set of $\delta \in \Gamma_{>0}$, then f is a polynomial. Every fixed equation has a finite set of possible positive weighted degrees (Corollary 16.6).*

Proof. If $d_P = 0$, then P is a nonzero polynomial in x alone and there is no solution; assume $d_P \geq 1$. Choose $n_* \geq 0$ with $\chi(n, X) \neq 0$ for every integer $n > n_*$. Since $f = \sum_\alpha a_\alpha x^\alpha$ is nonpolynomial and bounded-weight sets are finite, choose $a_\mu \neq 0$ of weight $n_\mu = \text{wt}(\mu) > n_*$ so large that $(d_P - d_\nu)n_\mu + w_P - w_\nu > 0$ for $d_\nu < d_P$. Take $\delta > 0$ satisfying the finitely many inequalities

$$\begin{aligned} (n_\mu - \text{wt}(\alpha))\delta &> v(a_\mu) - v(a_\alpha) & (\text{wt}(\alpha) \leq n_*, a_\alpha \neq 0), \\ (w_P - w_\nu)\delta &> \beta_P - v(c_\nu) & (d_\nu = d_P, w_\nu < w_P), \\ ((d_P - d_\nu)n_\mu + w_P - w_\nu)\delta &> \beta_P - v(c_\nu) + (d_P - d_\nu)v(a_\mu) & (d_\nu < d_P); \end{aligned}$$

as before they can be met in every nonzero ordered group and persist on a final segment. Suppose the joint family is strongly summable at such a scale, and put

$$F(X) = f(t^{-q_1\delta} X_1, \dots, t^{-q_m\delta} X_m) \in \mathcal{S}_m, \quad \gamma = \gamma(F), \quad g = t^{-\gamma} F, \quad \phi = \rho(g).$$

Then $\gamma \leq v(a_\mu) - n_\mu\delta$, and the first inequalities show that ϕ has no monomial of weight at most n_* ; its greatest weight N exceeds n_* . Let $\phi_N \neq 0$ be its weight- N homogeneous part. The Euler operator commutes with the scaling, so $(\vartheta_q f)(t^{-q_1\delta} X_1, \dots, t^{-q_m\delta} X_m) = t^\gamma \vartheta_q g$. Normalizing the substituted equation by $t^{-\beta_P - d_P\gamma + w_P\delta}$, the scalar computation (15.6) shows that only corner terms survive the residue, whence

$$0 = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{\alpha_\nu} \phi^{i_\nu} (\vartheta_q \phi)^{j_\nu}. \quad (16.5)$$

The largest possible weight on the right is $d_P N + w_P$. Since the weight- N part of $\vartheta_q \phi$ is $N\phi_N$, the homogeneous part of that weight is

$$\phi_N(X)^{d_P} \chi(N, X), \quad (16.6)$$

a product of two nonzero polynomials, hence nonzero. This contradicts (16.5). $\square$

The proof uses *joint* strong evaluation, not a scalar expression obtained by first setting $x_1 = \dots = x_m$ and cancelling: a factor $x_1 - x_2$ can make a diagonal scalar expression zero while the ungrouped monomial family is not summable. Keeping $X_1, \dots, X_m$ in (16.6) avoids that error.

Corollary 16.6 (Weighted-degree candidates). *With κ_P defined by (15.11) from the Euler data d_ν, w_ν , every nonconstant strongly entire solution of $P(x, f, \vartheta_q f) = 0$ has greatest weight in the finite set*

$$\{n \in \mathbb{Z}_{>0} : n \leq \kappa_P\} \cup \{n \in \mathbb{Z}_{>0} : I(n, X) \equiv 0\}.$$

For a fixed equation all strongly entire solutions are therefore described by finitely many polynomial equations in the coefficients of a bounded-weight polynomial ansatz.

Proof. Polynomiality is Theorem 16.5. For a polynomial of greatest weight $N > \kappa_P$, the terms of lower differential degree have smaller total weight, and the top-weight part of the equation is $f_N^{d_P} I(N, X)$, which must vanish; so $I(N, X) = 0$. The finite-root argument after (16.4) applies to I , and a bounded-weight ansatz has finitely many coefficients. $\square$

Example 16.7 (An exactly described solution space). For $N \geq 0$ the solutions of $\vartheta_q f = Nf$ in $\mathbb{K}[[x]]$ are exactly the weighted homogeneous polynomials of weight N , since coefficient comparison gives $(\text{wt}(\alpha) - N)a_\alpha = 0$. For $q = (1, 2)$ this space has dimension $\lfloor N/2 \rfloor + 1$ over $\mathbb{K}$, with basis $x_1^{N-2j} x_2^j$, $0 \leq j \leq \lfloor N/2 \rfloor$. The nonlinear equation $f(\vartheta_q f - Nf) = 0$ has the same solutions, because $\mathbb{K}[[x]]$ is an integral domain.

Why positive weights are needed. Let $\mathbb{K} = \mathbb{C}((t^{\mathbb{Q}}))$ and $G(u) = \sum_{n \geq 0} t^{n^2} u^n$, which is strongly entire (Corollary 17.1). The function $f(x, y) = G(y)$ is nonpolynomial and strongly entire but satisfies $x\partial_x f = 0$, so a zero weight invalidates the theorem. For the mixed weights $(1, -1)$, the nonpolynomial strongly entire function $f(x, y) = G(xy)$ satisfies $(x\partial_x - y\partial_y)f = 0$; at a fixed point its nonzero joint evaluation terms are exactly $t^{n^2} (xy)^n$, which are strongly summable. Positivity is therefore not a cosmetic substitute for nonzero weights, and a general first-order partial differential equation need not have the rigidity of a positive Euler operator.

Relation to Theorem C. Theorem C assumes that *all* mixed partial derivatives span a finite-dimensional space, and Section 6 notes that a single differential equation in several variables cannot force polynomiality in general, since it may ignore variables; the zero-weight example above is an instance. Theorem 16.5 instead assumes one algebraic relation between f and $\vartheta_q f$ with all weights positive, with no finite-dimensional-span hypothesis. These are different input criteria and different proof routes. Within the strongly entire class, each criterion forces a polynomial, which then satisfies both criteria: its mixed derivatives span a finite-dimensional space, and the relation $Y - f(x) = 0$ is an algebraic Euler relation.

17 Nonlinear consequences, hypotheses and predecessors

17.1 Two-jet independence of explicit series

With the surreal embedding of Section 2.4, for instance $\mathbb{C}((\omega^{-\mathbb{Q}}))$ is a set-sized surcomplex workspace to which Theorem D applies, and its coefficients and admissible arguments are actual elements of $\mathbf{No}[i]$. The independence conclusions, however, concern *formal functions*: each value $f(x)$ lies in $\mathbb{K}$ and is of course not transcendental over $\mathbb{K}$.

Corollary 17.1 (Two independent jets of the quadratic-exponent series). *Over $\mathbb{K} = \mathbb{C}((t^{\mathbb{Q}}))$ let $G(z) = \sum_{n \geq 0} t^{n^2} z^n$. Then G is nonpolynomial and strongly entire, and G and $D_z G$ are algebraically independent over $\mathbb{K}(z)$.*

Proof. For $x \neq 0$ with $v(x) = \sigma \in \mathbb{Q}$ the n th evaluated member has valuation $n^2 + n\sigma$, which eventually exceeds every rational bound, so Lemma 2.3 (a) gives strong summability even when x has an arbitrary infinite Hahn tail; at $x = 0$ the family is finite. All coefficients are nonzero. Theorem D gives the independence. $\square$

The series is not new: it is the rank-one example of the rank-one-Berkovich report [14], Example `found:ex:internal` of the foundations report [15], and a rescaled partial theta series: $G(z) = \Theta_{t^2}(tz)$ in the notation of (8.1). The contribution of source 10 is the nonlinear two-jet consequence only.

Corollary 17.2 (An example without an order unit). *In K_{Γ_∞} , with Γ_∞ as in (11.1), the strongly entire series $F = \sum_{n \geq 0} t^{e_n} z^n = \sum_{n \geq 0} \omega^{-\omega^n} z^n$ of (11.2) satisfies: F and $D_z F$ are algebraically independent over $K_{\Gamma_\infty}(z)$.*

Proof. Strong entireness is Theorem 11.2; source 10 proves it again by the cofinal-valuation route recorded in the second paragraph of that proof. Apply Theorem D. $\square$

F and its linear rigidity are in Theorem 11.2; the new statement is the two-jet strengthening. It answers the first-order case of the last sentence of Question 19.1 as posed by sources 08 and 09 (recorded in its status paragraph) with no coefficient restriction, and says nothing about relations of order two or more; Theorem 22.1 (source 12) answers that sentence in every order, again with no coefficient restriction.

Extension of scalars versus preservation of entireness. The degree classification is stable under the Hahn-field extensions of Corollary 15.6, but entireness of a nonpolynomial series need not be. Enlarge $\mathbb{Q}$ to $\mathbb{Q} + \mathbb{Q}\omega$ with its surreal order and view G over the larger field. At $z = t^{-\omega}$ the member exponents are $n^2 - n\omega$, with successive differences $(2n + 1) - \omega < 0$, so the support union contains an infinite descending sequence and the evaluation is not strong. The same computation appears in the rank-one-Berkovich report [14], and for F in Section 11.2. No theorem here claims that G or F is entire on the proper class $\mathbf{No}[i]$; hypotheses and derivative are relative to a specified set-sized workspace.

17.2 Which hypotheses are used

Example 17.3 (Positive characteristic destroys the conclusion). Let p be prime and $\mathbb{K} = \mathbb{F}_p((t^{\mathbb{Q}}))$. The nonpolynomial series $U(z) = \sum_{n \geq 0} t^{n^2} z^{pn}$ is strongly entire: at a point of valuation σ its

member valuations are $n^2 + pn\sigma$, and Lemma 2.3 (a) applies, its proof being independent of the coefficient field. Yet $D_z U = 0$.

So the conclusion fails in positive characteristic even for one linear equation. The characteristic-zero requirements of the proof are substantive: ordinary integer eigenvalues stay distinct, nonzero falling factorials stay nonzero, and a nonzero polynomial cannot vanish at all ordinary integers.

The hypothesis $\Gamma \neq 0$ is needed for the scale argument; the degenerate case $\Gamma = 0$ is treated in Section 12. Under ordinary complex analytic summation, by contrast, $\exp z$ is a nonpolynomial entire function with $f' = f$; these are different notions of infinite sum, and the classical exponential does not contradict the Hahn theorem. The coefficient field may be $\mathbb{R}$, $\mathbb{C}$ or any field of characteristic zero; neither algebraic nor real closedness is used, no completeness of a real-valued norm is used, and Γ needs neither an order unit nor a countable cofinal subset. Strong summability supplies exactly the finite initial layer the argument needs.

17.3 An exact finite workflow

For first-order input P the reduction is as follows. Combine like monomials and remove zero coefficients. If no Y or Y' remains, decide the polynomial identity directly. Otherwise compute d_P , w_P , I_P and κ_P from the finite term list, find the positive integer roots of I_P in the chosen exact representation of the coefficients, take the largest candidate degree, substitute a generic polynomial of that degree, and solve the resulting finite algebraic system. This procedure is complete for *entire* solutions whenever its finite coefficient operations and algebraic solving are available. It is not a general algorithm for arbitrary surreal names, it does not enumerate the local formal solutions, and it does not decide arbitrary strong-support membership. A negative answer after an exhaustive exact solve is a genuine nonexistence certificate for entire solutions; failure of a numerical solver is not.

For a nonpolynomial candidate formal solution a second workflow builds the certificate (15.2)–(15.4) from one sufficiently high nonzero coefficient and finitely many low-index ones; a claimed certificate must supply their exact leading valuations. The shipped program checks finite algebraic identities and sample certificates; it does not decide infinitary summability.

17.4 Predecessors, repository comparison and priority

Inherited ideas. Hahn fields, the positive-support summation lemma, Conway normal forms, Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning are established ideas and are not relabelled as new. The organizing method is related in spirit to maximum-term and central-index arguments in the Wiman–Valiron tradition, surveyed by Hayman [18]; here the finite residue polynomial gives an exact identity rather than an analytic approximation, and no real-valued growth estimate is imported. Source 10 consulted Hayman’s survey through its publisher metadata and abstract only, and uses no theorem from its full text.

Newton-polygon methods for nonlinear equations have a substantial history. Cano and Fortuny Ayuso construct initial and indicial polynomials for nonlinear q -difference equations and obtain support and growth results for generalized-series solutions [19]. Their operator and setting are not those of a formal z -series with arbitrary-rank Hahn *coefficients* evaluated strongly at every workspace scale. The conceptual precedent is important, and no invention of differential Newton polygons is claimed.

Non-Archimedean value distribution already restricts nonlinear functional and differential equations. Hu and Luan work over a complete algebraically closed field of characteristic zero with a nontrivial non-Archimedean absolute value and prove Clunie-, Malmquist- and Mokhon’ko-type results [20]; source 10 consulted the parsed text of that paper for its field hypotheses and Theorems 1.1, 1.3 and 1.5, reports that a remote screenshot service failed so that no visual inspection of the PDF is claimed, and reproduces no figure or table from it. Hu and Yang’s monograph [21] is a broader predecessor; only its publisher metadata and description were checked, so no claim is made that all rank-one special cases were previously unknown. Giansiracusa and Mereta develop differential enhancements of non-Archimedean norms and a tropicalization of differential equations [22]; the finite corner here records only the data this obstruction needs, and no equivalence to their framework and no new tropical fundamental theorem is asserted. Mantova and Matusinski’s survey [17] was consulted for normal forms.

Repository comparison at source 10’s pin. Source 10 compared its results with the collection at 048b72c; the comparison, with the corrections needed now, is as follows.

Material	Existing result or scope	Difference in the nonlinear part
This report, Theorem A	Strongly entire D-finite series are polynomials; the nonlinear case was the open Question 15.1, now Question 19.1.	All first-order algebraic equations, a higher-order corner criterion, and solution-dependent finite certificates.
This report, Theorem C	Finite-dimensional span of all mixed derivatives forces polynomiality.	One algebraic relation in f and $\vartheta_q f$ suffices for positive weights; no finite-dimensional span.
Tail-spans report [16]	Particular Galois-supported constructions have all-order differential independence, for scalar or fixed-domain analytic derivations over specified smaller base fields.	A two-jet theorem for every nonpolynomial strongly entire function over its own full Hahn field.
Documentation catalogue docs/README.md	Distinguishes workspaces, strong summability, coefficient rings and scalar derivations.	Those distinctions are kept; no global proper-class analytic claim is added.

The tail-span examples are neither weakened nor rediscovered: that report proves stronger all-order independence for particular constructions under different coefficient and function-class hypotheses, while the present universal statement assumes strong entireness and concludes only two-jet independence. (The coefficient-field part, from sources 11 and 12, now concludes all-order independence for groups without an order unit; its comparison with the tail-span report is in Section 22.2.) Source 10 cited the quadratic-exponent series as occurring in this report and in the catalogue; within the collection it occurs as the rank-one-Berkovich example [14], as `found:ex:internal` [15], and as a case of (8.1). This is the targeted comparison recorded at the pin, not an absence claim about every report in the current collection. The Riccati statement `diff:cor:riccati` of the differential-equations report concerns a scalar derivation (compare Proposition 15.7).

Source 10’s repository audit covered the root README, the catalogue, contents listings, this report’s definitions, theorems and nonlinear-question passages, and the README files of the tail-span, analytic-geometry and product-birthday reports (the last two for context only). It was a targeted comparison, not a line-by-line search of every report, Lean file, archive, branch or the Git history, and no absence claim rests on a truncated preview.

Priority and proof status. Source 10’s literature search, made on 22 September 2026, did not locate the exact all-rank strong-family first-order theorem, the finite exclusion certificate, or the positive-weight Euler formulation. Some searches were poorly targeted, and no exhaustive MathSciNet, zbMATH, full-book or citation-network review was made, so priority is provisional. The safest description is a proposed new theorem package and a proved first-order answer to a repository question, which is not a named historical conjecture; it is not a certified first discovery of nonlinear non-Archimedean rigidity. The written proofs are the evidence. The finite checks of Section 18 establish no arbitrary-support theorem; the main nonlinear rigidity theorems remain unmapped in the current implementation ledger. No independent human referee report accompanies the delivered source package.

18 The seven finite check suites

Each source manuscript shipped an independent exact-arithmetic suite. The suites of sources 08 and 09 were run on copies and passed, and neither supersedes the other: they cover overlapping but distinct finite facts of the linear theory, and they are kept side by side in `code/` for that reason. The suite of source 10 checks finite facts of the nonlinear part only; it was also run on a copy for this merge and reproduced its recorded counts exactly. The suites of sources 11 and 12 check finite facts of the coefficient-field part; both were run on copies for this merge and reproduced their records exactly, apart from the recorded interpreter and package versions. The suite of source 13 checks finite facts of the theta-descent part; it was run on a copy for this merge and reproduced its record apart from the interpreter version and the elapsed time. The suite of source 14 checks finite facts of the order-three-threshold part; it was run on a copy for this merge and reproduced its record apart from the interpreter version and the elapsed time. The suites of sources 08, 09, 10 and 12 use only the Python standard library, integers, and `fractions.Fraction`; the suites of sources 11, 13 and 14 use exact symbolic and rational arithmetic in SymPy, the only third-party package any suite needs. None uses floating-point mathematics or network access; those of sources 13 and 14 record their elapsed time.

code/08-finite-recurrences-verify.py — 3,705 checks, 0 failures. It emphasises the operator-to-recurrence conversions in all three formats and the higher-rank ordered-group arithmetic.

Finite calculation	Checks
Differential operator to coefficient recurrence	648
Dilation operator to coefficient recurrence	336
Mixed operator to coefficient recurrence	276
Unit residues, parity, and finite exceptional cancellation	400
Partial theta identities and finite quadratic estimates	552
Higher-rank lexicographic valuation examples	1,292
Sparse Airy recurrence and finite escape edges	201
Total	3,705

The operator tests compare the grouped recurrence formulas against direct finite polynomial differentiation, dilation and multiplication; the theta tests compare Laurent monomial exponents exactly, including the q^{-1} -to- q conversion of (8.7); the unit tests retain the exceptional index $n = 7$ rather than silently checking only generic values; and the ordered-group checks use lexicographic

integer tuples, not a real-valued surrogate norm. This script takes a *required* `--output` path and exits nonzero without one, which is deliberate: it prevents a re-run from overwriting the delivered record.

code/09-entire-hahn-holonomic-verification.py — 2,043 checks, 0 failures. It emphasises the boundary behaviour of the exclusion bounds and the exact theta domain.

Finite calculation	Checks
Theta homogeneous equation, positive and negative valuation	92
Theta first-order coefficient recurrence	45
Dilation and differential forward recurrence conversions	72
Unit orbit: parity, exceptional cancellation, residue polynomial	504
Rank-two noncofinal affine minimum	200
Theta excluded upper-scale argument	200
Theta included coarse boundary	170
Theta cofinal-scale growth	120
Finite-coordinate entire witness growth	480
Sparse recurrence exact identity	50
Escape boundary: constant leading exponent, strict exterior descent	50
Formal exponential recurrence	60
Total	2,043

Two of these rows exist only in this suite and check the two boundary phenomena that the report insists are different: “theta included coarse boundary” checks that a zero coarse valuation coexists with positive growth in the original lower-scale coordinate, which is Section 8.5, and “escape boundary” checks the constant-leading-exponent case that Lemma 2.2 rules out, which is what makes the excluded region in Corollary 3.5 closed. Its `--output` argument is optional. Seeds are recorded in the shipped records.

code/10-nonlinear-rigidity-verify.py — 2,113 checks, 0 failures. It checks the finite algebra of Sections 14–17.

Finite calculation	Checks
Finite Hahn initial layers and exclusion inequalities	450
Automatic first-order nondegeneracy of the corner	240
Univariate falling-factorial corner identities, orders 0 to 3	480
Lexicographic higher-rank ordered-group exclusion bounds	500
Weighted Euler homogeneous-part identities	360
Worked examples: Riccati, degree cutoff, resonance, Painlevé, vanishing corner, Euler solution space	83
Total	2,113

The group names in the record `data/10-nonlinear-rigidity-verification.json` are `finite_hahn_certificate`, `first_order_nondegeneracy`, `higher_order_corner`, `ordered_group_radi`
`us`, `weighted_euler` and `worked_examples`, in the order of the table; “radius” there is source 10’s

word for an exclusion bound. The default seed is 20260922. The `--output` argument is required, and the program refuses to overwrite an existing file. The vanishing-corner example is tested for the monomial degrees $0, \dots, 20$ only. The counts are program checks, not independently proved theorems.

code/11-coarsening-differential-rigidity-verify.py — 8,673 cases in 8 groups, 0 failures. It checks finite facts of source 11’s material and needs SymPy (data/11-coarsening-differential-rigidity-requirements.txt pins `sympy` $\geq 1.12, < 2$; the record was made with Python 3.13.5 and SymPy 1.14.0, the merge’s rerun with Python 3.14.4 and SymPy 1.14.0).

Group in data/11-coarsening-differential-rigidity-verification.json	Cases
singular_tail_coefficient_isolation: Taylor identity (21.4), orders 1–3	12
vanishing_corner_indicial_polynomial: (21.6) for $0 \leq m \leq 9$	10
painleve_local_formal_solutions: 24 coefficient equations, three initial conditions	72
common_factor_reduction_models: numerator and denominator reduction	3
binary_prefix_coding: code ranges and all pairs of length-seven branches	8,136
finite_reverse_lexicographic_tests: growth and loss after an exterior scale	434
generator_value_cancellation: Remark 20.8	1
boundary_equations: rational Riccati solution, characteristic p	5
Total	8,673

Most of the count is prefix-pair comparisons; it is not a count of independent theorems. The `--output` argument is optional and defaults to `verification.json` in the current directory, so a bare run can overwrite a file of that name; use an explicit new path.

code/12-coefficient-field-rigidity-verify.py — 9,308 checks, 0 failures. It checks finite facts of source 12’s material with the standard library only.

Category in data/12-coefficient-field-rigidity-verification.json	Checks
coefficient recovery; formal solution coefficients	387; 637
linearized recurrence identity; mixed derivative multiplier scaling	389; 59
derivative-before-dilation convention (20.1)	405
degenerate Euler multiplier (21.6)	1,500
exponential-polynomial annihilator	257
distinct monomial exponent vectors	256
exterior descending leading values; successive Archimedean levels	462; 1,400
partial-theta exponent identity; partial-theta finite escape bounds	200; 2,706
positive-characteristic derivative obstruction	500
torsion even-function identity	150
Total	9,308

The record also lists thirteen formal recurrence cases with their offsets and sampled zero multipliers; the sampled zeros are not an effective Skolem–Mahler–Lech bound. The `--output` argument is required and an existing file is refused unless `--force` is given.

code/13-theta-descent-verify.py — **1,622 checks, 0 failures.** It checks finite facts of the theta-descent part with exact rational arithmetic in SymPy (**data/13-theta-descent-requirements.txt** pins `sympy>=1.12,<2`; the record was made with Python 3.13.5 and SymPy 1.14.0 at precision p^{128} , the merge’s rerun with Python 3.14.4 and SymPy 1.14.0).

Group in data/13-theta-descent-verification.json	Checks
Chebyshev closed coefficients (23.3); Chebyshev reciprocal substitution (23.2)	30; 30
Taylor coefficient extraction: the coefficients of C_n entering (23.4)–(23.5)	495
Theta shift exponents (23.10)	81
Jacobi product p -coefficients (23.12)	128
Third-order ODE p -coefficients (24.7)	128
Exact jet identities: (24.8) and the system (24.18)	6
Root-factor cancellations; root and rounding leading terms (25.2)–(25.4)	16; 48
Finite local-linearization leading terms (23.15)	560
Lexicographic exclusion checks (Example 25.5)	100
Total	1,622

The record’s group names and its field **q_precision_exclusive** use source 13’s letter q for the nome called p here. The product and the cleared equation have zero residual in every p -coefficient below p^{128} , each checked as an exact polynomial in the variable; the Lambert constants use the nome p^2 . The `--output` argument is required and an existing file is overwritten, so pass a new path; `--precision` defaults to 64, and the record uses 128. For this merge the identities used by Lemmas 24.5 and 24.7 and Proposition 24.2, the expansions of Section 24.1, the rational forms of Theorems 24.1 and 24.10, and the cleared equation and product modulo p^{50} were also checked with an independent SymPy script, which is not shipped; it also confirms that with the nome p in place of p^2 the residual of the cleared equation is nonzero.

code/14-order-three-threshold-verify.py — 600 checks, 0 failures. It checks finite facts of the order-three-threshold part with exact rational and polynomial arithmetic in SymPy (`data/14-order-three-threshold-requirements.txt` pins `sympy>=1.13,<2`; the record was made with Python 3.13.5 and SymPy 1.14.0, the merge’s rerun with Python 3.14.4 and SymPy 1.14.0).

Group in <code>data/14-order-three-threshold-verification.json</code>	Checks
symbolic_jets: the coefficient $(z^2 - 4)^3 Y_0^4$ of Y_3^2 ; after (27.3), z -degree six, the form (27.4) and its residue (27.5); the binomial gap identity $(g^2 - 1)v^2$; a second-order endpoint example	6
endpoint_cancellation: removal of V^{i_0} , $0 \leq i_0 \leq 4$, and the resulting endpoint polynomial, as in Lemma 26.8	10
finite_initial_classification: all 242 nonzero polynomials of degree at most four with coefficients in $\{-1, 0, 1\}$ against Theorem 27.4	242
chebyshev: identities (23.2) and (23.3)	72
newton_corners: at $\delta_n = (2n + 1)\mu$, $0 \leq n \leq 80$, the active indices $\{n, n + 1\}$ and the value $-n(n + 1)\mu$ for $\mathcal{T}_p$	162
positive_characteristic: derivative zero of $\sum z^{pn}$ modulo $\mathfrak{p} \in \{2, 3, 5, 7, 11\}$	5
theta_product_truncation; theta_ode_truncation: the product (23.12) and the cleared equation (24.7) through p^{47}	48; 48
theta_coefficient_values: (23.6)	7
Total	600

The record’s group names use source 14’s word “corner” for a break and its letter q for the nome called p here; its field `cutoff_exclusive` is 48, so every p -coefficient below p^{48} of the product and of the cleared equation is compared as an exact polynomial in z . The initial-form enumeration admits 26 of the 242 polynomials and is not the proof of the classification. The script takes no arguments: it writes `data/verification.json` in the parent of its own directory. In this layout that is a new file beside the delivered record `data/14-order-three-threshold-verification.json`, which it does not touch, but a second run overwrites the first; run it on a copy.

What the checks do not do. No suite proves any infinite theorem. No finite prefix can prove strong summability of an infinite family, cofinality of (e_n) , noncofinality of $(n\epsilon)$, the existence of an infinite escape path, the eventual orbit statements, the generic-line theorem, or the absence of every possible annihilating equation. No infinite Hahn sum is evaluated, or numerically approximated, by any of the programs. In particular a finite truncation of the explicit series (11.2) is a polynomial and is D-finite, so no finite computation can witness its nonholonomicity; for the same reason no finite computation can witness the two-jet independence of Corollaries 17.1 and 17.2. Source 10’s suite in particular does not test arbitrary infinite supports, cofinality, all Hahn elements, the nonexistence of all annihilating equations, or historical novelty. The suites of sources 11 and 12 do not prove the coefficient-field theorem in general, rationality descent for infinite series, the Skolem–Mahler–Lech theorem, infinite support well ordering, cofinality, or any algebraic independence statement, and no finite computation can witness the independence of all jets of F . Source 13’s suite does not verify strong summability of infinite supports, minimal differential order or algebraic independence, the no-order-unit obstruction, completeness of the infinite zero set, or priority, as its record states; a finite truncation of $\mathcal{T}_p$ is a polynomial. Source 14’s suite does not establish the quantifiers over arbitrary Hahn supports, ordered groups, scales or formal solutions, and its enumeration of initial

polynomials is not the classification proof. What the suites can detect is sign, indexing, conversion and finite valuation-comparison mistakes. Everything else rests on the written arguments.

The mathematical dependency chain is short.

Conclusion	Load-bearing ingredients
Differential rigidity	Integer evaluation lemma; finite shift conversion; escape chain and exclusion criterion.
Unit-dilation rigidity	Residue polynomial; finite binomial leading terms; escape chain and exclusion criterion.
Noncofinal nonunit rigidity	Eventual order of finitely many affine values; noncofinality witness; escape chain and exclusion criterion.
Coarsened exclusion	The same affine values, projected to $\Gamma/\Delta_{ v(q) }$; strictness of the projected descent.
Positive dilation existence	Support monoid certificate and the eventual strict increase of h_n ; or, in the order-unit case, quadratic leading valuations and the cofinal-valuation criterion.
Several-variable rigidity	Countable polynomial avoidance; finite-block regrouping; rational differential system on a generic line.
First-order nonlinear rigidity	Hahn-family algebra and its residue homomorphism; large active degrees; exact corner identity; uniqueness of a corner term for each power of Y' .
Higher-order corner criterion	The same, with $\chi_P \neq 0$ assumed.
Positive-weight Euler rigidity	The same algebra in m variables; finite bounded-weight sets; the eigenvalue identity of ϑ_q .
All-order rigidity without an order unit	Minimal relation and its linearization; one finitely generated coefficient field; valuation-rank inequality; principal convex bound; exterior obstruction, or coarsened reduction.
Mixed-jet independence	The same, with an exponential-polynomial multiplier and the imported Skolem–Mahler–Lech theorem.
Rationality in $\text{Frac } \mathcal{E}_\Gamma(k)$	Finite coefficient field; common coarsened reduction of numerator and denominator; rationality descent.
Continuum family	Joint-independence criterion; independent scale monomials; binary-prefix codes; a separate cardinality bound.
Exact domain of $\mathcal{T}_p$	Chebyshev coefficients; quadratic positive-support certificate; descending leading exponents.
Order-three equation for $\mathcal{T}_p$	Classical triple product; elliptic pole cancellation; formal identity over $\mathbb{Q}$; ring maps to $\mathbb{K}$.
No equation of order below three	Laurent ring and substitution homomorphism; theta scaling law; fixed-field, additive-difference and u -adic obstructions; $\psi_3 \neq 0$.
Degree two in the third derivative	Three-jet independence; odd pole order of a cubic in the transcendental $\mathcal{A}$.
No linear equation for $\mathcal{T}_p$	Transcendence of ψ_1 over the elliptic field; highest power in normalized derivatives.
Omnific consequences	Linear-factor product; normal forms; local product linearization.

Conclusion	Load-bearing ingredients
Second-order rigidity (Theorem I)	All-scale criterion; cofinal divisible extension; local finiteness and late breaks; exterior homogeneous reduction by jet degree, then z -degree; nonzero dehomogenization; removal of V^{i_0} ; nonconstant logarithmic Euler derivative in characteristic zero; the finite endpoint lemma.
Minimum order three	Second-order rigidity; the entire witness $\mathcal{T}_p$ and its equation of order three.
Third-order gap certificate	Homogeneous logarithmic substitution; lowest transverse degree; expansions at 0 and ∞ .
Theta initial forms and eventual convexity	Residue form of the exterior equation; rational Riccati reduction; constants of rational differentiation; exclusion of skipped and triply tied indices; local finiteness.

All supports are sets. All operators are finite sums. All integer indices are ordinary nonnegative integers. The coefficient valuation is trivial in characteristic zero; the coefficient field is $\mathbb{C}$ in Sections 1–13 and any characteristic-zero field in the nonlinear, coefficient-field, theta-descent and order-three-threshold parts. The coefficient-field part imports one external recurrence theorem without proof, the Skolem–Mahler–Lech theorem, used only for relations involving at least two distinct dilations. The equations use the external formal variable derivative. None of the proofs asserts convergence in the fine topology of $\mathbf{No}[i]$, substitutes a real-valued norm for an arbitrary-rank valuation, or transports a fixed-workspace evaluation to every surcomplex argument.

19 Originality assessment and questions left open

19.1 Established background and proposed contribution

The D-finite/P-recursive correspondence [1], ordinary properties of Hahn fields, the Hahn–Neumann support lemmas and Higman’s finite-word theorem [6], finite polynomial root bounds, the Conway normal-form identification [9, 10], and partial theta series with their elementary functional identity [5] are established material. The companion report’s cofinality and scalar-extension work, including the group Γ_∞ and the distinction between order units and countable cofinality, is prior project material [12]. The rank-one quadratic-exponent entire example is already in the collection [14]. We claim to originate none of these, and we do not claim to have disproved any published theorem about a different analytic function class.

The proposed research contribution is their combination into the sharp arbitrary-rank dilation classification, Theorem B: in particular the necessity argument for all noncofinal scales, the unit case with root-of-unity residue, the uniform exterior obstruction applying to every solution of a fixed equation, the quantitative exclusion bounds with their two different boundary behaviours, and the exact theta evaluation domain for an arbitrary Hahn parameter. The mixed noncofinal extension (Theorem 10.2), generic-line rigidity (Theorem C), and the explicit series excluded from every nontorsion pure dilation equation (Theorem 11.2) are further consequences proved here.

The nonlinear part (source 10) inherits Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning, and is related to central-index, Newton-polygon, non-Archimedean value-distribution and tropical differential methods (Section 17.4). Its proposed contribution is the

precise arbitrary-rank strong-family formulation: Theorem D with its finite exclusion certificate (Theorem 15.1), the candidate degrees and the entire-solution locus, the higher-order corner criterion, the positive-weight Euler theorem, and the non-uniformity of the nonlinear exclusion bound. Its priority is provisional in the sense recorded there.

The coefficient-field part (sources 11 and 12) inherits classical coefficient recurrences, the finite-generation principle, the valuation-rank inequality and the Skolem–Mahler–Lech theorem (Section 22.6). Its proposed contributions are the all-order no-order-unit rigidity with its joint-dilation form (Theorem E), the coefficient-value-rank criterion and the order-unit dichotomy (Theorems F and G), the meromorphic coarsening descent (Theorem 21.11) and the full-constant-field continuum family (Theorem 22.6); its priority is equally provisional.

The theta-descent part (source 13) inherits the Chebyshev identity, Jacobi’s triple product and the theta–elliptic differential identity, all classical, the bilateral theta domain of the Hahn–Tate report, and the no-order-unit direction of Theorem E (Section 25.5). Its proposed contributions are the support-controlled descent to an entire series with its exact domain at every scale, the exact boundary of Theorem H with the minimal-order certificate, the differential field and non-D-finiteness, and the surreal and omnific consequences; its priority is equally provisional.

The order-three-threshold part (source 14) inherits the theta material of source 13, the no-order-unit direction of Theorem E, the Gauss initial polynomials of source 10 and classical logarithmic derivatives (Section 27.5). Its proposed contributions are second-order rigidity for every value group (Theorem I (a)) with its finite endpoint lemma, the exact least order three, the third-order endpoint–gap obstruction, the classification of theta initial polynomials and eventual Newton convexity; its priority is equally provisional.

19.2 A result-by-result attribution ledger

Result or ingredient	Status in this report
Hahn arithmetic, strong support sums, surreal normal forms	Standard background; references and support proofs supplied.
Higman’s finite-word theorem	External standard ingredient; alternatively import the classical Hahn–Neumann support lemma directly.
D-finite / polynomial recurrence conversion	Classical; credited to Stanley and derived explicitly for Hahn coefficients.
Partial theta series and functional identity	Classical; not a novelty claim.
Escape chain and exclusion criterion, with the local-finiteness boundary	Elementary support-theoretic lemma and quantitative formulation developed here.
Eventual affine and periodic orbit valuations	Proved here for arbitrary Hahn coefficients; the finite residue-polynomial treatment covers near-torsion units.
Exact fixed- q order-unit classification	Principal proposed contribution: arbitrary rank, arbitrary Hahn parameter, both signs of $v(q)$.
Exact theta evaluation domain	Support-controlled sharpness theorem; related in spirit to the companion report’s scalar-extension results, proved independently here.
Uniform exterior obstruction and common degree bound	Proved here; stronger than eventual termination of each solution separately.

Result or ingredient	Status in this report
Universal single-dilation nonholonomicity on no-order-unit workspaces	Consequence of the classification, with an explicit entire witness.
Mixed differential–dilation rigidity at noncofinal nonzero scales	Proved here; the unit-valuation mixed case is answered without an order unit by source 12 (Theorem E) and left open with one.
Several-variable rigidity and multivariate algebraic rigidity	Proved here; the generic line is an existence device.
Gauss initial forms, Euler eigenvalues, indicial polynomials, central-index and Newton-polygon ideas	Established (source 10); credited to Hayman, Cano and Fortuny Ayuso, Hu–Luan, Hu–Yang and Giansiracusa–Mereta as predecessors in different settings.
First-order nonlinear rigidity (Theorem D), finite exclusion certificate, candidate degrees and entire-solution locus	Proposed contribution of source 10, over any characteristic-zero coefficient field; priority provisional.
Higher-order corner criterion and positive-weight Euler rigidity	Proposed by source 10; sufficient conditions only; the vanishing-corner case is settled without an order unit by sources 11 and 12, and with one source 13’s equation of order three has vanishing residue corner and a nonpolynomial entire solution.
Solution-dependence of the nonlinear exclusion bound	Proved by source 10 with the explicit Riccati family of Theorem 15.10.
Two-jet independence of $\sum t^{n^2} z^n$ and of (11.2)	New consequences (source 10); both series are prior collection material.
Classical coefficient recurrences, finite generation, valuation-rank inequality, Skolem–Mahler–Lech	Established (sources 11 and 12); credited to Hurwitz, Laohakosol–Kongsakorn–Ubolsri, Bell and Lech; the last imported without proof.
All-order rigidity without an order unit, with dilations (Theorem E)	Proposed by sources 11 and 12 (part (b) and the dilations by source 12, part (c) by source 11); priority provisional.
Coefficient-value-rank criterion, order-unit dichotomy, rank-one family	Proposed by source 12; sufficient conditions and one equivalence, as stated.
Coarsening descent, rationality in $\text{Frac } \mathcal{E}_\Gamma(k)$, systems, Painlevé I in $\text{Frac } \mathcal{E}_\Gamma(k)$, joint-independence criterion, continuum family	Proposed by source 11; the polynomial half of the descent is the entire-function report’s obstruction, and neither the cardinal nor the almost-disjoint device is new.
All-order independence of (11.2)	New consequence (sources 11 and 12); the series is prior material.
Several multiplicatively independent dilations (Corollary 22.8)	The merge’s observation from source 12’s theorem; not claimed by source 12.
Mixed jets in the fraction field (Corollary 22.9)	Consequence added here from source 12’s coefficient theorem and source 11’s descent; neither source states it.

Result or ingredient	Status in this report
Chebyshev identity, Jacobi triple product, theta-elliptic differential identity	Classical (source 13); DLMF, Sebbar and Ohyama credited; no novelty claim for third-order theta relations.
Bilateral theta domain	The Hahn–Tate report’s theorem, specialized; reproved by source 13, not claimed anew.
Exact domain of $\mathcal{T}_p$, product and zeros, local valuations, omnific rounding	Proposed by source 13, derived from the classical product.
Exact boundary (Theorem H), minimal order three, differential field, non-D-finiteness	Proposed by source 13; the no-order-unit direction is Theorem E (a); priority provisional.
Substitution homomorphism, global proof of $\psi_3 \neq 0$, corner data of the cleared equation, clause (iv) of Theorem G, Corollary 25.2	Added by the merge from source 13’s results; the first two write out steps source 13 used.
Second-order rigidity for every Γ and exact least order three (Theorem I), endpoint lemma, endpoint–gap obstruction, theta initial forms, eventual Newton convexity, two-state systems and polynomial-denominator quotients	Proposed by source 14; the witness of order three is source 13’s and the no-order-unit direction is Theorem E (a); priority provisional.
Cofinal scalar extension, all-scale criterion, surreal full-class boundary	Lemma 26.2 and Proposition 27.6 are source 14’s; the criterion is Proposition 20.2, printed once.
Exterior form as the residue corner in Euler jets; three jets of G and Θ_p	Added by the merge (Remark 26.6, Corollary 26.13).
Named published open conjecture	None claimed solved; the nonlinear question, answered in order one, without an order unit in all orders, with an order unit by an example of order three, and in order two for every Γ , is a question of this report, not a historical conjecture.
Priority, peer review, formal verification	Priority and independent refereeing are not certified. Lean covers support and escape lemmas, exponential and partial-theta domains, polynomial-orbit valuations, constant-point avoidance, inward stability and torsion covariance. The full classifications and the nonlinear, coefficient-field, theta-descent and order-three-threshold theorem packages remain pending.

The targeted literature checks of sources 08 and 09 located the standard D-finite framework, classical complex q -difference growth theory, q -holonomic degree growth in a rational-function setting, and ultrametric q -difference theory, but did not establish whether an identical arbitrary-rank strong-Hahn formulation appears elsewhere. Several broad keyword searches returned irrelevant material and were not treated as evidence of absence; a repository content search for **holonomic** returning no matches is likewise not proof that no sentence anywhere in the repository can be reformulated into one of these results. Accordingly this is a proposed original result with proofs,

not a certified claim of bibliographic priority. Independent mathematical review is still needed; machine-checked scope is the partial coverage recorded in Section 13 and the formalization ledger. The corresponding records for the nonlinear, coefficient-field, theta-descent and order-three-threshold parts, with their own search limits, are Sections 17.4, 22.6, 25.5 and 27.5.

19.3 What the nonlinear part does not claim

Every limitation stated by source 10, in its manuscript or its two audit files, is kept; they are collected here. They describe source 10's results; where the coefficient-field or theta-descent part goes further, a parenthesis says so. The limitations of sources 11 and 12 are listed separately as (C1)–(C26) in Section 22.7, those of source 13 as (T1)–(T19) in Section 25.6, and those of source 14 as (O1)–(O18) in Section 27.6.

- (N1) Only the first-order case of Question 19.1 is answered, not the unrestricted higher-order question. (Sources 11 and 12 answer the case without an order unit in every order; source 13 shows that with an order unit a nonpolynomial strongly entire series can satisfy an equation of order three; source 14 answers order two for every Γ .)
- (N2) Theorem D does not say that a nonpolynomial entire function is differentially transcendental in all orders; independence of f and $D_z f$ does not exclude a relation involving $D_z^2 f$. (Theorem E (a) says it when Γ has no order unit, and Theorem H shows it fails with one; Theorem I adds $D_z^2 f$ to the independent jets for every Γ .)
- (N3) The criterion $\chi_P \neq 0$ is sufficient, not a complete invariant of a solution set, and the methods of source 10 do not decide equations of order at least two with vanishing residue corner. Proposition 16.4 is not a nonpolynomial entire solution and does not refute higher-order rigidity. (Source 13's order-three equation, with $\chi_P = 0$, does have one when Γ has an order unit; Proposition 24.2. Source 14 decides order two negatively for every Γ , Theorem I.)
- (N4) The corner is a finite selection, not a full differential Newton polygon; no invention of differential Newton polygons, no equivalence to tropical differential algebra, and no new tropical fundamental theorem is claimed.
- (N5) The exclusion bound (15.8) is not claimed optimal; the certificate has no converse; leading valuations alone do not decide evaluation at a boundary argument.
- (N6) Candidate degree sets are necessary, not realized; their finiteness is not a uniform algorithm for arbitrary coefficient names, and the workflow of Section 17.3 is not an algorithm for arbitrary surreal names, does not enumerate local formal solutions, does not decide strong-support membership, and accepts no numerical solver failure as a certificate.
- (N7) Corollary 15.6 does not say that the number of solutions is finite, is not a moduli statement over rings with nilpotents, and covers only the extension class it states.
- (N8) No theorem says that G , F or any nonpolynomial series is entire on the proper class **No**[i]; entireness and the derivative are relative to one set-sized workspace.
- (N9) The independence statements concern formal functions; no statement concerns algebraic independence of a value $f(x)$ from its own ambient field.
- (N10) D_z kills all Hahn coefficients and is not the Berarducci–Mantova derivation; no scalar-derivation result is claimed.

- (N11) The series $\sum t^{n^2} z^n$ and (11.2), and the linear rigidity of the latter, are prior material; only their nonlinear two-jet consequences are new. The tail-span report's all-order independence results are neither weakened nor rediscovered.
- (N12) Hahn fields, the positive-support summation lemma, Conway normal forms, Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning are established ideas, not relabelled as new; the positive-support lemma is imported by source 10, and proved in this report by Lemma 2.1 (2).
- (N13) The precise all-rank statements are proposed contributions; historical priority is not certified. The literature search of 22 September 2026 was targeted and not exhaustive; Hayman's survey and the Hu–Yang monograph were consulted through metadata only, and no claim is made that rank-one special cases were previously unknown.
- (N14) The repository comparison at 048b72c was targeted: no line-by-line audit of every report, Lean file, archive, branch or the Git history, and no absence claim rests on a truncated preview.
- (N15) The answered question is a question of this report, not a named historical conjecture.
- (N16) Source 10 supplied no Lean formalization or independent referee report for the nonlinear part. The 2,113 finite checks are program checks, not theorems; they do not test infinite supports, cofinality, all Hahn elements, nonexistence of annihilating equations or novelty, and no infinite surreal sum is numerically approximated. A successful PDF build is not a proof check.
- (N17) The sequence arguments behind the support calculus use ordinary set-theoretic choice.

19.4 Further mathematical questions

Question 19.1 (Differentially algebraic entire series of minimal order three). Suppose Γ has an order unit. By Theorem I no nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfies a nonzero algebraic differential equation over $K_\Gamma(z)$ of order at most two, and by Theorem H order three occurs. Which nonpolynomial strongly entire series have minimal differential order three? Are they all accounted for by support-controlled algebraic descents of elliptic or theta constructions, or are there fundamentally different families? More broadly, to what extent do finite algebraic descents of theta or abelian functions, combined with elementary operations preserving strong entireness, account for the nonpolynomial strongly entire series that are differentially algebraic over $K_\Gamma(z)$? The same questions are open over $\mathbb{K} = k((t^\Gamma))$ for every field k of characteristic zero.

Status. As posed by sources 08 and 09, the question asked about nonlinear algebraic differential equations of every order, for every Γ , and ended by asking under which hypotheses on Γ linear rigidity can be strengthened to differential algebraic rigidity in all orders, and whether the explicit series (11.2) avoids all equations of order at least two under a natural coefficient restriction. Source 10 answers its first-order case negatively, for every nonzero Γ and every characteristic-zero coefficient field (Theorem D): in order one no hypothesis on Γ beyond $\Gamma \neq 0$ is needed, and (11.2) avoids every first-order equation with no coefficient restriction (Corollary 17.2). It also settles the higher-order equations with $\chi_P \neq 0$ (Theorem 16.1) and the relations between f and a positive-weight Euler derivative (Theorem 16.5). After source 10, order at least two with a vanishing residue corner stayed open. Proposition 16.4 shows that such equations exist and can have polynomial solutions of unbounded degree, so the study cannot rely on a uniform degree bound for all polynomial solutions of an arbitrary higher-order equation; it is not a nonpolynomial entire solution and gives no negative answer. Source 10 suggests as a next step classifying the polynomial solutions of a vanishing-corner

residue equation beyond their leading monomial and determining how those constraints propagate between scales.

Sources 11 and 12 then answered the question, negatively, in every order for every Γ without an order unit, whatever the residue corner and for every characteristic-zero coefficient field (Theorem E (a)); source 11 extends the answer to quotients of entire series (Theorem E (c)). Their method needs no degree bound: along the solutions of (16.1) its linearization multiplier is $(T - m)^2 \neq 0$ (Example 21.6). The last sentence of the question as posed by sources 08 and 09 is answered positively for (11.2) with no coefficient restriction (Theorem 22.1, source 12; source 11 for $f_* = F - \omega^{-1}$). Its sentence on hypotheses is answered in one direction: the absence of an order unit suffices. The question was therefore re-scoped to groups with an order unit, in the wording recorded below, which merges source 11's Question 9.1 and source 12's Question 10.1; both sources leave it open and give no counterexample. There a finitely generated coefficient field can have values whose convex hull is all of Γ , and the argument of Section 21 stops. The partial theta series Θ_p and G are not counterexamples: Θ_p satisfies a dilation equation, not a known algebraic differential one, and only the first-order consequence is known for G (Corollary 17.1). Theorem G shows that the order unit is exactly the boundary of the *mixed* statement; whether it is also the boundary of the purely differential one was, at that stage, the question.

As re-scoped after sources 11 and 12, the question read: *Suppose Γ has an order unit. Can a nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfy a nonzero algebraic differential equation $P(z, f, D_z f, \dots, D_z^s f) = 0$ over $K_\Gamma(z)$ of order $s \geq 2$? By Theorems D and 16.1, such an f satisfies no nonzero equation of order at most one and no equation with $\chi_P \neq 0$; equivalently, can a nonpolynomial strongly entire function satisfy an equation whose residue corner polynomial vanishes identically, without satisfying any other equation to which Theorem 15.1 applies? More strongly, is every element of $\text{Frac } \mathcal{E}_\Gamma$ that is differentially algebraic over $K_\Gamma(z)$ rational? Is the absence of an order unit necessary for differential algebraic rigidity in all orders, or can linear rigidity be strengthened to it for every nonzero Γ ?* Source 13 (the theta-descent part) answers every sentence of it, for every characteristic-zero coefficient field, including $k = \mathbb{C}$. With an order unit μ and $p = t^\mu$, the series $\mathcal{T}_p$ of (1.4) is nonpolynomial and strongly entire and satisfies an explicit equation of order three and none of lower order (Theorems H, 24.1 and 24.10), so the answer to the first sentence is yes. Its equation has $\chi_P = 0$, and $\mathcal{T}_p$ satisfies no equation with $\chi_P \neq 0$ (Proposition 24.2, Remark 24.3), so the answer to the second is also yes. $\mathcal{T}_p$ is a differentially algebraic element of $\text{Frac } \mathcal{E}_\Gamma$ that is not rational, so the rationality question has a negative answer. The absence of an order unit is necessary: linear rigidity (Theorem A) cannot be strengthened to differential algebraic rigidity for any Γ with an order unit (Corollary 25.2). Source 13 leaves open whether order exactly two occurs and asks for a classification (its Questions 11.1 and 11.2); the question was re-scoped to these, in the wording recorded in the next paragraph. Neither Θ_p nor G is decided: whether either is differentially algebraic stays open.

As re-scoped after source 13, the question read: *Suppose Γ has an order unit. Can a nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfy a nonzero algebraic differential equation $P(z, f, D_z f, \dots, D_z^s f) = 0$ over $K_\Gamma(z)$ of order $s = 2$, that is, are f , $D_z f$ and $D_z^2 f$ algebraically dependent over $K_\Gamma(z)$ for some nonpolynomial $f \in \mathcal{E}_\Gamma$? By Theorem D no equation of order at most one is possible, and by Theorem H order three occurs, so the least order over all examples is two or three. More broadly, to what extent do finite algebraic descents of theta or abelian functions, combined with elementary operations preserving strong entireness, account for the nonpolynomial strongly entire series that are differentially algebraic over $K_\Gamma(z)$? The same questions are open over $\mathbb{K} = k((t^\Gamma))$ for every field k of characteristic zero.* Source 14 (the order-three-threshold part) answers the first sentence negatively, for every Γ , with or without an order unit, and for every characteristic-zero coefficient field, including $k = \mathbb{C}$: a strongly

entire solution of a nonzero equation of order at most two is a polynomial (Theorem I(a)), even when its residue corner vanishes. So the least order is exactly three (Theorem I(b)). Source 14 was written against the repository at [b895e86](#), before source 13 was merged, and answers source 13's question `q:order2`, which is the same question (Section 26.1). The classification sentence stays open; source 14 poses it again as its Research question 1, which is merged here, and the question is re-scoped to it, as printed above. For Θ_p and G the answer gives three independent jets (Corollary 26.13); whether either satisfies an equation of order at least three stays open. The eight further questions of source 14 are Questions 19.11–19.18 below.

The escape proof does not answer this, in any order. A nonlinear differential relation produces convolution sums in the coefficients rather than a bounded-step linear recurrence, and treating it as though it had the form (3.1) would be an invalid extension. The first-order answer uses the different route of Section 14, the no-order-unit answer the third route of Section 21, and the order-two answer a fourth, initial polynomials at infinitely many scales (Section 26.3).

Question 19.2 (Beyond a positive Euler operator). Which polynomial vector fields V have the property that every strongly entire f satisfying a nonzero equation $P(x, f, Vf) = 0$ is a polynomial?

Positive integer Euler weights give one family, and zero and mixed weights give explicit counterexamples (Section 16.3). A useful extension would need both a finite-dimensional filtration and a sufficiently nondegenerate action on its highest layers; diagonalization alone is not enough if infinitely many monomials share a bounded eigenvalue. (Source 10.)

Question 19.3 (Effective representations and minimal certificates). For which exact Hahn coefficient representations can the candidate degree set and an optimal excluded valuation cut be computed uniformly?

The bound (15.8) is always a valid choice from exact finite valuation data but can be far from minimal, and Theorem 15.10 shows that an optimal cut must in general depend on solution data. An algorithmic refinement must state how it receives those data and how it decides coefficient cancellation. (Source 10; the linear analogue is Question 19.5.)

Question 19.4 (Mixed operators at unit valuation, with an order unit). Suppose Γ has an order unit. For a nontorsion q with $v(q) = 0$, must every strongly entire solution of a nonzero mixed equation (10.1) be a polynomial?

As posed by sources 08 and 09 the question concerned every Γ . The unresolved point in the escape method is uniform control of $v(P(n, q^n))$ for a general bivariate P in the unit case. The pure q -difference proof controls $P(q^n)$ and the differential proof controls $P(n)$; neither statement by itself handles their combination. This is a question suggested by the argument, not asserted to be a previously catalogued open problem.

Status. When Γ has no order unit, source 12 answers it positively and more strongly: by Theorem E(b) with $\Lambda = \{q^i : i \in \mathbb{Z}\}$, a nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfies no nonzero algebraic relation, linear or not, among the $(D_z^j f)(q^i z)$, and a nonzero equation (10.1) is such a relation (Section 22.5). The method bounds the coefficient field instead of the valuations $v(P(n, q^n))$. The question is re-scoped to groups with an order unit, where it stays open: the witness of Theorem G uses a parameter of nonzero valuation, and source 13's series $\mathcal{T}_p$, which gives nonlinear mixed relations for every nontorsion q (Corollary 25.2), is not D-finite, and source 13 says nothing about linear mixed equations.

Question 19.5 (Optimal exterior regions and exact solution domains). For a fixed differential or rigid dilation operator, can one compute the largest common strong evaluation domain of its formal solution space from finite valued operator data, rather than only a sufficient exclusion bound? Can the coarse exclusion bound of Theorem 7.4 be supplemented by support hypotheses that give the exact domain?

The thresholds supplied here are uniform and explicit once the valuation certificates are available, but need not be optimal, and distinct solutions may have cancellations or terminating tails that change their individual domains. Proposition 5.3 shows that the uncoarsened bound can be exactly attained, and Theorem 8.1 exhibits one sufficient pattern for an exact answer: a common positive support monoid together with a controlled sequence of leading exponents. A full answer should retain arbitrary Hahn tails rather than use only a numerical radius.

Question 19.6 (Several independent dilations). When Γ has an order unit, what is the existence criterion for a nonpolynomial strongly entire solution constrained by equations using several multiplicatively independent dilations? Is an appropriate cofinality condition on their joint valuation data sufficient, or do compatibility relations impose additional obstructions? In the no-order-unit workspace (11.1), can one classify the finite linear relations among $F(q_1z), \dots, F(q_s z)$ when some quotient q_i/q_j is a root of unity?

Status. As posed by sources 08 and 09 the first two sentences concerned every Γ , and the last asked about parameters that are not powers of a common element. In a group without an order unit the merge observes (Corollary 22.8, from source 12's Theorem E (b); source 12 does not claim it) that no nonpolynomial strongly entire series satisfies any nonzero equation, linear or not, in finitely many derivatives at dilations from the group generated by multiplicatively independent parameters, and that $F(q_1z), \dots, F(q_s z)$ satisfy only the trivial linear relation whenever no quotient q_i/q_j is a root of unity, whether or not the q_i are powers of a common element. The question is re-scoped to what this leaves: groups with an order unit, and parameters with a root-of-unity quotient.

The single-dilation theorem identifies the exact obstruction when only one scale is repeatedly added. With several dilation directions, distinct monomials can have equal valuation slopes and their leading coefficients can produce nontrivial recurrence phenomena of their own. A meaningful formulation must also quotient out identities caused by repeated parameters and any genuine algebraic relations among them. No multidilation classification is implied by Theorem B.

The next three questions come from sources 11 and 12.

Question 19.7 (An intrinsic descent criterion; source 11). For a fixed Hahn field $\mathbb{K}$ and a subfield $L \subseteq \mathbb{K}$, can the equality $\text{Frac } \mathcal{E}_\Gamma(k) \cap L((z)) = L(z)$ be characterized solely in terms of the valued extension $\mathbb{K}/L$?

Containment of $v(L^\times)$ in a proper convex subgroup is sufficient (Theorem 21.11) and applies to many infinitely generated fields; it is not claimed necessary. A criterion using rational rank alone will not capture all cases: what matters to the reduction is whether the coefficient values reach every Archimedean scale of Γ .

Question 19.8 (A finite-rank replacement; source 12). Which additional properties of a finitely generated coefficient field, of its valuation, or of the recurrence (21.5) force a linear upper bound on the relevant nonzero coefficient values, and so exclude strong entireness even when those values are cofinal?

Bounding the multiplier $\Xi(n)$ is not enough: a nonlinear recurrence can involve cancellations among many products of earlier coefficients, and an extension must control them or replace them by another invariant. An answer would bear on Question 19.1. The series $\mathcal{T}_p$ of Theorem H is differentially algebraic and strongly entire, with a finitely generated coefficient field and coefficient values $m^2\mu$ that are cofinal and not linearly bounded; with an order unit, any such property must therefore fail for it (Section 25.2).

Question 19.9 (Torsion blocks; source 12). When some dilations have root-of-unity quotients, can the coefficient-field conclusion of Theorem 21.2 (b) be recovered separately on the finitely many residue classes of Taylor indices that survive the resonances?

The even-series example of Section 22.4 shows that no unconditional version for all coefficients exists; a blockwise theorem would need explicit conditions on which residue classes are constrained, not only on the nonzero part of the multiplier.

The next question comes from source 13 (its Question 11.3).

Question 19.10 (Effective strong domains for differential solutions). For an explicitly represented formal solution of an algebraic differential equation, can its maximal strong domain be recovered from a finite differential system and coefficient data?

For $\mathcal{T}_p$ the answer is the coarsened valuation ring $\mathcal{O}_{\Delta_\mu}$ (Theorem 23.5), whose boundary depends on the convex subgroup generated by one scale. General solutions can have much more complicated supports. An algorithm needs a representation of those supports and a decision procedure for the relevant coefficient cancellations; inspecting the leading differential monomials cannot suffice in the vanishing-corner case. Compare Questions 19.3 and 19.5.

The last eight questions come from source 14 (its Research questions 2–9; its Research question 1 is merged into Question 19.1).

Question 19.11 (Lifting admissible initial polynomials; source 14). Given an equation of order three and an unbounded sequence of initial polynomials satisfying its exterior form, what compatibility conditions guarantee that they are the initial polynomials of one strongly entire solution?

A criterion would have to control both coefficient lifting and global support: local formal solutions at separate scales need not assemble into one series. $\mathcal{T}_p$ is a fully worked positive case, with the initial polynomials $X^n(1+X)$ at its breaks.

Question 19.12 (The entire solutions of the theta equation; source 14). For the constants (24.2), determine all strongly entire solutions of (24.7). Which are scalar multiples, coordinate transforms or other members of a theta family, and which come from singular branches of the cleared equation?

Theorem 27.5 forces eventually full coefficient support and strictly increasing consecutive coefficient differences. The next step would use the lower exterior terms, discarded by the exterior form, to constrain those differences and the leading coefficients; the cleared form may have exceptional solutions not represented in logarithmic coordinates, which need separate treatment. Since (24.8) is homogeneous in the jets, every constant multiple of $\mathcal{T}_p$ is a solution.

Question 19.13 (Effective third-order exclusion; source 14). For which coefficient fields and equations can boundedness of the nonnegative-integral endpoint projection of $\Upsilon_{\mathfrak{h}_P}(T, G) = 0$ be decided? Can a finite library of classifications of residue polynomials extend the certificate of Corollary 27.3?

At order two the construction is finite algebra once the leading coefficient values are accessible: dehomogenize, substitute, remove a power of V and inspect the remainder on $V = 0$. Order three introduces a genuine two-integer constraint. Any algorithmic claim must include computability of the Hahn coefficients of the input and of their exact leading terms; compare Question 19.3.

Question 19.14 (Higher-order break invariants; source 14). Can the expansion transverse to constant logarithmic derivatives be organized into a hierarchy of gap and multiplicity spectra for orders four and above, with rigidity theorems at each level?

At higher orders several logarithmic derivatives vanish to the same first local order, with leading ratios determined by powers of an endpoint gap; secondary terms could record further support spacings. When the resulting constraints are complete is open.

Question 19.15 (Meromorphic second-order solutions; source 14). Beyond polynomial denominators, which algebraic differential equations of order two have nonrational quotients of strongly entire series as solutions? Can the break method bound the density or the valuation distribution of their poles?

Elliptic meromorphic functions suggest that polynomiality cannot simply be extended to every quotient of entire series when an order unit exists; the comparison involves separate Newton data for numerator and denominator, including cancellation and common zeros. Without an order unit, Theorem E (c) already makes every differentially algebraic element of $\text{Frac } \mathcal{E}_\Gamma(k)$ rational.

Question 19.16 (Partial differential systems; source 14). Is there a useful several-variable analogue of three-jet independence for series strongly summable at every point of $\mathbb{K}^d$? Which finite-dimensional rational differential systems force polynomiality of all strongly entire observables?

Restriction to lines may settle special cases, as for Theorem C, but algebraic relations among several partial derivatives do not in general reduce to a relation among the first two derivatives on every line; a several-variable analysis of initial forms is needed.

Question 19.17 (Positive residue characteristic; source 14). Can a replacement for second-order rigidity be formulated with Hasse or divided derivatives, after excluding series that are constant for the Frobenius and specifying the behaviour of integer valuations?

The example of Section 27.4 and Example 17.3 show that ordinary derivatives cannot suffice. A replacement must distinguish the characteristic of the field, that of the residue field and the notion of summation.

Question 19.18 (Omnific approximation of differential zero sets; source 14). For explicitly represented entire solutions of minimal order three, which zero sets admit uniform approximation by omnific integers, and how do endpoint gaps constrain the valuations of the rounding errors?

The zeros of $\mathcal{T}_{\omega-1}$ have exact rounding errors $\omega^{-(2m-1)}$ (Corollary 25.3). For other entire solutions the coefficient valuations are not enough: residue directions and higher terms can move individual zeros.

These questions are directions suggested by the proved statements, not claims that they were already posed in the published literature, and their formulations are deliberately narrower than a general theory of nonlinear surcomplex differential equations.

20 All-order rigidity without an order unit: setting and valuations

This section and the next two contain the material of sources 11 and 12, two manuscripts delivered together after the rest of this report was written. They answer Question 19.1 in every differential order when Γ has no order unit, including the equations with vanishing residue corner that Sections 14–17 leave open, and source 12 answers the same case of Question 19.4. The method is neither the escape chain of Section 3 nor the finite initial polynomial of Section 14. A finite relation forces all Taylor coefficients of a solution into one finitely generated field (Theorem 21.2); such a field has a value group of finite rational rank (Lemma 20.6), which is trapped in a proper convex subgroup when Γ has no order unit (Lemma 20.7); and strong entireness needs coefficient values at cofinally many scales (Proposition 20.2). The warning of Remark 5.8 stands: nothing here turns a nonlinear equation into a bounded-step linear recurrence with controlled valuations. The argument uses only that one scalar multiplier is eventually nonzero, and no valuation of it.

20.1 Sources, conventions and the merge's choices

Convention 20.1 (Coefficient field of the coefficient-field part). In Sections 20–22, k is an arbitrary field of characteristic zero with the trivial valuation, Γ is as in Convention 1.1, $\mathbb{K} = k((t^\Gamma))$, and v , strong summability, $\text{Dom}_{\text{str}}(f)$ and $\mathcal{E}_\Gamma(k)$ are as in Convention 14.1. Write $\text{Frac } \mathcal{E}_\Gamma(k)$ for the fraction field of $\mathcal{E}_\Gamma(k)$ inside the formal Laurent field $\mathbb{K}((z))$; the word *meromorphic* in this part means membership of $\text{Frac } \mathcal{E}_\Gamma(k)$, and nothing else. The derivative D_z kills $\mathbb{K}$ and extends to $\mathbb{K}((z))$ by the quotient rule. A series u is *differentially algebraic* over a differential subfield $B \subseteq \mathbb{K}((z))$ if $P(u, D_z u, \dots, D_z^r u) = 0$ for some finite r and some nonzero $P \in B[Y_0, \dots, Y_r]$, and *differentially transcendental* otherwise. For $\lambda \in \mathbb{K}^\times$ and $j \geq 0$ put

$$\mathcal{J}_{\lambda,j} f = (D_z^j f)(\lambda z) = \sum_{n \geq j} a_n(n)_{\underline{j}} \lambda^{n-j} z^{n-j} \quad \left(f = \sum_n a_n z^n \right), \quad (20.1)$$

differentiation *before* dilation. The same notation is used for $u \in \mathbb{K}((z))$, using formal Laurent substitution. Since $D_z^j \sigma_\lambda f = \lambda^j \sigma_\lambda D_z^j f$, the operator $D_z^r \sigma_q^\ell$ of (10.1), which dilates first, differs from $\mathcal{J}_{q^\ell, r}$ by the nonzero scalar $q^{\ell r}$. This changes no algebraic independence statement, but recurrence coefficients must use one convention throughout. An infinite family is algebraically independent when every finite subfamily is. For a subgroup $\Delta' \subseteq \Gamma$, $\text{rrk } \Delta' = \dim_{\mathbb{Q}}(\Delta' \otimes_{\mathbb{Z}} \mathbb{Q})$ is its rational rank; ordered groups are torsion free, so they embed in their divisible hulls, and no divisibility of Γ is assumed.

For $k = \mathbb{C}$ one has $\mathbb{K} = K_\Gamma$ and $\mathcal{E}_\Gamma(k) = \mathcal{E}_\Gamma$. As for the nonlinear part, the generality in k is recorded for this part only and is *not* transferred to Theorems A–C.

Two sources, one spine. Sources 11 and 12 were written independently and prove the same five steps.

- (1) Along an actual solution, a minimal relation is linearized. The evaluated partial derivatives have an offset θ and a nonzero *linearization multiplier* Ξ , the quadratic remainder has z -order at least $2(n - r) > n + \theta$, and so all Taylor coefficients lie in one field generated by finitely many of them and finitely many scalars (source 11's Theorem 4.1 and Corollary 4.3; source 12's Theorems 4.1 and 5.2).
- (2) The valuation-rank inequality (source 11's Lemma 5.1, source 12's Lemma 3.1).

- (3) A subgroup of finite rational rank lies in a principal convex subgroup Δ_β , which is proper when there is no order unit (source 11's Lemma 5.2, source 12's Lemma 3.2).
- (4) An exterior obstruction at $t^{-\eta}$ with $\eta > \Delta_\beta$ (source 11's Proposition 5.4, source 12's Proposition 3.4).
- (5) The same worked example: along the solutions az^m of the vanishing-corner equation (16.1) the linearization multiplier is a nonzero multiple of $(T - m)^2$.

Both assume only characteristic zero, $\Gamma \neq 0$ set-sized, no divisibility, and no order unit where that is used; neither has weaker hypotheses than the other. The spine is printed once, with both sources credited at each statement. **Source 12 is the base** of this part: its all-order theorem with dilations (Theorem E (b) here) contains, for the single dilation 1, source 11's statement about entire functions; it answers the no-order-unit case of Question 19.4; and its order-unit dichotomy (Theorem G) gives the exact boundary of the mixed statement. Source 11 contributes what source 12 lacks: coarsening descent for arbitrary, not necessarily finitely generated, coefficient fields (Theorem 21.11); rationality of every differentially algebraic element of $\text{Frac } \mathcal{E}_\Gamma(k)$ (Theorem E (c)); the rational-system and Painlevé corollaries; the joint-independence criterion (Theorem 22.2); and a continuum-sized independent family (Theorem 22.6). Two of source 11's proofs differ genuinely from source 12's and are kept as marked second routes: its choice of a minimal relation (Remark 21.3) and its proof of polynomiality by coarsened reduction (proof of Theorem E). Every proof taken from source 11 was re-read against the statement it is attached to: source 11 has no stronger standing hypothesis than source 12; its cardinal equality uses $k = \mathbb{C}$, while its lower bound holds for every k ; and its coefficient principle is relative to a Laurent field $k_0((z))$, a form kept because the joint-independence criterion needs it.

Renamed symbols. The following symbols of the two sources would collide with symbols of Sections 1–19 or of docs/NOTATION.md. They are renamed here, in the new material only.

Source 11	Source 12	Here	Reason
D	D	D_z	D is a scalar derivation in the notation guide; Definition 1.4.
$K, \mathcal{E}_\Gamma$	$K, \mathcal{E}$	$\mathbb{K}, \mathcal{E}_\Gamma(k)$	Convention 14.1; $\mathcal{E}_\Gamma$ is over $\mathbb{C}$.
$\mathcal{M}_\Gamma = \text{Frac } \mathcal{E}_\Gamma$	—	$\text{Frac } \mathcal{E}_\Gamma(k)$	$M(S)$ is a support monoid.
H, H_δ	H, H_h	Δ, Δ_β	(2.1); $H \in \Gamma$ in Section 11.3.
$w, \mathcal{O}_w, \mathbf{m}_w, L, \text{res}$	w_H	$v_\Delta, \mathcal{O}_\Delta, \mathbf{m}_\Delta, k_\Delta, \text{red}_\Delta$	w_P is a corner shift; res is the residue at v .
$s, A_j, C_j, Q(T)$	$\ell, S_{ij}, c_{ij}, B(T)$	$\theta, \Pi_{ij}, \xi_{ij}, \Xi(T)$	s is an order in Question 19.1, ℓ a dilation power in (10.1), B a threshold, Q_s a shift polynomial.
F (a subfield)	L, F	L, K_0, k_0	F is the series (11.2).
$F_\alpha, \sigma, c(\sigma)$	—	$\mathcal{F}_\alpha, b, \text{code}(b)$	F as above; σ_q is a dilation.
B_i	—	$k_{J,i}$	B is a threshold.
$\Gamma_\infty = \bigoplus_{m \geq 1} \mathbb{Q}e_m$	$\bigoplus_{n \geq 0} \mathbb{Q}e_n$	(11.1)	One group, indexed from 0.
f_*	F_ω	F	See below.
Θ over $\mathbb{Q}$	$\Theta_u, u = t^h$	G, Θ_p	Corollary 17.1; (8.1).
—	$\rho(f)$	$\text{cvr}(f)$	ρ is the residue of Lemma 14.2.
—	E_h	E_η	Proposition 5.3.
—	F_τ	R_τ	F as above.
—	s, λ, μ	m, λ, λ'	s is an order; μ a dilation valuation.
Theorems 1.1–1.3	Theorems A, B, C	Theorems 21.11, E, 22.6; E (b), F, G	Theorems A–D of this report are different theorems.

The shipped audit files `11-coarsening-differential-rigidity-*` and `12-coefficient-field-rigidity-SOURCE_AUDIT.md` keep the sources' notation and numbering.

Words with two meanings. *No order unit* is not a synonym of *non-Archimedean*, *higher rank* or *infinite rational rank* (Section 21.4). *Rank one* means Archimedean; the *rational rank* rrk is a $\mathbb{Q}$ -dimension. A *dilation* is always the argument dilation $f(z) \mapsto f(\lambda z)$ with $\lambda \in \mathbb{K}^\times$, never an automorphism of the scalar field. *Residue* keeps its meaning at v (res, ρ); the residue at a coarsened valuation is written red_Δ . The *linearization multiplier* Ξ depends on a solution; it is not the residue corner polynomial χ_P or the full corner polynomial I_P of (14.6), which depend on the equation alone, and not a shift polynomial Q_s or P_s of Sections 5–10. Source 12's lemma “escape of leading values is sufficient” is Lemma 2.3 (a), not the escape chain.

Results printed once. Sources 11 and 12 reprove several facts that the report already contains. They are printed once and cited: the definition of strong summability (Definition 1.2); the sufficiency half of the all-scale criterion (Lemma 2.3 (a)); closure of the entire class under derivatives and dilations (Proposition 2.7, whose proof uses no property of $\mathbb{C}$); the surreal normal-form embedding (Section 2.4, with the same convention $t^\gamma \mapsto \omega^{-\gamma}$); the group Γ_∞ and strong entireness of F (Theorem 11.2); the failure of that entireness after a dominating scale is adjoined (Section 11.2); the domain of the formal exponential (Proposition 5.3); and the failure in positive characteristic (Example 17.3). Everything else in the two sources is printed in Sections 20–22, with proofs.

Where each numbered result went. Source 11: Theorems 1.1–1.3 are Theorem 21.11, Theorem E (a),(c) and Theorem 22.6; Lemma 2.2 is Proposition 20.2, Remark 2.3 the paragraph after it, Proposition 2.4 and Lemma 2.5 are Proposition 20.4 and Lemma 20.5; Section 3 is Section 21.2;

Theorem 4.1 is Theorem 21.2 (a) with Remark 21.3, Remark 4.2 is Remark 21.4, Corollary 4.3 is Corollary 21.8, Example 4.4 is Example 21.6, and Example 4.5 follows it; Lemmas 5.1–5.2, Remark 5.3 and Proposition 5.4 are Lemmas 20.6–20.7, Remark 20.8 and Proposition 20.9; Corollaries 6.1–6.3 and Example 6.4 are Corollaries 21.15–21.17 and the paragraph after the last; Section 7 is Section 22.2; Section 8 is Section 22.3; Questions 9.1 and 9.3 are Questions 19.1 (re-scoped) and 19.7, and Example 9.2 is discussed in Section 21.4. Source 12: Theorems A–C are Theorems E (a),(b), F and G; Proposition 2.3, Corollary 2.4 and Proposition 2.5 are Proposition 20.2, Corollary 20.3 and Proposition 20.4; Lemmas 3.1–3.2, Remark 3.3, Proposition 3.4 and Corollaries 3.5–3.6 are Lemmas 20.6–20.7, Remark 20.8, Proposition 20.9 and Corollaries 20.10–20.11; Theorems 4.1 and 5.2 are Theorem 21.2 (a) and (b), Remarks 4.2 and 5.3 are Remarks 21.4 and 21.5, Corollary 4.3 is Theorem E (a), Corollary 4.4 is Corollary 21.18, Lemma 5.1 is Lemma 21.1 and Corollary 5.4 is Corollary 21.14; Proposition 6.1 and Theorem 6.2 are Proposition 21.19 and Theorem 21.20; Theorem 7.1 and Example 7.2 are Theorem 22.1 and the paragraph after it; Propositions 8.1 and 8.2 are Proposition 22.7 and the whole-class failure of Section 22.3; Section 9 is Section 22.4; Questions 10.1–10.3 are Questions 19.1 (re-scoped), 19.8 and 19.9.

The explicit series, and an indexing discrepancy. Source 12's F_ω is exactly F of (11.2), over the same group indexed from 0. Source 11 indexes the group from 1, maps e_m to ω^m , and writes $f_* = \sum_{n \geq 1} \omega^{-\omega^n} z^n$, calling it “this exact entire example” of the report. That is off by a constant: the term $n = 0$ of F is $t^{e_0} = \omega^{-1}$, so $F = f_* + \omega^{-1}$. The discrepancy is harmless, because F and f_* differ by an element of the field, their derivatives agree, and a polynomial $P(Y_0, Y_1, \dots)$ vanishes on the jets of F exactly when $P(Y_0 + \omega^{-1}, Y_1, \dots)$ vanishes on those of f_* . It is disclosed here and not corrected in the shipped audit files. Source 11's branch family uses only codes at least 2, so it is printed over the group (11.1) unchanged.

20.2 Strong entireness at every scale

Proposition 20.2 (All-scale coefficient criterion; sources 11 and 12). *For $f = \sum_n a_n z^n \in \mathbb{K}[[z]]$ the following are equivalent:*

- (i) $f \in \mathcal{E}_\Gamma(k)$;
- (ii) the family $(a_n t^{n\gamma})_{n \geq 0}$ is strongly summable for every $\gamma \in \Gamma$;
- (iii) for all $\gamma, \beta \in \Gamma$, only finitely many n with $a_n \neq 0$ satisfy $v(a_n) + n\gamma \leq \beta$.

Proof. (i) $\Rightarrow$ (ii) is immediate. For (ii) $\Rightarrow$ (iii) fix γ and suppose that infinitely many nonzero coefficients have $A_n := v(a_n) + n\gamma \leq \beta$. Strong summability at t^γ gives a common lower bound α for all A_n , the least element of the union of the supports. Choose any $\eta > 0$ in Γ and set $\epsilon = \max(\beta - \alpha, 0) + \eta$. Then $\epsilon > 0$ and $\epsilon > \beta - \alpha$, without division or an order-unit assumption. For two selected indices $n' < n$,

$$(A_n - n\epsilon) - (A_{n'} - n'\epsilon) \leq \beta - \alpha - (n - n')\epsilon < 0,$$

so at the more exterior argument $t^{\gamma-\epsilon}$ the leading exponents of the selected summands form a strictly decreasing sequence, contradicting well-ordered support. For (iii) $\Rightarrow$ (i), a nonzero x with $v(x) = \gamma$ gives summands $a_n x^n$ of valuation $v(a_n) + n\gamma$, and Lemma 2.3 (a), whose proof uses no property of $\mathbb{C}$, applies; at $x = 0$ only one summand is nonzero. $\square$

The sufficiency half is Lemma 2.3 (a). Source 11 describes the criterion as part of the collection's support framework and source 12 as part of its entire-function programme [12]; both prove it, and it

is printed here once, with the necessity half, which Sections 1–13 did not need. The necessity proof uses a second, more exterior argument: strong summability at one argument does not by itself give cofinal growth. In $k((t^{\mathbb{Q}}))$ the family $(t^{n/(n+1)})_{n \geq 1}$ is strongly summable although its valuations are bounded (source 12), and Remark 2.5 gives a higher-rank version. Replacing the all-scale hypothesis by evaluation on one ordinary complex disk would therefore be an error (source 11).

Corollary 20.3 (When the statements are nonvacuous; source 12). *A nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ exists if and only if Γ has countable cofinality.*

Proof. If f is nonpolynomial, (iii) with $\gamma = 0$ makes the values of its infinitely many nonzero coefficients a countable cofinal subset; a nonzero ordered abelian group has no largest element, so its cofinality is not finite. Conversely choose a nondecreasing cofinal sequence of positive δ_n , $n \geq 1$, and put $a_0 = 1$, $a_n = t^{n\delta_n}$. Given γ, β , eventually $\delta_n + \gamma > \max(\beta, 0)$, so $n(\delta_n + \gamma) > \beta$, and Proposition 20.2 applies. $\square$

The equivalence is background from the entire-function report [12] (Section 1.5); source 12's proof is printed because the next sections use it. Theorems E and G therefore have nonpolynomial instances exactly for groups of countable cofinality; in uncountable cofinality they remain true but are vacuous.

Proposition 20.4 (Differential algebra structure; source 11). *$\mathcal{E}_\Gamma(k)$ is a differential $\mathbb{K}$ -subalgebra of $\mathbb{K}[[z]]$, stable under every dilation σ_λ , $\lambda \in \mathbb{K}^\times$. Its fraction field $\text{Frac } \mathcal{E}_\Gamma(k)$ is a differential subfield of $\mathbb{K}((z))$.*

Proof. Closure under sums, products, D_z and σ_λ is Proposition 2.7, whose proof uses no property of $\mathbb{C}$. Through Proposition 20.2 it can also be read off the weighted values: for a product, if $A_i = v(a_i) + i\gamma$ and $B_j = v(b_j) + j\gamma$ have lower bounds m_A, m_B , then $A_i + B_j \leq \beta$ forces $A_i \leq \beta - m_B$ and $B_j \leq \beta - m_A$, which only finitely many pairs satisfy. If a nonzero product coefficient at degree n has weighted value at most β , one of its finitely many nonzero terms $a_i b_j$, $i + j = n$, must have weighted value at most β . Thus only finitely many degrees can occur, even when some terms cancel. A zero factor is immediate. For $\mathcal{J}_{\lambda,j} f$ the coefficient of z^n is $a_{n+j}(n+j)_j \lambda^n$, whose weighted value at γ is that of a_{n+j} at $v(\lambda) + \gamma$ minus the constant $j(v(\lambda) + \gamma)$. Here the falling factorial is a nonzero element of the trivially valued prime field, so it contributes no valuation (source 12). The quotient rule extends D_z to $\text{Frac } \mathcal{E}_\Gamma(k)$, and every nonzero power series is invertible in $\mathbb{K}((z))$, so $\text{Frac } \mathcal{E}_\Gamma(k)$ embeds there. $\square$

Lemma 20.5 (A minimum coefficient value; source 11). *If $0 \neq h = \sum_n h_n z^n \in \mathcal{E}_\Gamma(k)$, some h_j satisfies $v(h_j) \leq v(h_n)$ for every nonzero h_n .*

Proof. Take any nonzero coefficient value β . By Proposition 20.2 with $\gamma = 0$ only finitely many coefficients have value at most β ; the least of them is the minimum. $\square$

20.3 Value groups of finitely generated fields

Lemma 20.6 (Valuation-rank inequality; sources 11 and 12). *Let $k_0 \subseteq L$ be fields and v a valuation on L , with values in an ordered abelian group, trivial on $k_0^\times$. If $x_1, \dots, x_d \in L^\times$ have values linearly independent over $\mathbb{Q}$, they are algebraically independent over k_0 . Consequently*

$$\text{rrk } v(L^\times) \leq \text{trdeg}_{k_0} L. \quad (20.2)$$

Proof. For distinct $\alpha, \alpha' \in \mathbb{N}^d$ the monomials x^α and $x^{\alpha'}$ have different values, since equality would be an integer relation among the $v(x_i)$. A nonzero coefficient from k_0 has value zero. So the finitely many nonzero terms of a nonzero polynomial evaluated at the x_i have pairwise distinct values, and the unique term of least value cannot be cancelled. For the rank inequality, choose a rational basis of $v(L^\times) \otimes_{\mathbb{Z}} \mathbb{Q}$ from the images of elements of $v(L^\times)$, and lift those values to elements of $L^\times$. Every polynomial uses only finitely many of the lifts, so the preceding argument proves that the whole lifted family is algebraically independent. Its cardinality is the rational rank, giving the inequality also when either rank is infinite. $\square$

Source 12 states the lemma for $k \subseteq L \subseteq \mathbb{K}$, source 11 for $k_0 = \mathbb{Q}$ and any valuation trivial on $\mathbb{Q}$; both cases are used below.

Lemma 20.7 (A principal convex bound; sources 11 and 12). *Let $\Delta' \subseteq \Gamma$ be a subgroup of finite rational rank. Either $\Delta' = 0$, or $\Delta' \subseteq \Delta_\beta$ for some $\beta > 0$, where Δ_β is the convex subgroup (2.1); in the second case β can be chosen in Δ' , and Δ_β is then the smallest convex subgroup containing Δ' . The subgroup Δ_β is proper exactly when β is not an order unit, so it is proper for every β when Γ has no order unit.*

Proof. Choose $\delta_1, \dots, \delta_d \in \Delta'$ whose images form a basis of $\Delta' \otimes \mathbb{Q}$ and put $\beta = \sum_i |\delta_i| > 0$, an element of Δ' . For $\delta \in \Delta'$, clearing denominators gives $m\delta = \sum_i n_i \delta_i$ with a positive integer m and integers n_i , so $|\delta| \leq m|\delta| \leq (\sum_i |n_i|)\beta$. Hence $\Delta' \subseteq \Delta_\beta$, and Δ_β , the smallest convex subgroup containing $\beta \in \Delta'$, lies in every convex subgroup containing Δ' . The last assertion is the definition of an order unit. $\square$

No division in Γ occurs. Source 11 takes $\delta = \eta + \sum_i |\delta_i|$ with an arbitrary $\eta > 0$, which covers $\Delta' = 0$ by the same formula.

Remark 20.8 (Field values, not generator values; sources 11 and 12). The bound (20.2) concerns the value group of the whole field. It would be false to use the values of a chosen list of generators: for $0 < \alpha < \beta$ with α, β rationally independent, both generators of $\mathbb{Q}(t^\alpha, t^\alpha + t^\beta)$ have value α , but their difference has value β (source 11). Conversely, one generator $b \in \mathbb{K}$ may have infinitely many rationally independent exponents in its support without contradicting $\text{rrk } v(k(b)^\times) \leq 1$: the field $k(b)$ contains rational functions of b , not the truncations of its Hahn expansion (source 12). This is why the coefficients of the equations below may be arbitrary Hahn elements.

For a convex subgroup $\Delta \subseteq \Gamma$ the quotient Γ/Δ is an ordered abelian group, and

$$v_\Delta(a) = v(a) + \Delta \in \Gamma/\Delta \quad (a \in \mathbb{K}^\times), \quad v_\Delta(0) = +\infty,$$

is a valuation on the same field $\mathbb{K}$, a coarsening of v , as in Theorem 7.4; it is not a coefficientwise map to a smaller Hahn field. Write $\mathcal{O}_\Delta = \{a : v_\Delta(a) \geq 0\}$, $\mathfrak{m}_\Delta = \{a : v_\Delta(a) > 0\}$, $k_\Delta = \mathcal{O}_\Delta/\mathfrak{m}_\Delta$ and $\text{red}_\Delta : \mathcal{O}_\Delta \rightarrow k_\Delta$. No splitting of $\Gamma \rightarrow \Gamma/\Delta$ and no identification of k_Δ with a Hahn field is used. Write $\eta > \Delta$ when $\eta > \delta$ for every $\delta \in \Delta$; by convexity every positive element outside Δ satisfies this, and $v_\Delta(x) < 0$ means exactly $v(x) = -\eta$ with $\eta > \Delta$.

Proposition 20.9 (Coarsened exterior obstruction; sources 11 and 12). *Suppose all nonzero coefficients of a nonpolynomial $f = \sum_n a_n z^n \in \mathbb{K}[[z]]$ have values in a proper convex subgroup Δ . Then*

$$\text{Dom}_{\text{str}}(f) \subseteq \{0\} \cup \{x \in \mathbb{K}^\times : v_\Delta(x) \geq 0\}; \quad (20.3)$$

in particular $t^{-\eta} \notin \text{Dom}_{\text{str}}(f)$ for every $\eta > \Delta$.

Proof. Let $v(x) = -\eta$ with $\eta > \Delta$. For nonzero coefficients at indices $m > n$, $v(a_m x^m) - v(a_n x^n) = (v(a_m) - v(a_n)) - (m - n)\eta < 0$, because the first difference lies in Δ . The leading exponents of the infinitely many nonzero summands therefore strictly decrease, and Lemma 2.2 applies. $\square$

Source 11 states this for $x = t^{-\eta}$ and coefficients in a field whose values lie in Δ ; source 12 states (20.3). The inclusion is necessary, not sufficient: points with $v_\Delta(x) \geq 0$ need not lie in $\text{Dom}_{\text{str}}(f)$. For $k = \mathbb{C}$ it is consistent with inward stability (Lemma 2.4).

Corollary 20.10 (Finite transcendence degree; source 12). *Assume Γ has no order unit. If all coefficients of $f \in \mathcal{E}_\Gamma(k)$ lie in a subfield $L \supseteq k$ of $\mathbb{K}$ with $\text{trdeg}_k L < \infty$, then $f \in \mathbb{K}[z]$.*

Proof. By Lemmas 20.6 and 20.7, $v(L^\times)$ lies in $\{0\}$ or in some Δ_β , and both are proper. Proposition 20.9 excludes a nonpolynomial f . $\square$

For $f = \sum_n a_n z^n$ define the *coefficient-value rank*

$$\text{cvr}(f) = \dim_{\mathbb{Q}} \left(\text{span}_{\mathbb{Q}} \{v(a_n) \otimes 1 : a_n \neq 0\} \right). \quad (20.4)$$

The span is taken in $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$, and $\text{cvr}(f)$ is its dimension. It is the rational rank of the values of the Taylor coefficients, not the number of support exponents of any single coefficient.

Corollary 20.11 (Necessary infinite coefficient-value rank; source 12). *If Γ has no order unit and $f \in \mathcal{E}_\Gamma(k)$ is not a polynomial, then $\text{cvr}(f) = \aleph_0$ and $\text{trdeg}_k k(a_0, a_1, \dots) = \aleph_0$.*

Proof. If $\text{cvr}(f)$ were finite, the group generated by the coefficient values would lie in a proper Δ_β by Lemma 20.7, contradicting Proposition 20.9. There are countably many coefficients, which bounds the rank; Lemma 20.6 gives the lower bound for the transcendence degree and countable generation the upper bound. $\square$

21 The coefficient-field theorem and its consequences

21.1 One finitely generated field for every finite relation

Coefficient recurrences for algebraic differential equations go back to Hurwitz [23], and Lemma 3 of Laohakosol, Kongsakorn and Ubolsri [24] isolates a high coefficient through a polynomial in its index; Krattenthaler and Rivoal treat the arithmetic of such coefficients [25], and Ait El Manssour, Sattelberger and Teguia Tabuguia the general framework of D-algebraic functions [26]. Only the field-theoretic consequence is needed here. Both sources supply a self-contained formal linearization proof over arbitrary characteristic-zero fields that allows singular solutions, and neither advertises the finite-generation principle or the valuation-rank inequality as a discovery.

For several dilations one classical theorem is imported and not proved: the *Skolem–Mahler–Lech theorem*, that the zero set of a linear recurrence sequence over a field of characteristic zero is a finite set together with finitely many infinite arithmetic progressions. Source 12 uses it in the general-field form of Bell’s Theorem 1.1 [27], credited there to Lech [28]; no generalized dynamical form of the theorem is used.

Lemma 21.1 (Nondegenerate exponential polynomials; source 12). *Let K_0 be a field of characteristic zero and let $\lambda_1, \dots, \lambda_m \in K_0^\times$ with $\lambda_i/\lambda_{i'}$ not a root of unity for $i \neq i'$. If $p_1, \dots, p_m \in K_0[T]$ are not all zero, then*

$$\Xi(n) = \sum_{i=1}^m \lambda_i^n p_i(n) \quad (21.1)$$

is nonzero for all sufficiently large $n \in \mathbb{N}$.

Proof. Discard the zero p_i , and let E be the shift $(Eu)(n) = u(n+1)$. The sequence $\lambda_i^n p_i(n)$ is annihilated by $(E - \lambda_i)^{\deg p_i + 1}$, so the product of these annihilators kills Ξ : they commute because they are polynomials in the same shift E . Thus Ξ satisfies a linear recurrence with constant coefficients in K_0 , and by the imported theorem its zero set is a finite set together with finitely many infinite arithmetic progressions. No infinite progression can occur. If $\Xi(a + bu) = 0$ for all $u \geq 0$, with $b \geq 1$, then $\sum_i (\lambda_i^b)^u \lambda_i^a p_i(a + bu) = 0$ for all u , and the bases $\alpha_i = \lambda_i^b$ are pairwise distinct by the root-of-unity hypothesis. Polynomial-exponential sequences with pairwise distinct nonzero bases are linearly independent: to isolate the base α_i , apply $\prod_{i' \neq i} (E - \alpha_{i'})^{d_{i'} + 1}$, where $d_{i'}$ is the degree of the polynomial at $\alpha_{i'}$; this kills the other terms, and on $\alpha_i^u p(u)$ each factor keeps the degree and multiplies the leading coefficient by $\alpha_i - \alpha_{i'} \neq 0$, leaving a nonzero power times a nonzero polynomial, which cannot vanish at every natural number in characteristic zero. Hence every $p_i(a + bT)$, and so every p_i , is zero, a contradiction. $\square$

For $m = 1$ the lemma is the finiteness of the roots of a nonzero polynomial, since $\lambda_1^n \neq 0$. When only the $j = 0$ jets occur and $\lambda_i = q^{m_i}$ with distinct integers m_i and a nontorsion q , the multiplier is a nonzero Laurent polynomial evaluated at the distinct elements q^n , again without the imported theorem. The general mixed-dilation argument uses the imported theorem.

Theorem 21.2 (Finite coefficient field; sources 11 and 12). *Let $k_0 \subseteq K_0$ be fields of characteristic zero and $u = \sum_{n \geq 0} a_n z^n \in K_0[[z]]$, with D_z killing K_0 .*

- (a) (Sources 11 and 12.) *If u is differentially algebraic over $k_0((z))$, there is $N \geq 0$ with $a_n \in k_0(a_0, \dots, a_{N-1})$ for every n .*
- (b) (Source 12.) *Let $\lambda_1, \dots, \lambda_m \in K_0^\times$ have pairwise quotients that are not roots of unity. If finitely many of the series $\mathcal{J}_{\lambda_i, j} u$ are algebraically dependent over $K_0(z)$, then all a_n lie in one subfield $L \subseteq K_0$ finitely generated over k_0 , generated by the λ_i , the finitely many coefficients of a suitable polynomial relation, and finitely many a_n .*

Proof. Treat both parts at once, first choosing a relation involving derivative orders at most r . In case (a) put $m = 1$, $\lambda_1 = 1$ and $R = k_0[[z]]$, and multiply the relation by a sufficiently large power of z to remove the finitely many Laurent poles of its coefficients. In case (b) put $R = K_0[z]$ and clear a common polynomial denominator of its finitely many rational-function coefficients. In either case there is a nonzero polynomial over R in variables Y_{ij} , $1 \leq i \leq m$, $0 \leq j \leq r$, vanishing at $Y_{ij} = \mathcal{J}_{\lambda_i, j} u$. Among these relations choose P of least total degree in the Y_{ij} . Its degree is positive, since a nonzero element of R does not vanish in $K_0[[z]]$. Put

$$\Pi_{ij} = \frac{\partial P}{\partial Y_{ij}}((\mathcal{J}_{\lambda_i, j} u)_{i, j}) \in K_0[[z]].$$

A nonzero polynomial partial has nonzero evaluation, by minimality of the total degree, and characteristic zero gives at least one. Put

$$\theta = \min_{\Pi_{ij} \neq 0} (\text{ord}_z \Pi_{ij} - j), \quad \xi_{ij} = [z^{\theta+j}] \Pi_{ij}, \quad (21.2)$$

with $\xi_{ij} = 0$ when $\Pi_{ij} = 0$ or $\theta + j < 0$, so $\theta \geq -r$ and some $\xi_{ij} \neq 0$, and put

$$\Xi(n) = \sum_{i,j} \xi_{ij} \lambda_i^{n-j} (n)_{\underline{j}} = \sum_i \lambda_i^n p_i(n), \quad p_i(T) = \sum_j \xi_{ij} \lambda_i^{-j} (T)_{\underline{j}}. \quad (21.3)$$

Falling factorials of distinct degrees are linearly independent, so $p_i \neq 0$ whenever some $\xi_{ij} \neq 0$, and Lemma 21.1 makes $\Xi(n) \neq 0$ for all large n . In case (a), $\Xi(T) = \sum_j \xi_{1j} (T)_{\underline{j}}$ is a nonzero polynomial, and the lemma is not needed.

Let $T_n = \sum_{m' < n} a_{m'} z^{m'}$ and $g_n = u - T_n$. By (20.1), $\text{ord}_z \mathcal{J}_{\lambda_i, j} g_n \geq n - j \geq n - r$ for $n \geq r$. Taylor expansion of P at the jets of u gives

$$P((\mathcal{J}_{\lambda_i, j} T_n)_{i,j}) = - \sum_{i,j} \Pi_{ij} \mathcal{J}_{\lambda_i, j} g_n + R_n, \quad \text{ord}_z R_n \geq 2(n - r), \quad (21.4)$$

because every term of R_n has at least two factors $\mathcal{J}_{\lambda_i, j} g_n$ and its other factors have nonnegative order; this holds even when the evaluated partials vanish to high order. For $n > \theta + 2r$ one has $2(n - r) > n + \theta$, so R_n contributes nothing in degree $n + \theta$. For each nonzero Π_{ij} the coefficient contributed by $a_{m'}$ to degree $n + \theta$ is

$$a_{m'} (m')_{\underline{j}} \lambda_i^{m'-j} [z^{n+\theta-m'+j}] \Pi_{ij}, \quad m' \geq n.$$

If $m' > n$, the displayed exponent is below $\theta + j$ and hence below $\text{ord}_z \Pi_{ij}$, or is negative, so this coefficient is zero. At $m' = n$ it is $a_n (n)_{\underline{j}} \lambda_i^{n-j} \xi_{ij}$. Terms with $\Pi_{ij} = 0$ contribute nothing. Thus only the stated multiplier of a_n remains:

$$a_n \Xi(n) = -[z^{n+\theta}] P((\mathcal{J}_{\lambda_i, j} T_n)_{i,j}). \quad (21.5)$$

Put $N_0 = \theta + 2r \geq r$. Let $L_0 = k_0$ in case (a); in case (b), let L_0 be generated over k_0 by the λ_i and the finitely many scalar coefficients of P . Replacing u by T_{N_0+1} changes every jet of order at most r only in degrees at least $N_0 + 1 - r = \theta + r + 1$. Polynomial substitution with coefficients in R preserves this congruence. Since ξ_{ij} is a coefficient in degree $\theta + j \leq \theta + r$, it is determined by that truncation; thus $\xi_{ij} \in L_0(a_0, \dots, a_{N_0})$.

Choose $N > N_0$ so that $\Xi(n) \neq 0$ for every $n \geq N$, and put $L = L_0(a_0, \dots, a_{N-1})$. Its generators contain the required prefix, so $\Xi(n) \in L^\times$ for $n \geq N$. If all $a_{m'}$ with $m' < n$ already lie in L , then the right side of (21.5) lies in L : in case (a) every coefficient of the finitely many power series occurring as coefficients of P lies in k_0 , and in case (b) all scalar coefficients of P lie in L_0 . Division by $\Xi(n)$ gives $a_n \in L$, completing the induction. In case (a), $L = k_0(a_0, \dots, a_{N-1})$; in case (b), L is finitely generated over k_0 . $\square$

Remark 21.3 (Second route: source 11's minimal relation). For part (a), source 11 chooses a relation of least order r and then least degree in Y_r . Its separant $\partial P / \partial Y_r$ is a nonzero polynomial in characteristic zero, and its evaluation is nonzero, since otherwise it would be a relation of smaller order, or of the same order and smaller degree in Y_r . It then defines the offset and multiplier from the nonzero evaluated partials exactly as in (21.2). Source 12's least total degree makes *every* nonzero partial evaluate to a nonzero series. The proof needs only one nonzero evaluated partial, so either choice works; they are the two sources' routes to the same recurrence.

Remark 21.4 (No nonsingular initial condition; sources 11 and 12). The theorem assumes no evaluated partial nonzero at $z = 0$, no equation solved for the highest derivative, no irreducibility and no nonzero residue corner. A nonzero evaluated partial is used as a formal series, not through its constant term; the offset θ absorbs its vanishing order, and the finitely many natural zeros of Ξ are placed in the initial segment. No implicit-function theorem at an ordinary point is invoked.

Remark 21.5 (Existential, not an algorithm; source 12). The invoked form of Skolem–Mahler–Lech supplies no effective bound on the last exceptional index, and no effective equality, leading-exponent access or computable representation of the coefficients of P is assumed. Part (b) is an existence statement. In part (a) the multiplier is explicitly a polynomial, but computational claims would still have to say how its scalar coefficients are represented.

Example 21.6 (Linearization along the vanishing-corner example; sources 11 and 12). For the equation (16.1), $P = z^2 Y_0 Y_2 + z Y_0 Y_1 - z^2 Y_1^2$, whose residue corner polynomial vanishes identically, take the solution $u = az^m$ with $a \neq 0$ and $m \geq 0$ (Proposition 16.4). Writing Π_j for Π_{1j} , the evaluated partials are

$$\Pi_0 = am^2 z^m, \quad \Pi_1 = a(1 - 2m)z^{m+1}, \quad \Pi_2 = az^{m+2},$$

so $\theta = m$ (for $m = 0$ after discarding $\Pi_0 = 0$) and

$$\Xi(T) = a(m^2 + (1 - 2m)T + T(T - 1)) = a(T - m)^2 \neq 0. \quad (21.6)$$

Source 12 takes $a = 1$; source 11 carries a . The multiplier is nonzero although $\chi_P = 0$: it is computed along a solution, χ_P from the equation alone, and its only exceptional index $n = m$ moves with the solution. This is why the argument needs no degree bound for polynomial solutions, which this equation does not have. The displayed P is not of least degree for az^m , which also satisfies $zD_z u - mu = 0$; the computation shows that the recurrence (21.5) can be run for this P directly, since one of its evaluated partials is nonzero. Neither source infers the general theorem from the example.

Example 21.7 (The initial segment is not bounded by the order; source 11). For an integer $N \geq 0$, the equation $zD_z u - Nu = 0$ has the solution az^N with arbitrary a , and $\Xi(T) = T - N$. Over a field not containing a , the initial segment must reach degree N , so the differential order alone does not bound its length.

Corollary 21.8 (Finitely many scalar parameters; source 11). *Let K_0 be a field of characteristic zero. If $u \in K_0((z))$ is differentially algebraic over $K_0(z)$, all its Laurent coefficients lie in a subfield of K_0 finitely generated over $\mathbb{Q}$.*

Proof. Clear denominators in a relation; its finitely many scalar coefficients $c_1, \dots, c_q$ generate $k_0 = \mathbb{Q}(c_1, \dots, c_q)$. Write $u = z^{-m'}y$ with a power series y , taking $m' \geq 0$. Leibniz’s formula expresses the jets of u through order r as a triangular $k_0(z)$ -linear combination of the jets of y through that order, with nonzero diagonal $z^{-m'}$. This invertible substitution preserves nonzero polynomial relations, so y is differentially algebraic over $k_0(z) \subseteq k_0((z))$. Theorem 21.2 (a) puts the coefficients of y , hence of u , in k_0 with finitely many initial coefficients adjoined. $\square$

21.2 Coarsening descent

This subsection is source 11’s. Its polynomial half is a coefficient-field form of the entire-function report’s obstruction for noncofinal extensions of the workspace [12]; the fraction-field half is the part that needs a new argument.

Lemma 21.9 (Polynomial reduction of an entire series; source 11). *Let $\Delta \subsetneq \Gamma$ be convex. If $g = \sum_n g_n z^n \in \mathcal{E}_\Gamma(k)$ has all coefficients in $\mathcal{O}_\Delta$, its coefficientwise reduction $\bar{g} = \sum_n \text{red}_\Delta(g_n) z^n$ is a polynomial in $k_\Delta[z]$. Every nonzero $h \in \mathcal{E}_\Gamma(k)$ has a scalar multiple h/c with coefficients in $\mathcal{O}_\Delta$ and nonzero reduction.*

Proof. Because Δ is proper, choose a positive element η outside it; convexity gives $\eta > \Delta$. By Proposition 20.2 with $\gamma = 0$, eventually $v(g_n) > \eta$, so $v_\Delta(g_n) > 0$ and $\text{red}_\Delta(g_n) = 0$. For the second assertion take $c = h_j$ as in Lemma 20.5: all coefficients of h/c have nonnegative v , and the j th is 1. $\square$

Lemma 21.10 (Rationality needs no new constants; source 11). *For any extension of fields $L \subseteq L'$, $L'(z) \cap L((z)) = L(z)$ inside $L'((z))$.*

Proof. After multiplying by a power of z , let $u = \sum_{n \geq 0} u_n z^n \in L[[z]]$ lie in $L'(z)$. Being regular at zero, $u = p/q$ with $p, q \in L'[z]$ and $q(0) = 1$; put $d = \deg q$ and take $N \geq \max(d, \deg p)$. Then $u_n + q_1 u_{n-1} + \cdots + q_d u_{n-d} = 0$ for $n > N$. This is a linear system with coefficients in L in the finitely many unknowns $q_1, \dots, q_d$. The augmented rows lie in L^{d+1} , so finitely many of them span all the rows. Solving that finite subsystem is therefore equivalent to solving the whole system. Gaussian elimination uses coefficients in L , and cannot produce an inconsistent row because a solution exists in L' . Setting its free variables to zero gives a solution $b_1, \dots, b_d \in L$. Then $\tilde{q} = 1 + \sum_{j=1}^d b_j z^j \in L[z]$ satisfies $[z^n](\tilde{q}u) = 0$ for all $n > N$, so $\tilde{q}u \in L[z]$ and $u \in L(z)$. This includes $d = 0$, when there are no unknowns and u is already a polynomial. $\square$

Theorem 21.11 (Coarsening descent; source 11). *Let $L \subseteq \mathbb{K}$ be a subfield and $\Delta \subsetneq \Gamma$ a convex subgroup with $v(L^\times) \subseteq \Delta$. Then, inside $\mathbb{K}((z))$,*

$$\mathcal{E}_\Gamma(k) \cap L[[z]] = L[z], \quad \text{Frac } \mathcal{E}_\Gamma(k) \cap L((z)) = L(z). \quad (21.7)$$

Proof. Let $u = \sum_n u_n z^n \in \mathcal{E}_\Gamma(k) \cap L[[z]]$. Every nonzero u_n has value in Δ , below any fixed $\eta > \Delta$, so by Proposition 20.2 only finitely many u_n are nonzero.

Next let $u \in \text{Frac } \mathcal{E}_\Gamma(k) \cap L[[z]]$, say $u = g/h$ with $g, h \in \mathcal{E}_\Gamma(k)$ and $h \neq 0$. Normalize h by c as in Lemma 21.9. Since v_Δ vanishes on $L^\times$, red_Δ embeds L as a subfield $\bar{L} \subseteq k_\Delta$: every nonzero element of L is a unit of $\mathcal{O}_\Delta$, and so has nonzero residue. In the formal identity

$$g/c = (h/c)u \quad (21.8)$$

every coefficient on the right is a *finite* sum of products of elements of $\mathcal{O}_\Delta$, so g/c has coefficients in $\mathcal{O}_\Delta$; both g/c and h/c lie in $\mathcal{E}_\Gamma(k)$. By Lemma 21.9 their reductions are polynomials, the second nonzero, and reducing (21.8) coefficientwise gives $\overline{h/c} \bar{u} = \overline{g/c}$ in $k_\Delta[[z]]$. So $\bar{u} \in k_\Delta(z) \cap \bar{L}[[z]]$, and Lemma 21.10 gives $\bar{u} \in \bar{L}(z)$. Write $\bar{u} = \bar{p}/\bar{q}$ with $\bar{p}, \bar{q} \in \bar{L}[z]$ and $\bar{q} \neq 0$. Lift their finitely many coefficients through the isomorphism $L \rightarrow \bar{L}$ to $p, q \in L[z]$, with $q \neq 0$. Every coefficient of $qu - p$ belongs to L and has zero residue; injectivity on L therefore gives $qu = p$, hence $u \in L(z)$. This lifts a rational identity over the embedded field, not arbitrary residue coefficients from k_Δ . Finally, for $u \in \text{Frac } \mathcal{E}_\Gamma(k) \cap L((z))$ choose $m' \geq 0$ so that $z^{m'}u \in L[[z]]$, apply the power-series case, and divide by $z^{m'}$. The inclusions $\supseteq$ hold because all polynomials over L are strongly entire and their quotients belong to $\text{Frac } \mathcal{E}_\Gamma(k)$. $\square$

Remark 21.12 (Numerator and denominator together; source 11). The coefficients of g and h need not lie in L , and neither need be a polynomial. The proof works because their *common* coarsened reduction gives a polynomial equation for the quotient; asserting that $g, h \in L[[z]]$ would be unjustified.

Example 21.13 (The formal exponential; source 11). With $L = k$ and $\Delta = \{0\}$, Theorem 21.11 gives $\text{Frac } \mathcal{E}_\Gamma(k) \cap k((z)) = k(z)$ for every nonzero Γ . The formal series $\sum_n z^n/n!$ is not rational: if $r = p/q \neq 0$ with $p, q \in k[z]$ satisfied $D_z r = r$, then $p'q - pq' = pq$. The left side has degree at

most $\deg p + \deg q - 1$ (or is zero), while the right side has degree $\deg p + \deg q$. This is impossible, so the formal exponential is not a quotient of strongly entire series. This says nothing about the complex exponential, which belongs to a different analytic category.

21.3 Proofs of Theorems E–G and further consequences

Proof of Theorem F (source 12). A dependence uses finitely many $\lambda \in \Lambda$ and derivative orders. Theorem 21.2 (b) with $k_0 = k$ and $K_0 = \mathbb{K}$ puts every a_n in a subfield $L \subseteq \mathbb{K}$ finitely generated over k . By Lemma 20.6, $v(L^\times)$ has finite rational rank, and it contains every coefficient value, contradicting $\text{cvr}(f) = \infty$. $\square$

Proof of Theorem E. (b) (Source 12.) By Corollary 20.11, $\text{cvr}(f) = \aleph_0$, and Theorem F applies. For q of infinite order and $\Lambda = \{q^i : i \in \mathbb{Z}\}$, a quotient $q^{i-i'}$ is a root of unity only if $i = i'$.

(a) (Sources 11 and 12.) Take $\Lambda = \{1\}$ in (b). This case needs only Theorem 21.2 (a), with k_0 generated over k by the finitely many scalar coefficients of a relation, and Corollary 20.10; no Skolem–Mahler–Lech. An algebraic dependence among $f, D_z f, \dots, D_z^r f$ becomes, after clearing denominators, a nonzero equation $P(z, f, \dots, D_z^r f) = 0$ with $P \in \mathbb{K}[z, Y]$, and conversely. *Second route (source 11):* a differentially algebraic $f \in \mathcal{E}_\Gamma(k)$ has, by Corollary 21.8, its coefficients in a field L finitely generated over $\mathbb{Q}$; by Lemmas 20.6 (with $k_0 = \mathbb{Q}$) and 20.7, $v(L^\times)$ lies in a proper convex subgroup; and the polynomial half of Theorem 21.11 gives $f \in L[z]$.

(c) (Source 11.) Let $u \in \text{Frac } \mathcal{E}_\Gamma(k)$ be differentially algebraic over $\mathbb{K}(z)$. As in the second route, its Laurent coefficients lie in a field L with $v(L^\times)$ in a proper convex subgroup Δ , and the fraction-field half of Theorem 21.11 gives $u \in L(z) \subseteq \mathbb{K}(z)$. Conversely $\mathbb{K}(z) \subseteq \text{Frac } \mathcal{E}_\Gamma(k)$, since polynomials are entire, and rational functions are algebraic, hence differentially algebraic, over $\mathbb{K}(z)$. $\square$

Corollary 21.14 (A free differential algebra of dilates; source 12). *Under the hypotheses of Theorem E (b), the homomorphism from the differential polynomial algebra $\mathbb{K}(z)\{X_\lambda : \lambda \in \Lambda\}$, with $D_z(z) = 1$ and $D_z(\mathbb{K}) = 0$, to $\mathbb{K}((z))$ sending X_λ to $f(\lambda z)$ is injective. For $\Lambda = \{q^i : i \in \mathbb{Z}\}$ this gives countably many differentially algebraically independent functions.*

Proof. $D_z^j X_\lambda$ goes to $\lambda^j \mathcal{J}_{\lambda,j} f$, and multiplying members of an algebraically independent family by nonzero elements of the base field keeps it independent. $\square$

This is stronger than differential transcendence of each dilate separately: for $q = 2$, no nonzero polynomial over $\mathbb{K}(z)$ relates $f(z)$, $D_z f(z)$, $f(2z)$, $(D_z^2 f)(4z)$ and any further finite selection from the family.

Corollary 21.15 (Relative differential-algebraic closedness; source 11). *If Γ has no order unit, no element of $\text{Frac } \mathcal{E}_\Gamma(k) \setminus \mathbb{K}(z)$ is differentially algebraic over $\mathbb{K}(z)$.*

This is Theorem E (c) restated. It does not say that $\mathbb{K}(z)$ is differentially closed; it describes only the extension $\text{Frac } \mathcal{E}_\Gamma(k)/\mathbb{K}(z)$.

Corollary 21.16 (Finite rational differential systems; source 11). *Assume Γ has no order unit, and let $u_1, \dots, u_{m'} \in \text{Frac } \mathcal{E}_\Gamma(k)$ satisfy $D_z u_i = R_i(z, u_1, \dots, u_{m'})$ with $R_i \in \mathbb{K}(z)(Y_1, \dots, Y_{m'})$, each denominator nonzero after substitution. Then every u_i lies in $\mathbb{K}(z)$, and every strongly entire u_i is a polynomial.*

Proof. The field $\mathbb{K}(z)(u_1, \dots, u_{m'})$ is stable under D_z : the equations put each $D_z u_i$ in the field, and the quotient rule handles every rational expression in the u_i . Its transcendence degree over $\mathbb{K}(z)$ is at most m' , so the $m' + 1$ elements $u_i, D_z u_i, \dots, D_z^{m'} u_i$ are algebraically dependent. Apply Theorem E (c), and (a) for the entire ones. $\square$

Corollary 5.5 is the linear case for every $\Gamma \neq 0$; this corollary allows nonlinear rational systems and quotients of entire series, but needs the absence of an order unit.

Corollary 21.17 (Painlevé I in the fraction field; source 11). *If Γ has no order unit, the equation $D_z^2 u = 6u^2 + z$ has no solution in $\text{Frac } \mathcal{E}_\Gamma(k)$.*

Proof. A solution would be rational by Theorem E (c), and $u = 0$ fails. A nonzero rational u has a leading term cz^d at infinity. If $d \geq 1$, then $6u^2$ has degree $2d$, above the degree 1 of z and the degree at most $d - 2$ of $D_z^2 u$. If $d \leq 0$, then z has degree 1 while u^2 and $D_z^2 u$ have degree at most 0. No cancellation is possible. $\square$

Proposition 16.3 already excludes strongly entire solutions for every $\Gamma \neq 0$; the corollary extends this to quotients of entire series when there is no order unit. It is not a statement about local formal solutions, which exist for all $a_0, a_1 \in \mathbb{K}$ through $(n + 1)(n + 2)a_{n+2} = 6 \sum_{j=0}^n a_j a_{n-j} + \mathbf{1}_{n=1}$, nor about classical complex solutions. The conclusion cannot be strengthened from rational to constant: for every $c \in \mathbb{K}$, $u = 1/(z - c) \in \text{Frac } \mathcal{E}_\Gamma(k)$ satisfies $D_z u = -u^2$, and at $c = 0$ it has a pole at the origin, which is why the Laurent form of Theorem 21.11 is needed (source 11).

Corollary 21.18 (A solution-dependent exterior certificate; source 12). *Let $f \in \mathbb{K}[[z]]$ be a nonpolynomial formal series, differentially algebraic over $\mathbb{K}(z)$, and let $L \supseteq k$ be a field finitely generated over k that contains its coefficients, as supplied by Theorem 21.2 (a) with k_0 generated over k by the scalar coefficients of a relation. If the smallest convex subgroup Δ containing $v(L^\times)$ is proper, every x with $v_\Delta(x) < 0$ lies outside $\text{Dom}_{\text{str}}(f)$. If Γ has no order unit, Δ is always proper.*

Proof. Lemmas 20.6 and 20.7 make $\Delta = \{0\}$ or $\Delta = \Delta_\beta$ with $\beta \in v(L^\times)$; apply Proposition 20.9. $\square$

The certificate is structural, not a uniform algorithm for arbitrary Hahn names: it can need the values of rational functions of the finitely many generators of L , which their raw supports do not supply. It differs from Theorem 15.1, which needs $\chi_P \neq 0$ and finitely many leading valuations of the solution but no hypothesis on convex subgroups.

21.4 The order-unit boundary

Proposition 21.19 (A one-generator coefficient field; source 12). *Let $\mu > 0$ be an order unit of Γ and $p = t^\mu$. The partial theta series Θ_p of (8.1) is strongly entire and nonpolynomial, satisfies $\Theta_p(z) - z\Theta_p(pz) - 1 = 0$, has all its coefficients in $k(p)$, and p has infinite multiplicative order.*

Proof. All coefficients are nonzero and $v(p^{m'}) = m'\mu \neq 0$ for $m' \neq 0$. Given γ, β , choose positive integers A, B with $\gamma \geq -A\mu$ and $\beta \leq B\mu$; then $v(p^{n(n-1)/2}) + n\gamma \geq (n(n-1)/2 - An)\mu > \beta$ for large n , and Proposition 20.2 applies. The functional identity is (8.2). $\square$

For $k = \mathbb{C}$ this is also a case of Theorem 8.1; the short proof above works over any k .

Proof of Theorem G (source 12). Proposition 21.19 gives (i) $\Rightarrow$ (ii) and (i) $\Rightarrow$ (iii): the identity $\Theta_p(z) - z\Theta_p(pz) - 1 = 0$ is the nonzero polynomial $Y_0 - zY_1 - 1$ vanishing at $(\mathcal{J}_{1,0}\Theta_p, \mathcal{J}_{p,0}\Theta_p)$. If (ii) holds without an order unit, Corollary 20.10 contradicts nonpolynomiality, so (ii) $\Rightarrow$ (i). A dependence as in (iii) involves finitely many powers of q and derivative orders, and Theorem 21.2 (b) puts every coefficient in a field finitely generated over k , so (iii) $\Rightarrow$ (ii). $\square$

Theorem B and Corollary 11.3 detect an order unit by linear dilation equations with a cofinal parameter; Theorem G detects it by coefficient fields and by mixed jets, with the same partial theta witness. As proved by source 12, neither said that an order unit produces a nonpolynomial strongly entire *differentially* algebraic series: the witness's equation contains a dilation. That was Question 19.1 before source 13; clause (iv), proved in Section 25.1 from Theorem H, now says it, with the witness $\mathcal{T}_p$.

What “no order unit” does and does not mean (sources 11 and 12). It is not a synonym of non-Archimedean or higher rank: $\mathbb{Q} \oplus \mathbb{Q}$ ordered lexicographically, with the first coordinate dominant, has two Archimedean classes, and $(1, 0)$ is an order unit; so has every finite lexicographic product. An Archimedean group, $\mathbb{R}$ for instance, can have infinite rational rank and an order unit, and an infinite-rank group has an order unit when it has a largest Archimedean class. The increasing direct sum (11.1) has none. The proofs use exactly that every Δ_β is proper. With an order unit a finitely generated coefficient field can already have values whose convex hull is all of Γ ; then Lemma 20.7 forces no proper coarsening, and the argument stops for that definite reason. The series G of Corollary 17.1 over $\mathbb{C}((t^\mathbb{Q}))$ is the example of source 11: strongly entire, with coefficients in $\mathbb{Q}(t)$ whose value group $\mathbb{Z}$ has convex hull $\mathbb{Q}$. The collection already contains G and its first-order consequence, and neither source makes a new all-order claim for it.

Theorem 21.20 (An explicit rank-one family; source 12). *Let $k = \mathbb{C}$, $\Gamma = \mathbb{R}$, and let $\tau > 1$ be a transcendental real number. The series $R_\tau(z) = \sum_{n \geq 0} t^{\tau^n} z^n$ is strongly entire over $\mathbb{C}((t^\mathbb{R}))$, and for every $\Lambda \subseteq \mathbb{C}((t^\mathbb{R}))^\times$ with pairwise quotients that are not roots of unity, all $(D_z^j R_\tau)(\lambda z)$ are algebraically independent over $\mathbb{C}((t^\mathbb{R}))(z)$.*

Proof. For real γ , $\tau^n + n\gamma \rightarrow +\infty$, so Proposition 20.2 gives strong entireness. A nontrivial rational relation among distinct τ^n would be a nonzero polynomial equation for τ over $\mathbb{Q}$, so $\text{cvr}(R_\tau) = \infty$, and Theorem F applies. $\square$

The group $\mathbb{R}$ has an order unit, so Theorem E does not apply; the theorem uses the infinite rational rank of $\mathbb{R}$ through Theorem F. It says nothing about a series whose coefficient values span only $\mathbb{Q}\mu$, such as Θ_p or G , and it does not settle the purely differential problem over $\mathbb{C}((t^\mathbb{Q}))$. Through (2.4) its surreal realization is $\sum_{n \geq 0} \omega^{-\tau^n} z^n$ on the image of $\mathbb{C}((t^\mathbb{R}))$ in $\mathbf{No}[i]$, with independence over the full scalar field of that workspace.

22 Explicit families, surreal realizations and boundaries

22.1 The explicit series in all orders

Theorem 22.1 (Full mixed-jet independence of the explicit series; source 12). *Let F be the strongly entire series (11.2) over K_{Γ_∞} . For every $\Lambda \subseteq K_{\Gamma_\infty}^\times$ with pairwise quotients that are not roots of unity, the family $\{(D_z^j F)(\lambda z) : \lambda \in \Lambda, j \geq 0\}$ is algebraically independent over $K_{\Gamma_\infty}(z)$. In particular F satisfies no nonzero algebraic differential equation of any order over $K_{\Gamma_\infty}(z)$.*

Proof. Strong entireness is Theorem 11.2; source 12 reproves it by the highest-coordinate argument recorded there. The values e_n are rationally independent, so $\text{cvr}(F) = \infty$ and Theorem F applies; Theorem E gives the same, since Γ_∞ has no order unit. $\square$

Source 11 derives the purely differential part for $f_* = F - \omega^{-1}$ from Theorem E (a); by the remark on indexing in Section 20.1 this is the same statement. The theorem upgrades the linear rigidity of Theorem 11.2 and the two-jet independence of Corollary 17.2; the series itself is not new. For example, all formal series $(D_z^j F)(q^i z)$, $i \in \mathbb{Z}$, $j \geq 0$, for any one of $q = 2$, $q = 1 + t^{e_0}$ or $q = t^{e_0}$ are jointly algebraically independent over $K_{\Gamma_\infty}(z)$: the first two parameters have valuation zero, and the third has a positive valuation that is not an order unit (source 12).

22.2 Joint independence and a continuum-sized family

This subsection is source 11's.

Theorem 22.2 (Infinite independent coefficient supply; source 11). *Let $k_0 \subseteq K_0$ be fields of characteristic zero and $f_i = \sum_{n \geq 0} a_{i,n} z^n \in K_0[[z]]$ for $i \in I$. For every nonempty finite $J \subseteq I$ and $i \in J$ put $k_{J,i} = k_0(a_{i',n} : i' \in J \setminus \{i\}, n \geq 0)$, and suppose*

$$\text{trdeg}_{k_{J,i}} k_{J,i}(a_{i,n} : n \geq 0) = \infty \quad (22.1)$$

for all such J and i . Then the jets $D_z^j f_i$, $i \in I$, $j \geq 0$, are algebraically independent over $K_0(z)$.

Proof. Otherwise choose an inclusion-minimal nonempty finite J whose members have algebraically dependent jets, and fix $i \in J$. By minimality the jets of the other members are algebraically independent over $K_0(z)$. Take a nonzero relation, clear its denominators in z , and let $c_1, \dots, c_q \in K_0$ be its scalar coefficients. As a polynomial in the jets of f_i its coefficients are polynomials in z and in the jets of the other members, and they do not all vanish after substitution, because that substitution is injective by the independence just noted. So f_i satisfies a nonzero differential equation with coefficients in $k_{J,i}(c_1, \dots, c_q)[[z]]$, and Theorem 21.2 (a) puts all $a_{i,n}$ in $k_{J,i}(c_1, \dots, c_q, a_{i,0}, \dots, a_{i,N-1})$, of transcendence degree at most $q + N$ over $k_{J,i}$, contradicting (22.1). $\square$

The criterion does not require disjoint coefficient fields; finite overlaps are harmless. It assumes nothing about the Hahn supports of constants in a relation, which may be arbitrary elements of K_0 .

Lemma 22.3 (Independent scale monomials; source 11). *The monomials $t^{e_0}, t^{e_1}, \dots$ in $k((t^{\Gamma_\infty}))$ are algebraically independent over k .*

Proof. A polynomial in finitely many of them is a finite sum of monomials $t^{\sum_j m_j e_j}$; distinct exponent vectors give distinct elements of Γ_∞ , so no cancellation occurs. $\square$

For a binary word $b = (b_1, \dots, b_n)$ with $n \geq 1$ put

$$\text{code}(b) = 2^n + \sum_{j=1}^n b_j 2^{n-j} \in [2^n, 2^{n+1} - 1]. \quad (22.2)$$

Distinct words have distinct codes, codes strictly increase along an infinite branch, and two distinct branches share only finitely many prefixes, so their code sets meet in a finite set. The almost-disjoint

prefix device is standard and also appears, with a different coefficient construction, in the tail-span report [16]. For $\alpha \in \{0, 1\}^{\mathbb{N}_{>0}}$ put

$$\mathcal{F}_\alpha(z) = \sum_{n \geq 1} t^{e_{\text{code}(\alpha|n)}} z^n, \quad \alpha|n = (\alpha_1, \dots, \alpha_n). \quad (22.3)$$

Each nonzero coefficient is a Hahn monomial with coefficient 1 in the prime field.

Proposition 22.4 (Source 11). *Every $\mathcal{F}_\alpha$ is strongly entire over $k((t^{\Gamma_\infty}))$.*

Proof. Fix $\gamma, \beta \in \Gamma_\infty$, both supported on $\{e_0, \dots, e_{n_*}\}$. Once $\text{code}(\alpha|n) > n_*$, the largest nonzero coordinate of $e_{\text{code}(\alpha|n)} + n\gamma - \beta$ is at index $\text{code}(\alpha|n)$ and equals 1, so it is positive. Proposition 20.2 applies. $\square$

Proposition 22.5 (Source 11). *Over $K = k((t^{\Gamma_\infty}))$, all $D_z^j \mathcal{F}_\alpha$, with $\alpha \in \{0, 1\}^{\mathbb{N}_{>0}}$ and $j \geq 0$, are algebraically independent over $K(z)$.*

Proof. Given finitely many distinct branches and one of them, take the largest common-prefix length with the other branches, or 0 if there are none. Beyond that length every code of the chosen branch occurs on none of the others. So the chosen series has infinitely many coefficient monomials t^{e_c} indexed by codes absent from all other coefficient fields, and by Lemma 22.3 they are algebraically independent over the field generated by the other branches' monomials. Indeed, a putative polynomial relation uses only finitely many private monomials and rational functions of finitely many other monomials. Clearing denominators would contradict the algebraic independence of that finite combined family. Condition (22.1) holds with $k_0 = k$, and Theorem 22.2 applies. $\square$

Theorem 22.6 (Maximal differential transcendence in one workspace; source 11). *Let $k = \mathbb{C}$ and $K_{\Gamma_\infty} = \mathbb{C}((t^{\Gamma_\infty}))$. The family $(\mathcal{F}_\alpha)_{\alpha \in \{0, 1\}^{\mathbb{N}_{>0}}}$ of strongly entire series has all its jets jointly algebraically independent over $K_{\Gamma_\infty}(z)$, and*

$$\text{dtrdeg}_{K_{\Gamma_\infty}(z)} \text{Frac } \mathcal{E}_{\Gamma_\infty} = 2^{\aleph_0}. \quad (22.4)$$

Proof. Propositions 22.4 and 22.5 give a differentially independent family indexed by the $2^{\aleph_0}$ branches, so the differential transcendence degree is at least $2^{\aleph_0}$. Conversely Γ_∞ is countable and $|\mathbb{C}| = 2^{\aleph_0}$, so $|K_{\Gamma_\infty}| \leq |\mathbb{C}^{\Gamma_\infty}| = 2^{\aleph_0}$ and $|K_{\Gamma_\infty}[[z]]| = 2^{\aleph_0}$; no differentially independent subset of the fraction field can be larger. $\square$

The lower bound holds for every k ; the equality is stated, as by source 11, for $k = \mathbb{C}$. Equivalently, the free differential polynomial algebra $K_{\Gamma_\infty}(z)\{X_\alpha\}$ embeds in the fraction field by $X_\alpha \mapsto \mathcal{F}_\alpha$. This is cardinal maximality only: the family is not claimed maximal under inclusion, nor to be a differential transcendence basis.

The tail-span report already constructs continuum-sized differentially independent families [16], using algebraic coefficient extensions and Galois sign changes over smaller fields, and its analytic construction lives on a fixed ordinary disk rather than being strongly entire at every Hahn scale. Here the coefficients are exponent monomials, the functions are strongly entire on one full Hahn field, and the independence is over that whole field as field of constants. Neither the cardinal $2^{\aleph_0}$ nor the almost-disjoint combinatorics is claimed new.

22.3 Surreal realization, scalar extension, whole-class failure

Through (2.4), with $e_j \mapsto \omega^j$, the branch family becomes

$$\mathcal{F}_\alpha(z) = \sum_{n \geq 1} \omega^{-\omega^{\text{code}(\alpha|n)}} z^n,$$

and at every argument of the embedded workspace its evaluation is an actual surcomplex number. Everything is set-sized: the value group is countable, the field has the cardinality of the continuum, and no class-sized sum or unrestricted class quantification is used (source 11).

Proposition 22.7 (Independence survives extension of constants; source 12). *Let $E/\mathbb{K}$ be a field extension and (g_α) a family in $\mathbb{K}[[z]]$ algebraically independent over $\mathbb{K}(z)$. Under the coefficientwise inclusion $\mathbb{K}[[z]] \subseteq E[[z]]$ the family is algebraically independent over $E(z)$.*

Proof. A relation over $E(z)$ among finitely many g_α gives, after clearing denominators, a nonzero $Q \in E[z, X_1, \dots, X_{m'}]$ vanishing at them. Choose a $\mathbb{K}$ -basis $b_1, \dots, b_d$ of the $\mathbb{K}$ -span of its coefficients and write $Q = \sum_\nu b_\nu Q_\nu$ with $Q_\nu \in \mathbb{K}[z, X]$, some $Q_\nu \neq 0$. Each $Q_\nu(z, g)$ has coefficients in $\mathbb{K}$, so linear independence of the b_ν over $\mathbb{K}$ forces every $Q_\nu(z, g) = 0$, and a nonzero Q_ν is a relation over $\mathbb{K}(z)$. $\square$

The proposition concerns fixed formal functions and extensions of the constant field. It adjoins no new functional variable, says nothing about evaluation domains, and does not cover new dilation parameters taken from the extension. Entireness behaves quite differently. Adjoin a dominating level, $\Gamma' = \Gamma_\infty \oplus \mathbb{Q}e_*$ with $e_* > \Gamma_\infty$, realized by $e_* = \omega^\omega$; the new group has the order unit e_* . At $x = t^{-e_*} = \omega^{\omega^\omega}$ the evaluation family of F is not strongly summable (Section 11.2; source 12 repeats this computation), and neither is that of any $\mathcal{F}_\alpha$, whose leading exponents $e_{\text{code}(\alpha|n)} - ne_*$ strictly decrease since $e_{\text{code}(\alpha|n+1)} - e_{\text{code}(\alpha|n)} - e_* < 0$ (source 11). So the formal independences survive the larger field by Proposition 22.7, while entireness does not: two different kinds of stability. Nothing in this part is entire on the proper class $\mathbf{No}[i]$.

The derivative D_z kills every scalar, including every $\omega^{-\gamma}$, and $D_z(z) = 1$. The Berarducci–Mantova derivation acts on surreal scalars and is a different operation [8]. The tail-span report’s all-order independence results for particular constructions, other base fields and other derivations [16] are not restated here, and neither kind of statement is transferred to the other’s derivation. The independence statements concern formal functions: at a point $x \in \mathbb{K}$ every value $f(x)$ lies in $\mathbb{K}$ and cannot be transcendental over it, and at $z = 0$ the j th jet has the value $j! a_j \in \mathbb{K}$ (sources 11 and 12).

22.4 Boundary examples

Positive characteristic (sources 11 and 12). If k has characteristic $p > 0$, the series $\sum_{n \geq 0} t^{e_n} z^{pn}$ over $k((t^{\Gamma_\infty}))$ is strongly entire, by the highest-coordinate argument and Lemma 2.3 (a), whose proof does not use the characteristic, yet has D_z -derivative zero. This is the phenomenon of Example 17.3 in the no-order-unit workspace; characteristic zero is essential even for order-one linear equations, and the finite coefficient-field principle fails there. No claim about alternative derivations in positive characteristic is made.

Torsion-related dilations (source 12). Let $\zeta \in k$ be a root of unity of order $m' \geq 2$ and, over $k((t^{\Gamma_\infty}))$, let $g(z) = F(z^{m'}) = \sum_n t^{e_n} z^{m'n}$. The same highest-coordinate argument makes g strongly entire, and $\text{cvr}(g) = \infty$, yet $g(\zeta z) - g(z) = 0$; for $\zeta = -1$, an even series relates its dilates by 1 and -1 . The non-root-of-unity condition therefore cannot be dropped from Theorems E (b) and F; compare Section 9.

Local evaluation is not entireness (source 12). The formal exponential E_η of Proposition 5.3, whose proof works over any k , satisfies $D_z E_\eta = t^\eta E_\eta$, has its coefficients in $k(t^\eta)$, and is strongly evaluable exactly on $\{0\} \cup \{v(x) > -\eta\}$. In higher rank the leading exponents $nv(y)$ of its evaluation need not escape cofinally when $v(y) > 0$, which is why the positive-support lemma, not the cofinal criterion, decides that domain (Remark 2.5).

Finite coefficient-value rank is not a converse (source 12). Theorem F is a sufficient condition. A series with $\text{cvr}(f) < \infty$ may still be differentially transcendental, and finite $\text{cvr}(f)$ does not by itself put the coefficients in a finitely generated field, since residues and tails can carry further algebraic complexity. The implications proved are

$$\text{a mixed relation} \implies \text{a finitely generated coefficient field} \implies \text{cvr}(f) < \infty,$$

not their converses.

An explicit example, added here, is

$$f(z) = F(z) + \frac{1}{1-z} = \sum_{n \geq 0} (1 + t^{e_n}) z^n \quad \text{over } K_{\Gamma_\infty}.$$

Every coefficient has value zero, so $\text{cvr}(f) = 0$, while subtracting 1 recovers the algebraically independent monomials of Lemma 22.3; its coefficient field has infinite transcendence degree. This f is not strongly entire, by Proposition 20.2 with $\gamma = \beta = 0$. Nevertheless

$$(D_z^j f)(\lambda z) = (D_z^j F)(\lambda z) + \frac{j!}{(1 - \lambda z)^{j+1}}.$$

For every finite selection of jets the added terms belong to $K_{\Gamma_\infty}(z)$, so translation by them is a polynomial-algebra automorphism. The full mixed independence of Theorem 22.1 therefore holds for f with the same restriction on the dilation quotients, despite its zero coefficient-value rank.

22.5 Consequences for the questions of Section 19

Question 19.1. Theorem E (a) (sources 11 and 12) answers it for every Γ without an order unit, in every order: no nonpolynomial strongly entire series satisfies a nonzero algebraic differential equation over $\mathbb{K}(z)$, whatever its residue corner, for every characteristic-zero k , without divisibility, with arbitrary Hahn coefficients in the equation. Theorem E (c) (source 11) extends this to quotients of entire series. The last sentence of the question as posed by sources 08 and 09 is answered for F with no coefficient restriction (Theorem 22.1, source 12; source 11 for f_*). The question's sentence about hypotheses on Γ is answered in one direction: the absence of an order unit suffices. With an order unit it stayed open after sources 11 and 12 and was re-scoped to that case; source 13 has since answered it there (Theorem H), source 14 has excluded order two for every Γ (Theorem I), and it is now re-scoped to the classification of series of minimal order three in Section 19.

Question 19.4. Theorem E (b) with $\Lambda = \{q^i : i \in \mathbb{Z}\}$ (source 12) excludes, for every nontorsion q including those with $v(q) = 0$, every nonzero algebraic relation among the $\mathcal{J}_{q^i, j} f$ of a nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ when Γ has no order unit. Since $D_z^r \sigma_q^\ell f = q^{\ell r} \mathcal{J}_{q^\ell, r} f$, a nonzero equation (10.1) is such a relation, and so are nonlinear mixed relations. With an order unit the unit-valuation case stays open; the witness of Theorem G uses the parameter t^μ of nonzero valuation and says nothing there. Theorem 10.2 and Theorem E (b) do not contain each other: the first covers every $\Gamma \neq 0$ but only nonzero noncofinal $v(q)$, the second every nontorsion q but only groups without an order unit.

Question 19.6: a merge observation. Source 12 does not claim this question. The following consequence of its theorem was checked for this merge.

Corollary 22.8 (Several multiplicatively independent dilations; merge observation). *Suppose Γ has no order unit, and let $q_1, \dots, q_s \in \mathbb{K}^\times$ be multiplicatively independent, that is, $q_1^{m_1} \cdots q_s^{m_s} = 1$ with $m_i \in \mathbb{Z}$ only for $m_1 = \cdots = m_s = 0$. Let Λ be the subgroup of $\mathbb{K}^\times$ they generate. Then for every nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ the family $\{\mathcal{J}_{\lambda, j} f : \lambda \in \Lambda, j \geq 0\}$ is algebraically independent over $\mathbb{K}(z)$. In particular no nonpolynomial strongly entire series satisfies a nonzero equation, linear or not, in finitely many $(D_z^j f)(q_1^{m_1} \cdots q_s^{m_s} z)$. For the series F of (11.2), the only linear relation over $K_{\Gamma_\infty}(z)$ among $F(\lambda_1 z), \dots, F(\lambda_{s'} z)$ is the trivial one whenever $\lambda_1, \dots, \lambda_{s'} \in K_{\Gamma_\infty}^\times$ have pairwise quotients that are not roots of unity, for instance when they are distinct elements of a group generated by multiplicatively independent parameters.*

Proof. Λ is free abelian on the q_i , hence torsion free. For distinct $\lambda, \lambda' \in \Lambda$ the quotient λ/λ' is a nonidentity element of Λ , so no positive power of it is 1: it is not a root of unity. Apply Theorem E (b), and Theorem 22.1 for the last clause. $\square$

So the first sentence of Question 19.6 has, in a group without an order unit, the answer that no nonpolynomial strongly entire solution exists, and its last sentence has the answer that only the trivial relation occurs whenever no quotient of parameters is a root of unity. The existence criterion in groups with an order unit, and relations among parameters with a root-of-unity quotient, remain open; the question is re-scoped accordingly in Section 19.

Mixed jets in the fraction field. The coefficient theorem of source 12 and the fraction-field descent of source 11 also give the following consequence. Neither source states it; it is added here by combining their results.

Corollary 22.9 (Mixed rigidity in the fraction field). *Suppose Γ has no order unit, and let $u \in \text{Frac } \mathcal{E}_\Gamma(k) \setminus \mathbb{K}(z)$. If $\Lambda \subseteq \mathbb{K}^\times$ has no root-of-unity quotient between distinct elements, the Laurent series $\{(D_z^j u)(\lambda z) : \lambda \in \Lambda, j \geq 0\}$ are algebraically independent over $\mathbb{K}(z)$.*

Proof. Choose $m' \geq 0$ so that $y = z^{m'} u \in \mathbb{K}[[z]]$; also $y \in \text{Frac } \mathcal{E}_\Gamma(k)$. If the stated family were dependent, a nonzero relation would involve finitely many parameters and derivative orders. Adjoin the lower orders through the largest order r used, and for each parameter use Leibniz's formula, with differentiation before dilation:

$$(D_z^j u)(\lambda z) = \sum_{\ell=0}^j \binom{j}{\ell} (-m')_{\underline{j-\ell}} (\lambda z)^{-m'-j+\ell} (D_z^\ell y)(\lambda z).$$

For each λ this is a triangular change of jet variables over $\mathbb{K}(z)$ with nonzero diagonal $(\lambda z)^{-m'}$. The combined change is therefore an automorphism of the finite polynomial algebra of jet variables, so

it carries the relation to a nonzero relation among the jets of y . Clear its rational denominators and apply Theorem 21.2 (b), with $k_0 = k$ and $K_0 = \mathbb{K}$. All coefficients of y lie in a field $L \subseteq \mathbb{K}$ finitely generated over k . Lemmas 20.6 and 20.7 put $v(L^\times)$ in a proper convex subgroup. Since $y \in \text{Frac } \mathcal{E}_\Gamma(k) \cap L[[z]]$, the fraction-field part of Theorem 21.11 gives $y \in L(z)$. Thus $u = z^{-m'}y \in \mathbb{K}(z)$, a contradiction. $\square$

The excluded class is the rational functions: a nonpolynomial rational function still satisfies a polynomial relation over $\mathbb{K}(z)$. Taking $\Lambda = \{1\}$ recovers the purely differential statement in Theorem E (c). This consequence uses the same characteristic-zero and quotient hypotheses as the mixed coefficient theorem. The no-order-unit hypothesis supplies the proper convex subgroup needed for descent.

22.6 Predecessors, repository comparison and priority

Literature. Source 11 consulted the NUMDAM bibliographic page of Hurwitz’s paper [23] and its attribution in [24], without a complete fresh reading of Hurwitz’s proof; it read the parsed text of Laohakosol–Kongsakorn–Ubolsri and viewed its pages 404, 408 and 409, where Lemma 3 isolates a high coefficient through a polynomial in its index; it consulted the abstract of Krattenthaler–Rivoal [25] as arithmetic context, not as an imported theorem, and the framework of [26] for context, using none of its software. Source 12 read the general characteristic-zero form of the Skolem–Mahler–Lech theorem in Bell’s Theorem 1.1 [27], imports only that classical theorem and not Bell’s affine-variety generalization, and checked Lech’s paper [28] through publisher metadata only, without reconstructing its proof. Both consulted Hu–Luan [20], whose setting is an algebraically closed field of characteristic zero, complete for a nontrivial real-valued non-Archimedean absolute value; its hypotheses are not transferred to arbitrary-rank Hahn fields, its results are neither inferred nor replaced, and no rank-one question is claimed resolved. For normal forms source 11 used Mantova–Matusinski, Section 2.2 [17], and source 12 Bournez–Guilmant, Corollary 2.7 [29], with the same convention $t^a \mapsto \omega^{-a}$ as (2.4).

Repository comparison at the pin. Both sources compared their results with the collection at 465a54b, at which this report consisted of Sections 1–19 and the conclusion and was unchanged up to the merge. Their statements about it were checked and are accurate as of that pin: Question 19.1 left order at least two with vanishing residue corner open, and the README named $z^2ff'' + zff' - z^2(D_z f)^2 = 0$; Question 19.4 is the unit-valuation mixed case (source 12’s phrase “including unit-valued parameters” refers to exactly that); the source 10 audit file still says “Question 15.1”, as source 11 notes; the report already contained the sufficiency half of the all-scale criterion, the series F and its domain warning, first-order independence and the polynomial Painlevé exclusions. The entire-function report [12] contains the polynomial intersection obstruction $\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = K_\Gamma[Z]$ for noncofinal extensions, coarsened support methods and the distinction between countable cofinality and order units; the tail-span report [16] contains continuum-sized families built with binary-prefix almost-disjoint sets. Since the pin, the statements of this report that called the vanishing-corner case open have been re-scoped as recorded in Section 19.

Source 11 inspected the root README, the catalogue, directory listings, this report’s README, the source 10 audit, and the READMEs of the entire-function and tail-span reports; the `docs/new` listing then contained only a README. Source 12 inspected the root README, the catalogue, this report’s README (rereading its lines 131–182) and the tail-span README, and identified the

foundations and entire-function reports through the catalogue. Neither was a line-by-line audit of the manuscripts, Lean declarations, archived deliveries, branches or history.

Priority. Both literature searches were made on 22 September 2026, were targeted, and included no exhaustive MathSciNet, zbMATH, Zentralblatt, full-book or citation-network review; some searches returned irrelevant or inaccessible material, and no failed search is used as evidence of absence. The inspected sources did not show the meromorphic descent theorem with its no-order-unit consequence, the full-constant-field entire branch family, the all-order no-order-unit and joint-dilation conclusions, the coefficient-value-rank criterion or the order-unit dichotomy. These are proposed contributions with proofs, not certified historical firsts, and the answered questions are questions of this report, not named published conjectures.

22.7 What the coefficient-field part does not claim

Every limitation stated by source 11 or source 12, in its manuscript, README or audit files, is kept. The tag says which source states it.

- (C1) [11, 12] The purely differential case with an order unit is not settled, and no counterexample is given. Partial theta, Θ_p or G , is not a counterexample to purely differential rigidity: its equation contains a dilation, and no new all-order claim is made for it. (Source 13 settles the case with the series $\mathcal{T}_p$ of Theorem H; Θ_p and G remain undecided.)
- (C2) [11, 12] The delivered sources supply no Lean formalization of this theorem package or independent referee report. Source 11 performed no Lean build of the repository. The finite checks provide no computer-algebra certification of an infinite statement.
- (C3) [11, 12] Historical priority is not certified; the literature searches were targeted, with the access limits recorded in Section 22.6, and a failed search is not evidence.
- (C4) [11, 12] The repository comparison was targeted, not line by line; there is no blanket claim that no equivalent result occurs elsewhere in the collection.
- (C5) [11, 12] Not claimed new: classical coefficient recurrences (Hurwitz, Laohakosol–Kongsakorn–U bolsri; Krattenthaler–Rivoal as context), the finite-generation principle, the valuation-rank inequality, Hahn algebra and Hahn–Neumann support, Conway normal forms, the Skolem–Mahler–Lech theorem, partial theta series, the all-scale criterion, the polynomial half of the descent theorem, continuum cardinality and almost-disjoint combinatorics, and the series F (or f_*) with its two-jet consequence.
- (C6) [11, 12] *Entire* means strong evaluation on one set-sized field, not on the class $\mathbf{No}[i]$ and not ordinary convergence of constant coefficients; entireness fails after an exterior scale is adjoined, with the explicit witness $x = \omega^{\omega^\omega}$.
- (C7) [11, 12] D_z is not the Berarducci–Mantova derivation; nothing is said about scalar surreal differential equations, and nothing is transferred to the tail-span report’s derivations.
- (C8) [11, 12] Independence concerns formal functions, not point values.
- (C9) [11] Containment of $v(L^\times)$ in a proper convex subgroup is sufficient for descent, not claimed necessary (Question 19.7); a criterion using rational rank alone will not suffice.
- (C10) [11] Corollary 21.15 does not say that $\mathbb{K}(z)$ is differentially closed.

- (C11) [11] The continuum family is cardinally maximal only; it is not maximal under inclusion and not a differential transcendence basis.
- (C12) [11] Corollary 21.17 concerns $\text{Frac } \mathcal{E}_\Gamma(k)$; local formal solutions exist, and nothing is said about classical complex solutions.
- (C13) [11] “Meromorphic” means membership of $\text{Frac } \mathcal{E}_\Gamma(k)$, not a sheaf and not a fine-topology notion.
- (C14) [11, 12] Characteristic zero is essential, and no claim is made about alternative derivations in positive characteristic.
- (C15) [11, 12] There is no uniform algorithm for extracting a convex subgroup from Hahn names; Corollary 21.18 is structural and Theorem 21.2(b) existential.
- (C16) [11, 12] Hu–Luan’s rank-one hypotheses are not transferred; Hu–Luan is context only, and no rank-one question is claimed resolved.
- (C17) [11, 12] The finite checks prove no arbitrary-support, continuum, recurrence zero-set, cofinality or transcendence statement.
- (C18) [11] No font files or third-party texts are distributed.
- (C19) [12] The answered questions are questions of this research programme, not named historical or published conjectures.
- (C20) [12] The Skolem–Mahler–Lech theorem is imported through Bell’s Theorem 1.1; Bell’s affine generalization is not used, Lech’s proof was not reconstructed, and the zero multipliers sampled by the checks are not an effective bound.
- (C21) [12] Theorem F is sufficient only; $\text{cvr}(f)$ does not give a finitely generated coefficient field.
- (C22) [12] Proposition 22.7 covers extension of constants only, not new dilation parameters from the extension.
- (C23) [12] Theorem 21.20 does not settle the purely differential problem over $\mathbb{C}((t^\mathbb{Q}))$.
- (C24) [12] Dilations with a root-of-unity quotient are excluded (Question 19.9).
- (C25) [12] In groups of uncountable cofinality there is no nonpolynomial strongly entire series, so statements requiring one are vacuous (Corollary 20.3). The coefficient-value-rank criterion for arbitrary formal series has no such restriction.
- (C26) [merge] Corollary 22.8 is a consequence assembled from source 12. Corollary 22.9 is an additional consequence of sources 11 and 12 together. Neither is attributed to the delivered sources as a stated result, and no historical priority is claimed for either deduction.

23 Theta descent: an entire series with an equation of order three

This section and the next two contain the material of source 13, delivered after the coefficient-field part had been merged. They answer the order-unit case of Question 19.1 as it stood after that merge: when Γ has an order unit, a nonpolynomial strongly entire series satisfies a nonzero algebraic differential equation of order three and none of lower order (Theorem H). With Theorem E (a) this makes the order unit the exact boundary of differential algebraic rigidity. The ingredients are

classical: a bilateral theta series, Jacobi's triple product and the theta-elliptic differential identity. What source 13 adds is their descent to an ordinary power series by the substitution $z = u + u^{-1}$, the support control that makes the descended series strongly entire exactly when its scale is an order unit, and a difference-field proof that no equation of order below three exists. The method is neither the escape chain, nor the finite initial polynomial, nor the finite coefficient field; the last is used only for the converse direction, which is Theorem E (a).

23.1 Source, conventions and the merge's choices

Convention 23.1 (Setting of the theta-descent part). In Sections 23–25, k is an arbitrary field of characteristic zero with the trivial valuation, Γ is as in Convention 1.1, $\mathbb{K} = k((t^\Gamma))$, and v , lc , res , strong summability, $\text{Dom}_{\text{str}}(f)$, $\mathcal{E}_\Gamma(k)$, $\text{Frac } \mathcal{E}_\Gamma(k)$ and D_z are as in Conventions 14.1 and 20.1. Throughout, $\mu \in \Gamma$ is positive, $p = t^\mu$, and Δ_μ is the principal convex subgroup (2.1); μ is an order unit only where this is stated, and then $\Delta_\mu = \Gamma$. The power-series variable is z , while x and y denote elements of $\mathbb{K}$, and u is the variable of the Laurent series of Section 24.5. Write $\vartheta = zD_z$, as in the proof of Proposition 16.4, and $\vartheta_u = uD_u$. The elliptic constants below use the nome p^2 , not p .

For $k = \mathbb{C}$ one has $\mathbb{K} = K_\Gamma$ and $\mathcal{E}_\Gamma(k) = \mathcal{E}_\Gamma$. As for the two previous parts, the generality in k is recorded for this part only.

The source. Source 13 is *Theta Descent and the Exact Boundary of Differential Rigidity* (21 pages, 23 September 2026). It consulted the repository at

4cdeaec43ff1692ed2ad9f5bdf761a41872975a5,

at which this report was exactly the five-source text into which it is merged here; its references to Question 19.1 (`hol:q:nonlinear`), to its statement and status passage, to Theorem E and to the coefficient-field part were accurate there. It assumes characteristic zero, $\Gamma \neq 0$ set-sized, and, where stated, that the scale μ is an order unit. It uses no divisibility, no rank hypothesis, no real or algebraic closedness and no countable cofinal subset. Its standing hypotheses are therefore those of the coefficient-field part, and every proof taken from it was re-read against the statement it is attached to.

Renamed symbols. The following symbols of source 13 would collide with symbols of the earlier parts. They are renamed here, in the new material only.

Source 13	Here	Reason
variable x	z	Definition 1.4; x denotes points. (Source 13's $\mathbb{K} = k((t^\Gamma))$ and $\mathcal{E}_\Gamma(k)$ are this report's.)
$h, q = t^h$	$\mu, p = t^\mu$	h is a torsion order and q a dilation parameter; μ and $p = t^\mu$ are the notation of Theorem G.
$H_h, \mathcal{D}_h$	$\Delta_\mu, \mathcal{O}_{\Delta_\mu}$	(2.1); $\mathcal{D}_P$ is a candidate set (15.12); $\mathcal{O}_\Delta$ as in Section 20.3.
F_q	$\mathcal{T}_p$	F is the series (11.2).
Θ_q (bilateral)	Ψ_p	Θ_p is the partial theta series (8.1).
r_m, P_q, j_m, e, η	$x_m, \mathcal{P}_p, 2m - 1, y, \varpi$	r indexes derivative orders, P is a differential polynomial, e_n and η are elements of Γ .

Source 13	Here	Reason
$w_\gamma(F_q)$ $Q = q^2, s_r(Q), c$	γ_δ $p^2, \mathcal{Q}_r(p^2), c$	The coefficient Gauss value (14.1), with $\delta = -\gamma$. Q_s is a shift polynomial, s an order, c_ν a corner coefficient.
A_f, H_f, J_f, B_f	$\mathcal{A}[f], \mathcal{H}[f], \mathcal{K}[f],$ $\mathcal{B}[f]$	A_δ is a leading coefficient, $H \in \Gamma$, J indexes monomials in (14.4), B is a threshold, $\mathcal{J}_{\lambda,j}$ a jet.
$S_Q(w), W(z), a, b$	$\mathcal{Z}(w), \mathcal{W}(y), \mathbf{a}_2, \mathbf{a}_4$	z is the power-series variable; a_n are Taylor coefficients.
$\delta = u \, d/du, s = u - u^{-1}$ σ, τ	$\vartheta_u, \check{z}$ S, S_a	δ is an element of Γ ; s is an order. σ_q is an argument dilation; τ is the parameter of R_τ .
$\mathcal{A}, \mathcal{M} = \text{Frac } \mathcal{A}$ L, A, V (in u); Z $\ell, B = A', R(T), \mu = q^{-1}u^{-1}$	$\mathcal{U}_\mu, \text{Frac } \mathcal{U}_\mu$ $\psi_1, \psi_2, \psi_3; V$ $\mathcal{L}, \mathcal{A}_1, \Phi(T)$, written out	$\mathcal{A}[f]$ as above; $M(S)$ is a support monoid. L is a coefficient field; z as above. ℓ is a dilation power; R_τ is a series; μ is the scale.
$\vartheta = x \, d/dx; \mathbf{No}(i)$	$\vartheta = zD_z; \mathbf{No}[i]$	The same operator; not ϑ_q . The collection writes $\mathbf{No}[i]$.
Theorem 1.2	Theorem H	Theorems A–G of this report are different theorems.

The integer subscripts of the classical invariants g_2, g_3 distinguish them from the scaled series g_δ of (14.1). The shipped audit files `13-theta-descent-PROOF_AUDIT.md` and `13-theta-descent-SOURCES_AND_SCOPE.md` keep the source’s notation, including F_q, Θ, q, h, L, A and σ .

Words with two meanings. The *order* of an algebraic differential equation is the highest derivative occurring in it; an *order unit* is as in Definition 1.3; the *multiplicative order* of q is as in Theorem B; and ord_u is the u -adic order of a rational function. The *minimal order* of a series is the least order of a nonzero algebraic differential equation over $\mathbb{K}(z)$ that it satisfies. *Descent* in this part means the Chebyshev substitution $z = u + u^{-1}$, not the coarsening descent of Theorem 21.11. Three theta series occur and are not interchangeable: the partial series Θ_p of (8.1), which satisfies a dilation equation; the bilateral series Ψ_p of (23.8); and the Hahn–Tate report’s series $\sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n$ [30], of which $\Psi_p(u)$ is the value at the nome p^2 and the argument $-pu$. A *corner* keeps the meaning of Section 14.4: source 13’s “vanishing corner” is the vanishing of the residue corner polynomial χ_P , and the full corner polynomial I_P need not vanish with it (Proposition 24.2). A *zero* of $\mathcal{T}_p$ is a point $x \in \text{Dom}_{\text{str}}(\mathcal{T}_p)$ with $\mathcal{T}_p(x) = 0$.

Results printed once. Source 13 reproves several facts that the report already contains. They are printed once and cited: the definitions of an order unit and of strong summability (Definitions 1.3 and 1.2); its cofinal-valuation lemma, which is Lemma 2.3 (a); the positive-support lemma, which is Lemma 2.1 (2); the surreal normal-form embedding (2.4); the strong domain of the bilateral theta series, which is the Hahn–Tate report’s theta-domain theorem [30, `tate:thm:theta`] after the substitution just described; and its whole Section 7, the no-order-unit direction, which is Theorem E (a) with Theorem 21.2 (a), Lemmas 20.6 and 20.7 and Proposition 20.9; its Euler-derivative presentation of that argument is recorded in Remark 25.1. For $k = \mathbb{C}$ the non-D-finiteness of $\mathcal{T}_p$ is also a case of Theorem A; source 13’s separate proof, valid for every k , is kept (Proposition 24.11). Everything else in source 13 is printed in Sections 23–25, with its proofs.

Where each numbered result went. Theorem 1.2 is Theorem H; Lemma 2.3 is Lemma 23.4; Lemma 3.1, Proposition 3.2 and Theorem 3.3 are Lemma 23.2, Proposition 23.3 and Theorem 23.5; Theorem 4.1 and Proposition 4.2 are Theorem 23.6 and Proposition 23.7; Theorem 5.1 is Theorem 24.1, and Remark 5.2 is Remark 24.3; Lemma 6.1, Theorem 6.2, Corollary 6.3 and Theorem 6.4 are Lemma 24.6, Theorem 24.8, Corollary 24.9 and Theorem 24.10, and Section 6.5 is Proposition 24.11; Lemmas 7.1–7.2 and Theorem 7.3 are printed once as above; Corollary 8.1 is Corollary 25.3; Examples 9.1–9.3 and Corollary 9.4 are Examples 25.4–25.6 and Corollary 25.7; Questions 11.1 and 11.2 are merged into the re-scoped Question 19.1, and Question 11.3 is Question 19.10; Appendix A is (24.8) and the expansions after (24.2); Appendix B is merged into the dependency table of Section 18.

Two steps written out, and other choices. The review of source 13 for this merge found its main line correct and two steps used but not written. Both are supplied here, marked as the merge’s. First, the minimal-order proof substitutes $z = u + u^{-1}$ into strongly entire series and uses that this is a ring homomorphism into a Laurent ring, compatible with the derivations; Lemma 24.5 states and proves this. Second, source 13 proved $\vartheta_u \psi_2 \neq 0$ from “formal meromorphic germs” at the simple zero $u_0 = -p$ of Ψ_p , a local expansion inside $\text{Frac} \mathcal{U}_\mu$ that it did not define; Lemma 24.7 replaces this by the global identity $\Psi_p^3 \vartheta_u \psi_2 = p(u - u^{-1}) + O(p^2)$. The passage from the complex elliptic identity to $\mathbb{K}$ is written as three ring maps (Section 24.3), which is the content of source 13’s “support-controlled specialization”. The merge also computes the corner data of the cleared equation (Proposition 24.2), adds clause (iv) to Theorem G, and records the consequences for Theorem E (Corollary 25.2). Source 13’s theorems, proofs and non-claims are otherwise printed as delivered, in the renamed notation.

23.2 Chebyshev descent and the exact Taylor coefficients

The monic Chebyshev polynomials $C_n \in \mathbb{Z}[z]$ are defined by

$$C_0 = 2, \quad C_1 = z, \quad C_{n+1} = zC_n - C_{n-1} \quad (n \geq 1), \quad (23.1)$$

and $\mathcal{T}_p$ is the series (1.4). In ordinary notation $C_n(z) = 2T_n(z/2)$; the normalization $C_0 = 2$ is why the constant term of $\mathcal{T}_p$ is written separately.

Lemma 23.2 (Chebyshev identities; source 13). *For $n \geq 1$, C_n is monic of degree n and*

$$C_n(u + u^{-1}) = u^n + u^{-n}, \quad (23.2)$$

$$C_n(z) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \frac{n}{n-j} \binom{n-j}{j} z^{n-2j}. \quad (23.3)$$

All the coefficients in the second formula are integers.

Proof. The first identity follows from the initial values and the same three-term recurrence on both sides. The recurrence also proves monicity and integrality. Substituting the displayed coefficients into the recurrence and using Pascal’s identity proves (23.3); the initial cases are $C_1 = z$ and $C_2 = z^2 - 2$. $\square$

Thus

$$\mathcal{T}_p(z) = 1 + pz + p^4(z^2 - 2) + p^9(z^3 - 3z) + p^{16}(z^4 - 4z^2 + 2) + p^{25}(z^5 - 5z^3 + 5z) + \cdots$$

Proposition 23.3 (Exact Taylor coefficients; source 13). *For every $\mu > 0$ write $\mathcal{T}_p = \sum_{m \geq 0} a_m z^m$. Then*

$$a_0 = 1 + 2 \sum_{j \geq 1} (-1)^j p^{4j^2}, \quad (23.4)$$

$$a_m = \sum_{j \geq 0} (-1)^j \frac{m+2j}{m+j} \binom{m+j}{j} p^{(m+2j)^2} \quad (m \geq 1), \quad (23.5)$$

and

$$v(a_0) = 0, \quad v(a_m) = m^2 \mu, \quad \text{lc}(a_m) = 1 \quad (m \geq 1). \quad (23.6)$$

No coefficient vanishes, so $\mathcal{T}_p$ is not a polynomial.

Proof. For each m collect the terms $n = m + 2j$ of (23.3). Their exponents $(m + 2j)^2 \mu$ increase strictly, so the family is strongly summable for every $\mu > 0$ (Lemma 2.3 (b) with $S = \{\mu\}$) and each exponent carries one term. For $m \geq 1$ the first coefficient is one and every later term has strictly larger valuation. The constant term follows from $C_{2j}(0) = 2(-1)^j$ and $C_{2j+1}(0) = 0$. $\square$

23.3 The exact strong domain

Lemma 23.4 (A quadratic support certificate; source 13). *Let $\mu > 0$ and $x \in \mathbb{K}$ with $x = 0$ or $v(x) \geq -N\mu$ for some $N \in \mathbb{N}$. The family*

$$(p^{n^2} x^{n-2j})_{n \geq 1, 0 \leq j \leq \lfloor n/2 \rfloor}$$

is strongly summable, and so is every family obtained from it by multiplying members by ordinary integers or omitting members.

Proof. For $x = 0$ only the members with $n - 2j = 0$ are nonzero, and their exponents $n^2 \mu$ are distinct. Otherwise write

$$x = a t^{-N\mu+\beta} (1 + \varepsilon_x), \quad a \in k^\times, \quad \beta \geq 0, \quad \varepsilon_x = 0 \text{ or } v(\varepsilon_x) > 0.$$

The support of a member lies in $[n^2 - N(n - 2j)]\mu + (n - 2j)\beta + M(\text{supp } \varepsilon_x)$. For $n > N$ the integer $c_{n,j} = n^2 - N(n - 2j) \geq n(n - N)$ is positive and tends to infinity with n , uniformly in j . Apply Lemma 2.1 (2) to the well-ordered positive alphabet $S = \{\mu\} \cup (\{\beta\} \setminus \{0\}) \cup \text{supp } \varepsilon_x$. After the binomial expansion of $(1 + \varepsilon_x)^{n-2j}$, every support element of a member with $n > N$ is represented by a word with at least $c_{n,j}$ letters μ . A fixed exponent has only finitely many words, so the number of their letters μ is bounded, which bounds n and hence j ; each of the finitely many remaining members is one Hahn element. The members with $n \leq N$ are finitely many. The same argument shows that the union of the supports is well ordered. Multiplication by ordinary integers and omission enlarge neither supports nor multiplicities. No cofinality of $(n\mu)$ is assumed. $\square$

Theorem 23.5 (Exact strong domain; source 13). *For every $\mu > 0$, not necessarily an order unit,*

$$\text{Dom}_{\text{str}}(\mathcal{T}_p) = \mathcal{O}_{\Delta_\mu} = \{0\} \cup \{x \in \mathbb{K}^\times : v(x) \geq -N\mu \text{ for some } N \in \mathbb{N}\}, \quad (23.7)$$

the valuation ring of the coarsened valuation v_{Δ_μ} . Consequently $\mathcal{T}_p \in \mathcal{E}_\Gamma(k)$ if and only if μ is an order unit of Γ .

Proof. For $x \in \mathcal{O}_{\Delta_\mu}$, Lemma 23.4 applies to the full expansion $(p^{n^2} c_{n,j} x^{n-2j})_{n,j}$, with $c_{n,j}$ the integers of (23.3). Regrouping it by powers of x gives exactly $(a_m x^m)_{m \geq 0}$. Here and below we use the standard associativity of strong sums: if a strongly summable family is partitioned into arbitrary subfamilies, each subfamily is strongly summable, and the family of their sums is strongly summable with the same sum, because the supports of the partial sums lie in the union of the original supports and only finitely many subfamilies contain a member contributing at a fixed exponent.

If $x \notin \mathcal{O}_{\Delta_\mu}$, put $\gamma = v(x)$, so $\gamma < -N\mu$ for every N . By (23.6) the members with $m \geq 1$ have leading exponents $m^2\mu + m\gamma$, whose consecutive differences $(2m+1)\mu + \gamma$ are negative. They strictly decrease, and Lemma 2.2 excludes strong summability.

A coset in Γ/Δ_μ is nonnegative exactly when its representative lies in Δ_μ or is positive and outside it, that is, when it is bounded below by some $-N\mu$; this identifies the valuation ring. It equals $\mathbb{K}$ exactly when every negative value is so bounded, that is, when $\Delta_\mu = \Gamma$. $\square$

When μ is an order unit there is the shorter route: for fixed γ the values $m^2\mu + m\gamma$ escape cofinally, and Lemma 2.3 (a) applies. The support proof is needed for the noncofinal boundary, where strong summability and convergence in the full valuation topology differ (Remark 2.5). When μ is not an order unit, the coefficient values of $\mathcal{T}_p$ lie in the proper convex subgroup Δ_μ , and Theorem 23.5 shows that the necessary inclusion of Proposition 20.9 is then an equality for this series.

23.4 The bilateral series, the product and the zeros

Put

$$\Psi_p(u) = \sum_{n \in \mathbb{Z}} p^{n^2} u^n. \quad (23.8)$$

At $u \in \mathbb{K}^\times$ this family is strongly summable exactly when $v(u) \in \Delta_\mu$: both tails are covered by the argument of Lemma 23.4 when $|v(u)| \leq N\mu$, and otherwise one of the sequences $n^2\mu \pm nv(u)$ strictly decreases. Since $\Psi_p(u) = \sum_n (-1)^n (p^2)^{n(n-1)/2} (-pu)^n$, this is the Hahn–Tate theta domain [30, `tate:thm:theta`] at the nome p^2 and the argument $-pu$, printed once; source 13 describes it as the symmetric specialization of that domain. On this domain, and as identities of formal series in Section 24.3, reindexing gives

$$\Psi_p(u^{-1}) = \Psi_p(u), \quad (23.9)$$

$$\Psi_p(p^2 u) = p^{-1} u^{-1} \Psi_p(u), \quad (23.10)$$

$$\mathcal{T}_p(u + u^{-1}) = \Psi_p(u). \quad (23.11)$$

For (23.10) the exponent identity is $n^2 + 2n = (n+1)^2 - 1$. For (23.11), $x = u + u^{-1}$ has $v(x) \geq -|v(u)|$, so Lemma 23.4 applies at x ; grouping the full expansion by n gives $(p^{n^2} C_n(x))_n = (p^{n^2} (u^n + u^{-n}))_n$ by (23.2), and grouping it by powers of x gives the Taylor family. No square root or inverse cosine is used to define $\mathcal{T}_p$, and no branch is chosen.

Jacobi's triple product, in the normalization of DLMF 20.5.9 [32], is

$$\sum_{n \in \mathbb{Z}} q^{n^2} u^n = \prod_{m \geq 1} (1 - q^{2m})(1 + q^{2m-1}u)(1 + q^{2m-1}u^{-1}).$$

It is classical, not a new theorem about theta functions. Pairing the reciprocal factors gives the following form.

Theorem 23.6 (Product and complete zero set; source 13). *Let $\mu > 0$ and $\mathcal{P}_p = \prod_{m \geq 1} (1 - p^{2m})$. For every $x \in \mathcal{O}_{\Delta_\mu}$,*

$$\mathcal{T}_p(x) = \mathcal{P}_p \prod_{m \geq 1} (1 + p^{2m-1}x + p^{4m-2}), \quad (23.12)$$

where the products are strong products, that is, strong sums of their finite-subset expansions. The scalar $\mathcal{P}_p$ is a unit of valuation zero and leading coefficient one. The zeros of $\mathcal{T}_p$ in its domain are exactly

$$x_m = -(p^{2m-1} + p^{-(2m-1)}) \quad (m = 1, 2, \dots), \quad (23.13)$$

and every zero is simple. The conclusion is already split over $k((t^\Gamma))$; no algebraic closedness is used.

Proof. First let p be a formal variable. At each p -degree both sides of the triple product have Laurent-polynomial coefficients in u , and the identity holds in $\mathbb{Q}[u, u^{-1}][[p]]$. Since $(1 + p^{2m-1}u)(1 + p^{2m-1}u^{-1}) = 1 + p^{2m-1}(u + u^{-1}) + p^{4m-2}$ and the substitution $z \mapsto u + u^{-1}$ is injective on $\mathbb{Q}[z]$ (the leading power u^d of a polynomial of degree d cannot cancel), the paired identity $1 + \sum_{n \geq 1} p^{n^2} C_n(z) = \mathcal{P}_p \prod_m (1 + p^{2m-1}z + p^{4m-2})$ holds in $\mathbb{Q}[z][[p]]$.

Now let $p = t^\mu$ and $v(x) \geq -N\mu$. For $2m-1 > N$ both $p^{2m-1}x$ and p^{4m-2} have positive valuation, and their supports, with the factor labels retained, lie in a positive support monoid generated by μ and the support of the normalized x as in Lemma 23.4; every nonempty choice from a late factor uses at least one letter μ , so a fixed exponent bounds the number and the indices of the late factors involved. Lemma 2.1 (2) therefore makes the finite-subset expansion of the tail product strongly summable, and the finitely many early factors multiply it by a finite family. The same holds for $\mathcal{P}_p$. Expand both sides of the paired identity at x into families of terms $e p^k x^j$ with $e \in \mathbb{Z}$: the Taylor side is strongly summable by Lemma 23.4, the product side by the argument just given. Grouping either family by the formal monomial $p^k z^j$ gives the same family, by the identity in $\mathbb{Q}[z][[p]]$, so the two sums agree. The tail product lies in $1 + \mathfrak{m}_v$ and is nonzero.

Each factor of (23.12) equals $p^{2m-1}(x - x_m)$, and the roots have the distinct valuations $-(2m-1)\mu$. At a fixed point all sufficiently late factors form a nonzero unit product, so a zero comes from one of finitely many early factors; there are no other zeros. For simplicity, split off the m th factor $\Pi_m(z) = 1 + p^{2m-1}z + p^{4m-2}$: the formal identity gives $\mathcal{T}_p = \Pi_m \cdot G_m$ in $\mathbb{K}[[z]]$, with G_m the image of the complementary product, which is strongly evaluable on $\mathcal{O}_{\Delta_\mu}$ by the same argument. Differentiating and evaluating at the nonzero point x_m , where Π_m vanishes, gives $(D_z \mathcal{T}_p)(x_m) = p^{2m-1} G_m(x_m)$, a nonzero strong product. $\square$

For $k = \mathbb{R}$ every zero is real and negative; for $k = \mathbb{C}$ there are no further nonreal zeros. These conclusions come from the displayed factorization, not from a claim that non-Archimedean entire functions split in general.

Valuations at the zeros. Write $\mathcal{T}'_p = D_z \mathcal{T}_p$. At $x = x_m$ the factors with $k < m$ have valuation $-2(m-k)\mu$ and leading coefficient -1 , and those with $k > m$ have valuation zero and leading coefficient one. Hence

$$v(\mathcal{T}'_p(x_m)) = (-m^2 + 3m - 1)\mu, \quad \text{lc}(\mathcal{T}'_p(x_m)) = (-1)^{m-1}, \quad (23.14)$$

which is also a direct certificate that the derivative does not vanish.

Proposition 23.7 (Exact linearization on the nearest-root ball; source 13). *If $y \neq 0$ and $v(y) > -(2m-1)\mu$, then*

$$\mathcal{T}_p(x_m + y) = \mathcal{T}'_p(x_m) y (1 + \varpi), \quad \varpi = 0 \text{ or } v(\varpi) > 0. \quad (23.15)$$

Therefore $v(\mathcal{T}_p(x_m + y)) = (-m^2 + 3m - 1)\mu + v(y)$.

Proof. For $k \neq m$ the valuation of $x_m - x_k$ is the smaller of the two root valuations and is at most $-(2m-1)\mu$, so $v(y/(x_m - x_k)) > 0$. Dividing (23.12) at $x_m + y$ by $\mathcal{T}'_p(x_m) y$ gives

$$\frac{\mathcal{T}_p(x_m + y)}{\mathcal{T}'_p(x_m) y} = \prod_{k \neq m} \left(1 + \frac{y}{x_m - x_k}\right).$$

The finitely many early factors lie in $1 + \mathfrak{m}_v$. In the tail, $1/(x_m - x_k)$ has valuation $(2k-1)\mu$; expanding after normalizing its leading monomial gives the same positive support certificate as for the original product. The strong product therefore lies in $1 + \mathfrak{m}_v$. $\square$

Newton profile. In the notation (14.1), at a scale δ with $t^{-\delta} \in \text{Dom}_{\text{str}}(\mathcal{T}_p)$, that is $\delta \leq N\mu$ for some N , the coefficient Gauss value is $\gamma_\delta = \min_{n \geq 0} (n^2\mu - n\delta)$. The minimum exists because the members with $n > N$ exceed the member $n = 0$, and consecutive differences are $(2n+1)\mu - \delta$. There is one active index n when $(2n-1)\mu < \delta < (2n+1)\mu$ (with the evident change for $n = 0$), and two, n and $n+1$, when $\delta = (2n+1)\mu$; the root x_{n+1} has exactly the valuation $-(2n+1)\mu$. The quadratic Newton profile and the linear-factor product give the same complete shell classification, without a general Newton-polygon theorem.

24 An equation of order three, and none of lower order

24.1 The cleared third-order equation

For $r \in \{1, 3, 5\}$ put

$$\mathcal{Q}_r(p^2) = \sum_{n \geq 1} \frac{n^r p^{2n}}{1 - p^{2n}} = \sum_{N \geq 1} \left(\sum_{d|N} d^r \right) p^{2N}, \quad (24.1)$$

and

$$\mathfrak{c} = \frac{1}{12} - 2\mathcal{Q}_1(p^2), \quad g_2 = \frac{1}{12} + 20\mathcal{Q}_3(p^2), \quad g_3 = -\frac{1}{216} + \frac{7}{3}\mathcal{Q}_5(p^2). \quad (24.2)$$

The powers are supported on positive multiples of 2μ , and each exponent has finitely many divisors, so the series exist for every $\mu > 0$. They begin

$$\begin{aligned} \mathfrak{c} &= \frac{1}{12} - 2p^2 - 6p^4 - 8p^6 - 14p^8 + O(p^{10}), \\ g_2 &= \frac{1}{12} + 20p^2 + 180p^4 + 560p^6 + 1460p^8 + O(p^{10}), \\ g_3 &= -\frac{1}{216} + \frac{7}{3}p^2 + 77p^4 + \frac{1708}{3}p^6 + \frac{7399}{3}p^8 + O(p^{10}), \end{aligned}$$

where $O(p^{10})$ denotes a formal series divisible by p^{10} , not a numerical error bound. Put

$$\Phi(T) = 4(\mathfrak{c} - T)^3 - g_2(\mathfrak{c} - T) - g_3. \quad (24.3)$$

For a nonzero element f of a differential field extending $\mathbb{K}(z)$ with derivation D_z , put

$$\mathcal{A}[f] = (z^2 - 4) \left(\frac{D_z^2 f}{f} - \frac{(D_z f)^2}{f^2} \right) + z \frac{D_z f}{f}, \quad (24.4)$$

$$\mathcal{H}[f] = (z^2 - 4)(f D_z^2 f - (D_z f)^2) + z f D_z f, \quad (24.5)$$

$$\mathcal{K}[f] = f D_z \mathcal{H}[f] - 2 D_z f \mathcal{H}[f], \quad \mathcal{B}[f] = \mathfrak{c} f^2 - \mathcal{H}[f], \quad (24.6)$$

so that $\mathcal{A}[f] = \mathcal{H}[f]/f^2$ and $D_z \mathcal{A}[f] = \mathcal{K}[f]/f^3$. These are rational differential expressions; no logarithm of a field element is used.

Theorem 24.1 (Cleared third-order equation; source 13). *For every $\mu > 0$ the series $f = \mathcal{T}_p$ satisfies, in $\mathbb{K}[[z]]$,*

$$(z^2 - 4)\mathcal{K}[f]^2 - 4\mathcal{B}[f]^3 + g_2\mathcal{B}[f]f^4 + g_3f^6 = 0. \quad (24.7)$$

Its left side is $P(z, f, D_z f, D_z^2 f, D_z^3 f)$ for the nonzero polynomial

$$P = (z^2 - 4)K^2 - 4B^3 + g_2BY_0^4 + g_3Y_0^6 \in \mathbb{K}[z, Y_0, Y_1, Y_2, Y_3], \quad (24.8)$$

where $H = (z^2 - 4)(Y_0Y_2 - Y_1^2) + zY_0Y_1$, $H_1 = (z^2 - 4)(Y_0Y_3 - Y_1Y_2) + 2z(Y_0Y_2 - Y_1^2) + Y_0Y_1 + z(Y_1^2 + Y_0Y_2)$ (the total derivative of H), $K = Y_0H_1 - 2Y_1H$ and $B = \mathfrak{c}Y_0^2 - H$. The polynomial P is homogeneous of degree six in $Y_0, \dots, Y_3$, and its coefficient of Y_3^2 is $(z^2 - 4)^3Y_0^4$, so no values of $\mathfrak{c}$, g_2 , g_3 make it zero. In $\mathbb{K}((z))$ the equivalent rational form is

$$(z^2 - 4)(D_z \mathcal{A}[f])^2 = \Phi(\mathcal{A}[f]). \quad (24.9)$$

Both are formal identities. The cleared one can be evaluated at every point of $\text{Dom}_{\text{str}}(\mathcal{T}_p)$, including the zeros of f and $z = \pm 2$, since evaluation is a ring homomorphism on series evaluable at a common point and no division occurs.

The proof occupies the next two subsections. The coefficient of $D_z^3 f$ in $\mathcal{K}[f]$ is $(z^2 - 4)f^2$, which gives the leading coefficient in Y_3 .

24.2 The classical elliptic identity

Source 13 supplies the normalization calculation because a sign or nome convention changes all three constants. Theta-elliptic differential identities are classical [33, 37], and Tate coordinates give another route to the same normalization [35, 30].

In this subsection only, p is an ordinary real number with $0 < p < 1$, and $\Psi(u) = \sum_{n \in \mathbb{Z}} p^{n^2} u^n$ for $u \in \mathbb{C}^\times$. Let $\psi_1 = \vartheta_u \Psi / \Psi$ and $\psi_2 = \vartheta_u \psi_1$. Differentiating the product away from its zeros gives

$$\psi_2 = -\mathcal{Z}(-pu), \quad \mathcal{Z}(w) = \sum_{n \in \mathbb{Z}} \frac{p^{2n} w}{(1 - p^{2n} w)^2}. \quad (24.10)$$

For negative n the summand can be rewritten through $w/(1 - w)^2 = w^{-1}/(1 - w^{-1})^2$, so this is a normally convergent meromorphic series on $\mathbb{C}^\times$, invariant under $w \mapsto p^2 w$ and $w \mapsto w^{-1}$. Put

$$\mathcal{W}(y) = \mathcal{Z}(e^y) - 2\mathcal{Q}_1(p^2) + \frac{1}{12}.$$

It is meromorphic with periods $2\pi i$ and $\log p^2$, and its poles are exactly the points of the period lattice, all double. Near $y = 0$ the summand $n = 0$ is

$$\frac{e^y}{(1 - e^y)^2} = y^{-2} - \frac{1}{12} + \frac{y^2}{240} - \frac{y^4}{6048} + O(y^6),$$

and pairing the summands n and $-n$ for $n \geq 1$ gives $2 \sum_{d \geq 1} dp^{2nd} \cosh(dy)$. Consequently

$$\mathcal{W}(y) = y^{-2} + \mathfrak{a}_2 y^2 + \mathfrak{a}_4 y^4 + O(y^6), \quad \mathfrak{a}_2 = \frac{1}{240} + \mathcal{Q}_3(p^2), \quad \mathfrak{a}_4 = -\frac{1}{6048} + \frac{1}{12} \mathcal{Q}_5(p^2). \quad (24.11)$$

The elliptic function $(\mathcal{W}')^2 - 4\mathcal{W}^3 + 20\mathfrak{a}_2\mathcal{W} + 28\mathfrak{a}_4$ has no pole: its y^{-6} , y^{-2} and constant terms cancel by (24.11), and periodicity treats every lattice point. It is bounded on a compact period parallelogram, hence constant, and its limit at 0 is zero. As $20\mathfrak{a}_2 = g_2$ and $28\mathfrak{a}_4 = g_3$,

$$(\mathcal{W}')^2 = 4\mathcal{W}^3 - g_2\mathcal{W} - g_3.$$

With $e^y = -pu$ one has $\vartheta_u = d/dy$ and $\mathcal{W} = \mathfrak{c} - \psi_2$, so, writing $\psi_3 = \vartheta_u \psi_2$,

$$\psi_3^2 = \Phi(\psi_2) = 4(\mathfrak{c} - \psi_2)^3 - g_2(\mathfrak{c} - \psi_2) - g_3. \quad (24.12)$$

The proof uses only the classical product, a Laurent expansion and the constancy of an elliptic function without poles.

24.3 From the complex identity to the Hahn field

Now let p be an indeterminate again, and work in the ring $\mathbb{Q}[u, u^{-1}][[p]]$ of power series in p whose coefficients are Laurent polynomials in u . The series $\Psi = \sum_n p^{n^2} u^n = 1 + p(u + u^{-1}) + \dots$ is a unit there, so ψ_1, ψ_2, ψ_3 belong to it, and $\mathfrak{c}, g_2, g_3 \in \mathbb{Q}[[p]]$. For u in the annulus $\frac{1}{2} \leq |u| \leq 2$ and $0 < p < \epsilon_0$, with ϵ_0 small, all these formal expansions converge absolutely to the complex functions of Section 24.2: $|\Psi - 1| < \frac{1}{2}$, so $1/\Psi$ is a convergent geometric series. Hence, for each such u , the power series in p obtained from $\psi_3^2 - \Phi(\psi_2)$ vanishes on $(0, \epsilon_0)$, so all its coefficients vanish there; each coefficient is a Laurent polynomial vanishing on an annulus, hence zero. Thus (24.12) is an identity in $\mathbb{Q}[u, u^{-1}][[p]]$, over $\mathbb{Q}$, and not merely for particular complex nomes.

From u to z . The map $\iota : \mathbb{Q}[z][[p]] \rightarrow \mathbb{Q}[u, u^{-1}][[p]]$, $\sum_k P_k(z)p^k \mapsto \sum_k P_k(u + u^{-1})p^k$, is an injective ring homomorphism. With $\check{z} = u - u^{-1}$,

$$\vartheta_u(u + u^{-1}) = \check{z}, \quad \vartheta_u \check{z} = u + u^{-1}, \quad \check{z}^2 = \iota(z^2 - 4), \quad \vartheta_u(\iota g) = \check{z} \iota(D_z g), \quad (24.13)$$

the last by the chain rule, coefficientwise in p . The formal series $\mathcal{T} = 1 + \sum_{n \geq 1} p^{n^2} C_n(z) \in \mathbb{Q}[z][[p]]$ has $\iota(\mathcal{T}) = \Psi$ by (23.2). From (24.13) one computes $\iota(\mathcal{H}[\mathcal{T}]) = \Psi \vartheta_u^2 \Psi - (\vartheta_u \Psi)^2 = \Psi^2 \psi_2$, then $\check{z} \iota(\mathcal{K}[\mathcal{T}]) = \Psi^3 \psi_3$ and $\iota(\mathcal{B}[\mathcal{T}]) = \Psi^2(\mathfrak{c} - \psi_2)$. Hence ι maps the left side of (24.7) for $\mathcal{T}$ to $\Psi^6(\psi_3^2 - \Phi(\psi_2)) = 0$, and injectivity gives (24.7) in $\mathbb{Q}[z][[p]]$.

From $\mathbb{Q}[[p]]$ to $\mathbb{K}$. For every $\mu > 0$ the map $s_\mu : \mathbb{Q}[[p]] \rightarrow \mathbb{K}$, $\sum_k c_k p^k \mapsto \sum_k c_k t^{k\mu}$, is an injective ring homomorphism: the support lies in the well-ordered set $\{k\mu\}$, each exponent comes from one k , and the Cauchy product is preserved. Applied coefficientwise it is a ring homomorphism $\mathbb{Q}[[p]][[z]] \rightarrow \mathbb{K}[[z]]$ commuting with D_z , and $\mathbb{Q}[z][[p]] \subseteq \mathbb{Q}[[p]][[z]]$ as subrings of $\mathbb{Q}[[p, z]]$. It maps the formal $\mathcal{T}$ to $\mathcal{T}_p$, whose coefficients are (23.4)–(23.5), and the formal constants to (24.2). This proves (24.7) in $\mathbb{K}[[z]]$ for every $\mu > 0$. Dividing by f^6 in $\mathbb{K}((z))$ gives (24.9).

Inverses are taken only in $\mathbb{Q}[u, u^{-1}][[p]]$, where Ψ is a unit, and only the cleared identity is specialized. This avoids the incorrect claim that the p -adic geometric expansion of $1/\Psi_p(u)$ converges at every Hahn argument, a point source 13 makes explicitly; in a Hahn function field the inverse is taken as a fraction.

24.4 The corner of the cleared equation

Proposition 24.2 (Corner data of the cleared equation; merge). *Let P be the polynomial (24.8) with the constants (24.2), written in the combined-monomial form (14.4). In the notation of Section 14.4, $d_P = 6$, $w_P = 0$, $\beta_P = 0$, and*

$$I_P(T) = -\Phi(0) = -(4\mathfrak{c}^3 - g_2\mathfrak{c} - g_3), \quad \chi_P(T) = 0.$$

The constant $4\mathfrak{c}^3 - g_2\mathfrak{c} - g_3$ equals $-4p^2 - 48p^4 - 144p^6 - 448p^8 - 600p^{10} + O(p^{12})$; so I_P is a nonzero constant of valuation 2μ and leading coefficient 4, while the residue corner polynomial vanishes identically.

Proof. Every monomial of P has total degree six in the Y_j , so $d_P = 6$. For the weight $w_\nu = k_\nu - \sum_j j m_{\nu j}$ of (14.5), every term of H has weight at most 0, with weight-zero part $H_0 = z^2(Y_0Y_2 - Y_1^2) + zY_0Y_1$ (the terms $-4(Y_0Y_2 - Y_1^2)$ have weight -2). The total derivative lowers the weight of each term by exactly one, so the terms of H_1 have weight at most -1 , with weight -1 part the total derivative H'_0 of H_0 . Hence every monomial of P has weight at most 0, and its weight-zero part is

$$P_0 = z^2(Y_0H'_0 - 2Y_1H_0)^2 - 4(\mathfrak{c}Y_0^2 - H_0)^3 + g_2(\mathfrak{c}Y_0^2 - H_0)Y_0^4 + g_3Y_0^6.$$

The substitution $z \mapsto 1$, $Y_j \mapsto (T)_j$ is a ring homomorphism $\mathbb{K}[z, Y] \rightarrow \mathbb{K}[T]$ sending a monomial of weight 0 to the product of falling factorials in (14.6). It sends H_0 to $T(T-1) - T^2 + T = 0$ and $H'_0 = z^2(Y_0Y_3 - Y_1Y_2) + 2z(Y_0Y_2 - Y_1^2) + Y_0Y_1 + z(Y_1^2 + Y_0Y_2)$ to

$$T(T-1)(T-2) - T^2(T-1) + 2(T(T-1) - T^2) + T + T^2 + T(T-1) = 0,$$

so P_0 goes to $-4\mathfrak{c}^3 + g_2\mathfrak{c} + g_3 = -\Phi(0)$. From $\mathfrak{c} = \frac{1}{12} - 2p^2 + O(p^4)$, $g_2 = \frac{1}{12} + 20p^2 + O(p^4)$ and $g_3 = -\frac{1}{216} + \frac{7}{3}p^2 + O(p^4)$ one finds $4\mathfrak{c}^3 = \frac{1}{432} - \frac{1}{6}p^2 + O(p^4)$ and $g_2\mathfrak{c} = \frac{1}{144} + \frac{3}{2}p^2 + O(p^4)$, so $4\mathfrak{c}^3 - g_2\mathfrak{c} - g_3 = -4p^2 + O(p^4)$; the further terms were computed with SymPy. In particular the coefficient of the weight-zero monomial Y_0^6 in P is $-\Phi(0) \neq 0$, so $P_0 \neq 0$, $w_P = 0$, and $I_P = -\Phi(0)$. The monomial $z^6Y_0^4Y_3^2$ of P_0 has coefficient 1, and every coefficient of P lies in $\mathbb{Z}[\mathfrak{c}, g_2, g_3] \subseteq s_\mu(\mathbb{Q}[[p]])$, which has nonnegative valuation; so $\beta_P = 0$. Since res is a ring homomorphism on elements of nonnegative valuation and the falling-factorial products have integer coefficients, χ_P is obtained by applying res to the coefficients of I_P , and $\text{res}(-\Phi(0)) = 0$ because $v(\Phi(0)) = 2\mu > 0$. $\square$

The corner consists of 19 monomials (SymPy). The computation has a transparent form: the weight-zero part H_0 of $\mathcal{H}$ is exactly the left side of the vanishing-corner equation (16.1), whose residue corner polynomial is the identity (16.2).

Remark 24.3 (The vanishing corner; source 13 and merge). The corner criterion of Theorem 16.1 is a sufficient rigidity criterion, not a theorem for every higher-order equation, and the cleared equation necessarily lies outside it. Source 13 records the cancellation directly: on a trial monomial $f = z^N$ the top-degree part of $\mathcal{A}[f]$ vanishes, because $z^2(-N/z^2) + z(N/z) = 0$, so retaining only separate derivative degrees and ignoring their cancellation would miss the example. Proposition 24.2 makes this exact and adds three consequences. (1) The equation is consistent with Theorems 15.1 and 16.1: they require $\chi_P \neq 0$, and here $\chi_P = 0$. (2) When μ is an order unit, $\mathcal{T}_p$ satisfies no nonzero equation with $d_P \geq 1$ and $\chi_P \neq 0$ at all (and a nonzero P with $d_P = 0$ is a nonzero element of $\mathbb{K}[z]$, which no series satisfies), since by Theorem 16.1 such an equation has only polynomial strongly entire solutions; so a nonpolynomial strongly entire series can satisfy an equation whose residue corner vanishes identically without satisfying any equation to which Theorem 15.1 applies. (3) The

hypothesis $\chi_P \neq 0$ of those theorems cannot be weakened to $I_P \neq 0$: here the full corner polynomial is a nonzero constant of positive valuation, while the residue corner vanishes. The containment (14.7) and Proposition 16.2, which assume $\chi_P \neq 0$, are unaffected. None of this contradicts the first-order or D-finite rigidity theorems.

24.5 The Laurent ring and the substitution $z = u + u^{-1}$

In this subsection and the next two, μ is an order unit. This supplies a common Laurent function field for all nonzero arguments; the lower bound concerns formal functions, not the algebraic independence of values in their own ambient field.

Let $\mathcal{U}_\mu$ be the set of formal Laurent series $b = \sum_{m \in \mathbb{Z}} b_m u^m$ with $b_m \in \mathbb{K}$ such that, for all $\gamma, \beta \in \Gamma$, only finitely many m with $b_m \neq 0$ satisfy $v(b_m) + m\gamma \leq \beta$. This is source 13's condition " $v(b_m) + m\gamma \rightarrow +\infty$ as $|m| \rightarrow \infty$ ", written in the form of Proposition 20.2 (iii). For each γ the weighted values $v(b_m) + m\gamma$ of the nonzero coefficients then have a least element, attained by finitely many m .

Lemma 24.4 (The Laurent ring; source 13). (a) $\mathcal{U}_\mu$ is an integral domain containing $\mathbb{K}[u, u^{-1}]$, with the product $(\sum b_m u^m)(\sum c_m u^m) = \sum_m (\sum_{m_1+m_2=m} b_{m_1} c_{m_2}) u^m$, the inner sums being strong sums.

(b) $\vartheta_u(\sum b_m u^m) = \sum m b_m u^m$ is a derivation of $\mathcal{U}_\mu$ vanishing on $\mathbb{K}$, and $S(\sum b_m u^m) = \sum b_m p^{2m} u^m$ is a ring automorphism fixing $\mathbb{K}$ and commuting with ϑ_u . Both extend to $\text{Frac} \mathcal{U}_\mu$.

(c) Ψ_p and all $\vartheta_u^j \Psi_p$ belong to $\mathcal{U}_\mu$, and $\Psi_p \neq 0$.

Proof. (a) With $\gamma = 0$ the coefficient values of b and c are bounded below and escape cofinally, so for fixed m the family $(b_{m_1} c_{m-m_1})_{m_1}$ is strongly summable by Lemma 2.3 (a). For fixed γ , if the product coefficient $d_m \neq 0$ has $v(d_m) + m\gamma \leq \beta$, some term has $(v(b_{m_1}) + m_1\gamma) + (v(c_{m_2}) + m_2\gamma) \leq \beta$, and the lower bounds of the two weighted families leave finitely many pairs, as in the proof of Proposition 20.4; so the product lies in $\mathcal{U}_\mu$. Commutativity, associativity and distributivity follow from the associativity of strong sums applied to the families $(b_{m_1} c_{m_2} e_{m_3})$. For integrality, let w_b and w_c be the least coefficient values of nonzero b and c and in (b) $\sum_{v(b_m)=w_b} \text{lc}(b_m) u^m \in k[u, u^{-1}]$, a nonzero Laurent polynomial. The coefficient of $t^{w_b+w_c} u^m$ in bc is the coefficient of u^m in in(b) in(c), because every term has value at least $w_b + w_c$. Since $k[u, u^{-1}]$ is a domain, $bc \neq 0$.

(b) Multiplying coefficients by ordinary integers preserves the defining condition, and Leibniz's rule holds coefficientwise since $m = m_1 + m_2$. For S , $v(b_m p^{2m}) + m\gamma = v(b_m) + m(\gamma + 2\mu)$, the inverse uses p^{-2m} , and $p^{2m} = p^{2m_1} p^{2m_2}$ with Lemma 2.1 (3) gives multiplicativity. Both operators act diagonally on u^m , so they commute. An automorphism and a derivation of a domain extend to its fraction field.

(c) If $|\gamma| \leq N\mu$ and $\beta \leq B\mu$, then $v(p^{n^2}) + n\gamma \geq (n^2 - N|n|)\mu > \beta$ for all but finitely many n ; multiplying coefficients by n^j changes no valuation. $\square$

Lemma 24.5 (Substitution of $u + u^{-1}$; merge). For $f = \sum_n a_n z^n \in \mathcal{E}_\Gamma(k)$ and $m \in \mathbb{Z}$ the family $(a_n \binom{n}{n-m}/2)_{n \geq |m|, n \equiv m(2)}$ is strongly summable. Write b_m for its sum and $f^\natural = \sum_m b_m u^m$. Then:

(a) $f^\natural \in \mathcal{U}_\mu$;

(b) $f \mapsto f^\natural$ is a $\mathbb{K}$ -algebra homomorphism $\mathcal{E}_\Gamma(k) \rightarrow \mathcal{U}_\mu$, equal to the substitution $z \mapsto u + u^{-1}$ on $\mathbb{K}[z]$;

(c) $\vartheta_u(f^\natural) = \check{z}(D_z f)^\natural$, where $\check{z} = u - u^{-1}$;

(d) $\mathcal{T}_p^\natural = \Psi_p$.

Proof. Write $\beta_{n,m} = \binom{n}{(n-m)/2}$ when $n \geq |m|$ and $n \equiv m \pmod{2}$, and $\beta_{n,m} = 0$ otherwise; this is the coefficient of u^m in $(u + u^{-1})^n$. By Proposition 20.2 (iii) with $\gamma = 0$, the coefficient values of f escape cofinally, and the nonzero integers $\beta_{n,m}$ have value zero, so Lemma 2.3 (a) gives strong summability.

(a) Fix γ, β and put $\gamma' = -|\gamma|$. If $b_m \neq 0$ and $v(b_m) + m\gamma \leq \beta$, some $n \geq |m|$ with $a_n \neq 0$ has $v(a_n) + m\gamma \leq \beta$, and then $v(a_n) + n\gamma' \leq v(a_n) + m\gamma \leq \beta$ because $n\gamma' \leq |m|\gamma' \leq m\gamma$. By Proposition 20.2 (iii) at γ' only finitely many n qualify, and each serves only the m with $|m| \leq n$.

(b) Linearity over $\mathbb{K}$ holds because strong sums are linear. For multiplicativity fix m and consider the family $(a_i c_j \beta_{i,m_1} \beta_{j,m_2})$ indexed by (i, j, m_1, m_2) with $m_1 + m_2 = m$, where $g = \sum c_j z^j \in \mathcal{E}_\Gamma(k)$. The double family $(a_i c_j)$ is strongly summable, as in the proof of Lemma 14.2; each pair (i, j) carries finitely many (m_1, m_2) , and nonzero integer factors change neither supports nor multiplicities. Grouping by (m_1, m_2) and then by m_1 gives the coefficient of u^m in $f^\natural g^\natural$. Grouping by (i, j) gives $a_i c_j \beta_{i+j,m}$, because $(u + u^{-1})^i (u + u^{-1})^j = (u + u^{-1})^{i+j}$, and grouping those by $n = i + j$ gives $(\sum_{i+j=n} a_i c_j) \beta_{n,m}$, whose sum is the coefficient of u^m in $(fg)^\natural$. By associativity of strong sums the two totals agree. On polynomials all families are finite.

(c) Since $\vartheta_u(u + u^{-1})^n = n(u + u^{-1})^{n-1}(u - u^{-1})$, one has $m\beta_{n,m} = n(\beta_{n-1,m-1} - \beta_{n-1,m+1})$. The coefficient of u^m in $\vartheta_u(f^\natural)$ is $mb_m = \sum_n a_n m\beta_{n,m}$, and that of $\check{z}(D_z f)^\natural$ is $\sum_n n a_n (\beta_{n-1,m-1} - \beta_{n-1,m+1})$; they agree termwise, and $D_z f \in \mathcal{E}_\Gamma(k)$ by Proposition 20.4.

(d) With $c_{N,j}$ the integers of (23.3), the coefficient of u^m in $\mathcal{T}_p^\natural$ is the sum of the family $(p^{N^2} c_{N,j} \beta_{N-2j,m})_{N,j}$, grouped by $n = N - 2j$ (the constant term included with $N = 0$). Its exponents $N^2 \mu$ each come from finitely many j , so it is strongly summable. Grouping instead by N gives p^{N^2} times the coefficient of u^m in $C_N(u + u^{-1}) = u^N + u^{-N}$, which is nonzero only for $N = |m|$. The sum is p^{m^2} . $\square$

Injectivity of $f \mapsto f^\natural$ is not needed. By (c) and Proposition 20.4, $(D_z^r f)^\natural = (\check{z}^{-1} \vartheta_u)^r f^\natural$ in $\text{Frac } \mathcal{U}_\mu$, which equals $\check{z}^{-r} \vartheta_u^r f^\natural$ plus a combination of lower $\vartheta_u^j f^\natural$ with coefficients in $\mathbb{Q}(u)$. For $\mathcal{T}_p$,

$$\Psi_p = \mathcal{T}_p^\natural, \quad \vartheta_u \Psi_p = \check{z}(D_z \mathcal{T}_p)^\natural, \quad \vartheta_u^2 \Psi_p = \check{z}^2 (D_z^2 \mathcal{T}_p)^\natural + (u + u^{-1})(D_z \mathcal{T}_p)^\natural. \quad (24.14)$$

24.6 Three independent theta quantities

Lemma 24.6 (Three rational-function obstructions; source 13). *Let E be a field of characteristic zero and $a \in E^\times$ not a root of unity, and let S_a be the automorphism of $E(u)$ with $S_a u = au$ fixing E .*

(a) *If $S_a r = r$, then $r \in E$.*

(b) *For $d \in E^\times$ there is no $r \in E(u)$ with $S_a r - r = d$.*

(c) *On $E(u, V)$, with V transcendental, extend S_a by $S_a V = V - 1$. Then S_a preserves the u -adic order ord_u of every nonzero element.*

Proof. Expand a rational function as a formal Laurent series at $u = 0$. For (a), each coefficient of degree $n \neq 0$ is multiplied by $a^n \neq 1$, so invariance leaves only degree zero. For (b), the coefficient

of u^0 in $r(au) - r(u)$ is zero, also when r has a pole at 0. For (c), a lowest nonzero coefficient in $E(V)((u))$ is sent to the nonzero coefficient obtained by $V \mapsto V - 1$, multiplied by a nonzero power of a ; its degree is unchanged. $\square$

In $\text{Frac}\mathcal{U}_\mu$ put $\Psi = \Psi_p$, $\psi_1 = \vartheta_u \Psi / \Psi$, $\psi_2 = \vartheta_u \psi_1$ and $\psi_3 = \vartheta_u \psi_2$; the field $\mathbb{K}(u)$ lies in it and is preserved by S . By (23.10),

$$S\Psi = p^{-1}u^{-1}\Psi, \quad S\psi_1 = \psi_1 - 1, \quad S\psi_2 = \psi_2. \quad (24.15)$$

Lemma 24.7 (Transfer, and a global certificate for $\psi_3 \neq 0$; merge). (a) *For every polynomial Π in four variables with coefficients in $\mathbb{Q}[[p]]$ and every $m \in \mathbb{Z}$, the coefficient of u^m in*

$$\Pi(\Psi, \vartheta_u \Psi, \vartheta_u^2 \Psi, \vartheta_u^3 \Psi),$$

computed in $\mathcal{U}_\mu$ with the coefficients of Π mapped by s_μ , is the image under s_μ of the same coefficient computed in $\mathbb{Q}[u, u^{-1}][[p]]$. In particular $\psi_3^2 = \Phi(\psi_2)$ in $\text{Frac}\mathcal{U}_\mu$.

(b) *In $\mathcal{U}_\mu$,*

$$\Psi^3 \psi_3 = \mathcal{N} := \Psi^2 \vartheta_u^3 \Psi - 3\Psi \vartheta_u \Psi \vartheta_u^2 \Psi + 2(\vartheta_u \Psi)^3,$$

and the coefficient of u in $\mathcal{N}$ has valuation μ and leading coefficient 1. In particular $\psi_3 \neq 0$.

Proof. (a) The four series $\vartheta_u^j \Psi = \sum_n n^j p^{n^2} u^n$ lie in $\mathcal{U}_\mu$ (Lemma 24.4). By associativity of strong sums, the coefficient of u^m on the $\mathcal{U}_\mu$ side is the strong sum of the family of terms $e t^{(k+n_1^2+\dots+n_r^2)\mu}$, $e \in \mathbb{Z}$, indexed by a monomial of Π of degree r , a term ep^k of its coefficient, and a tuple $(n_1, \dots, n_r)$ with $n_1 + \dots + n_r = m$. The exponent $K\mu$ is carried by the finitely many members with $k + \sum n_i^2 = K$, whose integer coefficients add up to the coefficient of $u^m p^K$ on the formal side. This is the assertion. Applied to the cleared form of (24.12), $\Psi^6(\psi_3^2 - \Phi(\psi_2))$, which is a polynomial in the four series with coefficients in $\mathbb{Z}[\mathfrak{c}, g_2, g_3]$ and vanishes in $\mathbb{Q}[u, u^{-1}][[p]]$ (Section 24.3), it gives the identity in $\text{Frac}\mathcal{U}_\mu$, because $\Psi \neq 0$ there.

(b) The identity is the quotient rule applied twice. By (a), the coefficient of u in $\mathcal{N}$ is the image under the injective ring homomorphism s_μ of the coefficient of u in the same expression computed in $\mathbb{Q}[u, u^{-1}][[p]]$. There $\Psi = 1 + p(u + u^{-1}) + O(p^4)$, $\vartheta_u \Psi = \vartheta_u^3 \Psi = p(u - u^{-1}) + O(p^4)$ and $\vartheta_u^2 \Psi = p(u + u^{-1}) + O(p^4)$, so

$$\mathcal{N} = p(u - u^{-1}) + O(p^2).$$

The coefficient of u is $p + O(p^2)$; SymPy gives $p - 8p^3 + 20p^5 - 70p^9 + 64p^{11} + O(p^{12})$. Its image $t^\mu + \dots$ is nonzero, so $\mathcal{N} \neq 0$, and since $\mathcal{U}_\mu$ is a domain and $\Psi \neq 0$, $\psi_3 = \mathcal{N}/\Psi^3 \neq 0$. $\square$

Source 13's proof of this step expanded ψ_1 at the simple zero $u_0 = -p$ of Ψ as $u_0/(u - u_0) + O(1)$ and read off $\psi_3 = 2u_0^3/(u - u_0)^3 + O((u - u_0)^{-2})$; the lemma replaces that local argument, which used expansions in $\text{Frac}\mathcal{U}_\mu$ that were not defined. Neither route uses a transcendence theorem for complex theta values.

Theorem 24.8 (Three independent theta quantities; source 13). *Let μ be an order unit. The elements Ψ, ψ_1, ψ_2 of $\text{Frac}\mathcal{U}_\mu$ are algebraically independent over $\mathbb{K}(u)$. Equivalently, $\Psi, \vartheta_u \Psi, \vartheta_u^2 \Psi$ are algebraically independent over $\mathbb{K}(u)$.*

Proof. ψ_2 is transcendental over $\mathbb{K}(u)$. Otherwise let $\pi(T)$ be its monic minimal polynomial. Since $S\psi_2 = \psi_2$ and S preserves $\mathbb{K}(u)$, applying S to the coefficients gives the same monic minimal polynomial, so by Lemma 24.6 (a), with $E = \mathbb{K}$ and $a = p^2$, all its coefficients lie in $\mathbb{K}$. Applying ϑ_u to $\pi(\psi_2) = 0$ gives $\pi'(\psi_2)\psi_3 = 0$. By separability in characteristic zero $\pi'(\psi_2) \neq 0$, contradicting Lemma 24.7.

ψ_1 is transcendental over $\mathbb{K}(u, \psi_2)$. This field is $E(u)$ with $E = \mathbb{K}(\psi_2)$ fixed by S . Suppose the monic minimal polynomial of ψ_1 over it has degree $d \geq 1$ and begins $T^d + bT^{d-1} + \dots$. Applying S and then replacing T by $T - 1$ gives another monic polynomial vanishing at ψ_1 , because $S\psi_1 = \psi_1 - 1$; its next coefficient is $Sb - d$. Uniqueness gives $Sb - b = d$, contradicting Lemma 24.6 (b).

Ψ is transcendental over $\mathbb{K}(u, \psi_2, \psi_1)$. Let its hypothetical monic minimal polynomial have degree d and constant coefficient b_0 , nonzero because $\Psi \neq 0$. Applying S , substituting $p^{-1}u^{-1}T$ for T and dividing by $(p^{-1}u^{-1})^d$ gives another monic polynomial vanishing at Ψ , so $Sb_0 = (p^{-1}u^{-1})^d b_0$. By the first two steps the field is the rational function field $\mathbb{K}(\psi_2)(u, \psi_1)$, on which S acts as in Lemma 24.6 (c) with $V = \psi_1$. The left side has the u -adic order of b_0 , the right side that order minus d , which is impossible.

Finally $\vartheta_u \Psi = \psi_1 \Psi$ and $\vartheta_u^2 \Psi = (\psi_2 + \psi_1^2) \Psi$, and these formulas and their rational inverses identify the fields generated by the two triples. $\square$

Corollary 24.9 (Exact minimal order; source 13). *Let μ be an order unit. Then $\mathcal{T}_p$, $D_z \mathcal{T}_p$ and $D_z^2 \mathcal{T}_p$ are algebraically independent over $\mathbb{K}(z)$, and the least order of a nonzero algebraic differential equation for $\mathcal{T}_p$ over $\mathbb{K}(z)$ is three.*

Proof. Suppose that a nonzero polynomial P in Y_0, Y_1, Y_2 with coefficients in $\mathbb{K}[z]$, obtained after clearing denominators, satisfies $P(z, \mathcal{T}_p, D_z \mathcal{T}_p, D_z^2 \mathcal{T}_p) = 0$. Apply Lemma 24.5 (b): the image of this identity in $\mathcal{U}_\mu$ is $P^\natural(\Psi, (D_z \mathcal{T}_p)^\natural, (D_z^2 \mathcal{T}_p)^\natural) = 0$, where $P^\natural$ has the coefficients of P with $u + u^{-1}$ substituted for z . By (24.14) the three u -jets are obtained from $\Psi, \vartheta_u \Psi, \vartheta_u^2 \Psi$ by a triangular linear change with diagonal $1, \check{z}^{-1}, \check{z}^{-2}$, invertible over $\mathbb{K}(u)$ because $\check{z} \neq 0$. The polynomial $P^\natural$ is nonzero, because $z \mapsto u + u^{-1}$ is injective on $\mathbb{K}[z]$, so this is a nonzero relation among $\Psi, \vartheta_u \Psi, \vartheta_u^2 \Psi$ over $\mathbb{K}(u)$, contradicting Theorem 24.8. This proves the lower bound; Theorem 24.1 gives the upper bound. $\square$

24.7 The differential field and non-D-finiteness

Let μ be an order unit and write, in $\mathbb{K}((z))$,

$$f = \mathcal{T}_p, \quad \mathcal{L} = D_z f / f, \quad \mathcal{A} = \mathcal{A}[f], \quad \mathcal{A}_1 = D_z \mathcal{A}.$$

The first three are algebraically independent over $\mathbb{K}(z)$, being an invertible rational change of $f, D_z f, D_z^2 f$.

Theorem 24.10 (Differential-field presentation; source 13). *Let μ be an order unit. The differential field generated by $\mathcal{T}_p$ over $\mathbb{K}(z)$ is*

$$\mathbb{K}(z)\langle \mathcal{T}_p \rangle = \mathbb{K}(z)(f, \mathcal{L}, \mathcal{A}, \mathcal{A}_1), \quad (z^2 - 4)\mathcal{A}_1^2 = \Phi(\mathcal{A}). \quad (24.16)$$

It has transcendence degree three over $\mathbb{K}(z)$ and degree two over the rational function field $\mathbb{K}(z)(f, \mathcal{L}, \mathcal{A})$. Its derivation is

$$D_z f = \mathcal{L}f, \quad D_z \mathcal{L} = \frac{\mathcal{A} - z\mathcal{L}}{z^2 - 4}, \quad (24.17)$$

$$D_z \mathcal{A} = \mathcal{A}_1, \quad D_z \mathcal{A}_1 = \frac{-12(\mathfrak{c} - \mathcal{A})^2 + g_2 - 2z\mathcal{A}_1}{2(z^2 - 4)}. \quad (24.18)$$

Consequently the minimal polynomial of $D_z^3 \mathcal{T}_p$ over $\mathbb{K}(z)(\mathcal{T}_p, D_z \mathcal{T}_p, D_z^2 \mathcal{T}_p)$ has degree two, and (24.7) is that minimal relation up to a nonzero factor.

Proof. Regard $\Phi(\mathcal{A})/(z^2 - 4)$ as a rational function of the transcendental $\mathcal{A}$ over $\mathbb{K}(z)(f, \mathcal{L})$. Its order at $\mathcal{A} = \infty$ is -3 , an odd integer, so it is not a square, and adjoining $\mathcal{A}_1$ is a quadratic extension. In particular $\mathcal{A}_1 \neq 0$, as also follows from $\Phi \neq 0$ and the transcendence of $\mathcal{A}$.

The formulas for $D_z f$ and $D_z \mathcal{L}$ follow from the definitions, since $\mathcal{A} = (z^2 - 4)D_z \mathcal{L} + z\mathcal{L}$. Differentiate $(z^2 - 4)\mathcal{A}_1^2 = \Phi(\mathcal{A})$ and use $D_z \mathcal{A} = \mathcal{A}_1$ and $\Phi'(T) = -12(\mathfrak{c} - T)^2 + g_2$; cancelling $\mathcal{A}_1 \neq 0$ gives (24.18). So all derivatives of the four generators remain in the field, and conversely the generators are rational in $f, D_z f, D_z^2 f, D_z^3 f$; the field is the differential field generated by f . Finally $\mathcal{A}_1 = \mathcal{K}[f]/f^3$ is affine linear in $D_z^3 f$ with nonzero leading coefficient $(z^2 - 4)/f$, so the degree-two assertion for $\mathcal{A}_1$ is the same assertion for $D_z^3 f$, and clearing powers of f introduces only a nonzero factor. $\square$

The poles at $z = \pm 2$ of this rational system describe a coordinate chart, not poles of the entire f ; the globally meaningful equation is the cleared identity (24.7). The cubic in the elliptic coordinate $\mathfrak{c} - \mathcal{A}$ accounts for the algebraic part of the differential field, while $\mathcal{L}$ and f carry the logarithmic and multiplicative theta information.

Proposition 24.11 (Non-D-finiteness; source 13). *Let μ be an order unit. Then $\mathcal{T}_p$ satisfies no nonzero linear differential equation over $\mathbb{K}(z)$.*

Proof. Minimal algebraic order three does not by itself exclude a linear equation of higher order, so a separate argument is needed. In the u -coordinate, (24.12) holds in $\text{Frac } \mathcal{U}_\mu$, $\psi_3^2 = \Phi(\psi_2)$, and, after differentiation and cancellation of $\psi_3 \neq 0$, $\vartheta_u \psi_3 = \frac{1}{2}\Phi'(\psi_2)$ (Lemma 24.7). Hence $\mathbb{K}(\psi_2, \psi_3)$ is closed under ϑ_u and algebraic over $\mathbb{K}(\psi_2)$, and by Theorem 24.8 ψ_1 is transcendental over $\mathbb{K}(u, \psi_2, \psi_3)$. Inductively,

$$\frac{\vartheta_u^n \Psi}{\Psi} = \psi_1^n + \text{a polynomial of degree less than } n \text{ in } \psi_1 \text{ over } \mathbb{K}(\psi_2, \psi_3) :$$

apply ϑ_u , add multiplication by ψ_1 , and use $\vartheta_u \psi_1 = \psi_2$. A nontrivial $\mathbb{K}(u)$ -linear combination of the $\vartheta_u^n \Psi$, divided by Ψ , has a nonzero highest power of the transcendental ψ_1 , so it does not vanish. A nonzero linear operator $\sum_{r \leq R} \rho_r(z) D_z^r$ with $\rho_R \neq 0$ becomes, through Lemma 24.5 and the triangular formula after it, a $\mathbb{K}(u)$ -linear combination of $\vartheta_u^r \Psi$, $r \leq R$, whose coefficient of $\vartheta_u^R \Psi$ is $\rho_R(u + u^{-1})\tilde{z}^{-R} \neq 0$. $\square$

For $k = \mathbb{C}$ the proposition is also a case of Theorem A, since $\mathcal{T}_p$ is then a nonpolynomial element of $\mathcal{E}_\Gamma$; the argument above works for every k and uses no escape chain.

25 The exact boundary, surreal consequences and scope

25.1 Proof of Theorem H, and Theorems E and G

Proof of Theorem H. (ii) $\Rightarrow$ (i) is the contrapositive of Theorem E (a) (sources 11 and 12; source 13 proves it again, Remark 25.1). For (i) $\Rightarrow$ (iii) let μ be an order unit and $p = t^\mu$. Theorem 23.5 gives $\mathcal{T}_p \in \mathcal{E}_\Gamma(k)$ and Proposition 23.3 nonpolynomiality; Theorem 24.1, Corollary 24.9 and Theorem 24.10 give an equation of order three, algebraic independence of $\mathcal{T}_p, D_z \mathcal{T}_p, D_z^2 \mathcal{T}_p$ and degree two of $D_z^3 \mathcal{T}_p$

over the field they generate. (iii) $\Rightarrow$ (ii) is immediate. Non-D-finiteness is Proposition 24.11. Finally $\mathcal{T}_p$ is not in $\mathbb{K}(z)$, since it is transcendental over $\mathbb{K}(z)$. $\square$

Remark 25.1 (Source 13's proof of the no-order-unit direction). Section 7 of source 13 proves (ii) $\Rightarrow$ (i) self-contained, so that its classification does not rest on a citation. It is the argument of Theorem E(a) and is printed once there. Source 13 writes a relation with the Euler operator $\vartheta = zD_z$, which is a triangular rational change of the ordinary derivatives, chooses it of least total degree in the jets, and linearizes along the solution: if $\pi_j = \partial P / \partial Y_j$ evaluated at $(f, \vartheta f, \dots)$, θ' is the least z -order of a nonzero π_j and $I(T) = \sum_j [z^{\theta'}] \pi_j T^j$, then, because ϑ does not lower z -order, $I(n)a_n = -[z^{n+\theta'}]P(z, f_{<n}, \vartheta f_{<n}, \dots)$ for $n > \theta'$. With the Euler operator the linearization multiplier is a polynomial in the index. Its Lemmas 7.1 and 7.2 are Theorem 21.2(a) and Lemmas 20.6–20.7, and its Theorem 7.3, via the exterior obstruction of Proposition 20.9, is Theorem E(a). Source 13 credits this direction to the coefficient-field part and claims no new finite-generation principle.

Proof of Theorem G(iv). (i) $\Leftrightarrow$ (iv) is (i) $\Leftrightarrow$ (ii) of Theorem H; with the proof of Theorem G given in Section 21.4, all four conditions are equivalent. $\square$

Corollary 25.2 (Theorem E is sharp; merge). *For Γ as in Convention 1.1:*

- (a) *every nonpolynomial $f \in \mathcal{E}_\Gamma(k)$ is differentially transcendental over $\mathbb{K}(z)$ if and only if Γ has no order unit;*
- (b) *the elements of $\text{Frac } \mathcal{E}_\Gamma(k)$ that are differentially algebraic over $\mathbb{K}(z)$ are exactly those of $\mathbb{K}(z)$ if and only if Γ has no order unit.*

If Γ has an order unit μ , then for $f = \mathcal{T}_p$ and every $q \in \mathbb{K}^\times$ of infinite multiplicative order, including $v(q) = 0$, the family of Theorem G(iii) is algebraically dependent over $\mathbb{K}(z)$.

Proof. (a) is Theorem H, (i) $\Leftrightarrow$ (ii). For (b), Theorem E(c) gives the equality without an order unit; with one, $\mathcal{T}_p \in \mathcal{E}_\Gamma(k) \subseteq \text{Frac } \mathcal{E}_\Gamma(k)$ is differentially algebraic and not in $\mathbb{K}(z)$. The last assertion holds because the subfamily with $i = 0$ is already dependent (Theorem 24.1). $\square$

25.2 Consistency with the rest of the report

The new part contradicts no earlier statement; the points of contact are these.

- *Theorem E.* Without an order unit $\mathcal{T}_p$ is not entire: $\text{Dom}_{\text{str}}(\mathcal{T}_p) = \mathcal{O}_{\Delta_\mu} \neq \mathbb{K}$ (Theorem 23.5). Theorem E(a) is thus not contradicted, and by Corollary 25.2 its hypothesis is necessary.
- *Theorem D and the corner criterion.* The minimal order is three, not one; and $\chi_P = 0$ for the cleared equation, as Theorem 16.1 requires of an equation with a nonpolynomial entire solution, while $I_P \neq 0$ (Proposition 24.2, Remark 24.3).
- *Theorem F.* The coefficient values $m^2\mu$ span $\mathbb{Q}\mu$, so $\text{cvr}(\mathcal{T}_p) = 1$ and Theorem F does not apply; it is a sufficient condition only (Section 22.4). Theorem 21.20 says nothing about series whose coefficient values span $\mathbb{Q}\mu$, and $\mathcal{T}_p$ is such a series with an algebraic differential relation.
- *Theorem G.* $\mathcal{T}_p$ is a second witness for (ii) and (iii), and a stronger one for (iii): its jets are already dependent at $i = 0$, so (iii) holds for every q of infinite order, including $v(q) = 0$. Its coefficients lie in a field finitely generated over k , by Theorem 21.2(a) with $k_0 = k(\mathfrak{c}, g_2, g_3)$, as (ii) requires.

- *Theorem 21.11.* The coefficient values lie in $\mathbb{Z}\mu$, whose smallest containing convex subgroup Δ_μ is all of Γ exactly when μ is an order unit; the descent theorem needs a proper convex subgroup and does not apply.
- *The partial theta series.* Θ_p and G (Proposition 21.19, Corollary 17.1) are untouched: source 13 explains why a bilateral descent is used instead, since the dilation equation of Θ_p is not an algebraic differential equation. Whether Θ_p or G is differentially algebraic stays open.
- *Question 19.4.* It concerns *linear* mixed equations at unit valuation. $\mathcal{T}_p$ is not D-finite, and source 13 says nothing about linear mixed equations, so the question stays open; the nonlinear analogue is settled by Corollary 25.2.
- *Question 19.8.* $\mathcal{T}_p$ is differentially algebraic and strongly entire, with a finitely generated coefficient field whose values $m^2\mu$ are cofinal and grow faster than any linear bound. With an order unit, therefore, no property shared by the coefficient fields of all differentially algebraic series can force a linear bound. The question asks for additional properties and stays open.
- *The Hahn–Tate report.* Ψ_p is that report’s theta series at the nome p^2 and the argument $-pu$; its domain and product (`tate:thm:theta`, `tate:prop:theta-product`, the latter for $0 \leq v(u) < v(q)$) are specialized here, not claimed anew [30].

25.3 Surreal, surcomplex and omnific consequences

Through (2.4) take $\Gamma = \mathbb{R}$ and $\mu = 1$. Then $\mathcal{T}_t$ is an actual nonpolynomial strongly entire function on the set-sized surcomplex field $\mathbb{C}((\omega^{-\mathbb{R}})) \subseteq \mathbf{No}[i]$, with coefficients in the smaller field $\mathbb{Q}((\omega^{-1}))$; the same holds on $\mathbb{C}((\omega^{-\mathbb{Z}}))$ and $\mathbb{C}((\omega^{-\mathbb{Q}}))$, and algebraic closedness of the workspace is not needed. The derivative D_z still kills the coefficients, including $p = \omega^{-1}$, so this is a statement about functions whose values and coefficients are surreal or surcomplex, not about the scalar derivation of $\mathbf{No}$.

Not entire on the class. With $p = \omega^{-1}$ the same coefficient formula can be evaluated at every surreal or surcomplex point whose leading exponent is bounded above by an ordinary integer; the support proof is pointwise and uses only set-sized supports. It fails at $x = \omega^\omega$: the leading exponent of $a_m x^m$ in Conway’s convention is $m\omega - m^2$, and these increase strictly, so they are not the decreasing well-ordered support of a strong surreal sum. The example is therefore not strongly entire on $\mathbf{No}$ or $\mathbf{No}[i]$, as for F in Section 11.2. More generally, enlarging Γ to Γ' while keeping $p = t^\mu$ changes the domain exactly to $\{x : v(x) \geq -N\mu \text{ for some } N\}$ in the larger field: an extension can turn an order unit into a non-order-unit and destroy entireness without changing a coefficient or the equation.

Omnific integers. The omnific integers are $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, where Π consists of the surreals with purely positive Conway exponents [31, `odg:def:rings`]: a nonzero term with negative exponent excludes membership. $\mathbf{Oz}$ is discretely ordered with 1 the least positive element, and a surreal that is not an omnific integer lies between two consecutive ones, its omnific floor and ceiling; for the numbers below this is read off their normal forms.

Corollary 25.3 (Zeros next to omnific integers; source 13). *For $p = \omega^{-1}$ the zeros of $\mathcal{T}_p$ in its strong domain are $x_m = -\omega^{2m-1} - \omega^{-(2m-1)}$, $m \geq 1$. None is an omnific integer, and*

$$\lfloor x_m \rfloor_{\mathbf{Oz}} = -\omega^{2m-1} - 1, \quad \lceil x_m \rceil_{\mathbf{Oz}} = -\omega^{2m-1}. \quad (25.1)$$

So $\mathcal{T}_{\omega^{-1}}(x) = 0$ has no omnific-integer solution in its strong domain, although it has infinitely many simple real surreal solutions there.

Proof. The zeros are those of Theorem 23.6. The last summand of x_m has negative exponent and nonzero coefficient, so $x_m \notin \mathbf{Oz}$. Since $0 < \omega^{-(2m-1)} < 1$, x_m lies strictly between the two consecutive omnific integers of (25.1), and there is no other zero. $\square$

This concerns a strongly evaluated infinite-series equation, not a polynomial Diophantine statement: the differential equation of $\mathcal{T}_p$ is polynomial in its jets, but $\mathcal{T}_p(x) = 0$ is not a polynomial equation in x .

Rounding can magnify an infinitesimal error. Let $x_m^+ = -\omega^{2m-1}$ and $x_m^- = x_m^+ - 1$, the ceiling and floor of x_m . Then $x_m^+ - x_m = \omega^{-(2m-1)}$ is infinitesimal and $x_m^- - x_m = -1 + \omega^{-(2m-1)}$ is finite with leading coefficient -1 , and both satisfy the hypothesis of Proposition 23.7. With the valuation normalized by $v(\omega^{-1}) = 1$,

$$v(\mathcal{T}_p'(x_m)) = -m^2 + 3m - 1, \quad \text{lc}(\mathcal{T}_p'(x_m)) = (-1)^{m-1}, \quad (25.2)$$

$$v(\mathcal{T}_p(x_m^+)) = -m^2 + 5m - 2, \quad \text{lc}(\mathcal{T}_p(x_m^+)) = (-1)^{m-1}, \quad (25.3)$$

$$v(\mathcal{T}_p(x_m^-)) = -m^2 + 3m - 1, \quad \text{lc}(\mathcal{T}_p(x_m^-)) = (-1)^m. \quad (25.4)$$

m	$v(\mathcal{T}_p'(x_m))$	$v(\mathcal{T}_p(x_m^+))$	$v(x_m^+ - x_m)$
1	1	2	1
2	1	4	3
3	-1	4	5
4	-5	2	7
5	-11	-2	9
6	-19	-8	11

For every $m \geq 5$, $\mathcal{T}_p(x_m^+)$ is infinite although x_m^+ differs from a genuine zero by an infinitesimal. This violates no local continuity: the derivative at these increasingly large roots is itself large. The exact local formula records both scales; an argument based only on the small input difference would lose the dominant effect.

25.4 Examples

Example 25.4 (The simplest discrete field; source 13). For $\mathbb{K} = \mathbb{Q}((t))$ and $p = t$, the series $1 + tz + t^4(z^2 - 2) + t^9(z^3 - 3z) + \cdots$ is strongly entire, nonpolynomial, differentially algebraic of minimal order three and not D-finite. It needs no uncountable constant field, no real closure and no divisible exponents.

Example 25.5 (One group, two scales; source 13). Let $\Gamma = \mathbb{Z} \oplus_{\text{lex}} \mathbb{Z}$ with the first coordinate dominant. The scale $\mu = (1, 0)$ gives a strongly entire witness; $\mu = (0, 1)$ does not: its strong domain is the valuation ring for the projection to the first coordinate, and $t^{(-1, 0)}$ is excluded. The existence classification quantifies over the scale; it does not say that every positive μ works.

Example 25.6 (Countable cofinality without an order unit; source 13). Order $\Gamma = \bigoplus_{j \geq 0} \mathbb{Z}e_j$ by the largest nonzero coordinate. The sequence (e_j) is cofinal, but every fixed $\mu > 0$, and every ordinary

multiple of it, is dominated by some e_N . By Theorem H, or Theorem E (a), no nonpolynomial strongly entire series is differentially algebraic. This does not exclude nonpolynomial strongly entire series; it says that any such series has no algebraic differential relation over the full coefficient field. The group is the integer analogue of (11.1), over which the series F of (11.2) is strongly entire.

Corollary 25.7 (Stability under affine changes; source 13). *Let μ be an order unit, $a \in \mathbb{K}^\times$ and $b \in \mathbb{K}$. Then $g(z) = \mathcal{T}_p(az + b)$ is strongly entire and has minimal order three, and its zeros are exactly the $(x_m - b)/a$, all simple.*

Proof. For a fixed input, $v(ax + b)$ is bounded below by a multiple of $-\mu$, and the quadratic support bound, expanded binomially, justifies the Taylor coefficients of g and their evaluation. By the regrouping argument of Lemma 24.5 (b), $f \mapsto f(az + b)$ is a $\mathbb{K}$ -algebra endomorphism of $\mathcal{E}_\Gamma(k)$ with inverse $f \mapsto f((z - b)/a)$, and $D_z^j(f(az + b)) = a^j(D_z^j f)(az + b)$. A nonzero affine substitution is an automorphism of $\mathbb{K}(z)$ and scales the successive derivatives by nonzero powers of a , so it preserves the algebraic independence of the first three jets and the order of an equation. The zeros follow by substitution. $\square$

There are therefore many explicit entire nonlinear solutions, not one isolated series. The argument classifies neither all differentially algebraic entire series nor the possible minimal orders. Order two, which source 13 left open for other series, is excluded for every series by Theorem I (source 14); the classification is Question 19.1.

25.5 Predecessors, repository comparison and priority

Literature. Source 13 consulted the NIST DLMF product and elliptic sections [32, 33]; the Tate-curve lecture notes hosted by M. Woodbury [35], whose §16 (printed page 32, the Tate equation and invariant differential) it inspected as a PDF image and whose normalization agrees with the constants (24.2), the hosting attribution not being a claim that Woodbury wrote every part; the publisher abstract and bibliography of Sebbar’s paper [37], whose abstract explicitly discusses classical third-order theta differential relations, without its full text; the publication record of Ohyama’s paper [36], not used in place of a proof; and publication records of Mantova–Matusinski [17] for normal forms. Several broad searches returned irrelevant results and are not evidence of absence; no exhaustive MathSciNet or zbMATH review, survey of book treatments or citation-network review was made. In particular no claim is made that Sebbar or Ohyama lacks a related minimal-order theorem.

Repository comparison at the pin. Source 13 read the repository at 4cdeaec through a connector: the root and documentation inventories, this report’s README, its nonlinear corner criterion and limitations, the vanishing-corner example, the first-order limitation, its attribution and dependency sections, and the question and status passage of Question 19.1 (source lines 4135–4202 of the article; its audit file gives the read range as about 4125–4290); and the README of the Hahn–Tate report, not its full article. Its statements were checked for this merge and are accurate at the pin: Question 19.1 asked for existence with an order unit, for rationality of differentially algebraic elements of $\text{Frac } \mathcal{E}_\Gamma$, and whether the absence of an order unit is necessary; its status settled the case without an order unit and gave no counterexample with one; and the Hahn–Tate README contains the bilateral theta domain on U_q . A repository content search that returned no matches was followed by direct file reads and not used as proof of absence. This was a targeted comparison, not an audit of every file, archived manuscript, branch or commit. Since the pin the

statements of this report that called the order-unit case open have been updated, and Question 19.1 has been re-scoped, as recorded in Section 19.

Priority. The triple product, the Chebyshev identity and the theta-elliptic identity are classical, and no novelty is claimed for a third-order theta relation; the counterexample is a classical function, not a new theta identity. The no-order-unit direction is Theorem E (a), and the bilateral theta domain and uniformization belong to the Hahn–Tate report. The proposed contribution of source 13 is the support-controlled descent to an entire series, its exact domain at every scale, the existence classification of Theorem H, the minimal-order certificate with the differential field and non-D-finiteness, and the stated surreal and omnific consequences. Whether the arbitrary-rank formulation, the descent as an entire-function counterexample or the three-stage minimal-order proof already occurs in the literature needs a more exhaustive bibliographic review; the answered question is a question of this report, not a named historical conjecture.

25.6 What the theta-descent part does not claim

Every limitation stated by source 13, in its manuscript, README, audit files or verification record, is kept.

- (T1) No Lean formalization, no independent referee report and no certified bibliographic priority accompany source 13; its compiled and visually inspected PDF is no evidence of mathematical correctness.
- (T2) The classical theta, triple-product, Chebyshev and elliptic identities are not new, and no novelty is claimed for third-order theta differential relations; the counterexample is a classical function.
- (T3) The no-order-unit direction is prior repository material (Theorem E (a)), re-proved by source 13 and not claimed.
- (T4) The bilateral theta domain and the Hahn–Tate uniformization are not claimed anew.
- (T5) $\mathcal{T}_p$ is not strongly entire on **No** or **No**[i]; it fails at $x = \omega^\omega$. Entireness is relative to one specified Hahn field, and an extension of the value group can destroy it.
- (T6) D_z kills every Hahn coefficient, including $p = \omega^{-1}$, and is not the Berarducci–Mantova derivation.
- (T7) The finite checks do not verify strong summability of infinite or arbitrary supports, transcendence, algebraic independence or minimal order, the no-order-unit obstruction or the classification, completeness of the infinite zero set, novelty, or correctness in general; they are not a formal verification.
- (T8) No classification of strongly entire differentially algebraic series is given, and whether minimal order two occurs for some other series is open; a lower-order equation for $\mathcal{T}_p$ itself is excluded, which is a different assertion. (Source 14 has since excluded order two for every series, Theorem I; the non-claim is kept as source 13’s statement, and the classification stays open.) The geometric route suggested by Theorem 24.10 is a research direction, not a theorem.
- (T9) “No omnific zero” concerns the strong domain only; it is not a polynomial Diophantine statement, and no values are assigned outside the domain.
- (T10) The local valuation formulas are not uniform across the roots; a small input error does not imply a small output error.

- (T11) No transcendence theorem for numerical theta values is used or claimed; the independence statements, and all fields of meromorphic functions in the proofs, concern formal functions, not values at a point.
- (T12) The minimal-order, differential-field and non-D-finiteness arguments assume that μ is an order unit. The parameter is the monomial $p = t^\mu$; no result is generalized to a parameter with an arbitrary higher-order tail.
- (T13) Sebbar’s paper and Ohyama’s paper were not read in full, so no absence claim is made about them; broad searches are not evidence of absence, and no exhaustive MathSciNet or zbMATH review was made.
- (T14) The repository comparison at `4cdeaec` was targeted, not a full audit of every file, archived manuscript, branch or commit; the Hahn–Tate report was compared through its README.
- (T15) The answered question is a question of this report; no named historical conjecture is claimed solved.
- (T16) The existence classification quantifies over the scale: not every positive μ gives an entire series (Example 25.5).
- (T17) Without an order unit the conclusion excludes differentially algebraic nonpolynomial entire series only, not nonpolynomial entire series (Example 25.6).
- (T18) The poles of the rational system at $z = \pm 2$ are those of a coordinate chart; the globally meaningful equation is the cleared one.
- (T19) *[merge]* Lemmas 24.5 and 24.7, Proposition 24.2, clause (iv) of Theorem G, Corollary 25.2 and the consistency statements of Section 25.2 are the merge’s. The first two write out steps that source 13 used; no historical priority is claimed for any of them.

26 Second-order rigidity: finite endpoint spectra at exterior breaks

This section and the next contain the material of source 14, delivered after the theta-descent part had been merged. They answer the question left open by source 13 and re-scoped in Section 19 after it: no nonpolynomial strongly entire series satisfies a nonzero algebraic differential equation of order at most two, for every value group and every coefficient field of characteristic zero (Theorem I(a)). With Theorem H this makes the least order exactly three when Γ has an order unit (Theorem I(b)). The method is the finite initial polynomial of Sections 14–17, applied not at one scale but at infinitely many scales where the initial polynomial is not a monomial, and read through logarithmic Euler derivatives; it succeeds exactly where the corner criterion of Theorem 16.1 is silent, since an equation of order two may have $\chi_P = 0$ (Proposition 16.4). It is neither the escape chain, nor the finite coefficient field, nor a difference-field argument.

26.1 Source, conventions and the merge’s choices

Convention 26.1 (Setting of the order-three-threshold part). In Sections 26–27, k is an arbitrary field of characteristic zero with the trivial valuation, Γ is as in Convention 1.1, $\mathbb{K} = k((t^\Gamma))$, and v , lc , res , strong summability, $\text{Dom}_{\text{str}}(f)$, $\mathcal{E}_\Gamma(k)$, $\text{Frac } \mathcal{E}_\Gamma(k)$, D_z , the coefficient family algebra $\mathcal{S}_1$ with its residue homomorphism ρ , and the Gauss data γ_δ , ϕ_δ , N_δ of (14.1) are as in Conventions 14.1 and 20.1

and Section 14.3. Write $\vartheta = zD_z$, as in Convention 23.1, and $\vartheta_X = X d/dX$ on $k(X)$. Where a statement needs a divisible group it says so; it is then applied to the divisible hull $\Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathbb{K}_{\mathbb{Q}} = k((t^{\Gamma_{\mathbb{Q}}}))$ (Lemma 26.2). U, V, W, T and G are indeterminates of polynomial rings over k in this part, except that in Corollary 26.13 G is the series of Corollary 17.1; $E_0, \dots, E_s$ are Euler-jet variables.

For $k = \mathbb{C}$ one has $\mathbb{K} = K_{\Gamma}$ and $\mathcal{E}_{\Gamma}(k) = \mathcal{E}_{\Gamma}$. As for the three previous parts, the generality in k is recorded for this part only.

The source. Source 14 is *The Order-Three Threshold for Entire Hahn–Surcomplex Functions: Second-order rigidity, Newton-corner spectra, and sharp theta examples* (26 pages, 23 September 2026). It consulted the repository at

b895e8672990e8b5a56f97dd7246f9dd5f86c771.

At that pin the article and README of this report were exactly the five-source text (sources 08–12); source 13’s audit files, programs and records had been placed in this directory but its manuscript had not yet been merged. Source 14 read this report’s README there, and took source 13 from the author’s own file library as an unpublished manuscript, with its baseline `4cdeaec` and its question `q:order2`, “whether a different example could have order two”. Its statements about this report (first-order nonlinear rigidity and the all-order obstruction without an order unit) were accurate at the pin. Since then source 13 has been merged as Sections 23–25, and `q:order2` is its Question 11.1, merged into Question 19.1; the question source 14 answers is therefore the one recorded there, and source 14’s description of source 13 as a private project source that is not a published paper describes the state at its pin. Source 14 assumes characteristic zero and an arbitrary ordered abelian group; it passes to the divisible hull by a cofinal extension and uses no divisibility, rank, algebraic or real closedness or order-unit hypothesis in its main theorem. Its standing hypotheses are therefore those of the coefficient-field and theta-descent parts, and every proof taken from it was re-read against the statement it is attached to.

Renamed symbols. The following symbols of source 14 would collide with symbols of the earlier parts. They are renamed here, in the new material only.

Source 14	Here	Reason
$D = d/dz, f', f''$	$D_z, D_z f, D_z^2 f$	The notation of Convention 14.1. (Source 14’s $\mathbb{K} = k((t^{\Gamma}))$, $\mathcal{E}_{\Gamma}(k)$ and $\vartheta = zD_z$ are this report’s.)
radius r ; $\alpha_f(r)$; ϕ_r ; $I_f(r)$; $m_f(r)$, $N_f(r)$ Newton corner	scale δ ; γ_{δ} ; ϕ_{δ} ; Act_{δ} ; $\underline{N}_{\delta}$, N_{δ} break (Definition before Lemma 26.3)	The Gauss data (14.1); r indexes derivative orders, and I_P is the full corner polynomial. A <i>corner</i> is the finite term selection of Section 14.4.
restricted ring $\mathcal{O}\langle X \rangle_v$, red order bound ℓ ; blocks $P_{d,j}$; μ , M , C	S_1^+ , ρ s ; $P_{d,w}$; β , n_1 , C_1	Lemma 14.2; red_{Δ} is a coarsened residue. ℓ is a dilation power; μ is a scale; $M(S)$ is a support monoid.
H (exterior homogeneous polynomial)	$\mathfrak{h}_P$	$H \in \Gamma$, and $\mathcal{H}[f]$ is a theta expression.
$Q(U, V)$, $Q(U, V, W)$; s , R ; Q_s	$\mathfrak{q}$; i_0 , $\mathfrak{r}$; $\mathfrak{q}_{i_0}$	Q_s is a shift polynomial, s an order, R an order bound.
S_H	$\text{Sp}(\mathfrak{h})$	S is a support alphabet.

Source 14	Here	Reason
$u = \vartheta_X \phi / \phi, V_\phi, W_\phi;$ $h = W/V, r$	$\mathbf{u}_\phi, \mathbf{v}_\phi, \mathbf{w}_\phi; \mathbf{k}, \mathbf{r}$	u is the Laurent variable of Section 24.5.
$\chi_H(T, G)$	$\Upsilon_{\mathfrak{h}}(T, G)$	χ_P is the residue corner polynomial.
upper gap g , lower gap e	$\mathbf{g}^+, \mathbf{g}^-$	g_2, g_3 are theta invariants, $e_n \in \Gamma$.
$h, q = t^h, Q = q^2, s_r(Q), c$	$\mu, p = t^\mu, p^2, \mathcal{Q}_r(p^2),$ $\mathbf{c}$	As for source 13 (Section 23.1).
$F_q; H_f, J_f, B_f$	$\mathcal{T}_p; \mathcal{H}[f], \mathcal{K}[f], \mathcal{B}[f]$	(1.4), (24.5), (24.6).
$r_n = v(a_{n+1}) - v(a_n)$	δ_n	It is a break scale.
$F = g/p$; characteristic p	$F_0 = g/\mathbf{p}; \mathbf{p}$	F is the series (11.2); $p = t^\mu$.
differential field L ; state variables u, v	$\mathfrak{L} \subseteq \mathfrak{F}; u_1, u_2$	L is a coefficient subfield, $M(S)$ a support monoid.
Stirling numbers (implicit)	$\varsigma_{j,i}$	s is an order.
$\Gamma_{\mathbb{Q}}; \mathbf{No}(i)$	$\Gamma_{\mathbb{Q}}; \mathbf{No}[i]$	The collection writes $\mathbf{No}[i]$.
Theorems 1.1 and 1.3	Theorem I (a) and (b)	Theorems A–H of this report are different theorems.

The shipped audit file `14-order-three-threshold-SOURCE_AUDIT.md` keeps the source's notation.

Words with two meanings. A *break* is a scale δ at which the initial polynomial ϕ_δ is not a monomial; source 14 calls it a Newton corner. A *corner* keeps the meaning of Section 14.4, and the residue corner polynomial χ_P of a differential polynomial is not the endpoint-gap polynomial $\Upsilon_{\mathfrak{h}}$ of a homogeneous one; Remark 26.6 gives their exact relation. *Exterior* means, as elsewhere in this report, large δ , that is arguments $t^{-\delta}$ of large absolute valuation. The *endpoint spectrum* $\mathrm{Sp}(\mathfrak{h})$ is a finite set of nonnegative integers, not the spectrum of an operator. *Order* and *minimal order* are as in Section 23.1: the order of an equation is the highest derivative in it. A *gap* is a difference of exponents of ϕ at one end. $\mathbf{u}_\phi, \mathbf{v}_\phi, \mathbf{w}_\phi$ are rational functions of X attached to a polynomial ϕ , while U, V, W are indeterminates.

Results printed once. Source 14 reproves several facts that the report already contains. They are printed once and cited: the definitions of an order unit and of strong entireness (Definitions 1.3 and 1.2); its all-scale coefficient criterion (Lemma 2.2), which is Proposition 20.2, proved there by the same two-scale argument; the restricted-series ring and its reduction homomorphism, which are $\mathcal{S}_1^+$ and ρ of Lemma 14.2, with the commutation of Euler derivatives and reduction in Lemma 14.4; the first assertion of its Lemma 3.2, which is Lemma 14.5; its Proposition 5.2, whose equation $z^2(ff'' - (f')^2) + zff' = 0$ is (16.1) and whose classification is Proposition 16.4; its Corollary 6.4 on Painlevé I and II, which is Proposition 16.3 (source 14's second route is recorded in Remark 26.12); its Section 8, the theta example, whose entireness, coefficients, cleared equation and derivation are Theorem 23.5, Proposition 23.3, Theorem 24.1 and Sections 24.2–24.3, except that its minimality argument is a genuinely different route and is kept (Remark 26.14); its Section 10, the no-order-unit direction, which is Theorem E (a) in the Euler presentation already recorded in Remark 25.1 (its Lemmas 10.1–10.2 and Theorem 10.3 are Theorem 21.2 (a), Lemmas 20.6–20.7 and Proposition 20.9); its Examples 10.4 and 10.5, which are Example 25.6 and, over $\mathbb{R} \oplus_{\mathrm{lex}} \mathbb{R}$ instead of $\mathbb{Z} \oplus_{\mathrm{lex}} \mathbb{Z}$, Example 25.5; the surreal embedding (2.4); the surreal witness, product, zeros and omnific rounding of its Section 11.3, which are Section 25.3, Theorem 23.6 and Corollary 25.3; its Example 12.1 in positive characteristic, which is Example 17.3 with the scale 1 of $\mathbb{Q}$ replaced by an order unit of

Γ ; and the $\Gamma = 0$ remark, which is in Section 12. Source 14 credits the theta material and the no-order-unit direction as prior and claims neither.

Where each numbered result went. Theorem 1.1 is Theorem I(a), and Theorem 1.3 is Theorem I(b); Lemma 2.4 is Lemma 26.2; Lemma 3.1 is Lemma 26.3; the second assertion of Lemma 3.2 is Lemma 26.4; Lemma 4.1 and Remark 4.2 are Lemma 26.5 and Example 26.7; Lemma 5.1 is Lemma 26.8; Corollaries 6.1–6.3 are Corollaries 26.9–26.11; Definition 7.1, Theorem 7.2 and Corollary 7.3 are Definition 27.1, Theorem 27.2 and Corollary 27.3; Theorems 9.1 and 9.2 are Theorems 27.4 and 27.5; Proposition 11.1 is Proposition 27.6; Research question 1 is merged into the re-scoped Question 19.1, and Research questions 2–9 are Questions 19.11–19.18; Appendix A is (24.8) with (27.3); Appendix B and the proof checkpoints of Section 14.1 are merged into the dependency table of Section 18.

The merge’s choices. The review of source 14 for this merge found its proofs correct. Three points are written out here and marked as the merge’s. First, the existence of a late break (Lemma 26.4) is proved by naming the break scale, the least of finitely many quotients, rather than by the phrase “a change of the minimum occurs at an equality”. Second, the exterior form $\mathfrak{h}_P$ of Lemma 26.5 is identified with the residue corner of Section 14.4 written in Euler jets, and its values on monomials are χ_P (Remark 26.6); this is what connects source 14’s endpoint lemma to the corner criterion of source 10. Third, Corollary 26.9 and Corollary 26.10 are stated inside an explicit differential field containing $\mathbb{K}((z))$, which is how source 14’s phrase “every $f \in L \cap \mathcal{E}_\Gamma(k)$ ” must be read, and Corollary 26.13 applies Theorem I to the series G and Θ_p of earlier parts. Source 14’s theorems, proofs and non-claims are otherwise printed as delivered, in the renamed notation.

26.2 Breaks and the exterior form

Lemma 26.2 (Cofinal scalar extension; source 14). *Let $\Gamma \subseteq \Lambda$ be an extension of ordered abelian groups with Γ cofinal in Λ . Every $f \in \mathcal{E}_\Gamma(k)$ is strongly entire over $k((t^\Lambda))$. In particular this holds for $\Lambda = \Gamma_\mathbb{Q}$, in which Γ is cofinal.*

Proof. Write $f = \sum_n a_n z^n$ and use Proposition 20.2 (iii) over Λ . Given $\gamma, \beta \in \Lambda$, choose $\gamma' \in \Gamma$ with $\gamma' \leq \gamma$ and $\beta' \in \Gamma$ with $\beta' \geq \beta$; since Γ is cofinal in the group Λ , it is also coinital. If $n \geq 0$ and $v(a_n) + n\gamma \leq \beta$, then $v(a_n) + n\gamma' \leq \beta'$, which holds for only finitely many n with $a_n \neq 0$. For $\Lambda = \Gamma_\mathbb{Q}$, the torsion-free group Γ embeds in $\Gamma \otimes_\mathbb{Z} \mathbb{Q}$, and an element g/m with $g \in \Gamma$ and $m \geq 1$ satisfies $|g/m| \leq |g|$. $\square$

A series is a polynomial over $\mathbb{K}_\mathbb{Q}$ exactly when it is one over $\mathbb{K}$, and a polynomial relation over $\mathbb{K}(z)$ remains a nonzero relation over $\mathbb{K}_\mathbb{Q}(z)$. So a polynomiality conclusion proved over $\mathbb{K}_\mathbb{Q}$ descends to $\mathbb{K}$. This is the only use of divisibility in the part.

Let $f = \sum_n a_n z^n \in \mathcal{E}_\Gamma(k)$ be nonzero. For every $\delta \in \Gamma$ the Gauss value γ_δ and the initial polynomial ϕ_δ of (14.1) exist. Put

$$\text{Act}_\delta = \{n : a_n \neq 0, v(a_n) - n\delta = \gamma_\delta\}, \quad \underline{N}_\delta = \min \text{Act}_\delta, \quad N_\delta = \max \text{Act}_\delta, \quad (26.1)$$

so that $\underline{N}_\delta = \text{ord}_X \phi_\delta$, the X -adic order, and $N_\delta = \deg \phi_\delta$. A scale δ is a *break* of f if $\underline{N}_\delta < N_\delta$, that is, if ϕ_δ is not a monomial. Lemma 14.2 gives $\rho(t^{-\gamma_\delta}(\partial^j f)(t^{-\delta} X)) = \partial_X^j \phi_\delta$ for every j , because $\partial^j f$ scales like f with each coefficient multiplied by the integer n^j , and Euler operators commute with ρ . A derivative on the right may vanish; no equality of individual valuations is asserted.

Lemma 26.3 (Local finiteness of active indices; source 14). *Let $f \in \mathcal{E}_\Gamma(k)$ be nonzero and $\delta_1 \leq \delta_2$ in Γ . Only finitely many indices lie in Act_δ for some $\delta \in [\delta_1, \delta_2]$. If $\delta < \delta'$, $n \in \text{Act}_\delta$ and $n' \in \text{Act}_{\delta'}$, then $n \leq n'$.*

Proof. Fix $a_j \neq 0$. On the interval its weighted value $v(a_j) - j\delta$ is at most $C = v(a_j) - j\delta_1$. By Proposition 20.2 (iii) at $\gamma = -\delta_2$ all but finitely many nonzero a_n have $v(a_n) - n\delta_2 > C$, and then $v(a_n) - n\delta \geq v(a_n) - n\delta_2 > C$ for $\delta \leq \delta_2$, so such n is never active. For the monotonicity, the two minimality inequalities $v(a_n) - n\delta \leq v(a_{n'}) - n'\delta$ and $v(a_{n'}) - n'\delta' \leq v(a_n) - n\delta'$ add to $(n' - n)(\delta' - \delta) \geq 0$. $\square$

Lemma 26.4 (Late breaks; source 14). *Let Γ be divisible and $f \in \mathcal{E}_\Gamma(k)$ nonpolynomial. For every $\delta_* \in \Gamma$ and every $B \in \mathbb{N}$ there is a break $\delta \geq \delta_*$ of f with $\underline{N}_\delta > B$.*

Proof. By Lemma 14.5 there is δ_B such that every active index at every $\delta \geq \delta_B$ exceeds B . Let $\delta_1 = \max(\delta_*, \delta_B)$. If δ_1 is a break we are done; otherwise $\text{Act}_{\delta_1} = \{N\}$. Lemma 14.5 again gives $\delta_2 > \delta_1$ at which every active index exceeds N . By Lemma 26.3 the set F of indices active somewhere on $[\delta_1, \delta_2]$ is finite, and on that interval γ_δ is the minimum of the finitely many affine functions $v(a_n) - n\delta$, $n \in F$. For $n \in F$ with $n > N$ put

$$\delta^{(n)} = \frac{v(a_n) - v(a_N)}{n - N} \in \Gamma,$$

using divisibility; then $v(a_N) - N\delta \leq v(a_n) - n\delta$ exactly when $\delta \leq \delta^{(n)}$. Since N is active at δ_1 , every $\delta^{(n)} \geq \delta_1$, and since some $n > N$ in F is active at δ_2 while N is not, $\delta^{(n)} < \delta_2$ for that n . Let δ^* be the least $\delta^{(n)}$ (the merge names this scale). For $n \in F$ with $n < N$, activity of N at δ_1 and $\delta^* \geq \delta_1$ give $v(a_N) - N\delta^* \leq v(a_n) - n\delta^*$; for $n > N$ the same holds because $\delta^* \leq \delta^{(n)}$, with equality for an index attaining δ^* . Since the least weighted value on the interval is attained in F , N and that index are both active at δ^* , which is a break, and $\delta^* \geq \delta_1 \geq \delta_B$ makes every active index there exceed B . $\square$

The proof uses only finite lower envelopes on bounded intervals: no completeness of Γ , no real-valued norm, no countable cofinal sequence and no order unit.

Let $P \in \mathbb{K}[z, Y_0, \dots, Y_s]$ be a nonzero differential polynomial of order at most s . Since $z^j D_z^j = \vartheta(\vartheta - 1) \cdots (\vartheta - j + 1)$, the substitution

$$Y_j \longmapsto z^{-j} \sum_{i=0}^j \varsigma_{j,i} E_i \quad (0 \leq j \leq s), \quad (26.2)$$

with $\varsigma_{j,i}$ the signed Stirling numbers of the first kind, so that $\sum_i \varsigma_{j,i} T^i = (T)_j$, rewrites $P(z, f, D_z f, \dots)$ as a Laurent polynomial in z in the Euler jets $E_i = \vartheta^i f$. After multiplication by a power of z it becomes $\tilde{P}(z, f, \vartheta f, \dots, \vartheta^s f)$ with $0 \neq \tilde{P} \in \mathbb{K}[z, E_0, \dots, E_s]$; the change is invertible and preserves total jet degree. Decompose

$$\tilde{P} = \sum_{d,w} z^w P_{d,w}(E_0, \dots, E_s),$$

with $P_{d,w}$ homogeneous of total degree d in the jets, choose the largest d occurring, then the largest w with $P_{d,w} \neq 0$, let β be the least valuation of a coefficient of $P_{d,w}$, and put

$$\mathfrak{h}_P = \text{res}(t^{-\beta} P_{d,w}) \in k[Y_0, \dots, Y_s], \quad (26.3)$$

applying res to coefficients and renaming E_i as Y_i . It is a nonzero homogeneous polynomial, the exterior form of P .

Lemma 26.5 (Exterior homogeneous reduction; source 14). *Let $f \in \mathcal{E}_\Gamma(k)$ be nonpolynomial with $P(z, f, D_z f, \dots, D_z^s f) = 0$ for a nonzero P of order at most s . Then $\mathfrak{h}_P$ has positive total degree, and there is $\delta_0 \in \Gamma_{>0}$ such that for every $\delta \geq \delta_0$*

$$\mathfrak{h}_P(\phi_\delta, \vartheta_X \phi_\delta, \dots, \vartheta_X^s \phi_\delta) = 0 \quad \text{in } k[X].$$

No divisibility of Γ is used.

Proof. If $d = 0$, then $\tilde{P}$ is a nonzero element of $\mathbb{K}[z]$, which cannot vanish as a series; so $d \geq 1$. Because f is nonpolynomial there is n_1 with $a_{n_1} \neq 0$ and $(d - d')n_1 + w - w' > 0$ for every block $P_{d',w'} \neq 0$ with $d' < d$; put $C_1 = v(a_{n_1})$, so that $\gamma_\delta \leq C_1 - n_1\delta$ for every δ .

Evaluate at $z = t^{-\delta}X$ and normalize by $t^{\beta+d\gamma_\delta-w\delta}$. Each factor $(\vartheta^i f)(t^{-\delta}X)$ equals t^{γ_δ} times an element of $\mathcal{S}_1^+$, so a monomial of a block $z^{w'}P_{d',w'}$ with coefficient $b \neq 0$ becomes $t^{v(b)-w'\delta+d'\gamma_\delta}$ times an element of $\mathcal{S}_1^+$. Relative to the normalization its exponent is at least

$$v(b) - \beta - (d - d')\gamma_\delta + (w - w')\delta \geq v(b) - \beta - (d - d')C_1 + ((d - d')n_1 + w - w')\delta. \quad (26.4)$$

For $d' < d$ the coefficient of δ is a positive integer by the choice of n_1 ; for $d' = d$ and $(d', w') \neq (d, w)$ it is $w - w' > 0$. There are finitely many monomials, so the right side is positive for all $\delta \geq \delta_0$ once $\delta_0 > 0$ exceeds the absolute values of the finitely many constants; no division occurs. The monomials of $P_{d,w}$ itself with coefficients of value above β also lose their residue. Applying the ring homomorphism ρ of Lemma 14.2 to the normalized identity $0 = \tilde{P}(\dots)$ leaves $X^w \mathfrak{h}_P(\phi_\delta, \vartheta_X \phi_\delta, \dots) = 0$, and X^w cancels in $k[X]$. $\square$

Remark 26.6 (The exterior form is the residue corner in Euler jets; merge). Write P in the combined-monomial form (14.4). By (26.2) the monomial $c_\nu z^{k_\nu} \prod_j Y_j^{m_{\nu j}}$ becomes $c_\nu z^{w_\nu} \prod_j (\sum_i \varsigma_{j,i} E_i)^{m_{\nu j}}$, with the weight w_ν of (14.5). The substitution $Y_j \mapsto \sum_i \varsigma_{j,i} Y_i$ is a unitriangular, hence invertible, linear change of variables over $\mathbb{Z}$ and preserves total degree, so the selection of Lemma 26.5 picks $d = d_P$ and the z -power of weight w_P : its block is the image of $\sum_{\nu \in \mathcal{C}_P} c_\nu \prod_j Y_j^{m_{\nu j}}$. Its least coefficient value is β_P , because the image of the nonzero residue vector $(\bar{c}_\nu)$ under the invertible integral change is nonzero. Hence

$$\mathfrak{h}_P(Y_0, \dots, Y_s) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu \prod_{j=0}^s \left(\sum_{i=0}^j \varsigma_{j,i} Y_i \right)^{m_{\nu j}}, \quad \mathfrak{h}_P(1, T, T^2, \dots, T^s) = \chi_P(T),$$

the second because $\sum_i \varsigma_{j,i} T^i = (T)_j$. On a monomial $\phi = X^N$ one has $\vartheta_X^i X^N = N^i X^N$, so $\mathfrak{h}_P(X^N, \vartheta_X X^N, \dots) = \chi_P(N) X^{d_P N}$: the monomial test of the exterior form is the residue corner polynomial, and the corner criterion of Theorem 16.1 uses exactly this test. When $\chi_P = 0$ the exterior form vanishes on every monomial but is still a nonzero polynomial, and source 14 keeps the whole initial polynomial at breaks. For the equation (16.1), $\mathfrak{h}_P = Y_0 Y_2 - Y_1^2$; for the cleared theta equation (24.7), $\beta_P = 0$ and the dehomogenized exterior form is (27.5) below, consistent with Proposition 24.2.

Example 26.7 (The cancellation that must not be ignored; source 14). For $\mathfrak{h} = Y_0 Y_2 - Y_1^2$ one has $\mathfrak{h}(X^N, \vartheta_X X^N, \vartheta_X^2 X^N) = 0$ for every N , so substituting a trial monomial and assuming that the resulting polynomial in N is nonzero proves nothing. This $\mathfrak{h}$ is the exterior form of (16.1),

whose formal solutions are exactly 0 and the monomials cz^n (Proposition 16.4, which is source 14's Proposition 5.2). Lemma 26.8 below gives $\text{Sp}(\mathfrak{h}) = \emptyset$ for it: no nonmonomial initial polynomial is permitted at any exterior break, which is stronger than the identically vanishing monomial test and consistent with unbounded monomial solutions. Polynomiality is therefore not a uniform degree bound: a fixed equation can have polynomial solutions of every degree.

26.3 The endpoint lemma and the proof of Theorem I (a)

For a nonzero $\phi \in k[X]$ put $\mathfrak{u}_\phi = \vartheta_X \phi / \phi$, $\mathfrak{v}_\phi = \vartheta_X \mathfrak{u}_\phi$ and $\mathfrak{w}_\phi = \vartheta_X \mathfrak{v}_\phi$ in $k(X)$. Then

$$\frac{\vartheta_X^2 \phi}{\phi} = \mathfrak{u}_\phi^2 + \mathfrak{v}_\phi, \quad \frac{\vartheta_X^3 \phi}{\phi} = \mathfrak{u}_\phi^3 + 3\mathfrak{u}_\phi \mathfrak{v}_\phi + \mathfrak{w}_\phi.$$

Lemma 26.8 (Finite endpoint spectrum; source 14). *Let $0 \neq \mathfrak{h} \in k[Y_0, Y_1, Y_2]$ be homogeneous. There is a finite set $\text{Sp}(\mathfrak{h}) \subseteq \mathbb{N}$ such that every nonmonomial $\phi \in k[X]$ with $\mathfrak{h}(\phi, \vartheta_X \phi, \vartheta_X^2 \phi) = 0$ has both its X -adic order $\text{ord}_X \phi$ and $\deg \phi$ in $\text{Sp}(\mathfrak{h})$.*

Proof. Put

$$\mathfrak{q}(U, V) = \mathfrak{h}(1, U, U^2 + V). \quad (26.5)$$

It is nonzero: setting $Y_0 = 1$ is injective on homogeneous polynomials of a fixed degree, since the missing power of Y_0 is determined by the other exponents, and $Y_2 \mapsto U^2 + V$ is a triangular automorphism. Let V^{i_0} be the largest power of V dividing $\mathfrak{q}$ and write

$$\mathfrak{q} = V^{i_0} \mathfrak{r}, \quad \mathfrak{r}(U, 0) \neq 0. \quad (26.6)$$

By homogeneity $\mathfrak{q}(\mathfrak{u}_\phi, \mathfrak{v}_\phi) = 0$. The function $\mathfrak{u}_\phi$ is not constant: $\vartheta_X \phi = b\phi$ with $b \in k$ forces $n = b$ for every exponent n of ϕ , and in characteristic zero there is at most one such exponent. The constants of d/dX in $k(X)$ are k , so $\mathfrak{v}_\phi \neq 0$, and cancelling $\mathfrak{v}_\phi^{i_0}$ in the field $k(X)$ gives $\mathfrak{r}(\mathfrak{u}_\phi, \mathfrak{v}_\phi) = 0$. With $m = \text{ord}_X \phi$ and $N = \deg \phi$, the expansions at 0 and at ∞ are

$$\mathfrak{u}_\phi = m + O(X), \quad \mathfrak{v}_\phi = O(X); \quad \mathfrak{u}_\phi = N + O(X^{-1}), \quad \mathfrak{v}_\phi = O(X^{-1}),$$

and their constant terms give $\mathfrak{r}(m, 0) = \mathfrak{r}(N, 0) = 0$. The nonzero polynomial $\mathfrak{r}(T, 0)$ has finitely many roots in k , and distinct nonnegative integers are distinct in k ; let $\text{Sp}(\mathfrak{h})$ be its set of roots in $\mathbb{N}$. $\square$

Proof of Theorem I(a). Let $f \in \mathcal{E}_\Gamma(k)$ satisfy a nonzero relation of order at most two over $\mathbb{K}(z)$; clearing denominators gives a nonzero $P \in \mathbb{K}[z, Y_0, Y_1, Y_2]$. Suppose f is nonpolynomial. By Lemma 26.2, f is strongly entire over $\mathbb{K}_\mathbb{Q}$ and satisfies the same relation there, so we may assume Γ divisible. Lemma 26.5 gives $\mathfrak{h}_P \neq 0$ and δ_0 . Choose $B \in \mathbb{N}$ larger than every element of $\text{Sp}(\mathfrak{h}_P)$ ($B = 0$ if it is empty). Lemma 26.4 gives a break $\delta \geq \delta_0$ with $\underline{N}_\delta > B$; its initial polynomial is nonmonomial, satisfies $\mathfrak{h}_P(\phi_\delta, \vartheta_X \phi_\delta, \vartheta_X^2 \phi_\delta) = 0$, and has $\text{ord}_X \phi_\delta = \underline{N}_\delta \notin \text{Sp}(\mathfrak{h}_P)$, contradicting Lemma 26.8. So f is a polynomial over $\mathbb{K}_\mathbb{Q}$, hence over $\mathbb{K}$. The equivalent formulation is the same statement for the algebraic dependence of $f, D_z f, D_z^2 f$ over $\mathbb{K}(z)$. $\square$

The proof uses characteristic zero twice: nonzero integer multipliers keep valuation zero under differentiation, and infinitely many degree indices stay distinct in the residue field. The finite endpoint set is what the corner criterion lacks when $\chi_P = 0$: at order two the degenerate part of an equation leaves only the finitely many roots of $\mathfrak{r}(T, 0)$ for both endpoints of every nonmonomial initial polynomial, while a nonpolynomial entire series has breaks with arbitrarily large endpoints.

26.4 Consequences and the proof of Theorem I (b)

Corollary 26.9 (Three-jet transcendence; source 14). *For nonpolynomial $f \in \mathcal{E}_\Gamma(k)$,*

$$\text{trdeg}_{\mathbb{K}(z)} \mathbb{K}(z)(f, D_z f, D_z^2 f) = 3.$$

More generally, let $\mathfrak{F}$ be a differential field extension of $\mathbb{K}((z))$ whose derivation extends D_z , and $\mathfrak{L} \subseteq \mathfrak{F}$ a differential subfield containing $\mathbb{K}(z)$ with $\text{trdeg}_{\mathbb{K}(z)} \mathfrak{L} \leq 2$. Every $f \in \mathfrak{L} \cap \mathcal{E}_\Gamma(k)$ is a polynomial.

Proof. The first assertion is Theorem I(a). For the second, $f, D_z f, D_z^2 f \in \mathfrak{L}$ are algebraically dependent over $\mathbb{K}(z)$ in $\mathfrak{F}$, and a polynomial relation among elements of $\mathbb{K}((z))$ holds in $\mathfrak{F}$ exactly when it holds in $\mathbb{K}((z))$. $\square$

Corollary 26.10 (Rational two-state systems; source 14). *Let $\mathfrak{F}$ be as in Corollary 26.9 and let $u_1, u_2 \in \mathfrak{F}$ satisfy $D_z u_1 = R_1(z, u_1, u_2)$ and $D_z u_2 = R_2(z, u_1, u_2)$ with $R_1, R_2 \in \mathbb{K}(z, Y_1, Y_2)$, all denominators nonzero after substitution. Every element of $\mathbb{K}(z, u_1, u_2)$ that lies in $\mathcal{E}_\Gamma(k)$ is a polynomial.*

Proof. The system makes $\mathbb{K}(z, u_1, u_2)$ a differential field, by the quotient rule, of transcendence degree at most two over $\mathbb{K}(z)$; apply Corollary 26.9. $\square$

The state variables themselves need not be entire; the corollary restricts the strongly entire observables of the field. A nonpolynomial strongly entire function therefore cannot be generated by a rational nonautonomous system with two state variables over the full Hahn coefficient field, for every Γ . Corollary 21.16 (source 11) allows any number of states and quotients of entire series but needs the absence of an order unit, and Corollary 5.5 is the linear case. With an order unit, $\mathcal{T}_p$ is an observable of the rational system with the four states $f, \mathcal{L}, \mathcal{A}, \mathcal{A}_1$ of Theorem 24.10, so there the number two cannot be replaced by four; three states are not decided here (an observation of the merge). Arbitrary transcendental right-hand sides remove the finite differential-field hypothesis and are outside the corollary.

Corollary 26.11 (Polynomial-denominator meromorphic rigidity; source 14). *Let $g \in \mathcal{E}_\Gamma(k)$, $0 \neq \mathfrak{p} \in \mathbb{K}[z]$ and $F_0 = g/\mathfrak{p} \in \mathbb{K}((z))$. If F_0 satisfies a nonzero algebraic differential equation of order at most two over $\mathbb{K}(z)$, then $F_0 \in \mathbb{K}(z)$.*

Proof. The relations $g = \mathfrak{p}F_0$, $D_z g = D_z \mathfrak{p} F_0 + \mathfrak{p} D_z F_0$ and $D_z^2 g = D_z^2 \mathfrak{p} F_0 + 2D_z \mathfrak{p} D_z F_0 + \mathfrak{p} D_z^2 F_0$ form an invertible triangular linear change of jets over $\mathbb{K}(z)$, so g satisfies a nonzero relation of order at most two, and Theorem I(a) makes it a polynomial. $\square$

This concerns quotients with a polynomial denominator only; it does not assert that every object called meromorphic with finitely many poles in a Hahn analytic category has such a representation. For the fraction field $\text{Frac } \mathcal{E}_\Gamma(k)$ without an order unit, Theorem E(c) gives rationality in every order.

Remark 26.12 (Painlevé I and II: a second route; source 14). Source 14's Corollary 6.4 is Proposition 16.3, with the same degree comparison for polynomials. Source 10 reaches polynomiality through the nonzero corners $\chi_P = -6$ and $\chi_P = -2$; source 14 reaches it through Theorem I(a), since both equations have order two. The equations of DLMF §32.2 [34] are read as formal equations over $\mathbb{K}$; nothing is said about local formal solutions, which exist, or about classical complex Painlevé transcendents.

Corollary 26.13 (The explicit series of earlier parts; merge). *The series $G = \sum_{n \geq 0} t^{n^2} z^n$ over $\mathbb{C}((t^{\mathbb{Q}}))$ of Corollary 17.1, and, for an order unit μ and $p = t^\mu$, the partial theta series $\Theta_p \in \mathcal{E}_\Gamma(k)$ of Proposition 21.19, have three algebraically independent jets $f, D_z f, D_z^2 f$ over the respective $\mathbb{K}(z)$.*

Proof. Both are nonpolynomial and strongly entire (Corollary 17.1, Proposition 21.19); apply Theorem I(a). $\square$

Whether G or Θ_p satisfies an algebraic differential equation of order at least three is not decided; the corollary only raises the known lower bound from two jets (Corollary 17.1) to three.

Proof of Theorem I(b). Let μ be an order unit and $p = t^\mu$. By Theorem 23.5 and Proposition 23.3, $\mathcal{T}_p$ is a nonpolynomial element of $\mathcal{E}_\Gamma(k)$, and by Theorem 24.1 it satisfies a nonzero algebraic differential equation of order three. By part (a) no nonpolynomial element of $\mathcal{E}_\Gamma(k)$ satisfies one of order at most two; an equation of order zero is included, being of order at most two. So the least order is three. Without an order unit, Theorem E(a) excludes every finite order. $\square$

When $\Gamma = 0$ every strongly entire series is a polynomial, by strong summability at $x = 1$ (Section 12); source 14 includes this case in its statement, and Convention 1.1 excludes it here.

Remark 26.14 (A second route to minimal order three for $\mathcal{T}_p$; source 14). The lower bound of Corollary 24.9, that $\mathcal{T}_p, D_z \mathcal{T}_p, D_z^2 \mathcal{T}_p$ are algebraically independent over $\mathbb{K}(z)$ when μ is an order unit, is also a case of Theorem I(a), since $\mathcal{T}_p$ is nonpolynomial and strongly entire. Source 14 proves minimality this way, “without the separate difference-field minimality argument”. The two routes are genuinely different: source 13’s argument uses the Laurent ring, the substitution homomorphism and three difference-field obstructions for $u \mapsto p^2 u$, and also yields the degree-two statement of Theorem 24.10 and non-D-finiteness (Proposition 24.11), which Theorem I does not give; source 14’s argument uses no property of $\mathcal{T}_p$ beyond nonpolynomial strong entireness. Both are kept. The strict inequality of orders also shows again that $\mathcal{T}_p$ satisfies no equation to which Theorem 15.1 applies (Remark 24.3), but not conversely.

27 Order three: endpoint gaps, theta breaks and scope

27.1 The endpoint-gap polynomial

At order three the endpoint lemma has a precise analogue with one more integer parameter; this is what an equation of order three can use, and the theta equation uses it.

Definition 27.1 (Endpoint gaps; source 14). For a nonmonomial $\phi \in k[X]$ with highest exponent N and second-highest exponent $N - \mathbf{g}^+$, the positive integer $\mathbf{g}^+$ is its *upper gap*; with lowest exponent m and second-lowest exponent $m + \mathbf{g}^-$, the positive integer $\mathbf{g}^-$ is its *lower gap*. The pairs $(N, \mathbf{g}^+)$ and $(m, \mathbf{g}^-)$ are its endpoint-gap pairs.

Theorem 27.2 (Third-order endpoint-gap obstruction; source 14). *Let $0 \neq \mathfrak{h} \in k[Y_0, Y_1, Y_2, Y_3]$ be homogeneous and*

$$\mathfrak{q}(U, V, W) = \mathfrak{h}(1, U, U^2 + V, U^3 + 3UV + W). \quad (27.1)$$

Let i_0 be the least total degree in (V, W) of a nonzero term of $\mathfrak{q}$ and $\mathfrak{q}_{i_0}$ the sum of the terms of that degree. Then

$$\Upsilon_{\mathfrak{h}}(T, G) = \mathfrak{q}_{i_0}(T, 1, -G) \in k[T, G] \quad (27.2)$$

is nonzero, and every nonmonomial $\phi \in k[X]$ with $\mathfrak{h}(\phi, \vartheta_X \phi, \vartheta_X^2 \phi, \vartheta_X^3 \phi) = 0$ satisfies $\Upsilon_{\mathfrak{h}}(N, \mathfrak{g}^+) = 0$ and $\Upsilon_{\mathfrak{h}}(m, -\mathfrak{g}^-) = 0$ for its endpoint-gap pairs.

Proof. As in Lemma 26.8, $\mathfrak{q} \neq 0$. The polynomial $\mathfrak{q}_{i_0}$ is homogeneous of degree i_0 in (V, W) , and $V = 1, W = -G$ is injective on such polynomials, so $\Upsilon_{\mathfrak{h}} \neq 0$. By homogeneity and the identities before Lemma 26.8, $\mathfrak{q}(\mathfrak{u}_{\phi}, \mathfrak{v}_{\phi}, \mathfrak{w}_{\phi}) = 0$. At infinity write $\phi = aX^N(1 + bX^{-\mathfrak{g}^+} + \dots)$ with $ab \neq 0$; then, in $k((X^{-1}))$,

$$\mathfrak{u}_{\phi} = N - \mathfrak{g}^+ b X^{-\mathfrak{g}^+} + \dots, \quad \mathfrak{v}_{\phi} = (\mathfrak{g}^+)^2 b X^{-\mathfrak{g}^+} + \dots, \quad \mathfrak{w}_{\phi}/\mathfrak{v}_{\phi} = -\mathfrak{g}^+ + O(X^{-1}).$$

Writing each homogeneous part as $\mathfrak{q}_i(\mathfrak{u}_{\phi}, \mathfrak{v}_{\phi}, \mathfrak{w}_{\phi}) = \mathfrak{v}_{\phi}^i \mathfrak{q}_i(\mathfrak{u}_{\phi}, 1, \mathfrak{w}_{\phi}/\mathfrak{v}_{\phi})$, the parts with $i > i_0$ have X^{-1} -order above $i_0 \mathfrak{g}^+$, and the coefficient of $X^{-i_0 \mathfrak{g}^+}$ is $((\mathfrak{g}^+)^2 b)^{i_0} \mathfrak{q}_{i_0}(N, 1, -\mathfrak{g}^+)$; replacing $\mathfrak{u}_{\phi}$ by N changes only higher orders. So $\Upsilon_{\mathfrak{h}}(N, \mathfrak{g}^+) = 0$. At 0 write $\phi = aX^m(1 + bX^{\mathfrak{g}^-} + \dots)$: then $\mathfrak{u}_{\phi} \rightarrow m$, $\mathfrak{v}_{\phi}$ has order $\mathfrak{g}^-$ and $\mathfrak{w}_{\phi}/\mathfrak{v}_{\phi} \rightarrow \mathfrak{g}^-$, and the same argument gives $\mathfrak{q}_{i_0}(m, 1, \mathfrak{g}^-) = \Upsilon_{\mathfrak{h}}(m, -\mathfrak{g}^-) = 0$. $\square$

At order two the lowest transverse part is $A(U)V^{i_0}$, whose specialization is $A(T)$: no gap variable remains, which is Lemma 26.8. At order three a polynomial such as $G^2 - 1$ permits arbitrarily large endpoints while fixing the gap to one; this is what happens for the theta equation (Section 27.2).

Corollary 27.3 (A third-order rigidity certificate; source 14). *Let P have order at most three, with exterior form $\mathfrak{h}_P$ as in Lemma 26.5. If the set*

$$\{N \in \mathbb{N} : \Upsilon_{\mathfrak{h}_P}(N, \mathfrak{g}) = 0 \text{ for some integer } 1 \leq \mathfrak{g} \leq N\}$$

is bounded, every strongly entire solution of $P[f] = 0$ is a polynomial.

Proof. Pass to $\Gamma_{\mathbb{Q}}$ as in the proof of Theorem I(a). A nonpolynomial entire solution has breaks $\delta \geq \delta_0$ with $\underline{N}_{\delta}$, hence N_{δ} , arbitrarily large (Lemma 26.4), and the upper endpoint-gap pair $(N_{\delta}, \mathfrak{g}^+)$ of ϕ_{δ} has $1 \leq \mathfrak{g}^+ \leq N_{\delta}$; Theorem 27.2 and Lemma 26.5 contradict the boundedness. $\square$

This is a sufficient certificate, not a decision algorithm for third-order equations: deciding the integral zero set of a bivariate polynomial is a separate arithmetic problem, and an unbounded zero set of $\Upsilon_{\mathfrak{h}_P}$ is necessary for escaping the obstruction, not sufficient for the existence of an entire solution.

27.2 The exterior form of the theta equation

Let P be the polynomial (24.8) with the constants (24.2); it is homogeneous of degree six in the jets. In Euler jets E_j the substitution (26.2) reads

$$Y_0 = E_0, \quad Y_1 = E_1/z, \quad Y_2 = (E_2 - E_1)/z^2, \quad Y_3 = (E_3 - 3E_2 + 2E_1)/z^3, \quad (27.3)$$

and after multiplication by z^6 the highest z -power is six. Its block, dehomogenized by $E_0 = 1$, $E_1 = U$, $E_2 = U^2 + V$, $E_3 = U^3 + 3UV + W$, is

$$W^2 - 4(\mathfrak{c} - V)^3 + g_2(\mathfrak{c} - V) + g_3 = W^2 + 4V^3 - 12\mathfrak{c}V^2 + (12\mathfrak{c}^2 - g_2)V - 4\mathfrak{c}^3 + \mathfrak{c}g_2 + g_3. \quad (27.4)$$

Directly: in Euler jets the top part of $\mathcal{H}[f]$ is $f\vartheta^2 f - (\vartheta f)^2$, that is $E_0 E_2 - E_1^2$, which dehomogenizes to V ; the top part of $z\mathcal{K}[f]$ is $f\vartheta \mathcal{H}_0 - 2\vartheta f \mathcal{H}_0$ with $\mathcal{H}_0$ that part, which dehomogenizes to W ; so $\mathcal{B}[f]$ contributes $\mathfrak{c} - V$ and $(z^2 - 4)\mathcal{K}[f]^2$ contributes W^2 . The coefficient of W^2 is 1 and all coefficients lie

in $\mathbb{Z}[\mathfrak{c}, g_2, g_3]$, of nonnegative valuation, so $\beta = 0$ in (26.3). With the residues $\text{res } \mathfrak{c} = \frac{1}{12}$, $\text{res } g_2 = \frac{1}{12}$ and $\text{res } g_3 = -\frac{1}{216}$ the constant and linear terms cancel and the dehomogenized exterior form is

$$W^2 - V^2 + 4V^3. \quad (27.5)$$

Its value at $V = W = 0$ is 0, which is $\chi_P = 0$ in Proposition 24.2, by Remark 26.6. Its transverse part of least degree is $W^2 - V^2$, so Theorem 27.2 gives $\Upsilon(T, G) = G^2 - 1$: every exterior upper and lower gap of a solution is one. The Euler-jet conversion and (27.5) are checked symbolically by source 14's program.

27.3 Theta breaks and eventual Newton convexity

Theorem 27.4 (Polynomial solutions of the theta exterior equation; source 14). *Let $0 \neq \phi \in k[X]$. Then $\mathfrak{w}_\phi^2 - \mathfrak{v}_\phi^2 + 4\mathfrak{v}_\phi^3 = 0$ if and only if $\phi = bX^m$ or $\phi = bX^m(1 + aX)$ with $m \in \mathbb{N}$ and $a, b \in k^\times$.*

Proof. If $\mathfrak{v}_\phi = 0$, then $\mathfrak{u}_\phi$ is constant and ϕ is a monomial, as in the proof of Lemma 26.8. Suppose $\mathfrak{v}_\phi \neq 0$ and put $\mathfrak{k} = \mathfrak{w}_\phi/\mathfrak{v}_\phi$. The equation becomes $\mathfrak{k}^2 = 1 - 4\mathfrak{v}_\phi$. If $\mathfrak{k} = 0$ then $\mathfrak{v}_\phi = \frac{1}{4}$, impossible because the constant term of $\vartheta_X \mathfrak{u}_\phi$ in its Laurent expansion at 0 is zero. Applying ϑ_X and using $\mathfrak{w}_\phi = \mathfrak{k}\mathfrak{v}_\phi$ gives $2\mathfrak{k}\vartheta_X \mathfrak{k} = -4\mathfrak{k}\mathfrak{v}_\phi$, so $\vartheta_X \mathfrak{k} = (\mathfrak{k}^2 - 1)/2$; and $\mathfrak{k} \neq \pm 1$, since either would give $\mathfrak{v}_\phi = 0$. For $r = (\mathfrak{k} - 1)/(\mathfrak{k} + 1)$ a direct computation gives $\vartheta_X r = r$, so $d(r/X)/dX = 0$ and $r = b_1 X$ with $b_1 \in k^\times$. With $a = -b_1$,

$$\mathfrak{k} = \frac{1 - aX}{1 + aX}, \quad \mathfrak{v}_\phi = \frac{aX}{(1 + aX)^2}, \quad \vartheta_X \left(\mathfrak{u}_\phi - \frac{aX}{1 + aX} \right) = 0.$$

The constant is $m = \text{ord}_X \phi$, by evaluation at 0, so $\phi/(X^m(1 + aX))$ has zero logarithmic derivative and is a nonzero constant. Conversely a monomial has $\mathfrak{v}_\phi = \mathfrak{w}_\phi = 0$, and for $\phi = bX^m(1 + aX)$

$$\mathfrak{u}_\phi = m + \frac{aX}{1 + aX}, \quad \mathfrak{v}_\phi = \frac{aX}{(1 + aX)^2}, \quad \mathfrak{w}_\phi = \frac{aX(1 - aX)}{(1 + aX)^3},$$

which satisfy the equation. □

For a binomial with gap $\mathfrak{g}$, $\phi = bX^m(1 + aX^\mathfrak{g})$, the same computation with $aX^\mathfrak{g}$ in place of aX gives $\mathfrak{w}_\phi^2 - \mathfrak{v}_\phi^2 + 4\mathfrak{v}_\phi^3 = (\mathfrak{g}^2 - 1)\mathfrak{v}_\phi^2$: gap one is accepted and every larger binomial gap is rejected (source 14, Appendix A).

Theorem 27.5 (Eventual strict Newton convexity; source 14). *Let $\mathfrak{c}, g_2, g_3 \in \mathbb{K}$ have nonnegative valuation and the residues $\frac{1}{12}, \frac{1}{12}, -\frac{1}{216}$, and let a nonpolynomial $f = \sum_n a_n z^n \in \mathcal{E}_\Gamma(k)$ satisfy the cleared equation (24.7) with these constants. Then:*

- (i) *after passing to $\Gamma_\mathbb{Q}$, every break $\delta \geq \delta_0$ of f has initial polynomial $bX^m(1 + aX)$ with $a, b \in k^\times$;*
- (ii) *$a_n \neq 0$ for every sufficiently large n ;*
- (iii) *the differences $\delta_n = v(a_{n+1}) - v(a_n) \in \Gamma$ are strictly increasing for all sufficiently large n and tend cofinally to $+\infty$ in Γ .*

Proof. Pass to $\Gamma_\mathbb{Q}$ (Lemma 26.2). The exterior form of the equation, dehomogenized, is (27.5), by the computation of Section 27.2, which uses only the residues and the nonnegative valuations of the constants; so, beyond the δ_0 of Lemma 26.5, every initial polynomial satisfies $\mathfrak{w}^2 - \mathfrak{v}^2 + 4\mathfrak{v}^3 = 0$,

and Theorem 27.4 excludes the monomial case at a break, proving (i). Beyond δ_0 , therefore, a break has exactly two active indices n and $n + 1$, with no skipped index and no three-way tie, and between breaks the active index is single. By Lemma 26.3 only finitely many indices are active on a bounded interval, and a change of the active index happens only at a break (as in the proof of Lemma 26.4); so the active index increases by exactly one at each break, and by Lemma 14.5 it exceeds every bound. Every sufficiently large integer is therefore active at some scale, and its coefficient is nonzero, proving (ii). The break at which n is replaced by $n + 1$ is the solution of $v(a_n) - n\delta = v(a_{n+1}) - (n + 1)\delta$, namely $\delta = \delta_n$, which lies in Γ without any division. These scales increase by the monotonicity in Lemma 26.3, strictly because a repeated break would carry three active indices, and they are cofinal because an upper bound would put infinitely many active indices in one bounded interval. Nonvanishing of coefficients, inequalities between their values and cofinality in $\Gamma_{\mathbb{Q}}$ descend to Γ , since Γ is cofinal in $\Gamma_{\mathbb{Q}}$. $\square$

For $f = \mathcal{T}_p$ with μ an order unit, (23.6) gives the realization $\delta_n = (2n + 1)\mu$, $\gamma_{\delta_n} = -n(n + 1)\mu$ and $\phi_{\delta_n} = X^n(1 + X)$ for $n \geq 0$: at δ_n the weighted value of degree j is $(j^2 - (2n + 1)j)\mu$, least exactly at $j = n, n + 1$. This is the Newton profile of Section 23.4, and it makes the order-three escape visible without analytic asymptotics: unbounded degrees coexist with the fixed gap one. Theorem 27.5 is a necessary condition on solutions, not a classification of the entire solutions of the theta equation: it does not force $v(a_n) = n^2\mu$, does not determine coefficients from their values, and does not assert that every eventually strictly convex value sequence occurs.

27.4 Surreal consequences and boundaries

Through (2.4), Theorem I applies to every set-sized surreal or surcomplex field $k((\omega^{-\Gamma}))$ with $\Gamma \subseteq \mathbf{No}$ and $k = \mathbb{R}$ or $\mathbb{C}$: there no nonpolynomial strongly entire series satisfies an algebraic differential equation of order at most two, whatever the rank of Γ , and algebraic closedness of the coefficient field is not needed. For $\Gamma = \mathbb{Q}$ or $\mathbb{R}$ and $p = \omega^{-1}$ the series $\mathcal{T}_p$ of Section 25.3 has three algebraically independent jets over the field of rational functions and an explicit equation of order three. The following proposition, from source 14, states once for every series what Section 11.2, Section 25.3 and Corollary 17.1 show for particular ones.

Proposition 27.6 (No nonpolynomial series is entire on the class; source 14). *A power series $\sum_n a_n z^n$ with coefficients in $\mathbf{No}[i]$ that is strongly summable at every point of $\mathbf{No}[i]$ has only finitely many nonzero coefficients.*

Proof. Suppose infinitely many a_n are nonzero, and let $v(a)$ be the negative of the largest Conway exponent of $a \neq 0$. The values $v(a_n)$ of the nonzero coefficients and their pairwise differences form a set of surreals, so there is a positive surreal γ larger than the absolute value of every such difference. At $x = \omega^\gamma$ the leading values of the terms are $v(a_n) - n\gamma$, and along increasing indices with $a_n \neq 0$ they decrease strictly, since $v(a_{n'}) - v(a_n) < \gamma \leq (n' - n)\gamma$. They cannot lie in a well-ordered union of Hahn supports, contradicting strong summability at x . $\square$

This is a size-and-domain obstruction, not an instance of second-order rigidity: entireness is always relative to one set-sized field, $\mathcal{T}_p$ is entire on its field because μ is an order unit there, and enlarging to every surreal scale destroys that property. The zeros of $\mathcal{T}_p$ and their omnific floors and ceilings are those of Theorem 23.6 and Corollary 25.3, inherited; source 14's new contribution there is the universal lower bound for the function that produces them and the necessary Newton geometry of Theorem 27.5 for every other entire solution of its equation.

Boundaries recorded by source 14. In characteristic $\mathfrak{p} > 0$ and with an order unit μ , the series $\sum_{n \geq 0} t^{n^2 \mu} z^{\mathfrak{p}n}$ is nonpolynomial and strongly entire with derivative zero, so Theorem I fails already in order one; this is Example 17.3 with $\mathbb{Q}$ replaced by any group with an order unit. Mixed-characteristic $\mathfrak{p}$ -adic fields are not covered merely because they have characteristic zero: their residue characteristic and integer valuations differ from the hypotheses used here. The ordinary exponential $\sum z^n/n!$ solves $D_z f = f$ formally but is not strongly summable at $z = 1$, since infinitely many terms have exponent zero; allowing ordinary convergence inside a fixed coefficient layer would change the function class and need a different theorem. The proof classifies neither functions patched from local formulas nor analytic identities in the fine topology of $\mathbf{No}[i]$. It passes only to a cofinal divisible extension and never reduces the group to rank one; the no-order-unit argument uses finite rational rank of a coefficient field, not of the ambient group, and a group of high rank can have an order unit while one of countable cofinality can lack one (Examples 25.5 and 25.6).

27.5 Predecessors, repository comparison and priority

Literature. Source 14 consulted NIST DLMF §20.5 for the triple product and logarithmic derivatives [32], §23.3 for the Weierstrass differential equation [33], and §32.2 for the Painlevé equations [34]; the Mantova–Matusinski survey [17] for normal forms, not as a source for the second-order theorem; and the publisher abstract of Sebbar’s paper [37], which explicitly treats classical third-order theta relations, without its full text. It also consulted the Wikipedia article on surreal numbers for orientation only. General searches for second-order Hahn and non-Archimedean rigidity returned unrelated results, including Hahn difference operators; these are not evidence of absence. No exhaustive MathSciNet or zbMATH review and no full subscription-literature review was made. For this merge the DLMF sections and the Mantova–Matusinski survey were checked, and the bibliographic data of Sebbar’s paper were checked against its Crossref record; its content was not checked by this merge, and nothing in this part rests on it.

Repository comparison at the pin. Source 14 read the repository at `b895e86` through a connector: the root README and tree, `docs/README.md`, this report’s README and neighbouring report inventories. The formalization ledger did not return text, so no assertion about it is made, and the new proof is not represented as covered by the Lean build. It took source 13 from the author’s file library (`article(20260923-193210).tex`, baseline `4cdeaec`). Its statements were checked for this merge and are accurate at the pin, as recorded in Section 26.1. This was a targeted comparison, not an audit of every file, branch or archived draft.

Priority. The triple product, the theta–elliptic identity and third-order theta relations are classical; the Chebyshev descent of the theta series, its equation, zeros and omnific rounding are source 13’s; the no-order-unit obstruction and the finite-generation principle are the coefficient-field part’s. The proposed contribution of source 14 is universal second-order rigidity for every ordered value group and every characteristic-zero coefficient field (Theorem I(a)), with the endpoint lemma at degenerate breaks, the resulting exact minimum order (Theorem I(b)), the third-order endpoint–gap obstruction, the classification of theta initial polynomials and eventual Newton convexity, and the stated consequences for two-state systems and polynomial-denominator quotients. Whether an equivalent result exists in the literature needs a more exhaustive review; the answered question is a question of this report, not a named classical conjecture.

27.6 What the order-three-threshold part does not claim

Every limitation stated by source 14, in its manuscript, README, audit file or verification record, is kept.

- (O1) No Lean formalization, no independent referee report and no certified bibliographic priority accompany source 14; no new Lean file is presented as verified, and the repository’s build and axiom audit do not cover this part. Independent review of the exterior reduction and the endpoint lemma is the most useful next check.
- (O2) Only the question of this report is resolved; no named classical conjecture is claimed solved.
- (O3) The repository comparison at **b895e86** was targeted; the formalization ledger was not read, and no audit of every branch or archived draft was made.
- (O4) Entireness is relative to one specified Hahn field; $\mathcal{T}_p$ is not strongly entire on $\mathbf{No}[i]$, and no nonpolynomial series is (Proposition 27.6).
- (O5) D_z acts on the external variable and kills every Hahn coefficient; it is not the Berarducci–Mantova derivation.
- (O6) The theta product and the elliptic identity are classical, and the theta descent with its equation, zeros and omnific rounding and the no-order-unit obstruction are prior; source 14 reproves them for completeness and claims none. “No omnific zero” concerns the strong domain only.
- (O7) Theorem 27.5 is necessary only: it gives no sufficiency and no classification of entire solutions, does not force $v(a_n) = n^2\mu$, does not determine coefficients from their values, and does not assert that every eventually strictly convex sequence occurs.
- (O8) Corollary 27.3 is a sufficient certificate, not a decision algorithm for third-order equations; an unbounded zero set of $\Upsilon_{\mathfrak{h}_P}$ is necessary, not sufficient, for an entire solution.
- (O9) Corollary 26.11 concerns quotients $g/\mathfrak{p}$ with a polynomial denominator only; no global division theorem in a Hahn analytic category is asserted.
- (O10) Corollary 26.10 concerns rational systems; arbitrary transcendental right-hand sides are outside it.
- (O11) Nothing is claimed about local formal solutions or about classical complex Painlevé functions; the all-scale requirement, not local existence theory, does the work.
- (O12) Positive characteristic is excluded, with a counterexample, and mixed-characteristic $\mathfrak{p}$ -adic fields are not covered.
- (O13) Theorem I gives no uniform degree bound for polynomial solutions of a fixed equation (Example 26.7).
- (O14) The proof does not reduce every group to rank one; it classifies neither functions patched from local formulas nor identities in the fine topology, and a different summation convention is a different function class.
- (O15) The 600 finite checks, with the theta product and cleared equation compared modulo p^{48} (source 14’s q^{48}), do not certify the quantifiers over arbitrary supports, value groups, scales or formal solutions; the classification of Theorem 27.4 is proved for rational functions, not extrapolated from the finite enumeration, and no floating-point agreement is used as evidence.

- (O16) Sebbar’s paper was seen through its publisher abstract only, and broad searches are not evidence of absence; no exhaustive MathSciNet or zbMATH review was made.
- (O17) The research questions of source 14 (Questions 19.1 and 19.11–19.18) are not asserted solved.
- (O18) *[merge]* Remark 26.6, Corollary 26.13, the named break scale in the proof of Lemma 26.4, the reading of Corollaries 26.9 and 26.10 inside a differential field containing $\mathbb{K}((z))$, and the four-state observation after Corollary 26.10 are the merge’s; no historical priority is claimed for them.

28 Conclusion

Strong entireness over a surreal Hahn workspace imposes a global scale requirement that a finite linear differential equation cannot meet unless its solution terminates. A dilation equation can meet that requirement, but exactly when its valuation reaches every scale through integer multiples. The decisive distinction is therefore not between Archimedean and non-Archimedean fields, nor between rank one and higher rank; it is between *a cofinal dilation scale* and *a scale trapped below another one*.

The obstruction is finite and robust. A linear recurrence forces a later nonzero coefficient with a controlled valuation, and a large enough argument turns an infinite chain of such coefficients into a nonincreasing sequence of leading exponents. Descending supports and infinite reuse of one exponent are two distinct ways in which the evaluation fails, and keeping them separate is what makes one exclusion boundary closed and the other open. The mechanism works with arbitrary Hahn coefficient tails, arbitrarily high valuation rank, and sparse coefficient sequences, without replacing strong summability by topological convergence and without any hidden completeness assumption.

The countable-rank example makes the separation concrete. The series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is evaluable at every point of its full workspace, yet no single nontorsion dilation in that workspace supplies a finite linear equation for it. That gap between existence and finite linear describability is the main mathematical contribution proposed here.

The nonlinear part pushes the same global requirement past linear equations by a different mechanism. A finite algebraic relation between a function and its first derivative is incompatible with nonpolynomial strong entireness over any nontrivial characteristic-zero Hahn workspace: at one large admissible scale the whole series leaves a finite initial polynomial, whose top coefficient is controlled by a corner polynomial that is automatically nonzero in order one. In higher order an exact cancellation of that corner is a genuine obstruction to that method, and a fixed nonlinear equation can then have polynomial solutions of unbounded degree.

The coefficient-field part removes the order restriction when the value group has no order unit, by a third mechanism. A finite algebraic relation, linearized along its own solution, confines every Taylor coefficient to one finitely generated field, and such a field cannot supply independent valuations at cofinally many Archimedean levels; a nonpolynomial entire series needs them. So every nonpolynomial strongly entire series is then differentially transcendental over the full field of constants, jointly with its dilates, and every differentially algebraic quotient of entire series is rational, while one explicit workspace still carries continuum many entire series with jointly independent jets. With an order unit the mixed statement fails.

The theta-descent part shows that the purely differential statement fails too, so the order unit is the exact boundary. A bilateral theta series, descended along $z = u + u^{-1}$ by the Chebyshev identity, becomes an ordinary power series that is strongly entire exactly when its scale is an order unit, and the classical elliptic identity gives it an explicit algebraic differential equation of order three. Its residue corner vanishes, as it must. The shift of the theta series by $u \mapsto p^2u$ then excludes every equation of lower order, by three elementary difference-field arguments.

The order-three-threshold part shows that no other series does better. At the infinitely many scales where a nonpolynomial entire series has a nonmonomial initial polynomial, a fixed equation of order at most two reduces to one homogeneous polynomial over the residue field, and logarithmic Euler derivatives confine both endpoints of those initial polynomials to a finite set, even when the leading-monomial test vanishes identically. So order three is the least possible for every value group with an order unit. At order three one further integer, the gap next to an endpoint, survives, and the theta equation uses it with gap one. Which series attain order three is the open part of Question 19.1.

A Notation

Symbol	Meaning
Γ	A fixed nonzero set-sized ordered abelian group; divisible only where stated.
K_Γ	The complex Hahn field $\mathbb{C}((t^\Gamma))$.
$v(a), \text{lc}(a), \text{res}(a)$	Least support exponent, its coefficient, and the residue when $v(a) = 0$.
$\mathcal{E}_\Gamma, \text{Dom}_{\text{str}}(f)$	Series strongly evaluable at every point of K_Γ ; the exact set where a series is strongly evaluable.
D_z	Formal derivative in z , zero on every element of K_Γ .
σ_q	Argument dilation $f(z) \mapsto f(qz)$.
$M(S)$	The additive monoid generated by a well-ordered $S \subseteq \Gamma_{>0}$.
Δ_β	The smallest convex subgroup containing $\beta > 0$, (2.1).
B, B^+, B_{rec}	Exclusion thresholds in Γ ; B_{rec} divides by the jump length and needs Γ divisible.
$\overline{B}_{\text{rec}}^\flat, \overline{B}_{\text{rec}}$	Their coarsened analogues in $\Gamma/\Delta_{ v(q) }$.
μ, β	Usually a positive dilation valuation; an order unit only when stated.
$(X)_r$	The ordinary falling factorial polynomial of degree r .
Θ_p	The partial theta series (8.1).
Γ_∞	The increasing-rank direct sum (11.1).
<i>Nonlinear part, Sections 14–17</i>	
$k, \mathbb{K}, \mathcal{E}_\Gamma(k)$	Any characteristic-zero field; $k((t^\Gamma))$; its strongly entire series (Convention 14.1).
$\mathcal{S}_m, \mathcal{S}_m^+, \rho$	Series with strongly summable coefficient family; those with coefficients of nonnegative valuation; coefficientwise residue.
$\gamma_\delta, \phi_\delta, N_\delta$	Gauss value, initial polynomial and top active degree at the scale $t^{-\delta}$, (14.1).
$d_P, w_P, \mathcal{C}_P, \beta_P$	Corner data of a differential polynomial, (14.5).
χ_P, I_P	Residue and full corner polynomials, (14.6).
κ_P	The rational degree cutoff (15.11).
δ_0	A solution-dependent exclusion bound, (15.8).
ϑ_q, wt	Positive-weight Euler operator and weight.
<i>Coefficient-field part, Sections 20–22</i>	
$\text{Frac } \mathcal{E}_\Gamma(k)$	The fraction field of $\mathcal{E}_\Gamma(k)$ in $\mathbb{K}((z))$ (“meromorphic”).
$\mathcal{J}_{\lambda,j}f$	$(D_z^j f)(\lambda z)$, differentiation before dilation, (20.1).
$\text{rrk}, \text{trdeg}, \text{dtrdeg}$	Rational rank; transcendence degree; differential transcendence degree.
$\text{cvr}(f)$	Coefficient-value rank (20.4).
$\Delta, v_\Delta, k_\Delta, \text{red}_\Delta$	A convex subgroup, the coarsened valuation $v + \Delta$, its residue field and residue map.
$\Pi_{ij}, \theta, \xi_{ij}, \Xi$	Evaluated partials, offset, leading coefficients and linearization multiplier, (21.2)–(21.3).
$L, k_0 \subseteq K_0$	A coefficient subfield; a generic pair of characteristic-zero fields.
$\mathcal{F}_\alpha, \text{code}, R_\tau$	The branch family (22.3), its binary code, the rank-one family.

Symbol	Meaning
<i>Theta-descent part, Sections 23–25</i>	
μ, p, Δ_μ	A positive scale (an order unit only where stated), $p = t^\mu$, the convex subgroup (2.1).
$C_n, \mathcal{T}_p$	Monic Chebyshev polynomials (23.1); the descended series (1.4).
$\Psi_p, u, \tilde{z}$	The bilateral theta series (23.8); its variable; $u - u^{-1}$.
$\mathcal{O}_{\Delta_\mu}, x_m, \mathcal{P}_p$	The strong domain of $\mathcal{T}_p$; its zeros (23.13); $\prod_m (1 - p^{2m})$.
$\mathcal{Q}_r, \mathfrak{c}, g_2, g_3, \Phi$	Lambert series in the nome p^2 and the constants (24.1)–(24.3).
$\mathcal{A}[f], \mathcal{H}[f], \mathcal{K}[f], \mathcal{B}[f]$	Differential expressions (24.4)–(24.6).
$\mathcal{U}_\mu, f^\natural, \vartheta_u, \mathcal{S}$	The Laurent ring, the substitution $z = u + u^{-1}$, uD_u , and $u \mapsto p^2u$.
$\psi_1, \psi_2, \psi_3; \mathcal{L}, \mathcal{A}_1$	$\vartheta_u \Psi / \Psi$ and its two ϑ_u -derivatives; $D_z f / f$ and $D_z \mathcal{A}[f]$.
<i>Order-three-threshold part, Sections 26–27</i>	
$\Gamma_{\mathbb{Q}}, \mathbb{K}_{\mathbb{Q}}$	The divisible hull $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$ and $k((t^{\Gamma_{\mathbb{Q}}}))$.
$\text{Act}_\delta, \underline{N}_\delta$	Active indices at the scale δ and the least of them (26.1); a <i>break</i> is a scale with $\underline{N}_\delta < N_\delta$.
$\vartheta_X; \mathfrak{u}_\phi, \mathfrak{v}_\phi, \mathfrak{w}_\phi$	XD_X ; $\vartheta_X \phi / \phi$ and its two ϑ_X -derivatives.
$\mathfrak{h}_P$	The exterior form of P (26.3).
$\mathfrak{q}, i_0, \mathfrak{r}, \text{Sp}(\mathfrak{h})$	Dehomogenization, the power of V removed, the remainder, the finite endpoint spectrum (Lemma 26.8).
$\Upsilon_{\mathfrak{h}}, \mathfrak{g}^\pm, \delta_n$	Endpoint–gap polynomial (27.2); upper and lower gaps; the break scale $v(a_{n+1}) - v(a_n)$.

B Proof-dependency and hypothesis audit

The logical dependencies are short. Differential rigidity uses the integer-evaluation lemma, the coefficient conversion, and the escape chain with its exclusion criterion. Dilation necessity uses the geometric-orbit lemma or the unit-orbit theorem, the dilation conversion, and the same criterion. Dilation sufficiency uses only the partial theta identity and the support certificate. The mixed extension uses the bivariate orbit lemma. The several-variable theorem uses countable polynomial avoidance, finite-block regrouping and the one-variable theorem. The infinite-rank application uses a direct strong-entireness proof and the completed classification. The nonlinear part uses none of these linear arguments: first-order rigidity uses the Hahn-family algebra (with the support calculus of Section 2), the large-active-degree lemma, the corner identity and the order-one nondegeneracy of the corner; the higher-order criterion replaces the last step by the hypothesis $\chi_P \neq 0$; the Euler theorem uses the same algebra in several variables with finite bounded-weight sets. The coefficient-field part uses neither the escape chain nor the corner: its all-order theorem uses the linearized coefficient recurrence, the valuation-rank inequality, the principal convex bound and the exterior obstruction (or the coarsened reduction), with the cofinal criterion of Proposition 20.2; its mixed statements add the imported Skolem–Mahler–Lech theorem. The theta-descent part uses none of these for its existence direction: the exact domain uses the quadratic support certificate and the nonincreasing-exponent obstruction; the equation of order three uses the classical triple product and elliptic identity, proved as a formal identity over $\mathbb{Q}$ and transferred by ring homomorphisms; the lower bound uses the Laurent ring, the substitution homomorphism and three difference-field obstructions. Its necessity direction is Theorem E (a). The order-three-threshold part uses the all-scale criterion, a cofinal extension to the divisible hull, the Hahn-family algebra and its residue homomorphism, local finiteness of active indices and late breaks, the exterior reduction and the

finite endpoint lemma; its least-order statement adds the witness $\mathcal{T}_p$ and Theorem E (a), and its theta results the exterior form of the cleared equation and a rational Riccati reduction.

No unproved ring-theoretic conclusion from another report in this collection is used. The real-valued coarsening of [12, Lemma 9.1] is cited and not reproved, and no proof here depends on it. Higman’s theorem is an external standard ingredient; alternatively one may import the classical Hahn–Neumann support lemma directly. The surreal interpretation imports the normal-form field embedding, while every substantive function statement is already proved in the abstract Hahn field. The coefficient-field part imports one further external theorem without proof, the Skolem–Mahler–Lech theorem in the form of Bell’s Theorem 1.1 [27], and only its statements with several dilations use it. The theta-descent part imports the classical Jacobi triple product [32] and the constancy of an elliptic function without poles; it proves the elliptic identity it needs.

The potentially dangerous hypotheses, collected for review:

- Supports and the value group are sets; all operators are finite sums. Power-series degrees and derivative orders are ordinary nonnegative integers; dilation powers may be negative, as may the linearization offset θ . Formal Laurent series have only finitely many negative powers.
- $\Gamma \neq 0$, set-sized, ordered abelian. *Divisibility is not assumed* except in Corollary 3.5, the refined threshold (5.4), Theorem 7.4 (b), and Example 13.1, each of which states it, and in Lemma 26.4, which is applied only after the cofinal extension of Lemma 26.2 to the divisible hull.
- Series are ordinary $\mathbb{N}$ -indexed power series in one variable, except in the several-variable Sections 6 and 16.3, and in the formal Laurent fraction field $\text{Frac } \mathcal{E}_\Gamma(k)$. The main workspace rigidity equations have rational-function coefficients in the variable, with arbitrary Hahn scalar coefficients. The relative coefficient-field theorem (Theorem 21.2 (a)) also allows coefficients in $k_0((z))$; that generality is used in the joint-independence criterion.
- The residue field is characteristic zero and trivially valued. Algebraic closedness is not used. It is $\mathbb{C}$ in Sections 1–13 and an arbitrary characteristic-zero field in the nonlinear, coefficient-field, theta-descent and order-three-threshold parts; the latter generality is not transferred to the linear theorems (Convention 14.1). Characteristic zero is essential (Example 17.3).
- The single parameter q in the main dilation classification is nonzero and nontorsion. In the mixed-jet statements every λ is nonzero, and distinct parameters have non-root-of-unity quotients; an individual parameter may be 1. Section 9 explains the failure when this quotient condition is dropped.
- The derivative is d/dz , not a scalar surreal derivation.
- Entireness is always relative to one fixed workspace, never to $\mathbf{No}[z]$. Strong summability requires both well-ordering and local finiteness.
- The conclusions of Sections 1–13 concern *linear* equations. For every $\Gamma \neq 0$, nonlinear differential equations are treated only in order one, in higher order under $\chi_P \neq 0$, and for positive-weight Euler relations (the nonlinear part), and in order two (the order-three-threshold part). Nonlinear differential transcendence in all orders, and algebraic independence of derivatives at dilations with pairwise quotients that are not roots of unity, are proved only when Γ has no order unit, or for a single series with infinite coefficient-value rank (the coefficient-field part). No universal all-order rigidity theorem is claimed for groups with an order unit, and none can hold: for them the theta-descent part gives a nonpolynomial entire series with an algebraic differential equation of order three, and three is then the least order. No general mixed-jet independence theorem is claimed when a quotient is a root of unity.

Two boundary behaviours deserve separate emphasis because they point in opposite directions. The most delicate equality is in Corollary 3.5: weakly decreasing leading exponents already obstruct strong summability, so the excluded region is closed, and Proposition 5.3 shows the bound is attained. The most delicate strictness is in Theorem 7.4: equal *coarse* values need not obstruct summability in Γ , so that excluded region must be open, and Theorem 8.1 with Section 8.5 shows why.

Deliberate exclusions, recorded so that no reader supplies them by analogy: finite-order q is excluded from the main classification and has genuine periodic-support counterexamples; mixed operators at unit valuation are not classified when Γ has an order unit. For such Γ , the universal nonlinear rigidity results cover orders one and two, the nondegenerate higher-order class and positive-weight Euler relations; the infinite-coefficient-value-rank criterion gives additional all-order results for individual series, including Theorem 21.20, while Theorem H shows that universal differential algebraic rigidity fails for them, with least order three by Theorem I; which series have minimal order three is not decided. The nonlinear part keeps the non-claims of Section 19.3, the no-order-unit results keep those of Section 22.7, the theta-descent part keeps those of Section 25.6, and the order-three-threshold part those of Section 27.6; no complete algorithm for arbitrary surreal comparisons or cofinality decisions is claimed; the finite checks prove no infinite theorem. The full classifications and the nonlinear, coefficient-field, theta-descent and order-three-threshold theorem packages remain without Lean verification or independent refereeing. The formalization ledger records the proved support and escape lemmas, exponential and partial-theta domains, polynomial-orbit valuations, constant-point avoidance, inward stability and torsion covariance, including the partial-theta directions of the dilation classification.

C Reproducing the document and the finite checks

The directory contains this standalone L^AT_EX source, its compiled PDF, a README, eleven source audit documents at the top level, eleven files under `code/` (the seven source verification programs, the build scripts of sources 10, 11 and 13, and source 12's `Makefile`), and eighteen under `data/` (the seven recorded outputs, the source build and layout reports, and the requirements files of sources 11, 13 and 14). No source document from the inspected repository is redistributed, and no source manuscript is shipped. The bibliography is embedded: no BibTeX run, external graphics, shell escape, or repository checkout is required, and a conventional TeX Live or MiKTeX installation with the named packages suffices.

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python code/08-finite-recurrences-verify.py --output rerun-08.txt
python code/09-entire-hahn-holonomic-verification.py --output rerun-09.txt
python code/10-nonlinear-rigidity-verify.py --output rerun-10.json
pip install -r data/11-coarsening-differential-rigidity-requirements.txt
python code/11-coarsening-differential-rigidity-verify.py --output rerun-11.json
python code/12-coefficient-field-rigidity-verify.py --output rerun-12.json
pip install -r data/13-theta-descent-requirements.txt
python code/13-theta-descent-verify.py --precision 128 --output rerun-13.json
pip install -r data/14-order-three-threshold-requirements.txt
python code/14-order-three-threshold-verify.py
```

The `--output` argument is required by the scripts of sources 08, 10, 12 and 13, so that a rerun need not overwrite the record distributed with the report, and optional for those of sources 09 and 11;

the scripts of sources 10 and 12 refuse an existing output path (source 12's unless `--force` is given), source 13's overwrites one, and source 11's writes `verification.json` in the current directory when no path is given. Source 14's takes no arguments and always writes `data/verification.json` next to its own `code/` directory: here a new file beside the delivered record, overwritten by the next run, so it should be run on a copy. All use exact integer and rational arithmetic, those of sources 11, 13 and 14 through SymPy, and exit with a nonzero status if a check fails. The build script `code/10-nonlinear-rigidity-build.py` was written for source 10's own manuscript: run here it would typeset this report, leave auxiliary files in the directory and write `data/build_report.json`. The shipped `data/10-nonlinear-rigidity-build_report.json` and `data/10-nonlinear-rigidity-layout_audit.json` describe the 23-page source manuscript, not this report; use `latexmk` as above, on a copy. Likewise `code/11-coarsening-differential-rigidity-build.py` looks for an `article.tex` in `code/` and fails there, and `code/12-coefficient-field-rigidity-Makefile` names the delivery paths `article.tex` and `verify.py` and would rebuild `article.pdf` in place; the shipped `data/11-coarsening-differential-rigidity-build_audit.json` and `data/12-coefficient-field-rigidity-build_report.json` describe the 20- and 23-page source manuscripts, whose PDFs and sources, hashed in the first, are not shipped. Source 13's `code/13-theta-descent-build.py` typesets an `article.tex` in its own directory `code/`, where there is none, and its `--verify` option would call a `verify.py` there; run here it creates `code/build/`, writes a failed first-pass log into it and exits with an error. In its original delivery layout it writes only into `build/`. The shipped `data/13-theta-descent-build_report.json` describes the 21-page source manuscript, whose source and PDF, hashed there, are not shipped. Likewise `data/14-order-three-threshold-build_report.json` describes source 14's 26-page A4 manuscript, whose source and PDF, hashed there, are not shipped; source 14 shipped no build script.

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